Note: Please update this outline as we continue so it is up to date! Thank you!

Background

- A BRIEF HISTORY OF CHRONOBIOLOGY
- *(Jacob Li)*
    - Time-Keeping Throughout History
    - Ancient Chronobiology
    - The Birth of a New Field
    - Chronobiology in the Modern Day
- THE BASICS OF BIORHYTHMS: THE MOLECULAR CLOCK
- *(John Liu, Matthew Hernandez)*
    - The Molecular Clock and Circadian Rhythms
    - Serotonin
        - What Is Serotonin?
        - Serotonin's Daily Rhythm
        - Serotonin as Input to the Clock
        - Serotonin as Substrate of the Clock's Output
        - What Remains Unsettled
    - Central and Local Clocks in the Brain and Body
    - Genetic Mechanisms of the Molecular Clock
    - Clock Genes and Cyclic Behavior

Circadian Rhythms

- SLEEP, CAFFEINE, AND JET LAG
- *(Sarah Kim, Benjamin Shamash, Joseph Sultan, Sebert Supamongkolthavessuk)*
    - The Sleep/Wake Cycle
        - How Circadian Rhythms Influence Sleep
        - What Can Disrupt the Sleep/Wake Cycle?
        - Effects of a Disrupted Sleep/Wake Cycle
        - Maintaining a Healthy Sleep/Wake Cycle
    - Caffeine
        - Introduction
        - How Caffeine Affects the Brain and Body
        - Caffeine Timing and Sleep Quality
        - Caffeine in Everyday Life
    - What is Jet Lag?
        - What is Internal Desynchrony
            - Disruption of Human Circadian Rhythms
                - Sleep/Wake
            - Disruption of Human Ultradian Rhythms
            - Disruption of Human Infradian Rhythms
        - Flying Against the Biological Clock
            - How Flight Schedules Disrupt Circadian Timing
            - What Happens to the Pilot?
            - Designing Aviation around Human Biology

- HUNGER, DIGESTION, AND METABOLISM
- (*Will Protasowicki, Zoe Kessler, Clara Davis)*
    - Hunger and Digestion
        - How are Hunger and Digestion Related?
        - Hunger Signalling and the Nervous System
        - Eating Cycles in non-humans
        - Processes involved with digestion
        - Rhythm Disruption and Health Complications
        - Hunger/Metabolism-related diseases
    - Metabolism
        - Circadian Rhythm of Metabolism
        - Health Effects of Unstable Metabolism
        - Metabolism's Effect on Other Systems
        - How Alcohol Affects the Metabolism Cycle
- CHRONOTYPES
- *(Jack Lou)*
    - What is a chronotype?
    - How does chronotype affect physiology?
    - Chronotypes in athletics

Ultradian Rhythms

- CIRCULATION AND THE CARDIAC CYCLE
- *(Ada Langford, Ryan Nordahl)*
    - Mechanics of the Heart
        - How does the heart pump blood periodically?
    - Sympathetic vs Parasympathetic Control
    - Diurnal Blood Pressure/Heart Rate
    - How does long-term health affect the heart's clock?
        - Heart function with increased endurance training
    - How does disruption/failure of different parts of the heart affect its clock?
    - Can we artificially support normal cardiac rhythms?
- NEURONAL ACTIVITY: SPIKES AND LOCAL FIELD POTENTIALS
- *(Daniel Winter)*
    - What is Being Measured
    - Neural Activity is Rhythmic
    - Connection to Slower Body Rhythms
    - Getting a Signal from a Recording
    - From Biological to Artificial
- THE BREATHING RHYTHM
- *(Daniel Winter)*
    - Where Each Breath Begins
    - A Rhythm That Runs on Its Own
    - Where the Beat Comes From
    - Sighs, Calm, and Panic
    - When the Rhythm Stops

Infradian Rhythms

- IMMUNITY
- *(Ruben Hayrapetyan)*
    - What is the Immune System and how does it function?
    - Diseases during the rest vs active time
    - Seasonal Cycles within the Immune System
- HAIR GROWTH/SHEDDING
- *(Kaleb Yoon)*
    - Basics of Hair Growth and Shedding
    - Factors that Impact the Hair Cycle
    - Potential Treatments and Interventions
- BREEDING AND MIGRATION
- *(Sam Ding)*
    - How Annual Biological Clocks Work
    - Breeding Timeline Across Species
    - Animal Migration and Navigation
    - Modern Environmental Threats
    - Bird Safe Architecture
- PREDATOR-PREY POPULATION DYNAMICS
- *(Binqi Zhan)*
    - What are Predator-Prey Rhythms?
    - The Predator-Prey Cycle in the Ecosystem
    - What factors drive Predator-Prey periodicity?

UNIT AND CHAPTER

Unit: Background / Ultradian Rhythms / Circadian Rhythms / Infradian Rhythms

Chapter X: Title

CONTRIBUTORS

- James Cunningham
    - Subject Matter Expert – ‘Sleep and Cognition’
        - Sections 1, 2 and 4
    - Figures 1, 3, 5
- Semaj Cunningham
    - Copyeditor
    - Peer Reviewer

LEARNING OBJECTIVES

After reading this chapter, students should be able to …

1. Sample
2. Sample
3. Sample
4. Sample

CHAPTER OUTLINE

- Section
    - Subsection
- Section
    - Subsection

CHAPTER WRITTEN CONTENT

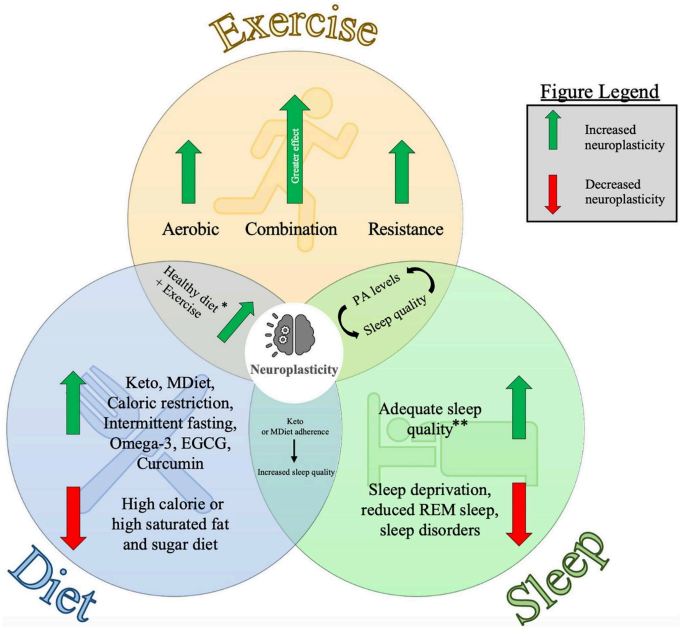
- Section
    - Subsection
        - This is sample text. When I want to refer to a figure, I will use the following (see Figure 1). Don’t actually put the figure here though.
- Section

- Subsection
  - Actual content

CHAPTER ADD-INS

Figure 1 - Diet, Exercise, and Sleep's Role in Neuroplasticity

- Author: James Cunningham



- Figure Description
- Attribution (if applicable)

GLOSSARY

Term	Definition

REFERENCES

UNIT AND CHAPTER

Unit: Background / Ultradian Rhythms / Circadian Rhythms / Infradian Rhythms

Chapter 1: A Brief History of Chronobiology

CONTRIBUTORS

- Jacob Li
    - Subject Matter Expert – 'A Brief History of Chronobiology'
- James Cunningham
    - Peer Reviewer
- Daniel Winter
    - Peer Reviewer
- Ada Langford
    - Peer Reviewer

LEARNING OBJECTIVES

After reading this chapter, students should be able to …

1. Compare and contrast ancient civilizations, relationship with time and chronobiology.
2. Construct a rudimentary time-keeping device.
3. Compare modern day understanding of chronobiology with ancient chronobiology.
4. Identify recurring themes that motivated the development of the field of chronobiology throughout history.

CHAPTER OUTLINE

A Brief history of Chronobiology

- Time-Keeping Throughout History (2)
    - This will give a general overview of humanity's relationship with time, as well as the current definition of the second that we have.

- Ancient Chronobiology (2)
    - Egypt (0.5) - Ancient Egyptians used stars to separate their calendar into 360 parts, demonstrating how ancient civilizations were able to use their surroundings to track time in a specific case. Furthermore, the 360 day calendar mostly persists today.
    - Mayans (0.5) - The famous Mayan calendar was based on the stars and other astronomical observations that made it possible to independently create their own calendar.
    - China (0.5) - Ancient Chinese physicians noticed a relationship between time and when certain bodily processes were most active, giving us a good example of how ancient civilizations had a rudimentary understanding of chronobiology applied in medicine.
        - Ancient Chinese physicians discovered that some bodily functions were more active at certain times than others and named this phenomena.
        - They applied this concept to treat their patients, give diagnoses, and when to give treatments
        - Developed a clock tower
    - Greeks (0.5) - Sundials and water-clocks helped the Greeks and Romans treat their patients as they were able to time intervals between symptoms and dosages. This may have also led to the length of the hour as we know today.
        - Also noticed how they could use chronobiological concepts to give correct treatments and diagnoses.
        - They noticed how symptoms such as fevers followed regular patterns.

        - Used sundials and water-clocks as well as created the modern time-interval that is the hour.
    - Design a (time-keeping device) Activity - Introduce the engineering design process. Reflect on the difficulties of telling time without modern technology as a short reflection.
- A Birth of a New Field (1 Page) - Mostly, this will talk about the shift from just general observation to the creation of chronobiology as an experimental science through the discovery of numerous scientists.
        - Introducing the Engineering Design Process.
        - Reflection on what limitations the created sundial has.
    - Jean-Jacques d'Ortous de Miran's Minosa Plant (0.5) - 1972 experiment by a French astronomer, discovering endogenous rhythms.
        - Describe how Miran noticed that the Mimosa plant still regularly continued opening their leaves as though it wanted to take in as much sunlight as possible despite being put in a dark area.
        - Proved that organisms have endogenous biological rhythms.
    - Colin Pittendrigh's Fruit Flies (0.5) - Scientist who experimented with fruit flies, demonstrating the shift of understanding in chronobiology from plants to animals.
        - Showed that there are biological rhythms in fruit flies too (a shift from plants to animals).
        - Finally establishing chronobiology as an experimental science under the field of biology.
- Chronobiology in the Modern Day (1.5 Pages)

- The Period Gene (0.5) - Discovery of the molecular clock gene, proving that biological rhythms are encoded into our genome and is not a learned behavior.
    - Describe the discovery of the period gene, demonstrating that biological rhythms are inherently encoded into organisms and not learned behaviors.
    - Gave a molecular basis for the field of chronobiology.
- Transcription-Translation Feedback loops (0.5) - This section will give a brief description on how the human body regulates biological rhythms using clock proteins. This result earned the associated scientists the Nobel prize in medicine for their work.
    - Clock genes -> mRNA -> translation into proteins -> proteins accumulate and inhibit production -> cycle repeats
- Chronobiology Today (0.5) - A description of modern chronobiology research as well as any modern applications of chronobiology.
    - Explain how there is ongoing research in fields of chronobiology such as sleep disorders, circadian disruption, and chronotherapy for applications in medicine.

CHAPTER WRITTEN CONTENT

Time-Keeping Throughout History

Without checking a clock, you may have a natural intuition for the time, day, or month based on numerous factors including the position of the sun, seasonality, sunshine duration, or moon phase. In the absence of these **exogenous cycles**, your body has many internal processes, called **endogenous cycles**. These endogenous cycles rely on **cyclical processes**, which can include metabolism, the sleep-wake cycle, the menstrual cycle, and hormonal release. Many of these cycles can be placed into three distinct groups: **ultradian**, **circadian**, and **infradian** rhythms. This is all under the field of **chronobiology,** a field of biology that studies cyclical phenomena in organisms. Through the course of history, the ability to measure time has proven to be a very useful one, and humans have always had numerous ways of measuring time even without the advent of mechanical time-keeping devices.

Primitive time-keeping methods involved following many of these exogenous cycles. Many ancient cultures, such as the Egyptians, Sumerians, Persians, and Chinese, used the lunar cycle to form a general lunar calendar. A remnant of the lunar calendar persists in the modern Gregorian calendar, as the month is roughly one lunar cycle long (Breasted, 1935). Many of the aforementioned cultures also measured time relative to the seasons, as the dependence on agriculture necessitated an understanding of the seasons. For smaller divisions of time, devices such as the sundial, hourglasses, water-wheels, and candle clocks were invented. Although the invention of these devices were major successes in time-keeping, they each had their disadvantages. For example, the sundial required a clear, sunny day to properly function and with limited precision.

In terms of precision, the invention of the pendulum clock in 1656 by Christiaan Huygens revolutionized precise time-keeping. The importance of the invention is one that cannot be

understated; the development decreased time error down to mere seconds per day, and gave scientists the much needed precision to measure tiny intervals of time. The pendulum clock remained in regular use as the most effective form of time-keeping up until the 1930s.

Modern day time-keeping is through the use of atomic clocks through the use of the number of oscillations in transitions of the cesium-133 atom, which occurs in highly consistent intervals. In fact, the current definition of a second is defined to be the length of time associated with 9,129,631,770 periods of these oscillations. Note that the official definition presented from the SI unit system can be measured from anywhere in the universe and will stay constant regardless of time. This is opposed to the legacy definition of exactly 1/86,400 of the time it takes for the Earth to complete one revolution, as it requires the observer to not only be near the Earth, but the time it takes for one earth revolution is not constant.



(When it was finished in 2012, the Three Gorges Dam in China slowed Earth's rotation and lengthened the day by 0.06 microseconds. This is the same principle that causes a figure skater to slow down when they stretch their arms out.)

Ancient Chronobiology

Learning Objective: Compare and contrast ancient civilizations, relationship with time and chronobiology.

Throughout ancient history, many civilizations have noticed the periodic nature of certain natural processes. Many of these civilizations have independently developed their own time-keeping system and calendars, as well as noticing how to take advantage of their own internal rhythms for medicinal purposes. Even before the development of the formal field of chronobiology, these concepts were used by many.

Egyptians

The ancient Egyptians developed their own lunar calendar at around 3000 BCE during the Early Dynastic Period. Through the use of 36 groups of stars called decans, they were able to track time even during night. The Egyptians were noted to have used the stars' "vibrations," which likely corresponded to the movement of the stars in the night sky, to calculate the approximate time during the night. Furthermore, they initially used these stars to subdivide their yearly calendar into 360 parts, which still persist in the modern calendar. However, when the Egyptians noticed that the star Sirius reappeared every 365 days, they appended those extra days into their calendar, creating the 365-day yearly cycle (Elgammal, 2022).

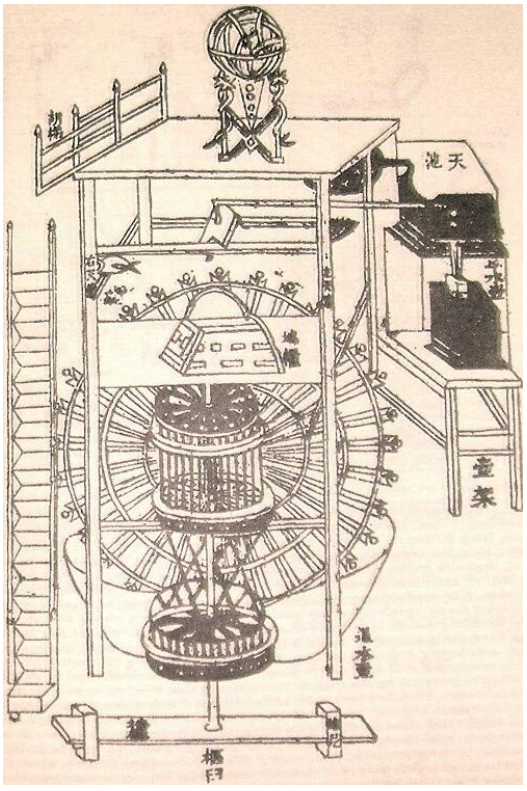
Mayans

Similar to the Egyptians, the Mayans independently created their own calendar system based on celestial bodies (Šprajc et al., 2019). Their calendar system involved two different systems. The first was the Tzolk'in, which involved 20 named days combined with 13 other numbers, creating a 260-day calendar used primarily for religious ceremonies. Notably, their second calendar independently landed on the 365-day civil calendar, but it was separated between 18 different months of 20 days plus an extra 5 days called the Uayed days. Compared to

the Egyptians, the Mayans were aware of the fact that the solar year was not exactly 365 days, estimating the year to be approximately 365.24 days, but they did not include the leap year into their calendar. Notably, this elaborate system was used for optimizing the cultivation of corn. Despite not explicitly knowing the internal rhythms of plants, they used exogenous astronomical observations to calculate seasonal agricultural periods, demonstrating ancient application of chronobiological concepts.

China

On the other hand, ancient Chinese physicians are reported to have observed a relationship between certain times of day and the activity of bodily functions, which Balzer notes that they called the Ying and the Yang (Balzer, 2009). Using this information, Chinese physicians would care for patients by timing treatments and predicting when symptoms would be most likely to appear or worsen. This understanding is formulated in the *Yellow Emperor's Classic of Medicine*, which describes how they believed 'Qi' flows through the human body over the course of the day: the lungs from 03:00-05:00, the large intestines from 05:00-07:00, the stomach from 05:00-07:00, and so on throughout the body in two-hour intervals (Balzer, 2009). For purposes such as timing the treatments, the ancient Chinese also constructed the Astronomical Clock Tower of Su Song, which used a water-wheel to track



time (Yan & Lin, 2002). With these developments, it is clear that the Chinese had some understanding of the principles of chronobiology and how to apply them in medicine.

Figure: the original diagram of Su's book showing the inner workings of his clocktower.

Greece and Rome

The Greeks and Romans also applied chronological concepts to treat their patients (Miller, 2023). Similar to Chinese physicians, they noticed that symptoms such as fevers would occur with regular patterns, and understood how the timing of the treatment would affect treatment effectiveness. Physicians would monitor the state of each patient, and they had logged general patterns on "critical days," on whether or not a patient's symptoms would either worsen or fade. To keep track of these measurements accurately, they adopted the sundial for measuring time in the sunlight and the clepsydra for using water to measure time in rooms and night-time. Clearly, Greek and Roman medicine was also heavily influenced by their understanding of endogenous rhythms, with their observations also informing their treatment plans.

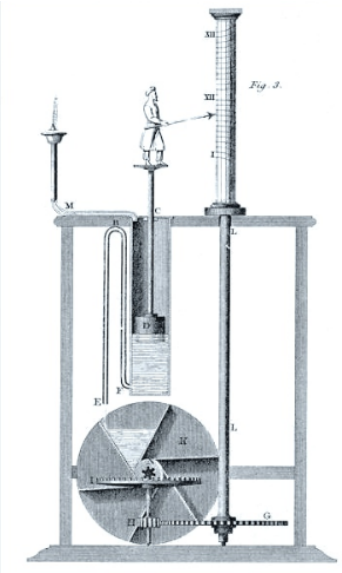


Figure: A 19th-century illustration of a clepsydra from the 3rd-century BCE.

Activity: Engineering a Sundial

Learning Objective #2: Construct a rudimentary time-keeping device.

Expected construction time: ~1 hour

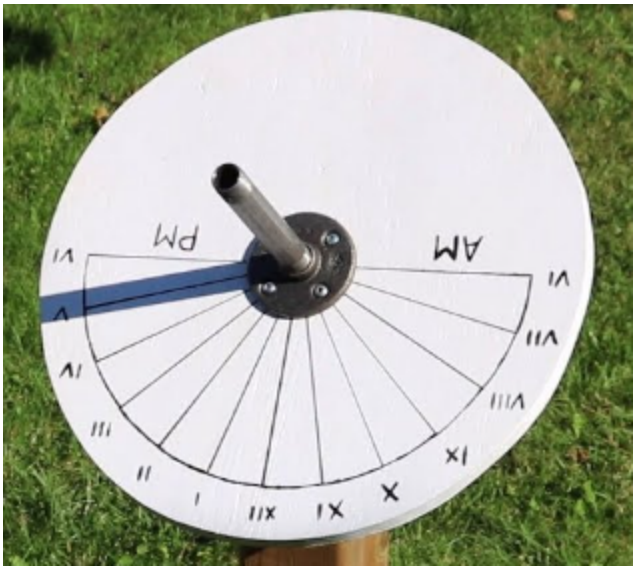
Expected materials needed: paper plate, pencil, ruler, protractor, compass, marker, tape, straw, scissors, sunlight

The sundial is one of the oldest methods of time-keeping, dating back to 1500 BCE in ancient Egypt. The sundial has been used for a large portion of human history and has been passed to numerous different cultures outside of ancient Egypt such as to the ancient Greeks and Romans. The sundial has two main components: the gnomon and the dial. The gnomon is the part of the sundial that casts the shadow onto the dial, and the dial is lined with numerous different angles corresponding to different times. Throughout the sky, the sun moves at a constant rate of 15 degrees per hour, changing the shadow cast by the gnomon, which will tell the time. We will try to construct a sundial that is accurate to within 20 minutes.

Instructions

1. Poke a hole into the paper plate. This will be the dial.
2. Push a straw out through the center hole. This will be the gnomon. The gnomon should be either vertical or be slightly slanted but stay still.
3. On a sunny day, set the sundial outside.
4. Using a compass, align the gnomon with True North (if in the northern hemisphere) or with True South (if in the southern hemisphere).

5. In the sunlight, at the start of every hour, mark the position the gnomon casts onto the dial, and label it with the time. Come back at the start of every possible hour to complete the dial.



Reflection

1. What are some limitations of the sundial?
2. What does the function of the sundial depend on?
3. Compare this with your own body. If you were locked in a room without windows, how would you be able to guess how much time has passed?
4. What does this tell you about endogenous versus exogenous rhythms?
5. The chapter describes how the advent of the pendulum clock reduced time-keeping error to seconds per day. What limitations does this address?
6. How would you track the time at night using the sky?

A Birth of a New Field

Learning Objective: Identify recurring themes that motivated the development of the field of chronobiology throughout history.

Jean-Jacques d'Ortous de Miran's Mimosa Plants

In 1729, French astronomer Jean Jacque d'Ortous de Mairan placed a mimosa plant into a dark cabinet and left it there. A mimosa plant regularly opens its leaves during the day and folds them at night. De Mairan discovered that the plant opened and closed its leaves in regular intervals even in the absence of light stimuli. At the time, de Mairan concluded that the mimosa plants were able "to sense the Sun without ever seeing it" (De Mairan, 1729). However, this finding would be the catalyst that would eventually spark the creation of the field of chronobiology. It was not until almost a century later that the Swiss botanist Augustin Pyramus de Candolle measured the regular intervals of leaf movements to have a period of approximately 22-23 hours, demonstrating a preliminary understanding of an endogenous circadian rhythm in plants (McClung, 2006).



Figure: A mimosa plant with leaves opened (left) and leaves folded (right).

Colin Pittendrigh's Fruit Flies

While de Mairan's experiment demonstrated the existence of biological rhythms in plants, Colin Pittendrigh would prove that such cycles exist in animals too. Pittendrigh found that fruit flies had a roughly 24 hour cycle even in the darkness, similar to the mimosa plant. Furthermore, Pittendrigh also found that by adjusting the times at which they were exposed to light, he was able to shift those cycles forward or backwards, demonstrating how these cycles respond to light (Pittendrigh & Minis, 1972). This finding eventually established the use of fruit flies as a useful organism to study biological rhythms on, and also they fully solidified the study of chronobiology as an experimental science that can be studied in both plants and animals.

Chronobiology in the Modern Day

Learning Objective: Compare modern day understanding and application of chronobiology with ancient chronobiology.

The Period Gene

While scientists knew about the existence of these biological rhythms by the 1960s, there was no understanding of what exactly causes them to have these rhythms. In 1971, while Ron Konopka and Seymour Benzer were breeding fruit flies, they discovered that fruit flies had variations in their circadian rhythms (Konopka & Benzer, 1971). Some fruit flies had rhythms longer than 24 hours, some had shorter, and some had no regular circadian rhythms. By this time, they had the field of genetics at their disposal, to which they were able to pinpoint the variation to a gene which they called *period*. This marks the very first time scientists discovered a gene that controls a biological clock, officially proving that biological rhythms are not a learned behavior or a passive response to stimuli. In the upcoming decades, a few scientists found that it produces a protein called PER, and that it builds up during the night and degrades in the day,, which is how the gene gives rise to a circadian rhythm.

Transcription-Translation Feedback loops

The clock gene demonstrates how biological rhythms are an inherent part of an organism, but how exactly does it work? In preliminary terms, the clock gene activates and produces mRNA, which then creates a clock protein. After many repeats of this, these proteins build up until the cell senses that there is enough. Eventually, these proteins break down, allowing more room for more of them to be produced. This self-sustaining cycle takes roughly 24 hours to

complete, giving the cell a 24 hour cycle. This negative feedback loop was discovered by 3 scientists: Hall, Rosbash, and Young. For this discovery, the three of them won the Nobel prize in medicine in 2017.

Chronobiology Today

Today, the study of chronobiology has many applications in medicine. Understanding sleep disorders such as insomnia, non-24-hour sleep-wake disorder, and shift-wake disorder is one such area of active research. When viewing these disorders through the lens of chronobiology, it asks what happens to the biological clocks of individuals who suffer from these disorders, and how we can treat them. Circadian disruption asks how events such as jet-lag and night-shifts impact the circadian rhythm and any health risks that may have. Another example is the development of chronotherapy, which is getting further developments since ancient times.

GLOSSARY

Term	Definition
Chronobiology	The scientific study of natural biological rhythms and how time affects living systems.
Endogenous Cycle	Rhythms that originate from within an organism and persist even without external environmental cues like light or dark.

Exogenous Cycle	Rhythms or behavioral shifts that are directly driven, forced, or influenced by external environmental factors.
Cyclical Processes	A self-sustaining, repeating biological rhythm that regulates the timing and duration of physiological and behavioral activities in living organisms.
Ultradian Rhythm	Biological cycles that operate on a cycle shorter than 24-hours.
Circadian Rhythm	Biological cycles that operate on a roughly 24-hour cycle.
Infradian Rhythm	Biological cycles that operate on a cycle longer than 24-hours.

REFERENCES

Balzer, H.-U. (2009). Chronobiology — As a foundation for and an approach to a new understanding of the influence of music. *Music That Works*, 25–82. https://doi.org/10.1007/978-3-211-75121-3_3

Breasted, J. H. (1935). The Beginnings of Time-Measurement and the Origins of Our Calendar. *The Scientific Monthly*, *41*(4), 289–304. JSTOR. https://www.jstor.org/stable/15981

Elgammal, K. (2022). Triangle decans. International Journal of Advanced Studies in World, 5(1), 89–106. https://ijaswa.journals.ekb.eg/article_227303_3cfd7f25991cdc1341b705eeb4e58ad9.pdf

JJ, DE M. (1729). *Observation botanique*. Histoire de l'Academie Royale Des Sciences Paris. https://cir.nii.ac.jp/crid/1573105975426983936

Konopka, R. J., & Benzer, S. (1971). Clock mutants of drosophila melanogaster. *Proceedings of the National Academy of Sciences*, *68*(9), 2112–2116. https://doi.org/10.1073/pnas.68.9.2112

McClung, C. R. (2006). Plant circadian rhythms. *The Plant Cell*, *18*(4), 792–803. https://doi.org/10.1105/tpc.106.040980

Miller, K. J. (2023). *Time and ancient medicine* (pp. 15–30). Oxford University Press.

Pittendrigh, C. S., & Minis, D. H. (1972). Circadian systems: Longevity as a function of circadian resonance in drosophila melanogaster. *Proceedings of the National Academy of Sciences*, *69*(6), 1537–1539. https://doi.org/10.1073/pnas.69.6.1537

Yan, H.-S., & Lin, T.-Y. (2002). A study on ancient chinese time laws and the time-telling system of Su Song's clock tower. *Mechanism and Machine Theory*, *37*(1), 15–33. https://doi.org/10.1016/S0094-114X(01)00059-3

Šprajc, I., Morales-Aguilar, C., & Hansen, R. D. (2019). Early Maya astronomy and urban planning at El Mirador, Peten, Guatemala. Anthropological Notebooks, 25(1), 79–98.

UNIT AND CHAPTER

Unit: Background

Chapter 2: THE BASICS OF BIORHYTHMS: THE MOLECULAR CLOCK

CONTRIBUTORS

- John Liu
  - Copyeditor
  - Peer reviewer
  - Subject Matter Expert - ‘Genetics of the Molecular Clock and Cyclic Behavior’
    - Sections:
      - The Molecular Clock and Circadian Rhythms;
      - Central and Local Clocks in the Brain and Body;
      - Genetic Mechanisms of the Molecular Clock;
      - Clock Genes and Cyclic Behavior
    - Figures x,y,z

- Matthew Hernandez
  - Assessment and Solutions Author
  - Peer reviewer
  - Subject Matter Expert - ‘Serotonin’s Roles in the Circadian Clock’
    - Section: Serotonin
    - Figures 1, 2, 3

LEARNING OBJECTIVES

After reading this chapter, students should be able to …

1. Relate the molecular clock to the production and regulation of circadian rhythms.
2. Define serotonin, identify the raphe nuclei as the main source of brain serotonin, and list the two enzymatic steps that build serotonin from tryptophan. (Remember / Understand)
3. Describe serotonin’s daily rhythm in the suprachiasmatic nucleus (SCN), and explain what its persistence in constant darkness reveals about what controls its timing. (Understand / Analyze)
4. Describe how serotonin acts as an input to the SCN and predict the direction of a serotonergic phase shift at a given circadian time. (Understand / Apply)

5. Trace the pathway by which serotonin becomes melatonin in the pineal gland, and explain how AANAT gives melatonin its daily rhythm. (Understand / Analyze)
6. Compare serotonin's effects on the clock across species, and evaluate why findings in brain slices (in vitro) and living animals (in vivo) sometimes disagree. (Analyze / Evaluate)
7. Compare the suprachiasmatic nucleus with local molecular clocks in the brain and body.
8. Explain how CLOCK, BMAL1, PER, and CRY interact to produce an approximately 24-hour rhythm.
9. Evaluate evidence connecting clock genes to cyclic behaviors such as sleeping, feeding, and physical activity.

CHAPTER OUTLINE

- Section: The Molecular Clock and Circadian Rhythms
    - Define circadian rhythms and give examples, including sleep, activity, body temperature, and feeding.
    - Introduce the molecular clock as a system of genes and proteins that helps produce these rhythms.
    - Explain that environmental signals such as light synchronize the molecular clock with the outside world.
    - Briefly discuss why genetics is important for understanding biological timing.
- Section: Serotonin
    - What Is Serotonin? (the molecule, its two-step synthesis, and the raphe nuclei)
    - Serotonin's Daily Rhythm (release in the SCN rises and falls on a clock-driven schedule)
    - Serotonin as Input to the Clock (The Clock and Its Inputs; Forward by Day; Gating by Night; The Receptor Involved)
    - Serotonin as Substrate of the Clock's Output (AANAT and the melatonin rhythm)
    - What Remains Unsettled (species and method differences)
- Section: Central and Local Clocks in the Brain and Body
    - Introduce the suprachiasmatic nucleus, or SCN, as the central circadian pacemaker.
    - Explain how the SCN receives light information and helps coordinate rhythms throughout the body.
    - Introduce local clocks in other brain regions.
    - Describe peripheral clocks in organs such as the liver, heart, pancreas, and skeletal muscles.
    - Compare the SCN's coordinating role with the specialized functions of local clocks.
    - Explain how different clocks work together as part of a larger circadian system.

- Section: Genetic Mechanisms of the Molecular Clock
    - Introduce CLOCK and BMAL1 as proteins that activate the expression of other clock genes.
    - Explain how the Per and Cry genes produce PER and CRY proteins.
    - Describe how PER and CRY eventually inhibit CLOCK and BMAL1.
    - Explain that the breakdown of PER and CRY allows the cycle to start again.\
    - Relate this feedback loop to the production of an approximately 24-hour rhythm.
    - Briefly introduce REV-ERB and ROR as additional regulators of the clock.
- Section: Clock Genes and Cyclic Behavior:
    - Define cyclic behavior and provide examples such as sleep-wake, feeding, and activity patterns.
    - Explain how researchers measure daily behavior in laboratory animals.
    - Introduce actograms as graphs that display activity over several days.
    - Discuss studies in which mutations or deletions of Clock or Bmal1 changed activity and feeding rhythms.
    - Explain that these findings connect clock genes to the timing of behavior.
    - Consider other factors, such as illness, metabolism, or reduced movement, that could affect the results.
- Section: Conclusion
    - Review how molecular clocks help produce circadian rhythms.
    - Summarize the roles of the SCN and local clocks.
    - Reconnect clock genes with cyclic behavior.
    - Emphasize that circadian timing depends on interactions among genes, the brain, the body, and the environment.

CHAPTER WRITTEN CONTENT - Mathew

Serotonin

1. What Is Serotonin?

Commonly known as the “happy chemical,” serotonin, or 5-hydroxytryptamine (5-HT), is a neurotransmitter: a chemical messenger that one neuron releases to signal another, either activating it or quieting it. The body must build serotonin from available resources, which it does by using the amino acid tryptophan obtained through food (see Figure 1). To do this, the enzyme

tryptophan hydroxylase (TPH) first converts tryptophan into an intermediate, or in-between, amino acid called 5-hydroxytryptophan (5-HTP). Then, a second enzyme called aromatic amino acid decarboxylase (AADC) converts the 5-HTP into serotonin. Of these two steps, the first is the slower one and therefore sets the pace of the whole process, which is why TPH is called the rate-limiting enzyme. Notably, TPH exists within the body in two forms, one found mainly in the gut and the other in the brain, with about 90% of serotonin made in the first (Akram et al., 2024). Moreover, since serotonin in the blood does not easily cross the blood-brain barrier that is the tight lining of the brain's blood vessels, the brain primarily relies on its own smaller supply of serotonin, and it is this supply that is being referenced in the context of the circadian clock. This supply within the brain comes mainly from clusters of neurons along the midline of the brainstem, called the raphe nuclei, of which two nuclei matter mainly: the dorsal raphe and the median raphe (see Figure 2). As the following sections show, it is these two nuclei that connect serotonin to the circadian clock.

2. Serotonin's Daily Rhythm

The supply that the dorsal raphe and median raphe nuclei provide, however, is not constant, and instead rises and falls over the course of each day. A clear example of this can be found from Dudley et al. (1998), where, by sampling the fluid around neurons in the suprachiasmatic nucleus (SCN), the brain's master circadian clock (see Central and Local Clocks in the Brain and Body), they found that serotonin release in living Syrian hamsters fell to its lowest level during the late midday, but increased sharply to its peak at the transition from light to darkness, and then declined through the night. This cycle persisted when the hamsters were kept in constant darkness without light to signal the time of day: the peak was at the start of their

subjective night, or rather began at the half of the cycle that the hamsters' own clocks treated as night. That is, the timing of serotonin release was set by the clock itself rather than by the typical signaling of light. The neurons that supply this serotonin may also keep track of time of their own. Specifically, in adult male rats, Strnad and Spencer (2026) found that core rhythmic clock genes rise and fall over roughly 24 hours within the dorsal and median raphe, and that the raphe's serotonin-producing neurons express these genes. Since this finding comes from a single recent study of one sex in one species, however, it should be treated with a degree of caution until it is further verified. Nonetheless, the hamster findings show how serotonin reaches the clock as a rhythmic supply rather than a background supply, which contextualizes the content for the next section, where what serotonin does depends on the time of day.

3. Serotonin as Input to the Clock

3.1 The Clock and Its Inputs

Left alone, the SCN maintains a rhythm of roughly 24 hours (about 24.2 hours in humans; Czeisler et al., 1999); external time cues such as sunlight and mealtimes, called zeitgebers, reset it regularly so that it stays aligned with Earth's approximately 24-hour day. Of these cues, the strongest is light, which reaches the SCN directly from the eyes (described in Central and Local Clocks in the Brain and Body), but serotonin serves as another input and can arrive by two routes. From what has been tested so far, using hamsters, serotonergic neurons of the median raphe project directly to the SCN, while those of the dorsal raphe project to the intergeniculate leaflet (IGL), a relay region in the thalamus that passes nonphotic, or non-light, information on to the SCN (Meyer-Bernstein & Morin, 1996; see Figure 2). In other words, the first route using the median raphe is direct and the other route using the dorsal raphe is relayed

by the IGL. However, the specifics of what serotonin does once it then arrives is dependent on the time of day, mainly separated by day and night.

3.2 Forward by Day

The dependence on the time of day regarding serotonin's results is described in terms of phase shifts; that is, its changes in the timing of a rhythm: a phase advance moves the circadian clock earlier, while a delay moves it later. The size of the shift that a stimulus produces, and whether it is an advance or a delay, is dependent on when within the cycle the stimulus is delivered. This relationship is often graphed or plotted and then is called a phase response curve (PRC). Conventionally, the time within the cycle is measured in circadian time (CT), which counts from the start of an organism's subjective day, where CT0 to CT12 marks this subjective day while CT12 to CT24 marks the subjective night. For serotonin, the most relevant window is the subjective day. In a study where hamsters were given 8-OH-DPAT, a synthetic drug that activates serotonin receptors, in the middle of their subjective day, their clocks advanced (Challet et al., 1998). Likewise, in mice, the same drug given at CT6 advanced the start of their daily wheel-running, which the activity researchers used to read the clock, by about 37 minutes (Shelton et al., 2014). The shifts can be induced, then, but the results yield that they are merely nudges rather than any meaningful shove.

3.3 Gating by Night

At night, this relationship of advancing the cycle switches: rather than the clock itself shifting as it does during the day, serotonin reduces the clock's sensitivity to light at night. Working again with hamsters, Rea, Glass, and Colwell (1994) found that 8-OH-DPAT weakened the phase advance that would otherwise be normally produced by a nighttime light pulse, in fact,

reducing it by 57 percent at a low dose and by 82 percent at a dose ten times higher. At this higher dose, the drug also blocked light-induced phase delays; that is to say, the circadian clock was not being moved so much as it was being made harder to move. Particularly within interest, the start of the night is also when serotonin release in the hamsters' SCN reaches its peak (as discussed in Section 2). Taken together, then, the daytime and nighttime findings suggest that serotonin acts as a forward signal during the day and as a gate at night.

3.4 The Receptor Involved

These effects of signaling and gating depend on which serotonin receptors are activated. 8-OH-DPAT acts on more than one type of serotonin receptor, but in mice, the evidence points to one of them, called 5-HT7, as the receptor mainly responsible for the daytime advance. When activated, the 5-HT7 receptor raises the level of cyclic AMP (cAMP), a messenger molecule that relays the receptor's signal to the machinery inside the cell and that will reappear in section 4. Pretreating mice with a drug that selectively blocks 5-HT7 receptors completely prevented the 8-OH-DPAT advance and also weakened the clock's shifts in response to light (Shelton et al., 2014). Definitive conclusions regarding if the daytime and nighttime effects share a singular mechanism, however, are not yet determined, but Section 5 will indeed return to the questions that remain open.

4. Serotonin as Substrate of the Clock's Output

Within a small pine-shaped gland near the center of the brain, accordingly named the pineal gland, serotonin serves a rather different role than its previously explored use as an input: instead of acting as a neurotransmitter, serotonin serves as a chemical reactant from which the hormone melatonin is made. This conversion consists of two main steps: first, serotonin is

acetylated to form N-acetylserotonin by the enzyme arylalkylamine N-acetyltransferase, abbreviated AANAT; then, N-acetylserotonin is methylated to form melatonin by a second enzyme. Of these two steps, the first, carried out by AANAT, is the one that pertains the most in relevancy for circadian timing. In particular, the activity of the AANAT enzyme rises and falls throughout the day and, in doing so, sets the daily cycle of melatonin release. Aptly nicknamed "the Timezyme" by Klein (2007), AANAT is essentially what turns the supply of serotonin into a hormonal signal useful for the circadian cycle; that is, the rhythm comes from the regulation of the enzyme rather than from the amount of serotonin available. Specifically, the regulation goes through cyclic AMP (cAMP), the same messenger from Section 3.4: at night, the rising cAMP switches the AANAT on and also protects it from being broken down (Klein, 2007). The size of this nighttime increase varies across species, but it is especially large in rats, in which AANAT messenger RNA (mRNA) rises more than 150-fold at night (Roseboom et al., 1996). The instructions to make melatonin, however, do not actually come from the pineal gland itself, but instead are relayed from the SCN through a chain of neurons that leaves the brain, runs down into the spinal cord and out to a cluster of nerve cells in the neck, and then climbs back up to the pineal gland, where its nerve endings release the chemical compound norepinephrine at night, raising cAMP and switching melatonin production on (Simonneaux & Ribelayga, 2003; see Figure 3). Thus, serotonin, the same input molecule that resets the circadian clock, reappears later as the substrate, to be then synthesized into melatonin, the final hormonal output.

5. What Remains Unsettled

Despite these aforementioned findings, serotonin's role in the clock remains nondefinitive, with much of the disagreement going back to which animals were studied and how they did so. For instance, the primary usage of rodents such as hamsters, mice, and rats in

scientific experiments generally and within neurobiological testing is of indubitable note. More specifically, for hamsters and mice, serotonergic drugs act as nonphotic signals, advancing the clock during the subjective day; by contrast, for rats, they behave more like light, shifting the clock at night, which has become a mismatch coined and known as the serotonergic paradox (Cuesta et al., 2009). To that end, the method of study matters as well. In another instance, serotonin applied directly to isolated slices of mouse SCN (in vitro) during the subjective day advances the clock by two to three hours (Prosser, 2003), whereas 8-OH-DPAT given to living mice (in vivo) produces the far smaller advance of about 37 minutes described in Section 3.2 (Shelton et al., 2014). Perhaps surprisingly, removing serotonin does not actually stop the clock in these experiments: tested adult mice that could no longer make serotonin in the brain still kept a daily rhythm, albeit a transformed one (Whitney et al., 2016; see Case Study). Serotonin, then, appears to be one of several signals that adjust the clock rather than the source of its rhythm, as counter-simplified as that may be.

○

CHAPTER WRITTEN CONTENT - John Liu

The Molecular Clock and Circadian Rhythms

The **molecular clock** is an internal cellular mechanism that helps create this timing. It is a cycle made of **genes** and **proteins** whose activity level changes throughout the day. These changes can affect neuronal activity, hormone release, and behavioral timing. Reviews of circadian genetics explain how interactions among clock genes and proteins produce daily rhythms and influence functions throughout the body (Kuo et al., 2026; Takahashi, 2017).
(*Note - Will extend this portion of the text with further research and background information)

Central and Local Clocks in the Brain and Body

Within the **hypothalamus**, sitting directly above the point where the optic nerves cross, sits a tiny structure known as the **suprachiasmatic nucleus (SCN)**. In mammals, the SCN acts as the master biological **circadian pacemaker**. Specialized cells in the **retina** receive environmental light and send signals directly to the SCN through a pathway called the **retinohypothalamic tract**. Through this process, the SCN synchronizes internal rhythms with the **light-dark cycle** and sends signals that help regulate sleep cycles, daily activity, body temperature, hormone release, and other daily functions (Das & Launico, 2026).

Individual **neurons** in the SCN contain molecular clocks and also display daily changes in activity. These daily changes in neuronal activity are regulated by internal molecular clocks. In turn, neuronal activity can influence the **expression of clock genes** (Colwell, 2011). Furthermore, the SCN contains many neurons that communicate with each other using **neurotransmitters** like **GABA** and **vasoactive intestinal polypeptide (VIP)**. VIP plays an important role in synchronizing the individual cellular clocks into a stronger and more coordinated timing signal. GABA also influences **synchronization** among SCN neurons and, depending on the conditions, may either support or reduce it. Together, this chemical and electrical communication allows the individual cellular clocks to contribute to the SCN's coordinated rhythm (Herzog et al., 2017).

However, the SCN is not the only internal timing system. Other internal molecular clocks are located in varying brain regions and organs such as the liver, heart, pancreas, and skeletal muscles. These clocks contribute to different functions within specific tissues. For example, clocks located in metabolic organs help regulate digestion and glucose levels. The main difference between the SCN and other **local clocks** is that the SCN uses light information as its main cue for timing regulation, while local clocks use a combination of stimuli. These stimuli include feeding, hormones, temperature, and signals from the nervous system; they not only

cue local clocks to regulate time, but also help keep these clocks connected (Takahashi, 2017; Kuo et al., 2026).

Genetic Mechanisms of the Molecular Clock

The molecular clock is driven by a **transcription-translation feedback loop**. **Transcription** is the process where information in **DNA** is copied into **RNA** messengers; **translation** is the process where the cell uses that RNA to build a protein. At the beginning of the cycle, **CLOCK** and **BMAL1**, two master regulators, form a pair inside cells to drive the body's 24-hour internal clock, known as the circadian rhythm. They join together in the cell's **nucleus** to create a protein complex called a **heterodimer** (Menet et al., 2014). Together, they act as a **transcription factor** by binding regulatory regions of DNA and turning on specific target genes, such as ***Per*** (Period) and ***Cry*** (Cryptochrome). Without them—especially BMAL1—the entire molecular clock completely breaks down. Importantly, since they have the ability to open up tightly packed **chromatin,** they act as **pioneer-like transcription factors**, making the DNA accessible so thousands of other genes can be expressed at specific times of the day.

After CLOCK and BMAL1 form a pair, they activate the transcription of the Per and Cry genes. The resulting messenger RNA provides instructions for producing PER and CRY proteins. These proteins slowly collect in the cell, form complexes, and enter the nucleus. They then inhibit the activity of CLOCK and BMAL1, causing the transcription of *Per* and *Cry* to slow. Since the products of the process inhibit the same process that produced them, this process is called **negative feedback**. Later, PER and CRY proteins are modified and degraded. As that happens, CLOCK and BMAL1 reactivate and resume transcription, causing the cycle to repeat. The entire process involves delays, making the cycle approximately 24 hours.

There are additional regulatory loops that help stabilize this cycle. **REV-ERB** and **ROR** proteins act as a molecular push-and-pull switch that turns the ***Bmal1*** gene on and off at exact times of day. REV-ERB is a protein that suppresses the *Bmal1* gene, while ROR is a protein that activates the *Bmal1* gene (Guillaumond et al., 2005). Because they compete for the same binding site, they act as an inversely proportional relationship: high levels of REV-ERB block transcription and lower BMAL1 levels, while high levels of ROR facilitate transcription and increase BMAL1 levels. The core clock proteins also regulate many **clock-controlled genes**. These genes do not create the clock itself, but allow the clock to influence other cell functions, enabling different tissues to use a similar clock mechanism while producing different rhythms, such as hormone release in one tissue and metabolic activity in another (Takahashi, 2017).

Although cells throughout the body use the same core negative feedback loop, the genes regulated by these clocks and their effects differ across tissues. For example, a liver cell may regulate genes involved in processing nutrients, while a neuron may regulate genes that affect electrical activity. Signals from the SCN, along with feeding and other daily cues, help align these local cycles. This connection between the shared genetic mechanism and different tissue functions displays how one circadian system can organize many biological rhythms (Kuo et al., 2026).

Clock Genes and Cyclic Behavior

The molecular clock influences many behaviors that follow a daily repeated pattern. Because these behaviours can be measured, researchers can use them to study how genes contribute to circadian timing. One method of studying this behaviour is to compare normal mice with mice that have a **mutation** in the **clock gene**. Turek and colleagues studied mice carrying a mutation in ***Clock***, one of the genes involved in the circadian system. The mutant mice displayed changes in daily timing of movement, feeding, and metabolism. They found that their

feeding rhythm was weaker than that of normal mice, and many developed obesity and other issues regarding metabolic processes. These findings suggest that *Clock* contributes to feeding rhythm and activity across the day (Turek et al., 2005).

However, this does not necessarily mean that *Clock* is the guiding gene that controls all feeding behaviors. The mutation also affects metabolic processes, which in turn, could change eating habits and appetite. Light can also directly influence activity, even when the internal clock is not functioning normally. Therefore, a daily pattern of activity under a light-dark cycle does not necessarily mean that the internal clock is producing that pattern. For this reason, researchers must investigate this behavior over several days and under many different environmental conditions.

Research involving *Bmal1* has revealed that clock genes contribute to **cyclic behavior**. Mice that lack functional *Bmal1* are found to have dysfunctional circadian pattern activity. However, removing the gene from the entire body can also cause developmental issues, creating issues independent from mutated genes. To combat this issue, some researchers removed *Bmal1* from only some particular parts of the body. Mieda and Sakurai studied mice lacking *Bmal1* in the nervous system. These mice had difficulty adjusting activity and feeding behavior. These findings suggest that clocks outside the SCN also contribute to the timing of food-related behaviors (Mieda & Sakurai, 2011).

Conclusion

The SCN and internal molecular clocks help produce circadian rhythms through interactions among environmental cues, genes, and proteins. Specifically, the CLOCK–BMAL1 and PER–CRY negative feedback loop generates an approximately 24-hour cycle, while additional regulators, such as feeding and environmental light, help maintain its timing. In mammals, the eyes receive environmental light information that is sent to the SCN, coordinating local clocks

throughout the brain and body. Although these clocks use similar molecular processes, their function regulation in tissues vary from tissue to tissue. Importantly, research involving mice with mutations to the Clock or Bmal1 gene reveal a connection between clock genes and daily activity and feeding patterns. However, overarching health, development, and metabolism changes must be considered when evaluating these behavioral changes before making conclusions. Together, these findings show how cellular clocks, signals from the SCN, and environmental cues work together to organize daily biological rhythms.

CHAPTER ADD-INS

Figure 1. How the molecular clock repeats

Author: John Liu

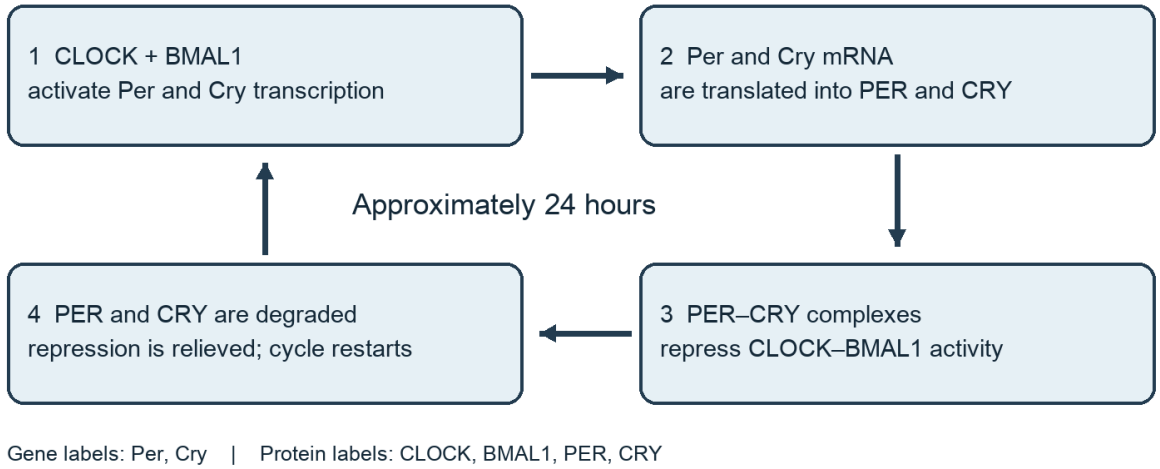


Figure description. This simplified diagram shows the negative feedback loop; REV-ERB and ROR provide additional regulation not pictured here. CLOCK and BMAL1 activate transcription of *Per* and *Cry* genes. Their mRNA is translated into PER and CRY proteins, which accumulate and form complexes that inhibit CLOCK–BMAL1 activity. Modification and degradation of these proteins relieve repression, allowing another cycle of transcription to begin. Delays during this process help produce an approximately 24-hour rhythm. The arrows show the chronology of events, not equal time intervals.

Attribution. Original schematic created on Canva.

Figure 2. How central and local clocks work together

Author: John Liu

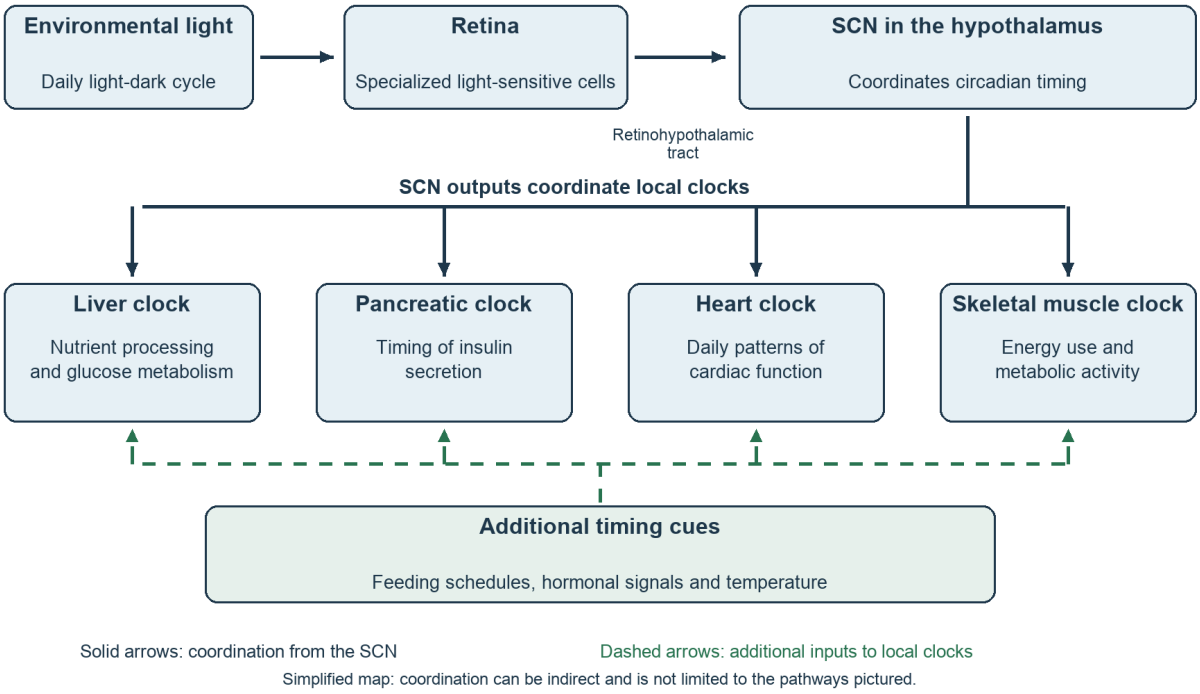


Figure description. Light detected by specialized cells in the retina reaches the suprachiasmatic nucleus (SCN) through the retinohypothalamic tract. The SCN uses this information to help align internal timing with the light-dark cycle. Its outputs, depicted above, coordinate clocks throughout the body through neural signals, hormonal signals, and daily changes in body temperature. The solid arrows represent this coordination.

Local clocks: Liver clocks help time nutrient processing and glucose metabolism; pancreatic clocks contribute to the timing of insulin secretion. Heart and skeletal-muscle clocks contribute to daily patterns of cardiac function and energy use.

Additional timing cues. The dashed arrows show that local clocks also respond to cues such as feeding schedules, hormones, and temperature. Some of these signals are influenced by the SCN, while others reflect behavior and the environment.

Attribution. Original schematic created on Canva.

GLOSSARY

Term	Definition
Circadian rhythm	A biological pattern that repeats approximately every 24 hours.
Molecular clock	A system of interacting genes and proteins that generates biological timing within a cell.
Gene	A segment of DNA containing instructions for producing a functional RNA or protein product.
Protein	A molecule made of amino acids that performs functions such as regulating genes, transmitting signals, or supporting cell structure.
Hypothalamus	A brain region involved in regulating functions such as body temperature, hunger, hormone release, and circadian timing.
Suprachiasmatic nucleus (SCN)	A region of the hypothalamus that receives light information from the eyes and coordinates circadian rhythms in mammals.
Circadian pacemaker	A biological timing system that generates circadian rhythms and helps coordinate their timing.
Retina	The light-sensitive tissue at the back of the eye that contains cells involved in vision and detecting light for circadian timing.
Retinohypothalamic tract	A nerve pathway that carries light information from specialized retinal cells to the SCN.
Light-dark cycle	The recurring pattern of light and darkness that helps synchronize biological clocks with the environment.
Neuron	A nerve cell that receives and transmits information through electrical and chemical signals.
Expression of clock genes	The process of using information in clock genes to produce RNA and, for protein-coding genes, clock proteins.
Neurotransmitter	A chemical messenger released by neurons that influences other cells.
GABA	Gamma-aminobutyric acid, a neurotransmitter that influences neuronal activity and synchronization within the SCN.
Vasoactive intestinal polypeptide (VIP)	A signaling peptide that helps coordinate the timing of individual cellular clocks in the SCN.
Synchronization	The coordination of rhythms so that they maintain a consistent timing

	relationship.
Local clock	A molecular clock in a particular brain region, organ, or tissue that helps time functions in that location.
Transcription-translation feedback loop	A cycle in which genes are transcribed into RNA, the RNA is translated into proteins, and those proteins influence further gene transcription.
Transcription	The process of copying information from DNA into RNA.
Translation	The process of using information in messenger RNA to build a protein.
DNA	Deoxyribonucleic acid, the molecule that stores genetic information.
Messenger RNA (mRNA)	An RNA molecule that carries instructions used by the cell to produce a protein.
Nucleus	The compartment in a cell that contains most of its DNA and is a major site of gene transcription.
Heterodimer	A complex formed by two different molecular subunits, such as CLOCK and BMAL1 proteins.
Transcription factor	A protein that helps regulate the transcription of particular genes.
Pioneer-like transcription factor	A transcription factor that can bind DNA in relatively inaccessible chromatin and help make that region accessible to other regulatory proteins.
Chromatin	DNA and its associated proteins, which package genetic material and influence access to genes.
CLOCK	A clock protein that partners with BMAL1 to activate transcription of genes involved in circadian timing.
BMAL1	A clock protein that partners with CLOCK or a related protein to activate genes involved in the molecular clock.
Clock	A mouse gene that encodes the CLOCK protein.
Bmal1	A mouse gene that encodes the BMAL1 protein.
Per	A family of Period genes that encode PER proteins involved in circadian negative feedback.
Cry	A family of Cryptochrome genes that encode CRY proteins involved in circadian negative feedback.
PER	Period proteins that help regulate circadian timing and participate with CRY proteins in negative feedback.

CRY	Cryptochrome proteins that repress CLOCK–BMAL1-driven transcription as part of the molecular clock's negative feedback loop.
Negative feedback	A process in which the products of a system reduce the activity that produced them.
REV-ERB	A family of nuclear receptor proteins that repress transcription of genes such as *Bmal1* and help regulate circadian timing.
ROR	A family of nuclear receptor proteins that activate transcription of genes such as *Bmal1* and help regulate circadian timing.
Clock-controlled gene	A gene whose expression is regulated by the circadian clock and helps produce timed cellular or bodily functions.
Clock gene	A gene whose product contributes to generating or regulating circadian timing.
Mutation	A change in a DNA sequence that may alter a gene's expression or the function of its product.
Feeding rhythm	A recurring pattern in the timing or amount of food consumption.
Cyclic behavior	A behavior, such as sleeping, feeding, or movement, that repeats in a regular pattern over time.

REFERENCES

Colwell, C. S. (2011). Linking neural activity and molecular oscillations in the SCN. *Nature Reviews Neuroscience, 12(10), 553–569.* https://doi.org/10.1038/nrn3086

Das, J. M., & Launico, M. V. (2026). Neuroanatomy, nucleus suprachiasmatic. In *StatPearls*. StatPearls Publishing. https://www.ncbi.nlm.nih.gov/books/NBK546664/

Guillaumond, F., Dardente, H., Giguère, V., & Cermakian, N. (2005). Differential control of Bmal1 circadian transcription by REV-ERB and ROR nuclear receptors. *Journal of Biological Rhythms*, 20(5), 391–403. https://doi.org/10.1177/0748730405277232

Haque, S. N., Booreddy, S. R., & Welsh, D. K. (2019). Effects of BMAL1 manipulation on the brain's master circadian clock and behavior. *Yale Journal of Biology and Medicine*, 92(2), 251–258. https://pmc.ncbi.nlm.nih.gov/articles/PMC6585533/

Herzog, E. D., Hermanstyne, T., Smyllie, N. J., & Hastings, M. H. (2017). Regulating the suprachiasmatic nucleus (SCN) circadian clockwork: Interplay between cell-autonomous and circuit-level mechanisms. *Cold Spring Harbor Perspectives in Biology*, 9(1), a027706. https://doi.org/10.1101/cshperspect.a027706

Kuo, T. T., Wang, J., & Kay, S. A. (2026). Time matters: Circadian genetics and the molecular logic of human health and disease. *Nature Reviews Genetics*, 27(10), 744–761. https://doi.org/10.1038/s41576-026-00962-2

Menet, J. S., Pescatore, S., & Rosbash, M. (2014). CLOCK:BMAL1 is a pioneer-like transcription factor. *Genes & Development,* 28(1), 8–13. https://doi.org/10.1101/gad.228536.113

Mieda, M., & Sakurai, T. (2011). Bmal1 in the nervous system is essential for normal adaptation of circadian locomotor activity and food intake to periodic feeding. *The Journal of Neuroscience*, 31(43), 15391–15396. https://doi.org/10.1523/JNEUROSCI.2801-11.2011

Takahashi, J. S. (2017). Transcriptional architecture of the mammalian circadian clock. *Nature Reviews Genetics*, 18(3), 164–179. https://doi.org/10.1038/nrg.2016.150

Turek, F. W., Joshu, C., Kohsaka, A., Lin, E., Ivanova, G., McDearmon, E., Laposky, A., Losee-Olson, S., Easton, A., Jensen, D. R., Eckel, R. H., Takahashi, J. S., & Bass, J. (2005). Obesity and metabolic syndrome in circadian Clock mutant mice. *Science*, 308(5724), 1043–1045. https://doi.org/10.1126/science.1108750

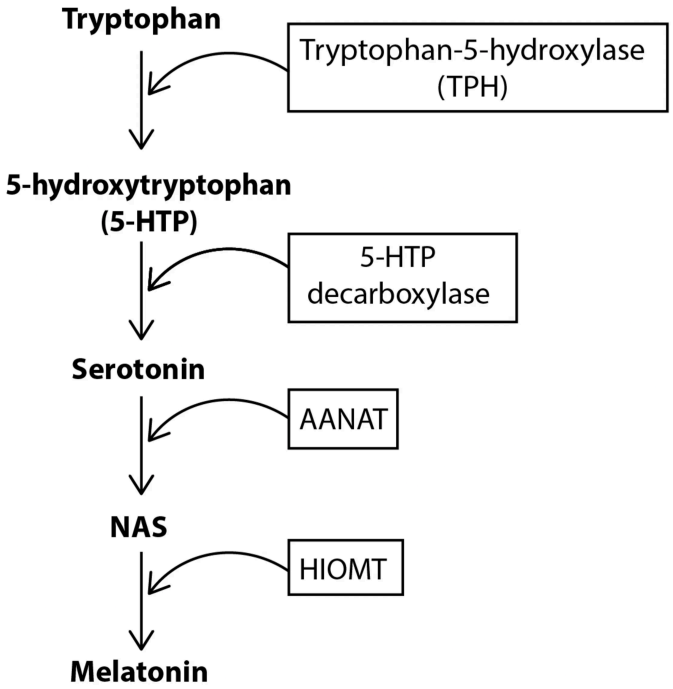
CHAPTER ADD-INS

Focus Question (placement: before Section 1)

How does serotonin both reset the body's master circadian clock while also serving as the chemical material that the clock uses to signal that it is night?

Figure 1 - From Tryptophan to Serotonin to Melatonin

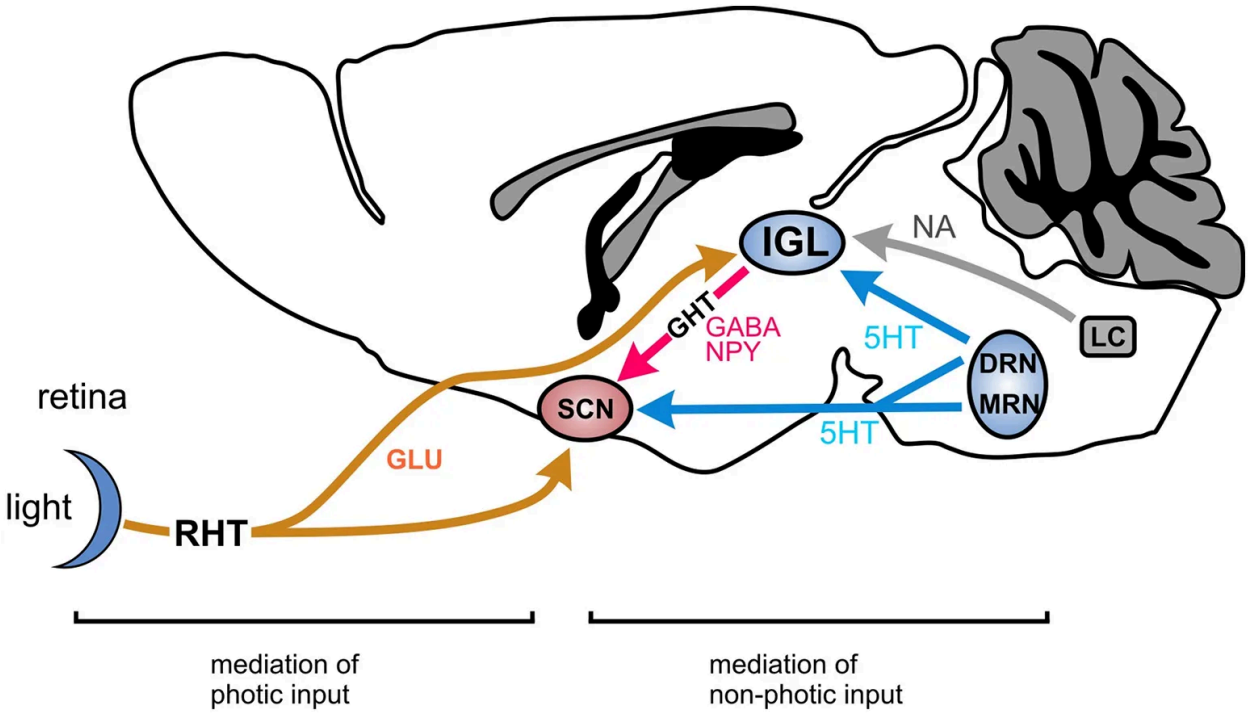
- Author: Lumsden, Clarkson, and Cakmak (2020); placement: Section 1



- Figure Description: Tryptophan is turned into 5-HTP, then serotonin, followed by NAS and ultimately melatonin. These reactions and syntheses will be explained more in-depth in Sections 1 and 4.
- Attribution: Lumsden et al. (2020), Figure 1: https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2020.00264/full

Figure 2 - Light and Serotonergic Inputs into the Circadian System

- Author: Stieglitz et al. (2018); placement: Section 3.1

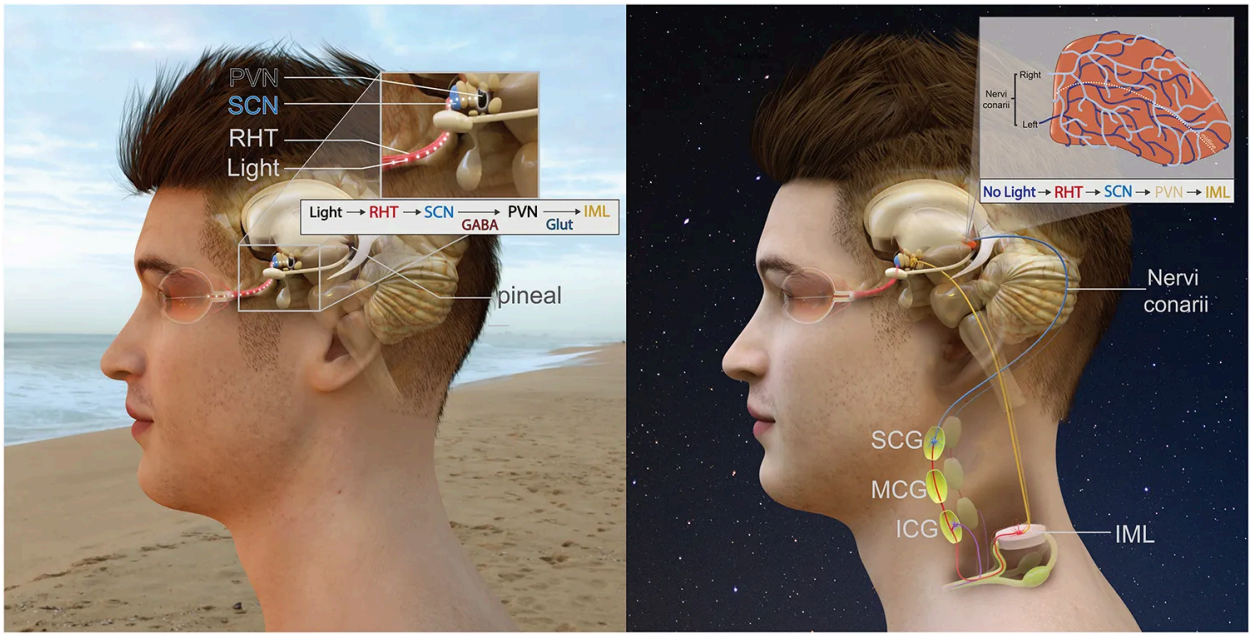


- Figure Description: The suprachiasmatic nucleus (SCN) is the master circadian clock and is shown in the bottom center in red. Inputs from the eyes reach the SCN through the retinohypothalamic tract (RHT), the orange pathway on the left. On the right, serotonin, or 5-hydroxytryptamine (5HT), arrives through the blue pathways: the median raphe nucleus (MRN) projects directly to the SCN, while the dorsal raphe nucleus (DRN) projects to the intergeniculate leaflet (IGL), which relays to the SCN (as mapped in hamsters). The gray pathway shows noradrenaline (NA) input from the locus coeruleus (LC). Other labels: GLU, glutamate; GHT, geniculohypothalamic tract; GABA, gamma-aminobutyric acid; NPY, neuropeptide Y.
- Attribution: Stieglitz et al. (2018), Figure 1: https://www.frontiersin.org/journals/molecular-neuroscience/articles/10.3389/fnmol.2017.00436/full

- Figure Interpretation Question (FIQ): (a) Which route has serotonin (5HT) go directly to the SCN, and (b) which instead uses a relay?

Figure 3 - The SCN's Signal Out to the Pineal Gland

- Author: Lumsden, Clarkson, and Cakmak (2020); placement: Section 4



- Figure Description: The left panel shows the day, and the right panel shows the night. During the day (left), light from the eye travels along the retinohypothalamic tract (RHT) to the suprachiasmatic nucleus (SCN), which then quiets the paraventricular nucleus (PVN) of the hypothalamus, stopping the signal there. At night (right), with no daylight of course, the signal goes from the SCN out to the pineal gland through the PVN, then to the intermediolateral cell column (IML) of the upper spinal cord, followed by going into the superior cervical ganglion (SCG) in the neck (in order), passing on its way up through the inferior and middle cervical ganglia (ICG, MCG) and finally reaching the pineal gland as the nervi conarii (top-right). Essentially, the signal to make melatonin leaves the

brain only to ultimately come back to it, and only at night; once there, its nerve endings release norepinephrine, which turns melatonin production on (Section 4). The boxed expression chains in each of the panels summarizes their respective routes, with GABA and glutamate (Glut) naming the chemical messengers at the SCN-to-PVN and PVN-to-IML steps. Although the pathway is drawn onto a human figure, much of it was actually first determined in rodents.

- Attribution: Lumsden et al. (2020), Figure 2: https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2020.00264/full
- Figure Interpretation Question (FIQ): (a) Which relay is the first one outside the brain, and (b) why does the pathway stay quiet during the day?

Case Study: A Brain Without Serotonin (placement: Section 5)

Whitney et al. (2016) asked what happens to daily behavior when an adult brain stops making serotonin. Working with genetically modified mice, the researchers switched off the gene for the brain's form of tryptophan hydroxylase (TPH, the rate-limiting enzyme from Section 1) in adulthood, nearly completely eliminating serotonin production by the brain's serotonin neurons. The mice then became notably hyperactive in both unfamiliar settings and their original cages. Their circadian rhythm also changed shape, with activity beginning earlier and ending later each day, and the "siesta," a rest period that normally interrupts mice nighttime activity, disappeared entirely as well.

Reflect: Section 3.2 described a serotonin-receptor drug shifting the mouse's clock by only about 37 minutes. Why might removing serotonin entirely then change its daily behavior so much more than just a single dose of that drug?

Myth vs. Nuance (placement: end of Section 5)

Myth	Nuance
Serotonin is simply the brain's "happy chemical."	About 90 percent of the body's serotonin is made in the gut (Akram et al., 2024), and in the brain it does far more than lift mood, including adjusting the timing of the circadian clock.
Melatonin is a sleeping pill or is only for signaling sleep.	Melatonin is a signal of biological darkness. Nocturnal animals such as rats also make it at night, when they are awake and active, so its message is "it is night," not "go to sleep" (Simonneaux & Ribelayga, 2003).
Serotonin always pushes the clock forward.	It depends on the hour and the animal. In hamsters, serotonin advances the clock by day but acts as a gate against light at night; in rats, serotonergic drugs act more like light, shifting the clock at night (Cuesta et al., 2009).

End-of-Chapter Questions (ECQs):

1.
2.
3.
4.
5.

GLOSSARY

Term	Definition

Circadian rhythm	A biological pattern that repeats approximately every 24 hours.
Molecular clock	A system of interacting genes and proteins that produces internal biological timing within a cell.
Suprachiasmatic nucleus (SCN)	A region of the hypothalamus that receives light information and coordinates circadian rhythms across the body.
Local clock	A molecular clock in a brain region, organ, or tissue that helps time functions in that location.
Transcription-translation feedback loop	A cycle in which genes produce proteins that later increase or decrease the activity of those genes.
Negative feedback	A process in which the products of a system reduce the activity that produced them.
Clock gene	A gene whose product helps generate, regulate, or transmit circadian timing.
Cyclic behavior	A behavior, such as sleeping, feeding, or movement, that repeats in a regular pattern over time.
Serotonin (5-HT)	A neurotransmitter made from the amino acid tryptophan. In the brain it comes mostly from the raphe nuclei, though most of the body's serotonin is made in the gut.
Raphe nuclei	Clusters of neurons along the midline of the brainstem that produce most of the brain's serotonin. The dorsal and median raphe are the two that connect to the circadian system.
Phase response curve (PRC)	A graph showing how far, and in which direction, a clock shifts when a stimulus is applied at different times in the circadian cycle.
Phase advance / phase delay	A shift of the circadian clock to an earlier time (advance) or a later time (delay).
Circadian time (CT)	A 24-part scale for position within the cycle. CT0 to CT12 is the subjective day and CT12 to CT24 the subjective night; for a nocturnal animal, CT12 marks the start of activity.
Zeitgeber	An external time cue, such as light, that resets (entrains) the circadian clock to the 24-hour day.
Nonphotic stimulus	A cue that resets the circadian clock but is not light, such as physical activity or a serotonergic drug.
AANAT	The pineal enzyme, nicknamed "the Timezyme," that converts

(arylalkylamine N-acetyltransferase)	serotonin into N-acetylserotonin. Because its activity rises at night, it sets the daily rhythm of melatonin.
Melatonin	A hormone made from serotonin in the pineal gland. Its levels rise at night, signaling biological darkness to the rest of the body.
Pineal gland	A small gland near the center of the brain that converts serotonin into melatonin, mainly at night.
In vitro / in vivo	In vitro means in an isolated preparation, such as a slice of brain tissue; in vivo means in the living animal.

REFERENCES

Akram, N., Faisal, Z., Irfan, R., Shah, Y. A., Batool, S. A., Zahid, T., Zulfiqar, A., Fatima, A., Jahan, Q., Tariq, H., Saeed, F., Ahmed, A., Asghar, A., Ateeq, H., Afzaal, M., & Khan, M. R. (2024). Exploring the serotonin–probiotics–gut health axis: A review of current evidence and potential mechanisms. *Food Science & Nutrition, 12*(2), 694–706. https://doi.org/10.1002/fsn3.3826

Challet, E., Scarbrough, K., Penev, P. D., & Turek, F. W. (1998). Roles of suprachiasmatic nuclei and intergeniculate leaflets in mediating the phase-shifting effects of a serotonergic agonist and their photic modulation during subjective day. *Journal of Biological Rhythms, 13*(5), 410–421. https://doi.org/10.1177/074873098129000237

Colwell, C. S. (2011). Linking neural activity and molecular oscillations in the SCN. *Nature Reviews Neuroscience*, 12, 553–569. https://doi.org/10.1038/nrn3086

Cuesta, M., Clesse, D., Pévet, P., & Challet, E. (2009). New light on the serotonergic paradox in the rat circadian system. *Journal of Neurochemistry, 110*(1), 231–243. https://doi.org/10.1111/j.1471-4159.2009.06128.x

Czeisler, C. A., Duffy, J. F., Shanahan, T. L., Brown, E. N., Mitchell, J. F., Rimmer, D. W., Ronda, J. M., Silva, E. J., Allan, J. S., Emens, J. S., Dijk, D.-J., & Kronauer, R. E. (1999). Stability, precision, and near-24-hour period of the human circadian pacemaker. *Science, 284*(5423), 2177–2181. https://doi.org/10.1126/science.284.5423.2177

Das, J. M., & Launico, M. V. (2026). Neuroanatomy, nucleus suprachiasmatic. In *StatPearls*. StatPearls Publishing. https://www.ncbi.nlm.nih.gov/books/NBK546664/

Dudley, T. E., DiNardo, L. A., & Glass, J. D. (1998). Endogenous regulation of serotonin release in the hamster suprachiasmatic nucleus. *Journal of Neuroscience, 18*(13), 5045–5052. https://doi.org/10.1523/JNEUROSCI.18-13-05045.1998

Guillaumond, F., Dardente, H., Giguère, V., & Cermakian, N. (2005). Differential control of Bmal1 circadian transcription by REV-ERB and ROR nuclear receptors. *Journal of Biological Rhythms*, 20(5), 391–403. https://doi.org/10.1177/0748730405277232

Haque, S. N., Booreddy, S. R., & Welsh, D. K. (2019). Effects of BMAL1 manipulation on the brain's master circadian clock and behavior. *Yale Journal of Biology and Medicine*, 92(2), 251–258. https://pmc.ncbi.nlm.nih.gov/articles/PMC6585533/

Herzog, E. D., Hermanstyne, T., Smyllie, N. J., & Hastings, M. H. (2017). Regulating the suprachiasmatic nucleus (SCN) circadian clockwork: Interplay between cell-autonomous and circuit-level mechanisms. *Cold Spring Harbor Perspectives in Biology*, 9(1), a027706. https://doi.org/10.1101/cshperspect.a027706

Klein, D. C. (2007). Arylalkylamine N-acetyltransferase: "The Timezyme." *Journal of Biological Chemistry, 282*(7), 4233–4237. https://doi.org/10.1074/jbc.R600036200

Kuo, T. T., Wang, J., & Kay, S. A. (2026). Time matters: Circadian genetics and the molecular logic of human health and disease. *Nature Reviews Genetics*, 27(10), 744–761. https://doi.org/10.1038/s41576-026-00962-2

Lumsden, S. C., Clarkson, A. N., & Cakmak, Y. O. (2020). Neuromodulation of the pineal gland via electrical stimulation of its sympathetic innervation pathway. *Frontiers in Neuroscience, 14*, 264. https://doi.org/10.3389/fnins.2020.00264

Menet, J. S., Pescatore, S., & Rosbash, M. (2014). CLOCK:BMAL1 is a pioneer-like transcription factor. *Genes & Development,* 28(1), 8–13. https://doi.org/10.1101/gad.228536.113

Meyer-Bernstein, E. L., & Morin, L. P. (1996). Differential serotonergic innervation of the suprachiasmatic nucleus and the intergeniculate leaflet and its role in circadian rhythm modulation. *Journal of Neuroscience, 16*(6), 2097–2111. https://doi.org/10.1523/JNEUROSCI.16-06-02097.1996

Mieda, M., & Sakurai, T. (2011). Bmal1 in the nervous system is essential for normal adaptation of circadian locomotor activity and food intake to periodic feeding. *The Journal of Neuroscience*, 31(43), 15391–15396. https://doi.org/10.1523/JNEUROSCI.2801-11.2011

Prosser, R. A. (2003). Serotonin phase-shifts the mouse suprachiasmatic circadian clock in vitro. *Brain Research, 966*(1), 110–115. https://doi.org/10.1016/S0006-8993(02)04206-3

Rea, M. A., Glass, J. D., & Colwell, C. S. (1994). Serotonin modulates photic responses in the hamster suprachiasmatic nuclei. *Journal of Neuroscience, 14*(6), 3635–3642. https://doi.org/10.1523/JNEUROSCI.14-06-03635.1994

Roseboom, P. H., Coon, S. L., Baler, R., McCune, S. K., Weller, J. L., & Klein, D. C. (1996). Melatonin synthesis: Analysis of the more than 150-fold nocturnal increase in serotonin

N-acetyltransferase messenger ribonucleic acid in the rat pineal gland. *Endocrinology, 137*(7), 3033–3044. https://doi.org/10.1210/endo.137.7.8770929

Shelton, J., Yun, S., Losee Olson, S., Turek, F., Bonaventure, P., Dvorak, C., Lovenberg, T., & Dugovic, C. (2014). Selective pharmacological blockade of the 5-HT7 receptor attenuates light and 8-OH-DPAT induced phase shifts of mouse circadian wheel running activity. *Frontiers in Behavioral Neuroscience, 8*, 453. https://doi.org/10.3389/fnbeh.2014.00453

Simonneaux, V., & Ribelayga, C. (2003). Generation of the melatonin endocrine message in mammals: A review of the complex regulation of melatonin synthesis by norepinephrine, peptides, and other pineal transmitters. *Pharmacological Reviews, 55*(2), 325–395. https://doi.org/10.1124/pr.55.2.2

Stieglitz, M. S., Fenske, S., Hammelmann, V., Becirovic, E., Schöttle, V., Delorme, J. E., Schöll-Weidinger, M., Mader, R., Deussing, J., Wolfer, D. P., Seeliger, M. W., Albrecht, U., Wotjak, C. T., Biel, M., Michalakis, S., & Wahl-Schott, C. (2018). Disturbed processing of contextual information in HCN3 channel deficient mice. *Frontiers in Molecular Neuroscience, 10*, 436. https://doi.org/10.3389/fnmol.2017.00436

Strnad, H. K., & Spencer, R. L. (2026). Daily rhythms in clock gene mRNA expression in serotonergic brain regions of adult male rats. *Journal of Biological Rhythms, 41*(4), 383–399. https://doi.org/10.1177/07487304261427042

Takahashi, J. S. (2017). Transcriptional architecture of the mammalian circadian clock. *Nature Reviews Genetics*, 18(3), 164–179. https://doi.org/10.1038/nrg.2016.150

Turek, F. W., Joshu, C., Kohsaka, A., Lin, E., Ivanova, G., McDearmon, E., Laposky, A., Losee-Olson, S., Easton, A., Jensen, D. R., Eckel, R. H., Takahashi, J. S., & Bass, J.

(2005). Obesity and metabolic syndrome in circadian Clock mutant mice. *Science*, 308(5724), 1043–1045. https://doi.org/10.1126/science.1108750

Whitney, M. S., Shemery, A. M., Yaw, A. M., Donovan, L. J., Glass, J. D., & Deneris, E. S. (2016). Adult brain serotonin deficiency causes hyperactivity, circadian disruption, and elimination of siestas. *Journal of Neuroscience, 36*(38), 9828–9842. https://doi.org/10.1523/JNEUROSCI.1469-16.2016

Circadian Rhythms: Sleep, Caffeine, and Jet Lag

Unit: Circadian Rhythms

Chapter X: Sleep, Caffeine, and Jet Lag

CONTRIBUTORS

- Sarah Kim
  - Graphic Designer
  - Peer Reviewer
  - Subject Matter Expert: Caffeine
- Benjamin Shamash
  - Copyeditor
  - Peer Reviewer
  - Subject Matter Expert: Sleep/Wake Cycle
- Joseph Sultan
  - Copyeditor
  - Peer Reviewer
  - Subject Matter Expert: Flying Against the Biological Clock
- Sebert Supamongkolthaveesuk
  - Copyeditor
  - Peer Reviewer
  - Subject Matter Expert: Jet lag & Internal Desynchrony

LEARNING OBJECTIVES

After reading this chapter, students should be able to …

Ben

1. Describe the sleep/wake cycle and explain its relationship to circadian rhythms
2. Explain how circadian rhythms regulate sleep through the internal biological clock, light exposure, and melatonin.
3. Analyze how factors such as artificial light, travel, and irregular schedules can disrupt circadian rhythms and affect daily functioning
4. Evaluate everyday sleep habits and identify strategies that can help maintain a healthy and consistent sleep/wake cycle.

Joseph

5. Apply chronobiological principles to explain fatigue risks created by aviation schedules
6. Interpret chronobiological data and experimental results from pilot fatigue research
7. Interrelate chronobiology with aviation medicine, human factors, and aviation safety
8. Explain how biological timing influences pilot physiology, alertness, and behavior

Sebert

1. Construct/interpret visuals of how sleep/wake cycles adjust from jet lag over time, using zeitgebers and other external cues
2. Relate how changes made to one circadian rhythm can indirectly alter the timing of other circadian rhythms
3. Differentiate between synchronized and desynchronized rhythms: how can you determine if a rhythm is mistimed and its underlying cause?
4. Critique the effectiveness of various intervention methods used by the body and exogenous methods in addressing and recovering from jet lag.

Sarah

1. Readers will be able to understand where caffeine is commonly found, explain its purpose, and describe its effects on the body.
2. Readers will be able to describe how caffeine affects the brain, body, and sleep-wake system.
3. Readers will be able to analyze how consuming caffeine at different times of the day affects sleep quality and the body's natural sleep-wake cycle.
4. Readers will be able to apply scientific findings about caffeine timing to real-world situations involving daily schedules and sleep habits and identify relationships between the timing of caffeine consumption and sleep quality.

CHAPTER OUTLINE (Ben) : The Sleep/Wake Cycle

Introduction: The Sleep/Wake Cycle (~0.5 page)

- Introduce the idea that sleeping and waking up follow a predictable daily rhythm rather than occurring randomly
- Define the sleep/wake cycle and explain its connection to circadian rhythms
- Briefly introduce the idea that the body uses both internal and external cues to determine when we feel awake or sleepy
- Transition into how the circadian rhythm controls this timing

Learning Objective addressed: 1

Paragraph 2: How Circadian Rhythms Influence Sleep (~1 Page)

1. The internal biological clock

- Explain how circadian rhythms act as an approximately 24-hour internal biological clock that organizes sleep and wakefulness while still being influenced by outside signals

2. Light and Darkness

- Explain how light and darkness influence circadian timing, including how evening light from devices can delay sleep and reduce next-morning alertness,
- Use Chang et al. study.
- **Chapter Add-In**: Graph showing how melatonin levels change across a 24-hour period in relation to light and darkness

3. Melatonin
- Explain how darkness increases melatonin signaling and how melatonin helps regulate the timing of sleep.

Leaning Objective 2 Addressed

Paragraph 3: What Can Disrupt the Sleep/Wake Cycle? (~1 Page)

1. Artificial Light and Screen Use
- Explain how nighttime light and screen use can suppress melatonin and shift normal circadian timing.
- **Chapter Add-In:** "Screen vs Book Before Bed" Case study based on the Chang et al. eReader Experiment
2. Jet Lag and Behavioral Jet Lag
- Explain how travel across time zones and differences between weekday and weekend schedules can create a mismatch between behavior and the biological clock.
3. School, Work, and Irregular Schedules
- Discuss how inconsistent bedtimes, wake times, school schedules, and work schedules can interfere with normal circadian timing.

Learning Objective Addressed: 3

Paragraph 4: Effects of a Disrupted Sleep/Wake Cycle (~1.5 Pages)

1. Tiredness and Alertness
- Explain how circadian disruption can lead to daytime tiredness, reduce alertness, and difficulty waking at appropriate times.
2. Concentration and Daily Performance
- Discuss how disrupted or poorly timed sleep can make concentration, learning, and everyday performance more difficult.
3. Mood and emotional Regulation
- Explain how poor sleep can influence mood, emotional control, and responses to stress
- Use Sikka and Gross source
4. Broader Health Effects
- Discuss how repeated disruption of normal circadian timing may affect health beyond just causing tiredness

Learning Objective Addressed: 3

Paragraph 5: Maintaining A Healthy Sleep/Wake Cycle (~1 Page)

1. Consistent Sleep and Wake times
- Explain how keeping similar sleep and wake times throughout the week can reduce behavioral jet lag.
2. Managing Light Exposure
- Explain how limiting bright light and screens before bed and getting appropriate daytime light can support normal circadian timing
3. Melatonin and Circadian Restoration
- Discuss how melatonin and light approaches may be used to help shift or restore disrupted circadian rhythms
4. Building a Healthy Sleep/Wake Routine

- **Chapter Add-In:** How-To Box on maintaining a healthy Circadian Rhythm through consistent sleep timing and light exposure

Learning Objective Addressed: 4

CHAPTER WRITTEN CONTENT (Ben) : The Sleep/Wake Cycle

Introduction

Every day, the body follows an internal schedule that helps determine when we feel awake, alert, and ready to sleep. The sleep/wake cycle is cyclical in nature and closely related to the circadian system. Circadian rhythms are produced internally by the body, but are often influenced and adjusted by external cues such as light exposure and daily routines. Understanding how circadian rhythms regulate the sleep/wake cycle helps explain why sleep disruptions occur and how healthy sleep patterns can be maintained.

How Circadian Rhythms Influence Sleep

There exists an internal clock within an individual that regulates the sleep/wake cycle throughout the day and generates circadian rhythms with a period of approximately 24 hours. External stimuli, particularly light, can also help regulate the circadian cycle to match the environment. Light during the day promotes wakefulness, while darkness in the evening facilitates sleep-related changes.

Exposure to light at night can disrupt this transition by signaling daytime to the body. In a study by Chang et al. (2015), subjects were asked to read either a printed book or an eReader that emitted light before bed. Subjects using the eReader showed lower alertness the next day, reduced melatonin secretion, delayed circadian phase, and longer time to sleep compared to those who read the printed book. This study shows that exposure to light at night can disrupt sleep at night as well as alter alertness during the day. These results illustrate the influence of external stimuli on the internal sleep/wake cycle.

Another important contributor to the management of the sleep/wake cycle is melatonin. Melatonin is a hormone that plays a role in informing the body when it's time to start preparing for sleep. The production of this hormone is related to exposure to light. As light levels decrease, melatonin production increases and tends to aid in sleep. Exposure to light can inhibit or postpone the production of melatonin secretion and consequently disrupt circadian phase (Vasey et al., 2021).

Melatonin also plays a significant role in aligning the body to the day/night cycle. If melatonin is secreted at the expected time, then the shift from wake to sleep occurs properly. If there is a shift in the secretion of melatonin there will also be a shift in the sleep/wake cycle. Disruptions in melatonin secretion timing are associated with problems such as jet lag and other circadian sleep

disturbances (Zisapel, 2018). Thus, melatonin is an important connection between the body's internal circadian system and natural sunlight conditions. Light, melatonin, and the circadian

Case Study: Screen vs Book Before Bed

Background

Consuming electronic devices such as a phone, tablet, or computer prior to bed is a fairly common occurrence. The light produced from these devices could have an influence on circadian rhythms. Chang et al. wanted to investigate whether reading from a light-emitting eReader to bed had an effect on sleep compared to reading from a printed book.

The Experiment

Twelve healthy young adults took part in the study. On five separate nights they read for around four hours prior to bedtime from a light-emitting eReader. On the same five nights they also read a printed book in a low-light environment. Alertness the next day, sleep onset, and melatonin release were all measured.

The Results

Participants who used the eReader demonstrated delayed circadian rhythm, longer sleep onset, lower levels of alertness the next day, and decreased melatonin release when compared to participants who read a printed book. This indicates that light emitted from electronic devices prior to bedtime can disrupt the natural sleep/wake cycle (Chang et al., 2015).

Think About It!

1. Why might reading from an eReader before bed make it harder for someone to fall asleep compared to reading a printed book?
2. How might the results of this experiment relate to students who use their phones or laptops before going to sleep?

Answer Key

1. Light emitted from the eReader can suppress melatonin secretion and delay circadian timing, making it harder for the body to prepare for sleep.
2. Students who use light-emitting devices before bed may experience delayed sleep timing and reduced alertness the following morning, potentially affecting their performance in school.

What Can Disrupt the Sleep/Wake Cycle?

Although circadian rhythms are generated within the body, modern lifestyles can disrupt the external signals that help maintain circadian timing. One example of this is artificial light. Phones, laptops, televisions, and similar devices can provide bright light at an unusual time of night. Chang et al. (2015) have found that exposure to bright light at an unusual time of night can shift the circadian phase and decrease melatonin. With repeated exposure like this, an individual's preferred sleep time may shift later, but they are still expected to wake up early due to school or work. As a result, they may get fewer hours of sleep than they need, leading to daytime sleepiness, reduced alertness, and difficulty concentrating throughout the day.

Travel may also cause disruption in the sleep/wake cycle. If an individual rapidly crosses time zones and their circadian clock remains in sync with the time in the previous location, they experience jet lag. For example, when arriving at a new location the time may be nighttime, but the individual's circadian clock may still indicate daytime. It will then take time and environmental signals, especially sunlight, for the circadian system to shift appropriately. There have also been studies on the use of melatonin to shift circadian phase during jet lag. The effectiveness of melatonin depends on when it is taken relative to the individual's internal clock. For example, taking melatonin at an appropriate time may help the body adjust to a new time zone, while taking it at the wrong time can shift the circadian clock in an unwanted direction (Zisapel, 2018).

Another form of disruption is behavioral jet lag, which involves a shift in behaviors such as sleeping, waking, or eating at markedly different times between weekdays and weekends. Xu et al. (2025) examined sleep patterns in young adults and found that larger differences in sleep between weekdays and weekends were correlated with more circadian disruption and later timing of melatonin. This could be especially important to students who wake up early during the week but stay up later and wake up several hours later on the weekend. Although they have not traveled across time zones, their changes in sleep schedule may still create a mismatch with their circadian timing.

Work and school schedules may also cause circadian disruption. An individual's circadian system may indicate that they are biologically ready to sleep or wake at a certain time, but their work obligations may require them to follow a different schedule. For instance, a night shift worker may need to stay awake during the biological night, when the body is naturally ready for sleep, and attempt to sleep during the biological day. In these cases, behavior, the environmental light-dark cycle, and circadian timing may not be in sync. This is known as circadian misalignment or circadian disruption (Sletten et al., 2020).

Tiredness at times when a person needs to be awake is one of the most noticeable effects of circadian disruption. When the internal clock doesn't match a person's sleep schedule, they may have difficulty falling asleep and waking up at the times they should. This can result in daytime sleepiness and reduced alertness. Over time, repeated sleep disruption may interfere with concentration, memory, and their ability to perform everyday tasks. Sletten et al. (2020) describe impaired attention, memory, and other aspects of cognition as consequences associated with circadian misalignment, particularly among people experiencing work shifts.

Sleep and circadian timing are also connected to emotional functioning. Sikka and Gross (2023) explain that emotional experiences vary across the wake-sleep cycle rather than remaining constant throughout the day and night. Circadian timing, sleep, daily activities, and other environmental factors can all contribute to these changes. Because emotional experiences are influenced by both sleep and circadian timing, disrupted or insufficient sleep can affect more than physical tiredness. It may also influence mood and the way people respond emotionally to events in their daily lives.

Studies have linked chronic circadian disruption to changes in the immune, cardiovascular, and metabolic systems (Sletten et al., 2020). For instance, glucose metabolism can be disrupted by consuming food during the biological night, and metabolic disorders have also been found in shift workers (Sletten et al., 2020). This indicates that circadian disruption can have consequences on health outside of fatigue and alertness.

Sleep also plays a role in mental and physical health. Good sleep has been linked to mood, immune function, cardiovascular function, hormone regulation, cognition, memory consolidation, and mental health (Baranwal et al., 2023). As such, disruption to the sleep/wake cycle can potentially disrupt a variety of functions within the body, such as hormone regulation, immune responses, and memory consolidation. This could have an influence on both short-term health (such as poor attention) due to poor sleep and long-term health due to chronic circadian and sleep disruption. However, one night of poor sleep does not necessarily have such consequences. The more detrimental effects are more likely to be seen with chronic or repeated circadian and sleep disruption.

Maintaining A Healthy Sleep/Wake Cycle

To help synchronize the circadian system to the environment, it is also beneficial to maintain a regular sleep/wake schedule across days of the week to minimize behavioral jet lag (Xu et al., 2025). Also, daily exposure to sunlight and minimizing device use prior to sleep can aid in maintaining proper circadian phase and minimize delays in melatonin onset (Chang et al., 2015). In cases of circadian disruption (e.g., jet lag), appropriately timed light and melatonin may assist in phase-shifting the circadian system to a normal phase (Zisapel, 2018). These strategies can aid in minimizing disruption to the sleep/wake cycle and promote improved sleep and performance.

How to Box: Maintaining A Healthy Sleep/Wake Cycle

Do	**Don't**
Keep a consistent sleep schedule	Avoid major changes in sleep timing
Get natural light during the day	Avoid excessive bright light before bed
Maintain a regular nighttime routine	Avoid irregular nighttime habits
Allow time to adjust when traveling	Don't expect your internal clock to adjust immediately
Understand the importance of melatonin timing	Don't assume taking melatonin at any time will improve sleep

Works Cited

Baranwal, N., Yu, P. K., & Siegel, N. S. (2023). Sleep physiology, pathophysiology, and sleep hygiene. *Progress in Cardiovascular Diseases, 77*, 59–69. https://doi.org/10.1016/j.pcad.2023.02.005

Chang, A. M., Aeschbach, D., Duffy, J. F., & Czeisler, C. A. (2015). Evening use of light-emitting eReaders negatively affects sleep, circadian timing, and next-morning alertness. *Proceedings of the National Academy of Sciences of the United States of America, 112*(4), 1232–1237. https://doi.org/10.1073/pnas.1418490112

Sikka, P., & Gross, J. J. (2023). Affect across the wake-sleep cycle. *Affective Science, 4*(3), 563–569. https://doi.org/10.1007/s42761-023-00204-2

Sletten, T. L., Cappuccio, F. P., Davidson, A. J., Van Cauter, E., Rajaratnam, S. M. W., & Scheer, F. A. J. L. (2020). Health consequences of circadian disruption. *Sleep, 43*(1), zsz194. https://doi.org/10.1093/sleep/zsz194

Vasey, C., McBride, J., & Penta, K. (2021). Circadian rhythm dysregulation and restoration: The role of melatonin. *Nutrients, 13*(10), 3480. https://doi.org/10.3390/nu13103480

Xu, Y. X., Li, J., Wan, Y. H., Su, P. Y., Tao, F. B., & Sun, Y. (2025). Association between behavioral jet lag with subjective and objective circadian rhythm among Chinese young adults. *Social Science & Medicine, 366*, 117689. https://doi.org/10.1016/j.socscimed.2025.117689

Zisapel, N. (2018). New perspectives on the role of melatonin in human sleep, circadian rhythms and their regulation. *British Journal of Pharmacology, 175*(16), 3190–3199. https://doi.org/10.1111/bph.14116

CHAPTER Outline (Joseph) : The Sleep/Wake Cycle

Unit: Circadian Rhythms
Chapter: Flying Against the Biological Clock
Learning Objectives:

- Apply chronobiological principles to explain fatigue risks created by aviation schedules
- Interpret chronobiological data and experimental results from pilot fatigue research
- Interrelate chronobiology with aviation medicine, human factors, and aviation safety
- Explain how biological timing influences pilot physiology, alertness, and behavior

Chapter Outline

Flying Against the Biological Clock

- Purpose: Introduce aviation as a real-world application of circadian biology rather than reteaching the basic circadian system.
    - Why aviation creates a unique conflict between biological and operational time
    - Overnight and early-morning flights
    - Irregular work schedules
    - Rapid movement across time zones
    - Difference between sleep deprivation and circadian disruption
    - Introduce the Window of Circadian Low (WOCL)

How Flight Schedules Disrupt Circadian Timing

- Overnight and Early-Morning Operations

    - Pilots may be required to work during their biological night
    - Early departures and late arrivals can interfere with normal sleep opportunities
    - Repeated schedule changes make maintaining a regular sleep schedule difficult
- Transmeridian Flight
    - Crossing several time zones rapidly
    - Internal biological time versus local time
    - Jet lag and incomplete circadian adjustment
- Short-Haul vs. Long-Haul Operations
    - Long-haul: time zone changes, long duty periods
    - Short-haul: repeated flights, early starts, irregular schedules, limited recovery
    - Research showing circadian disruption in real pilot schedules

What Happens to the Pilot?

- Sleepiness and Alertness
    - Increased sleepiness during the biological night
    - Reduced vigilance
    - Interaction between circadian timing and sleep pressure
- Cognitive and Behavioral Effects
    - Slower reaction time
    - Reduced attention and vigilance
    - Decision-making
    - Situational awareness
- Biological Time vs. Hours of Sleep
    - Fatigue is not determined only by the amount of sleep obtained
    - Circadian phase influences performance independently of sleep history
    - Interaction among sleep duration, time awake, and biological timing

Case Study: Colgan Air Flight 3407

- The Accident
    - February 12, 2009
    - Flight approaching Buffalo-Niagra International Airport
    - Aircraft entered an aerodynamic stall and crashed near Buffalo
    - All 49 people aboard and one person on the ground were killed
    - The National Transportation Safety Board (NTSB) determined the probable cause was the captain's inappropriate response to the stick shaker, leading to an aerodynamic stall from which the aircraft did not recover
- Pilot Sleep and Fatigue
    - Examine the captain's and first officer's sleep opportunities before the flight
    - Commuting and opportunities for adequate rest
    - Extended wakefulness and irregular sleep
    - Relate these conditions to sleep pressure and circadian timing
- Applying Chronobiology to the Accident
    - How inadequate sleep could interact with biological timing
    - Possible effects on vigilance, attention, and decision-making
    - Connect the case to experimental pilot fatigue research discussed earlier
- Broader Impact

    - Increased attention to pilot fatigue and commuting
    - Relationship between accident investigations and aviation policy
    - Development of stronger flight- and duty-time protections
    - Demonstrates how chronobiological research can influence transportation safety

Designing Aviation Around Human Biology

- Flight and Duty Scheduling
    - Flight-time and duty-time limitations
    - Rest requirements
    - Importance of considering when a flight occurs, not only how long pilots work
- Fatigue Risk Management
    - Fatigue Risk Management Systems (FRMS)
    - Using scientific and operational evidence to identify fatigue risks
    - Scheduling around known periods of reduced biological alertness when possible
- Countermeasures
    - Pre-light sleep and naps
    - Controlled rest
    - Strategic caffeine use
    - Light exposure
    - Scheduling and adequate recovery periods
- Limitations
    - Countermeasures can reduce but can't completely eliminate fatigue
    - Regulations based only on hours worked can't fully account for circadian timing
    - Aviation illustrates the need to design human systems around biological limitations

Key Vocabulary

- Window of Circadian Low (WOCL): The period of the biological night when alertness and performance are typically lowest and the drive for sleep is greatest
- Fatigue Risk Management System (FRMS): A science-based system used by aviation organizations to identify, monitor, and attempt to reduce fatigue-related safety risks
- Flight Duty Period (FDP): The regulated period beginning when a pilot reports for duty and ending after completion of the final flight, used in determining fatigue-related scheduling limits
- Controlled Rest: A planned, limited period of sleep on the flight deck used as an operational countermeasure to improve pilot alertness during appropriate phases of flight
- Augmented Flight Crew: A flight crew with additional qualified pilots beyond the minimum required, allowing pilots to rotate into scheduled in-flight rest during long operations

Sources

Flynn-Evans, E. E., Bathurst, N., Pradhan, S., Gregory, K. B., Barger, L. K., Lamp, A. C. M., Glaros, Z., Arsintescu, L., Jansen, R. A., Baumgartner, H. M., Nesthus, T. E., & Hilditch, C. J. (2026). Characterizing Pilot Fatigue During Circadian-Disrupted Short-Haul Aviation

Operations. *Aerospace Medicine and Human Performance*, *97*(8), 607–617. Aerospace Medical Association. https://doi.org/10.3357/AMHP.6898.2026

Gander, P. H., Mulrine, H. M., van den Berg, M. J., Smith, A. A. T., Signal, T. L., Wu, L. J., & Belenky, G. (2015). Effects of sleep/wake history and circadian phase on proposed pilot fatigue safety performance indicators. *Journal of Sleep Research*, *24*(1), 110–119. PubMed. https://doi.org/10.1111/jsr.12197

Hilditch, C. J., Gregory, K. B., Arsintescu, L., Bathurst, N. G., Nesthus, T. E., Baumgartner, H. M., Lamp, A. C. M., Barger, L. K., & Flynn-Evans, E. E. (2023). Perspectives on fatigue in short-haul flight operations from US pilots: A focus group study. *Transport Policy*, *136*. https://doi.org/10.1016/j.tranpol.2023.03.004

National Transportation Safety Board (NTSB). (2010). Loss of Control on Approach, Colgan Air, Inc., Operating as Continental Connection Flight 3407, Bombardier DHC-8-400, N200WQ, Clarence Center, New York, February 12, 2009 (Aircraft Accident Report NTSB/AAR-10/01). https://www.ntsb.gov/investigations/accidentreports/reports/aar1001.pdf

Wingelaar-Jagt, Y. Q., Wingelaar, T. T., Riedel, W. J., & Ramaekers, J. G. (2021). Fatigue in aviation: Safety risks, preventive strategies and pharmacological interventions. *Frontiers in Physiology*, *12*(712628). https://doi.org/10.3389/fphys.2021.712628

Yang, S. X., Cheng, S., Sun, Y., Tang, X., & Huang, Z. (2024). Circadian Disruption in Civilian Airline Pilots. *Aerospace Medicine and Human Performance*, *95*(7), 381–389. Aerospace Medical Association. https://doi.org/10.3357/AMHP.6316.2024

CHAPTER Written Content (Joseph) : The Sleep/Wake Cycle

Case Study: Colgan Air Flight 3407

The Accident and Preflight Circumstances

On February 12, 2009, Colgan Air Flight 3407 was approaching Buffalo-Niagara International Airport after departing Newark, New Jersey. As the aircraft's airspeed decreased during the approach, its stick shaker, a warning system that physically vibrates the pilots' control columns when the aircraft approaches an aerodynamic stall, activated. An aerodynamic stall occurs when the angle between an aircraft's wings and the oncoming airflow, known as the angle of attack, exceeds a critical threshold, causing airflow separation and a substantial reduction in lift. Rather than initiating the appropriate stall-recovery procedure, the captain pulled backward on the control column, increasing the aircraft's angle of attack and worsening the stall. The aircraft subsequently crashed into a residence in Clarence Center, New York, killing all 49 people aboard and one person on the ground. The National Transportation Safety Board (NTSB) identified the captain's inappropriate response to the stall warning as the probable cause of the accident (NTSB, 2010).

Beyond the immediate circumstances of the crash, Flight 3407 raised broader questions about pilot fatigue, particularly how sleep opportunities and activities before reporting for duty may influence operational readiness. Before the accident flight, both pilots had experienced limited opportunities for quality sleep. The captain had spent the previous night in the airline's crew room, an environment that was not conducive to restorative sleep due to interruptions, noise, activity, and light (NTSB, 2010). The first officer had begun a transcontinental commute from Seattle, Washington approximately 15 hours before her scheduled report time, traveling through Memphis, Tennessee before reaching Newark, New Jersey. According to the NTSB investigation, this commute did not provide an opportunity for an uninterrupted sleep period (NTSB, 2010). These circumstances illustrate an important principle of aviation chronobiology, in which the hours immediately spent flying are only part of a pilot's fatigue risk. Sleep obtained before duty, time already spent awake, commuting, and biological timing can all influence how alert a pilot is when entering the cockpit.

Sleep Pressure and Circadian Regulation

The crew's limited and irregular sleep opportunities are particularly relevant when examining the interaction between homeostatic sleep pressure and circadian timing. Sleep pressure increases the longer a person remains awake and decreases through restorative sleep. For both pilots, disrupted or limited sleep opportunities before reporting for duty may have prevented that pressure from fully dissipating. At the same time, the timing of sleep matters. Attempting to sleep during irregular hours or while traveling can conflict with the body's circadian signals for wakefulness and sleep. Together, elevated sleep pressure and circadian misalignment can impair alertness during flight operations, particularly when duty periods overlap with the biological night or follow insufficient restorative sleep.

The interaction between sleep pressure and circadian timing has important implications for aviation performance, where pilots must sustain attention while processing multiple sources of information. During the approach to Buffalo, the flight crew was responsible for simultaneously tracking airspeed, aircraft configuration, instrument readings, weather conditions, and approach procedures. However, the NTSB investigation identified the crew's failure to adequately monitor decreasing airspeed during the approach as a contributing factor in the accident (NTSB, 2010). Fatigue can affect functions involved in this task, including vigilance, attention, and decision-making. The circumstances surrounding Flight 3407 illustrate the operational relevance of research examining the relationship between sleep history, circadian timing, and cognitive performance. Researchers investigated how sleep-wake history and circadian phase influence proposed measures of pilot fatigue and safety performance. Their research provides a scientific framework for understanding why inadequate sleep and biological timing must be considered when evaluating the cognitive demands of flight operations (Gander et al., 2015).

Preflight Fatigue and Restorative Sleep

Although the captain and first officer reported for the same flight, their preceding sleep-wake histories differed substantially. The captain's overnight stay in the crew room and the first officer's extended commute illustrate how fatigue-related risks, such as insufficient sleep or irregular sleep schedules, can develop before a scheduled duty period begins. The NTSB found evidence that the captain had been awake at several points during the night and concluded that both pilots had experienced some degree of sleep loss. The investigation considered fatigue an important safety issue and examined how the pilots' activities before reporting for duty may have affected their performance, raising broader concerns about fatigue management in commercial aviation (NTSB, 2010).

The pilots' contrasting preflight experiences highlight a broader limitation of aviation scheduling, that time allocated for rest does not necessarily translate into sufficient biological

recovery. Although scheduled rest periods provide opportunities for sleep, their restorative value depends on whether individuals can obtain sleep of adequate duration and continuity. Effective recovery requires opportunities for sleep that are sufficiently long, continuous, and free from environmental disturbances. The first officer's commute illustrates this problem particularly well. Traveling overnight from Seattle toward Newark created opportunities to sleep, including while aboard an aircraft, but these opportunities differed from a consolidated sleep period in a normal sleeping environment. The captain's use of the crew room presented a different form of the same problem. Although he was not performing flight duties during the night, the environment did not provide ideal conditions for continuous sleep. Consequently, measuring only the time between scheduled duty periods would provide an incomplete description of either pilot's physiological state before the accident flight.

From Chronobiological Research to Aviation Safety

The relationship between homeostatic sleep pressure and circadian timing further complicates the distinction between scheduled rest and biological recovery. Although both processes influence sleepiness and alertness, they operate through different physiological mechanisms. Homeostatic sleep pressure accumulates during wakefulness and decreases during sleep, whereas the circadian system regulates daily fluctuations in the tendency toward sleep and wakefulness. Consequently, obtaining several hours of sleep may reduce accumulated sleep pressure without necessarily aligning an individual's internal biological timing with the demands of their environment. Conversely, the circadian promotion of wakefulness during the biological day is unable to fully compensate for substantial prior sleep loss. The circumstances surrounding Flight 3407 illustrate how these interacting processes must be considered when evaluating fatigue-related risks, particularly when individuals experience irregular sleep schedules or insufficient opportunities for restorative sleep.

Beyond Aviation: Implications for Everyday Life

Aviation operations depend on sustained vigilance, accurate information processing, and coordinated decision-making, particularly during demanding phases of flight. These cognitive functions are sensitive to sleep loss and circadian disruption, illustrating why fatigue management must account for human physiological limitations alongside operational requirements. Controlled laboratory studies allow researchers to examine how sleep history and circadian phase influence alertness and cognitive performance under systemically measured conditions. Aviation operations, however, introduce additional demands, including complex decision-making, environmental variability, and the need to maintain attention over extended periods. The circumstances surrounding Flight 3407 demonstrate the practical importance of studying these biological processes in operational settings, where reductions in vigilance may have serious safety implications. Experimental measures such as reaction time or sustained attention may appear small when considered by themselves, but aviation requires these functions to remain reliable for extended periods. The case therefore provides a bridge between chronobiological data and the practical study of human factors, the ways in which human abilities and limitations interact with equipment, environments, and procedures.

Although aviation provides a particularly consequential example of fatigue-related performance risks, the underlying biological mechanisms extend beyond the cockpit. University students studying late into the night, healthcare professionals working rotating shifts, and individuals traveling across multiple time zones experience similar interactions between accumulated sleep pressure and circadian timing. For example, a student who remains awake studying for an early exam may experience reduced vigilance and slower information processing because of elevated sleep pressure, even when motivated to remain attentive. Similarly, individuals working overnight may experience increased sleepiness when their schedules conflict with the circadian system's normal promotion of sleep during the biological

night. These examples demonstrate that the relationship between biological timing and cognitive performance is a fundamental feature of human physiology rather than a phenomenon unique to aviation.

Ultimately, the case of Colgan Air Flight 3407 demonstrates why sleep must be understood as a biological process, rather than simply a period of inactivity between scheduled responsibilities. Adequate recovery depends not only on the amount of time available for sleep but also on sleep continuity, prior wakefulness, and alignment with the body's circadian rhythms. Maintaining consistent sleep schedules and allowing sufficient opportunities for uninterrupted sleep can help support alertness and cognitive performance, although these practices are unable to eliminate every source of fatigue. The broader lesson of aviation chronobiology is that human performance cannot be evaluated solely through scheduled work hours or time spent resting. Instead, understanding the interaction between sleep pressure and circadian timing provides a scientific foundation for recognizing fatigue-related risks in aviation, education, healthcare, and other aspects of everyday life.

Chapter Add-In

The Psychomotor Vigilance Task (PVT) is a test of sustained attention, or the ability to maintain focus over time. Participants respond as quickly as possible when a visual stimulus appears on a screen. Researchers measure reaction time, the interval between the appearance of the stimulus and the participant's response. These measurements can be converted into response speed, expressed in responses per second (s^{-1}). Higher response speeds indicate faster reactions and generally better performance on the task.

Gander et al. (2015) examined PVT performance among commercial airline pilots during long-range and ultra-long-range flights. The researchers measured performance near the top of descent (TOD), when an aircraft begins descending toward its destination. Flights were grouped according to arrival time expressed in domicile time, the local time in the city where each trip began and ended. This served as an approximate indicator of circadian phase.

The researchers were particularly interested in whether the timing of flight operations was associated with differences in pilots' alertness and cognitive performance. This relationship is important, as circadian rhythms influence the body's natural tendency toward sleep and wakefulness throughout the day. During the biological night, when the circadian system promotes sleep, individuals may experience greater difficulty maintaining attention and responding quickly to stimuli. In aviation, these changes are especially relevant during demanding phases of flight, when pilots must continuously process information and respond to changing conditions. Examining PVT performance across different arrival times therefore provides insight into how biological timing may relate to the cognitive abilities required for safe flight operations.

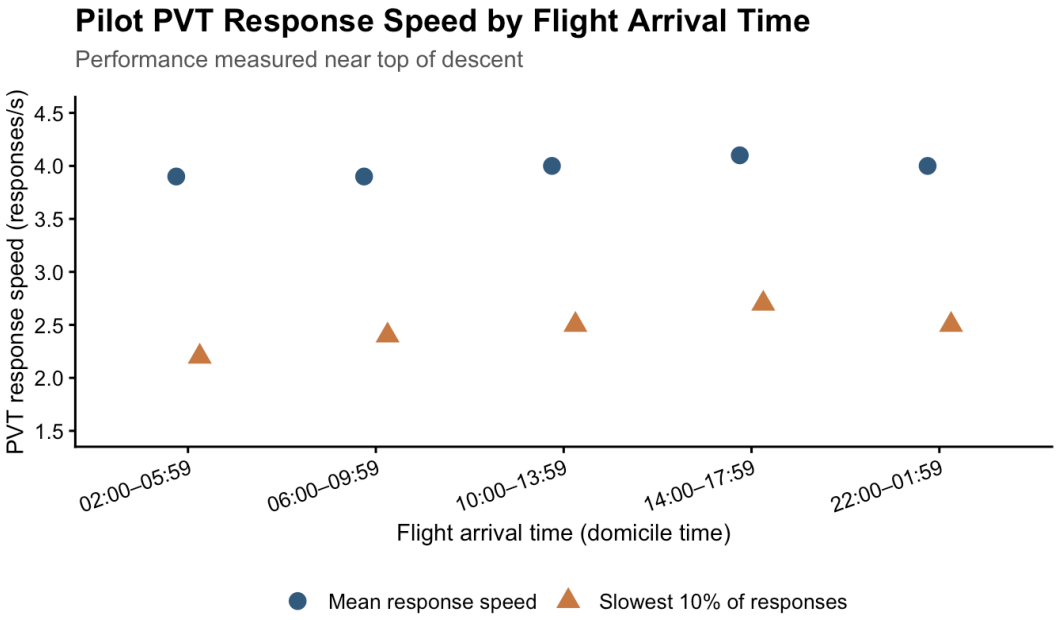


Figure 1. **Pilot reaction speed by flight time.** Psychomotor Vigilance Task (PVT) performance measured within 60 minutes before the top of descent, grouped by flight arrival time expressed in domicile time. The graph displays estimated mean response speed and response speed for the slowest of 10% of reactions. Higher values indicate faster responses. Mean response speed was significantly higher for flights arriving between 14:00 and 17:59 than for flights arriving between 02:00 and 05:59 (P = 0.0041) or 06:00 and 09:59 (P = 0.0050). The 18:00-21:59 interval was excluded because only eight flights fell within that category. Adapted from Gander et al. (2015), Figure 4. Values not explicitly reported in the article were estimated visually; standard-error bars are omitted.

Figure 1 shows that pilots on flights arriving between 14:00 and 17:59 demonstrated the fastest mean responses, approximately 4.1 responses per second. In contrast, mean response speed was approximately 3.9 responses per second for flights arriving between 02:00 and 09:59. These differences were statistically significant, meaning they provided evidence of differences between the groups beyond what would ordinarily be expected from random sampling variation. The slowest 10% of responses showed a similar pattern, ranging from 2.2 responses per second in the earliest arrival interval to 2.7 in the afternoon interval. These findings are consistent with circadian variation in alertness, although arrival time alone cannot establish the cause of the differences.

CHAPTER Outline (Sarah)

Outline of 5-page Contribution

1. Introduction to Caffeine- Learning Objective: Readers will be able to understand where caffeine is commonly found, explain its purpose, and describe its effects on the body.
    a. What is Caffeine and Where is it Found?
    b. Why Do People Consume Caffeine?

2. How Caffeine Affects the Brain and Body- Learning Objective: Readers will be able to describe how caffeine affects the brain, body, and sleep-wake system.
    a. Caffeine and the Central Nervous System
    b. Adenosine and Sleep Pressure
    c. Caffeine's Effects on the Sleep-Wake System
3. Caffeine Timing and Sleep Quality- Learning Objective: Readers will be able to analyze how consuming caffeine at different times of the day affects sleep quality and the body's natural sleep-wake cycle.
    a. The Circadian Rhythm and Biological Clock
    b. Caffeine Consumption in the Morning
    c. Caffeine Consumption in the Afternoon
    d. Caffeine Consumption in the Evening
4. Readers will be able to apply scientific findings about caffeine timing to real-world situations involving daily schedules and sleep habits and identify relationships between the timing of caffeine consumption and sleep quality.
    b. Findings from Scientific Studies
    c. Summary and Conclusion Connecting Back to Daily Life

5 Vocabulary Words

1. Caffeine: A widely consumed stimulant, found in coffee and other beverages, that can improve performance and alertness while reducing tiredness. (O'Callaghan et al., 2018).
2. Circadian Rhythm: A self-sustaining biological rhythm that follows an approximately 24-hour cycle and continues even without external time cues. It regulates processes such as sleep, wakefulness, body temperature, and hormone release. (Czeisler et al., 2019).
3. Sleep Efficiency: The percentage of time spent in bed that a person is actually asleep (O'Callaghan et al., 2018).
4. Sleep-Wake Cycle: An approximately 24-hour biological cycle that regulates periods of sleep and wakefulness, primarily influenced by the circadian pacemaker, the body's internal biological clock (Beersma et. al, 2007).
5. Adenosine: A chemical produced through cellular metabolism that acts as a neuromodulator in the nervous system and regulating sleep. Adenosine builds up during wakefulness, increasing feelings of sleepiness, and decreases during sleep (Antle et al., 2001).

CHAPTER WRITTEN OUTLINE (Sarah Kim): Caffeine

Name: Sarah Kim

Unit: Circadian Rhythms (~24 hours)

How Does the Timing of Caffeine Consumption Affect Sleep?

INTRODUCTION TO CAFFEINE

a. What is Caffeine and Where is it Found?

Caffeine, one of the most widely consumed stimulants worldwide, is a naturally occurring compound belonging to the methylated xanthine group, which also includes theophylline, found in tea, and theobromine, found in cocoa (Portillo et al., 2022). While most people think of coffee or energy drinks when they think of caffeine, caffeine comes from many other sources commonly found in people's daily lives. It can be found in more than 60 plant species, but the most common natural sources of caffeine include coffee beans, tea leaves, kola nuts, and cocoa beans (Tarka et al., 1998). After consumption, caffeine typically reaches its highest concentration in the bloodstream within approximately 45–60 minutes. Though, this number varies with each person (Ágoston et al., 2017). In the United States, coffee accounts for approximately 75% of total caffeine intake, while tea contributes 15% and caffeinated soda contributes 10%, respectively. In sources like chocolate and caffeine-containing medications, caffeine makes up a smaller portion, as the amount of caffeine depends on the plant source and type of product (Tarka et al., 1998). Caffeine is found in almost everyone's daily life, and people may consume it more than they think, which is why it is important to know how caffeine can affect your body and sleep.

b. Why Do People Consume Caffeine?

In a 2017 cross-sectional study, Ágoston et al. identified six main reasons why participants consumed caffeine: to increase alertness, out of habit, to improve mood, for social reasons, for its taste, and to manage symptoms. These motivations reflect caffeine's stimulating effects on the central nervous system, its role in daily routines, and its ability to temporarily reduce fatigue and sleepiness. Generally, caffeine is also known for reducing tiredness, improving alertness, and increasing reaction speed (Ágoston et al., 2017). A study by Orbeta et al. (2006) examined 15,686 American adolescents and found that people who reported consuming high amounts of caffeine were more likely to experience morning tiredness than those who consumed very little caffeine, even after accounting for socioeconomic factors. Although caffeine is commonly used to increase alertness and reduce the effects of fatigue, people who are already tired may consume more caffeine to feel awake. This creates a potential cycle in which caffeine is used to manage tiredness while also potentially contributing to the fatigue experienced by individuals (Orbeta et al. 2006).

HOW CAFFEINE AFFECTS THE BRAIN AND BODY

a. Caffeine and its effects on the Central Nervous System and Adenosine

When consumed in beverages, caffeine acts on the central nervous system by blocking adenosine receptors, reducing drowsiness (Portillo et al., 2022). As a central nervous system (CNS) stimulant, caffeine contributes to its psychostimulant effects and influences cognitive functioning (Fiani et al., 2021). Caffeine can affect cognitive abilities in both humans and animals, as caffeine is seen to be related to improved memory retention (Fiani et al., 2021). Caffeine may also decrease the availability of serotonin, which can reduce its calming effects on brain activity. This change can influence processes related to sleep, motor function, and the

regulation of blood vessels in the brain, all of which are affected by serotonin (Nehlig et al., 1992).

b. Caffeine's Effects on the Sleep-Wake System and the Circadian Rhythm

In human cells, caffeine causes the circadian period to become longer. Researchers found that this effect was primarily connected to adenosine receptors and cAMP signaling, which are involved in cellular communication and circadian regulation (Reichart et al. 2022). Beyond its effects on the body's internal clock, caffeine's impact on alertness may also vary depending on when it is consumed and how sleepy a person is. As someone stays awake longer, their need for sleep gradually increases, often leading to greater sleepiness around midday. Research suggests that caffeine's ability to increase alertness may vary depending on these factors. Its stimulating effects may be more noticeable around midday, when sleepiness is greater, than in the evening (Roehrsa & Rotha, 2017).

CAFFEINE TIMING AND SLEEP QUALITY

a. Caffeine Consumption in the Morning

Morning tiredness may lead people to consume more caffeine, which can then contribute to poorer sleep in the long run (Snel et al. 2011). Researchers have found that feeling tired in the morning can lead people to consume more caffeine, creating the “coffee cycle”, where one finds themselves in a loop where morning drowsiness leads to high caffeine intake. This then ruins nighttime sleep quality and causes even more exhaustion the next day (O'Callaghan et al. 2018). In a study done by Landolt et al. (1994), participants were given 200 mg of caffeine at 7 a.m. and were monitored using electroencephalography (EEG). Even though caffeine levels had

significantly decreased by the time participants went to sleep, both sleep efficiency and total sleep time (TST) were lower compared with the placebo condition. The researchers concluded that caffeine remaining in the central nervous system can reduce the gradual development of drowsiness that normally occurs after being awake for long periods of time (Landolt et al. 1994).

b. **Caffeine Consumption in the Afternoon**

Consuming caffeine even six hours before bed significantly disrupted sleep compared with a placebo (O'Callaghan et al. 2018). Caffeine also changed the timing of different sleep stages, causing rapid eye movement (REM) sleep to occur earlier in the night and stages 3 and 4 to occur later during the shortened sleep period. Many studies have focused on caffeine consumed during the daytime, with several hours between consumption and falling asleep. This time gap can make it difficult to separate caffeine effects from other factors that may influence sleep, such as food and dietary intake (O'Callaghan et al. 2018).

c. **Caffeine Consumption in the Evening**

A 100 mg dose of caffeine consumed immediately before sleep alters non-REM brain activity by reducing delta waves, which weakens deep restorative rest, while simultaneously increasing sleep spindles to block external disturbances. Although the brain attempts to maintain sleep stability through these bursts of activity, overall sleep quality is compromised (Landolt et al. 1995). Additionally, evening caffeine caused the melatonin rhythm to shift by about 40 minutes later compared with the placebo condition (Reichart et al. 2022). Therefore, consuming caffeine in the evening can change the timing of the body's circadian system. This potentially contributes to a later sleep schedule by delaying the biological clock.

APPLYING RESEARCH ON CAFFEINE TIMING TO SLEEP HABITS

a. Findings from A Scientific Study

Scientific studies have found that the effects of caffeine on sleep depend on both the amount consumed and the time it is consumed. In a randomized, placebo-controlled crossover trial, Gardiner et al. (2025) examined how consuming 100 mg or 400 mg of caffeine at different intervals before bedtime affected sleep. The researchers found that the 100 mg dose did not significantly affect sleep outcomes, while the 400 mg dose disrupted sleep even when consumed 12 hours before bedtime.

These disruptions included taking longer to fall asleep, reduced total sleep time, and changes in sleep structure, with more pronounced effects when caffeine was consumed closer to bedtime. The findings suggest that consuming larger amounts of caffeine can affect sleep long after consumption, even when people drink it earlier in the day. However, individual responses to caffeine may vary, and the study's small sample of 23 adult men limits how broadly the results can be generalized. Overall, this research demonstrates the importance of considering both caffeine dosage and timing when examining their relationship with sleep quality (Gardiner et al. 2025).

b. Conclusion Connecting Back to Daily Life

Although caffeine is often treated as a simple way to stay awake, its effects on sleep depend on more than just how much a person consumes. The research discussed in this chapter shows that caffeine can influence sleep even when it is consumed in the morning, several hours before bedtime, or immediately before going to sleep. Morning consumption may contribute to a cycle in which poor sleep leads to greater caffeine intake the following day, while afternoon consumption can disrupt sleep even when there is a six-hour gap before bedtime. In the evening,

caffeine may interfere with deep sleep and delay the body's internal clock, potentially shifting a person's natural sleep schedule. These findings show that the timing of caffeine consumption matters because its effects can continue long after a person initially feels more alert. In everyday life, this matters especially for students and others who rely on coffee to manage busy schedules, study, or combat daytime fatigue. Rather than focusing only on the amount of coffee consumed, individuals should also consider when they drink it and how it may affect their sleep later that night. Ultimately, understanding the relationship between caffeine timing and sleep can help people make more informed decisions about their daily routines and recognize the importance of balancing alertness with sufficient sleep.

CHAPTER ADD-INs

Sample Exercise: Coffee & Sleep Diary

Meet Alex: Alex is a college student who drinks coffee to stay awake during classes and while studying. Alex recorded his coffee habits and sleep for one week to see whether the time they drink coffee might be related to their sleep quality.

Day	Time of Coffee	Amount (Cups)	Bedtime	Sleep quality (1–5)
Monday	8:00 AM and 3:00 PM	2	11:00 PM	3
Tuesday	9:00 AM	1	11:30 PM	4
Wednesday	2:00 PM	1	11:15 PM	3
Thursday	3:00 PM and 5:00 PM	2	12:00 AM	2
Friday	8:30 AM	1	11:30 PM	5
Saturday	5:00 PM	1	12:30 AM	2

Sunday	10:00 AM	1	11:00 PM	4

After one week: Alex's reflections

1. What time did you usually drink coffee?

Alex usually drank coffee in the morning, but on Wednesday, Thursday, and Saturday, Alex drank it in the afternoon or evening.

2. Did your sleep quality seem to change depending on when you drank it?

Alex noticed that sleep quality ratings were generally higher on the days when coffee was consumed in the morning. On days when coffee was consumed later, the ratings were lower. Saturday had the lowest rating after coffee at 5:00 PM.

3. Did your observations match what you expected?

Alex expected evening coffee to have the biggest effect on sleep. The results seemed to support this prediction, although one week of observations cannot prove that coffee timing was the cause of the differences.

4. How do your results compare with the research from the chapter?

Alex's observations are consistent with research suggesting that caffeine consumed later in the day can interfere with sleep. Caffeine can remain in the body for several hours, potentially making it harder to fall asleep or affecting sleep quality. However, Alex's diary only shows a possible relationship, not proof of direct cause and effect.

Try It Yourself: Coffee & Sleep Diary Exercise

For one week, record:

Day	Time of Coffee	Amount (Cups)	Bedtime	Sleep Quality 1-5
Monday				
Tuesday				
Wednesday				
Thursday				
Friday				

Saturday				
Sunday				

After one week:

1. What time did you usually drink coffee?

2. Did your sleep quality seem to change depending on when you drank it?

3. Did your observations match what you expected?

4. How do your results compare with the research from the chapter?

CHAPTER OUTLINE (Sebert): Jet Lag and Internal Desynchrony

Subtitle 1: *What is Jet Lag and Internal Desynchrony, and How Are They Caused?* (~½ Page)

- Brief explanation of both what jet lag and internal desynchrony is.
    - **(VOCAB) Jet Lag (Circadian Desynchrony)**: temporary disturbance to the natural sleep/wake cycle (and other physiological processes discussed later) due to travel across time zones
    - **(VOCAB) Internal Desynchrony**: When your internal biological clocks are out of sync with each other and/or the environment

Subtitle 2: Disruption of Human Circadian Rhythms (~3 Pages)

- Brief explanation of why circadian rhythms are affected by jet lag and how it causes desynchrony between the master clock and peripheral clocks
    - **(VOCAB) Suprachiasmatic Nucleus**: the biological master clock of the human body, located in the hypothalamus of the brain

Subsection 1: The Sleep/Wake Cycle

- Brief introduction to the 24-hour sleep/wake cycle, and how it is controlled

    - Short discussion on the two-system model of sleep/wake cycle that continuously works in tandem to regulate sleep/wake cycle (Borbély, A. A., Daan, S., Wirz‐Justice, A., & Deboer, T. (2016). The two‐process model of sleep regulation: a reappraisal. *Journal of Sleep Research*, *25*(2), 131–143. https://doi.org/10.1111/jsr.12371):
        - **(VOCAB) Process S**: Homeostatic pressure input
        - **(VOCAB) Process C**: Circadian pacemaker
    - Function of Melatonin in controlling sleep/wake cycle
        - **(VOCAB) Melatonin**: a natural hormone produced in the pineal gland of the brain responsible for inducing drowsiness in preparation for sleep
    - Long-term impact on sleep architecture (Avidan, A., & Colwell, C. (2010). Jet lag syndrome: circadian organization, pathophysiology, and management strategies. *Nature and Science of Sleep*, *2*, 187. https://doi.org/10.2147/nss.s6683)
        - **(VOCAB) Sleep Architecture**: The cycling through different stages of sleep, critical to proper rest
- How these biological processes are disturbed by jet lag and hence cause desynchronization.
    - Mistimed hormonal releases and light cues that confuse your body, etc.
- **Learning Objective 1**: Construct visuals of how sleep/wake cycles adjust from jet lag over time. After learning more about the sleep/wake cycle, how it is impacted by jet lag, and how the body naturally readjusts over time, you should be able to hypothesize and construct visualizations of how sleep/wake cycles slowly readjusts, using zeitgebers and other external cues.

Subsection 2: Other Circadian Rhythms Affected (Probably will change this name

- Discuss various other circadian rhythms that are affected (which I won't go into as much depth, since sleep/wake is the primary one. However, there can be an included chapter add-in focusing on one of these other rhythms):
    - Metabolism and Digestion
    - Body Temperature
    - Cortisol Levels
    - Immune and Cardiovascular Function
- **Learning Objective 2**: Relate how changes made to one circadian rhythm can indirectly alter the timing of other circadian rhythms (Ex: how does a desynched sleep/wake cycle cascade downstream to the various peripheral biological cycles)

Subtitle 3: Disruptions to Human Ultradian Rhythms (~3 pages)

- Explore the disruptions made to the human ultradian rhythms; brief intro to infradian rhythms
    - **(VOCAB) Ultradian Rhythms**: Biological rhythms that periodically repeat and cycle over periods of time shorter than 24 hours

Subsection 1: Deep Dive: the Menstrual Cycle

- Introduction to the menstrual cycle and its cyclic use of hormones to coordinate and regulate its process
    - Estrogen, Progesterone, Follicle-Stimulating Hormone, Luteinizing Hormone
- Exploring how changing time zones affects biological processes that occur over time periods greater than just 24 hours
- Discuss other reproductive damage caused by jet lag, such as decreased reproductive success and increased risk for breast cancer (repeated, continuous desynchrony) (Mahoney, M. M. (2010).

Shift work, jet lag, and female reproduction. *International Journal of Endocrinology*, *2010*, 1–9. https://doi.org/10.1155/2010/813764)

- **Learning Objective 3**: Differentiate between synchronized and desynchronized rhythms: how can you determine if a rhythm is mistimed and its underlying cause? How can you describe it or identify it on a graph (ex: hormone cycles in the menstrual cycle when synchronized vs. unsynchronized)

Subsection 2: The Delayed Healing Tail

- Explore how seemingly minor changes/disruptions can cause prolonged healing stages, which is especially severe for infradian rhythms
- Using the gut microbiome as an example, it takes weeks for the microbiome to be replenished, so abrupt change can cause damage to the microbiome that may take weeks to recover (Bilinski, K., Wisniewski, K., Rafner, L., Witko, R., & Gawel-Dabrowska, D. (2026). *Travel-Induced Circadian and Microbiota Disturbances: Implications for Athlete Health and Performance: A Narrative Review*. Mdpi.Com. https://www.mdpi.com/2072-6643/18/10/1523)

Subtitle 4: Disruptions to Human Infradian Rhythms (~3 pages)

- Explore the disruptions made to the human infradian rhythms; brief intro to infradian rhythms
    - **(VOCAB) Infradian Rhythms**: Biological rhythms that periodically repeat and cycle over periods of time longer than 24 hours

Subsection 1: Deep Dive: the REM-NREM cycle

- Deeper dive into the cycles that happen within sleep itself, where the mind cycles through various stages of sleep every 90-120 minutes.
    - REM, N1, N2, N3
- Explore how sleep architecture is damaged due to jet lag, which fundamentally (though temporarily) alters and disrupts it
    - Suppressed REM, reduced deep sleep, sleep fragmentation

Subsection 2: Exploration of BRAC Cycle

- **(VOCAB) Basic Rest-Activity Cycle**: a natural 90-minute rhythm in the human body, where the body cycles between high alertness and mental rest
- Exploration of both short-term and long-term effects on the BRAC cycle due to jet lag

Subtitle 5: Natural Adaptation, Intervention Methods, and their Limitations (~3-4 pages)

- Explore how the body readjusts to a new time zone over time, whether naturally or through exogenous, artificial interventions. The limitations of each of these methods will also be discussed.

Subsection 1: Natural Adaptation and its Limitations

- Exploration of methods by which the body naturally recovers from jet lag and its symptoms
    - The body naturally recovers from jet lag and internal desynchrony through external stimuli.
        - Exploration of some of these natural environmental cues, such as sunlight and temperature fluctuations throughout the day
    - **(VOCAB) Zeitgebers**: German word that translates to “time-giver.” They are environmental cues that help the body synchronize with the 24-hour day cycle.
- Analysis of the limitations of zeitgebers on the readjustment of the body to a new time zone.

    - For example, the human body usually can only adjust 1.5 hours per day using environmental cues, which also has some variance depending on the direction you traveled
    - Organs and other body systems may remain out of sync due to them having their own internal clocks that remain desynchronized even if the master clock (SCN) recovers
    - Environmental conditions experience great variability, such as changes in weather conditions over the seasons, or the interference from artificial light sources that further confuses the body
    - (Choy, M., & Salbu, R. L. (2011). Jet Lag: Current and Potential Therapies. *Pharmacy and Therapeutics*, *36*(4), 221. https://pmc.ncbi.nlm.nih.gov/articles/PMC3086113/)

Subsection 2: Artificial Interventions and their Limitations

- ~~Exploration of how intervention methods can be used to expedite the recovery stage~~
    - ~~**(VOCAB) Exogenous**: things that originate from outside the body or body system~~
    - ~~Exogenous and artificial interventions can be split into three main categories, which will be discussed in this section (Arendt, J. (2018). Approaches to the Pharmacological Management of Jet Lag. *Drugs*, *78*(14), 1419–1431. https://doi.org/10.1007/s40265-018-0973-8):~~
        - ~~**(VOCAB) Chronobiotics**: substances/drugs that mimic/augment/diminish the body's natural signals to help the body correct its internal clock faster, such as melatonin~~
        - ~~**(VOCAB) Hypnotics**: sedative drugs that induce sleep to help anchor the master clock (SCN) to the proper day/night cycle~~
        - ~~**(VOCAB) Stimulants**: agents that, when used strategically, will induce wakefulness to help align the body with the new time zone's day/night cycle~~
- ~~Discussion of the limitations of these artificial methods of expediting the process of readjustment to new time zones.~~
    - ~~Each of the different categories of artificial intervention comes with its own shortcomings:~~
        - ~~Chronobiotics: the drugs must be used at the correct time and correct dosage to get the best effects. Additionally, it will only help with correcting the master clock, not the peripheral clocks~~
        - ~~Hypnotics: Can help push the master clock towards the target time zone, but don't synchronize the rest of the body. Can lead to grogginess and tiredness, which will be experienced in the next few days~~
        - ~~Stimulants: Often cause a crashing effect; once the stimulant wears off, the body is drastically more tired and can lead to more severe fatigue~~

CHAPTER WRITTEN CONTENT (Sebert): Jet Lag and Internal Desynchrony

UNIT: CIRCADIAN RHYTHMS

What is Jet Lag and Internal Desynchrony, and How Are They Caused?

Humans are highly influenced by time, which we keep track of using various biological clocks within our bodies. This tracking of time is highly precise and is controlled by the **suprachiasmatic nucleus (SCN)**, a portion of the hypothalamus that acts as the master clock of the human body. However, various problems can arise when this clock is out of sync with the rest of the body and/or the external environment – a phenomenon called **internal desynchrony**. One well-known form of desynchrony is commonly known as "**jet lag**," which occurs when the body rapidly moves across time zones and hence becomes confused as external cues do not align with the internal clocks. This mismatch of timing of internal biological processes with the actual time of day can lead to various symptoms, such as severely disrupted sleep cycles. Additionally, although many know that sleep schedules can get heavily altered due to jet lag, there are also many other other bodily functions and biological cycles which can be similarly disrupted, potentially taking weeks to months to fully readjust to normal.

Disruption of Human Circadian Rhythms

Circadian rhythms are biological rhythms that cycle in roughly 24-hour periods. The most famous of these is the human sleep/wake cycle, which is also the most well-known disruption due to jet lag. However, to understand how the sleep/wake cycle is affected by jet lag, we must first understand the underlying processes that control and regulate it. As discussed earlier, the suprachiasmatic nucleus is the master biological clock within the body, which is also directly responsible for regulating the sleep/wake cycle. Although the SCN directly controls this cycle, the way in which it does so is highly complex, with numerous different systems and hormones all playing intertwined roles in regulating the process.

The Sleep/Wake Cycle

Like most animals, all humans require sleep for proper functioning throughout the day. Hence, the process of sleeping (and wakefulness) is fine-tuned and regulated to ensure we receive adequate amounts of sleep when our bodies need it. This sleep/wake cycle is regulated by two main biological mechanisms that work in tandem: **Process S**, Sleep Homeostasis, and **Process C**, Circadian Pacemaker (Borbély et al., 2016).

Normally, Process S and Process C work perfectly together to regulate when we sleep, which boils down to promoting a sense of drowsiness to induce sleep. According to Borbely et al. from a paper in 2016, Process S, sleep homeostasis, is responsible for controlling sleep pressure (biological urge to sleep), which induces drowsiness in preparation for sleep. This process continuously builds sleep pressure the longer you stay awake, which ultimately leads to the amplified feeling of fatigue and drowsiness before sleep. The process occurs through the accumulation of adenosine, a natural organic compound formed as a byproduct of cell metabolism that occurs constantly to keep cells alive, inside the brain. This adenosine is. As the adenosine continues to accumulate inside the brain, it eventually reaches a particular "sleep threshold," beyond which the brain signals the body to become more tired and prepare for sleep by inhibiting and stopping the brain's arousal networks. During sleep, your body naturally recycles the adenosine in your brain, resetting the sleep pressure so you feel refreshed once you wake up. You may think of Process S as an internal stopwatch that keeps track of the amount of time you've been awake through the buildup of adenosine in the brain.

On the other hand, Process C, the circadian pacemaker, is independent of how long you've been awake. As the name suggests, it relies heavily on your body's circadian rhythm to dictate when to arouse drowsiness versus alertness throughout the day. This circadian pacemaker

is controlled by the SCN and uses a variety of hormones and chemical signals to regulate the sleep/wake cycle. When you first wake up, hormones like cortisol spike, increasing your stress and alertness in the morning in preparation for awakening (Sanchez and Dalley, 2025). Additionally, throughout the morning and day, your body manufactures hormones like serotonin to keep you active and awake. Serotonin is primarily responsible for maintaining alertness and keeping your brain active and responsive. However, as night approaches and the sun goes down, your body starts the process of turning serotonin into melatonin, a hormone responsible for inducing sleep and drowsiness. This conversion is controlled by the SCN, which receives cues and signals from environmental factors such as sunlight and darkness. As more melatonin is synthesized, it signals the body to start shutting down, relax, and prepare for sleep. *Visual graph would go here*

Although Process S and Process C work in sync to regulate our sleep/wake cycle, jet lag uncouples these two processes and thereby leads to chaos in the rhythm. Process S is dependent specifically on how long we remain awake and is mostly unaffected by changes in environmental cues and time zones. However, Process C relies heavily on the local time and the current state of the SCN. Hormone production gets misaligned with the current time zone, and chemical signals are released at the wrong time of day. Even though environmental cues help Process C slowly realign with the current time, there nonetheless remains a mismatch between the two processes.

For instance, if it is night in your new time zone but morning in your previous time zone, Process C would have produced a large amount of serotonin to promote wakefulness. Even if Process C was able to readjust quickly and started synthesizing melatonin, Process S is still convinced it is morning time. Because Process S works like a stopwatch that resets every morning, the adenosine levels in your brain are nowhere near enough to reach the “sleep

threshold" needed to induce sleepiness. Hence, the misalignment of these two biological mechanisms jams the regulation of your sleep/wake cycle, producing this iconic jet lag symptom.

[Learning Objective #1: Construct/explain visuals showing how the two processes that regulate the sleep/wake cycle changes due to jet lag, and the consequences of the mismatch. (Produce and recognize the reasons for why the graph looks the way it does, and understand what a mistimed system looks like)]

Sleep Architecture & Consequences

This damage to the sleep/wake cycle extends beyond the disrupted sleep schedule, however. Although the most noticeable observation of circadian misalignment is the mismatched sleeping time and reduced sleep duration, the quality of your sleep is also severely affected by jet lag. The quality of your sleep can be referred to as **Sleep Architecture**, the cycling through different stages of sleep, which is critical to getting proper rest. Sleep is divided into several stages, that being REM (Rapid Eye Movement) and NREM (Non-Rapid Eye Movement), the latter of which is split further into four stages of progressively deeper sleep (Gonnissen et al., 2013). A specific balance of each stage of sleep is required for a good night's rest. However, when suffering from jet lag or other circadian misalignments, the duration of time spent in each stage of sleep is severely affected, causing overall fragmentation in the onset of each stage of sleep. The stage of sleep most adversely affected by jet lag is REM sleep, which is deeply dependent on the circadian rhythm and Process C. Although the other stages of sleep are slightly more preserved, they nonetheless reduce in duration and can become mistimed.

Damage to the REM stage of sleep due to jet lag — becoming shortened, fragmented, or delayed — can even play a significant role in overall endocrine and metabolic functioning as well (Gonnissen et al., 2013). In the paper by Gonnissen et al., it was found that damage to REM sleep led to disrupted insulin activity and sensitivity. By using HOMA-IR, an index used to estimate insulin resistance, the researchers found that insulin resistance follows a circadian pattern, and the mismatch of timing due to reduced REM sleep and circadian misalignment caused the regulation of glucose and insulin to be suboptimally synchronized with food intake. The implications of this mismatch can be linked to increased rates of obesity and diabetes, though this would require systematic and regular onsets of circadian misalignments.

Reduced sleep quality and damaged sleep architecture in general also leads to detriments in cognitive functioning. According to Weingarten and Collop, cognitive abilities in attention and vigilance, working memory, short-term memory, and executive functioning are reduced (Weingarten and Collop, 2013). Reduction in sleep duration and quality inhibits proper mental functioning due to the lack of adequate rest and restoration, thereby producing these symptoms. People find that they have difficulty maintaining concentration and alertness while having an impaired memory and thought process. Even if you are able to wake up in the morning and feel physically awake, your mind has not received the proper rest and hence experiences these symptoms. Emotions are also impacted by reduced sleep, such as increased irritability, stronger emotional reactions, reduced regulation of emotions, and sudden changes in mood (Helm and Walker, 2014). Without your brain working with the sufficient amount and quality of sleep and rest, cognitive function and emotional regulation thereby become compromised and extend the repercussions of jet lag beyond simply having reduced sleep duration and quality.

[Learning Objective #2: Relate and connect disruptions to the sleep/wake cycle with damage to the architecture of sleep, and how damage to sleep can extend to other bodily systems and functions, such as cognitive and emotional function (analyze the downstream effects of sleep/wake cycle disturbance)]

CHAPTER ADD-INS

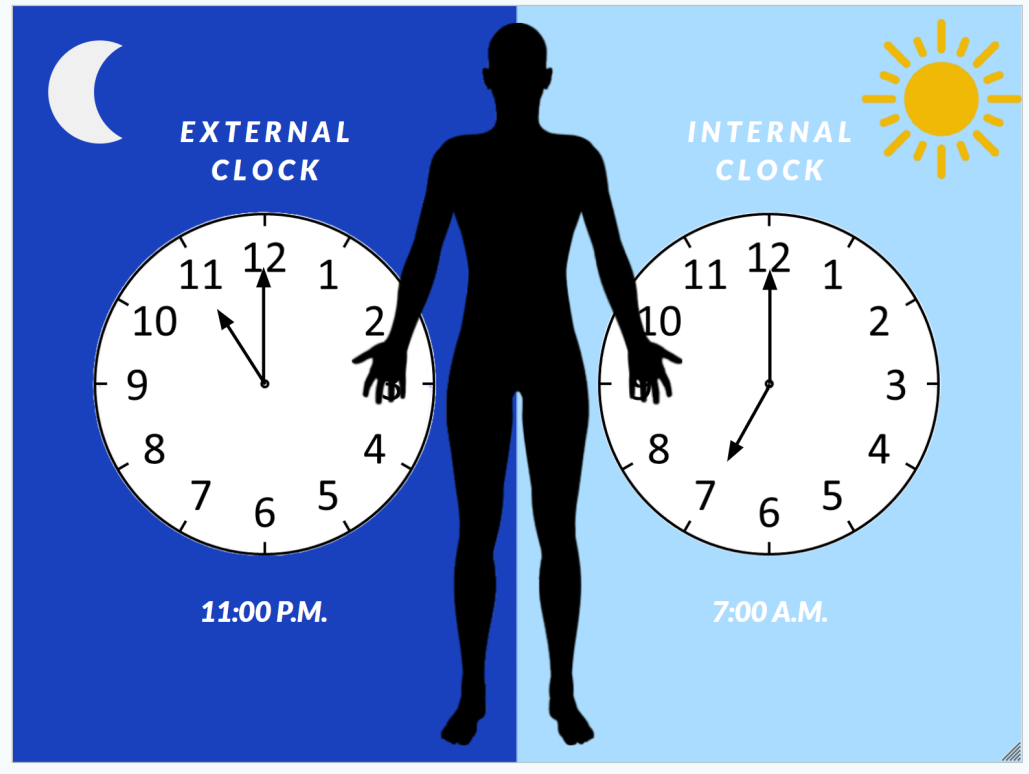


Figure 1: Illustration representative of circadian misalignment. Although misalignment of the circadian rhythm can be due to a variety of reasons and factors, jet lag is a prime example. Although you use clocks in the physical world to keep track of time, your body also subconsciously keeps track of time through an internal biological clock that operates on a near 24-hour cycle. However, if you move through time zones rapidly, the physical world (and the respective time) around you changes while your internal clock continues as if nothing happened.

This results in a conflict between the external, real time and the time your body believes it currently is. Hence, although it may be 11:00 P.M. and dark outside (and thus be the time you should be sleeping,) your body is still accustomed to the old time zone, believing that it's still 7:00 A.M. (and thus be the time you should be getting most alert to wake up). This conflict thus leads to the symptoms observed when you are experiencing jet lag.

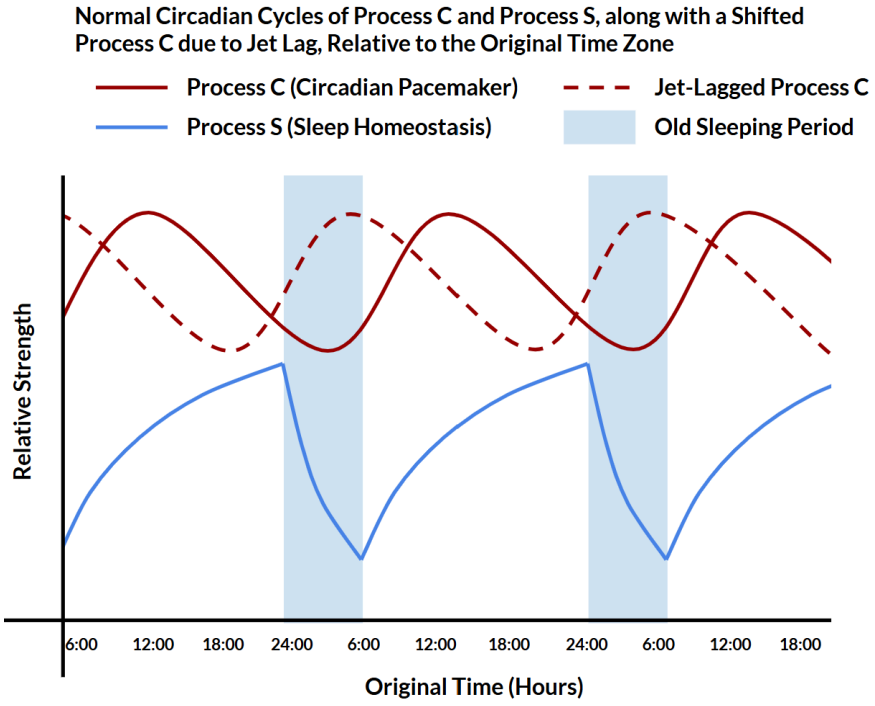
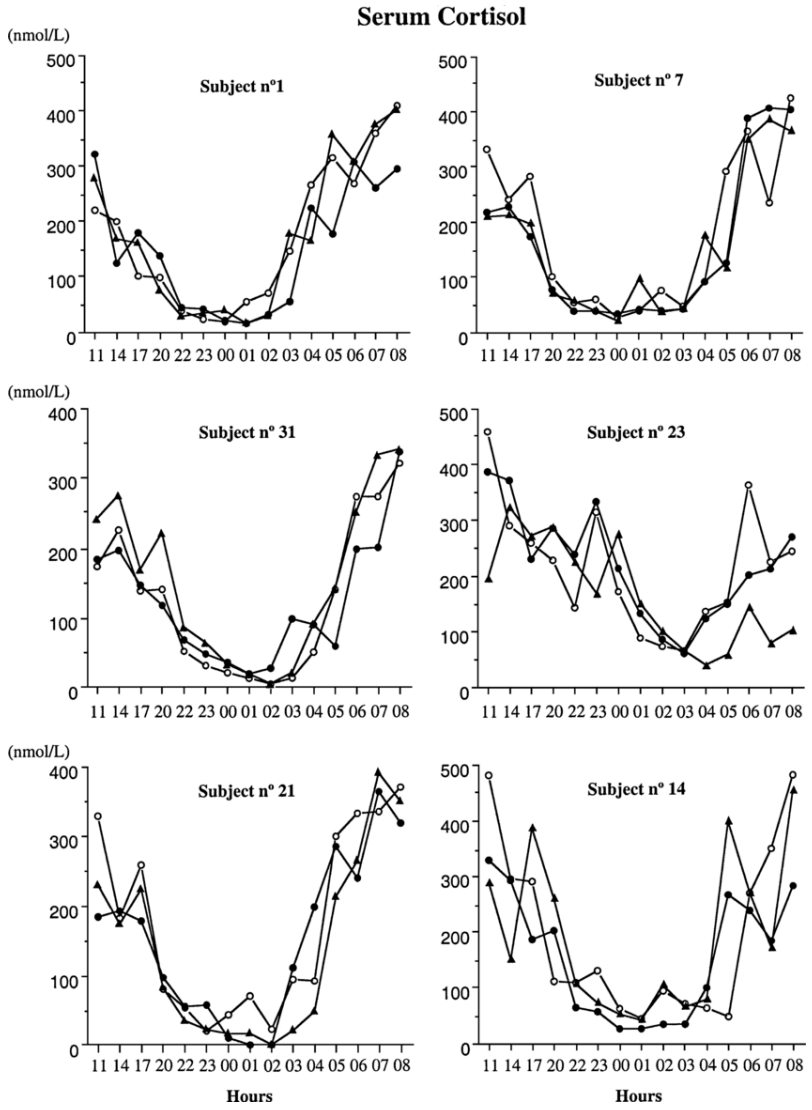


Figure 2: A graph showing the cyclic pattern of both Process S and Process C throughout the day to regulate sleep, and the resulting shift caused by jet lag. The solid lines represent the normal cyclic pattern of Process S and Process C, which cycles every 24 hours and overlap to induce sleep. The old sleeping period is shown by the blue shaded rectangles, which occur when Process S reaches its peak and when Process C reaches its minimum. Because Process S is sleep pressure which gradually increases through the accumulation of adenosine in the brain, the graph gradually builds up. During sleep, however, the adenosine is broken down and recycled, thus returning to the baseline level in the morning. Process C is the circadian pacemaker, which

gradually increases and decreases throughout the day in a sinusoidal pattern, where the troughs represent when drowsiness is induced. When jet lag occurs, only Process C is affected as the circadian cycle falls out of sync while the sleep pressure (which depends only on how long you've been awake) does not change as you travel across time zones.



(Honestly might not use this one)

Figure 3: Experimental Results from Yvan Touitou showing serum cortisol concentration in healthy subjects throughout the day to study circadian rhythms. Here, these results reveal that there is a strong correlation between cortisol concentration in the body and the time of day that

cortisol levels peak. With cortisol being an alertness hormone, these results suggest that it plays a role in regulating the sleep/w ake cycle, with its concentration being highest early in the morning to induce wakefulness while being lowest around midnight to promote drowsiness and sleep.

GLOSSARY

Term	Definition

REFERENCES

Circadian Rhythms: Hunger, Digestion, and Metabolism

Unit: Circadian Rhythms

Chapter X: Hunger, Digestion, and Metabolism

CONTRIBUTORS

- Clara Davis
- Will

Name: Will

Chapter: Meal Timing and the Circadian Rhythm

Learning objectives:

1. Explain the relationship between the major processes of the circadian rhythm and meal timing.
2. Identify the key zeitgebers (entrainment factors) of the hunger cycle and explain the mechanisms in which they function.
3. Explain the leptin/ghrelin cycle and how it connects to other hormones and nutrients that are processed in various parts of the digestive tract.
4. Predict the likely case-by-case affect(s) of circadian rhythm dysregulation and hormone dysregulation on the hunger cycle.
5. Interpret data regarding two species in order to compare and contrast entrainment factors and patterns regarding their meal timing.

Name: Clara Davis

Chapter: Alcohol's Effect on the Metabolic Cycle

Learning objectives:

- Learning Objective 1: list the biological steps of metabolism
    - Remember
    - Possible supplemental tie-in: How would the depletion of ADP affect metabolism?
- Learning Objective 2: Analyze alcohol vs metabolic circadian cycle graphs
    - Analyze
    - How is the metabolic system affected by the differing concentration of NADH?
- Learning Objective 3: Present hypothesis explaining the effect of alcohol on production of ATP.
    - Evaluate
    - If alcohol is consumed, how is the production of ATP affected?
- Learning Objective 4: Describe how liver damage is caused in an unstable molecular metabolic cycle
    - Understand
    - Explain the biological pathway if liver damage is caused by alcohol
- Learning Objective 5: How does unstable metabolism affect other systems in the body?
    - Understand
    - Explain the side effects caused by alcoholic damage to the metabolic system

Sections:

The basics of routine-based meal timing

- Food and Time of Day as a Zeitgeber
- Hunger Entrainment
- Basics of Hunger Hormones

Hormones and hunger signaling

- Leptin and Ghrelin - a Closer Look
- Additional Hormones (cortisol melatonin, GLP-1, gut microbiome byproducts)
- Food entrainable oscillator (FEO) and Peripheral Clocks
- Nervous System: SCN, AgRP/NPY neurons, POMC neurons, and gut-brain axis
- Insulin sensitivity throughout the day

Rhythm disruption and health implications

- Conflicting zeitgebers
- Obesity, Diabetes, and Eating Disorders - a Hormonal View
- GLP-1 Receptor Agonists
- Social Jetlag and Shift Work
- Night Eating Syndrome
- Hunger and Sleep

Comparative Chronobiology of Hunger

- Human
- Mammalian
- Non-mamilian
- Unicellular Organisms

Introduction to Metabolism

- Define
- Catabolism vs anabolism
- ATP

Metabolic Cycle through Molecular Biology

- Molecular Biology
- Four Main Stages

Circadian Rhythm of Metabolism

- Control through circadian rhythms
- Control through metabolism

Alcohol Breakdown through Molecular Biology

- Ethanol catabolism
- Biological pathway

The Issues Alcohol Causes in the Metabolic Cycle

- Build-up of NADH
- Slowed cycles
- Drop in blood sugar
- Less ATP production

Alcohol's Effect on Metabolism's Circadian Rhythm

- Liver's internal clock
- Sleep-wake cycle impaired
- Body temperature

Long-Term Health Effects

- Liver harmed
- Fat accumulation
- Weakened organs

Vocab:

Grhelin: A peptide hormone responsible for regulating energy intake and expenditure. Ghrelin synthesis in the body rises as the stomach empties and energy availability is low. The peptide also inhibits insulin secretion and promotes glucagon in order to increase blood sugar to provide energy to cells.

Leptin: A peptide hormone responsible for regulating energy intake and expenditure. Leptin is produced by adipose (fat) cells when levels of stored fat are high. Leptin also promotes usage of fat for energy.

Gut Microbiome: The ecosystem of microbial life (prokaryotes such as bacteria and archea) which can consist of both beneficial and pathogenic microorganisms inside of the digestive tract.

Zeitgeber: Any entertainment factor for the circadian clock. Examples include light exposure, food intake, and wake/sleep times.

Shift Work: The process of working at abnormal times such as night shifts or early morning shifts.

Metabolism: The process of breaking down energy in a cycle

ATP: A molecule that stores and transports energy in the cell

Citations:

Young ER, Jialal I. Biochemistry, Ghrelin. [Updated 2023 Jul 17]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK547692/

Dornbush S, Aeddula NR. Physiology, Leptin. [Updated 2023 Apr 10]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK537038/

Zeng Y, Wu Y, Zhang Q, Xiao X. Crosstalk between glucagon-like peptide 1 and gut microbiota in metabolic diseases. mBio. 2024 Jan 16;15(1):e0203223. doi: 10.1128/mbio.02032-23. Epub 2023 Dec 6. PMID: 38055342; PMCID: PMC10790698.

Wehrens SMT, Christou S, Isherwood C, Middleton B, Gibbs MA, Archer SN, Skene DJ, Johnston JD. Meal Timing Regulates the Human Circadian System. Curr Biol. 2017 Jun 19;27(12):1768-1775.e3. doi: 10.1016/j.cub.2017.04.059. Epub 2017 Jun 1. PMID: 28578930; PMCID: PMC5483233

Yeung AY, Tadi P. Physiology, Obesity Neurohormonal Appetite And Satiety Control. [Updated 2023 Jan 3]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2026 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK555906/

Rodin J. Insulin levels, hunger, and food intake: an example of feedback loops in body weight regulation. Health Psychol. 1985;4(1):1-24. doi: 10.1037//0278-6133.4.1.1. PMID: 3894001.

Di Rosa V, Frigato E, Negrini P ...
Sporadic feeding regulates robust food entrainable circadian clocks in blind cavefish
iScience, 2024; 27

Mistlberger RE. Food-anticipatory circadian rhythms: concepts and methods. Eur J Neurosci. 2009 Nov;30(9):1718-29. doi: 10.1111/j.1460-9568.2009.06965.x. Epub 2009 Oct 28. PMID: 19878279.
Liu C, Liu Y, Xin Y, Wang Y. Circadian secretion rhythm of GLP-1 and its influencing factors. Front Endocrinol (Lausanne). 2022 Dec 2;13:991397. doi: 10.3389/fendo.2022.991397. PMID: 36531506; PMCID: PMC9755352.

Boersma, P., Villarroel, M. A., & Vahratian, A. (2020). Heavy drinking among U.S. adults, 2018. National Center for Health Statistics. https://www.cdc.gov/nchs/products/databriefs/db374.htm

Breus, M. J. (2018, January 11). Alcohol and sleep: What you need to know. Psychology Today. https://www.psychologytoday.com/us/blog/sleep-newzzz/201801/alcohol-and-sleep-what-you-need-know

Davis, B. T., IV, Voigt, R. M., Shaikh, M., Forsyth, C. B., & Keshavarzian, A. (2018). Circadian mechanisms in alcohol use disorder and tissue injury. Alcoholism: Clinical and Experimental Research, 42(4), 668–677. https://doi.org/10.1111/acer.13612

Judge, A., & Dodd, M. S. (2020). Metabolism. *Essays in biochemistry*, *64*(4), 607–647. https://doi.org/10.1042/EBC20190041

Ma, M. A., & Morrison, E. H. (2023). Neuroanatomy, nucleus suprachiasmatic. In StatPearls. StatPearls Publishing. https://www.ncbi.nlm.nih.gov/books/NBK546664/

Meyrel, M., Rolland, B., & Geoffroy, P. A. (2020). Alterations in circadian rhythms following alcohol use: A systematic review. Progress in Neuro-Psychopharmacology and Biological Psychiatry, 99, Article 109831. https://doi.org/10.1016/j.pnpbp.2019.109831

Myers, R. D. (1981). Alcohol's effect on body temperature: Hypothermia, hyperthermia or poikilothermia? Brain Research Bulletin, 7(2), 209–220. https://doi.org/10.1016/0361-9230(81)90085-X

National Institute on Alcohol Abuse and Alcoholism. (n.d.). Alcohol metabolism. https://www.niaaa.nih.gov/publications/alcohol-metabolism

Sato, T., & Sato, S. (2023). Circadian regulation of metabolism: Commitment to health and diseases. Endocrinology, 164(7), bqad086. https://doi.org/10.1210/endocr/bqad086

Wasielewski, J. A., & Holloway, F. A. (2001). Alcohol's interactions with circadian rhythms: A focus on body temperature. Alcohol Research & Health, 25(2), 94–100. https://pmc.ncbi.nlm.nih.gov/articles/PMC6707125/

Overview

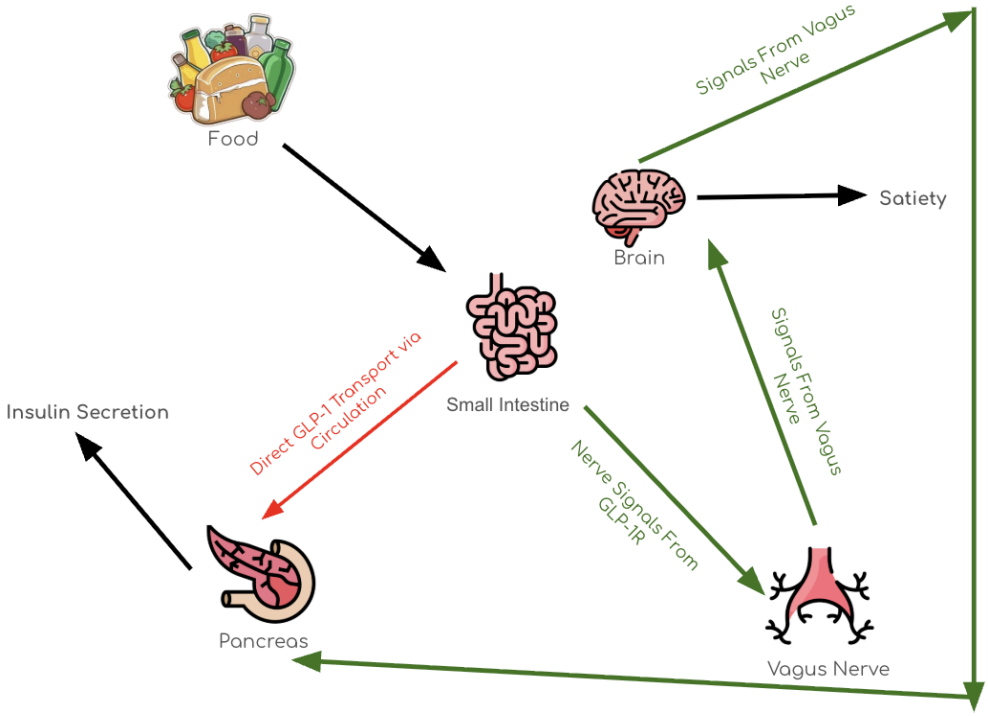
How do you know when you need food? How do you know when to finish eating? For most, the answer is quite simple: you eat when you feel hungry, and finish eating when you feel full. However, when looking at the biological reasoning behind appetite regulation and energy intake, it can get quite complicated very quickly. The body relies on a wide variety of feedback loops to maintain energy homeostasis. Meal timing is mostly independent of the main circadian rhythm and more dependent on other factors such as hormones produced in the body, time of day, and meal schedule.

The Essential Appetite Regulators

GLP-1

GLP-1 is critical in maintaining appetite on a rhythmic basis, and its main function is to signal to the pancreas to produce more insulin when blood sugar is high. It also signals to the central nervous system to indicate that food is currently being digested. GLP-1 is a peptide (small protein consisting of a short chain of amino acids) produced by L-cells in the intestinal lining when food is being digested and broken down by microorganisms. L-cells are specialized cells mainly found in the small intestine, specifically the ileum. Once produced, in order to enter the bloodstream, it must cross into capillaries and then reach the hepatic portal vein, which transports nutrients and hormones from the digestive tract to the heart. Once GLP-1 reaches the circulatory system, it binds to a receptor known as GLP-1R in the brain and the pancreas. In the pancreas, this stimulates the release of insulin to lower blood sugar and decrease hunger when blood sugar levels are high. The mechanisms of insulin will be further explained later on in this chapter. The interesting thing about GLP-1 is that around 80% of it is broken down before it reaches the hepatic portal vein by the enzyme DPP-IV. The remaining 20% will make if further, potentially to the brain, before it is broken down by DPP-IV within minutes. The whole idea of Ozempic, a drug used to treat diabetes, is to create a GLP-1-like molecule that does not break down as fast and continuously as GLP-1 so that blood sugar is constantly downregulated. Ozempic binds to the same GLP-1 receptors, hence the drug class name GLP-1 receptor antagonists (GLP-1RA). In the brain, GLP-1 binds to its receptors in multiple locations to slow down stomach emptying and increase satiety.

Figure 1: GLP-1 Signaling Pathway



Leptin

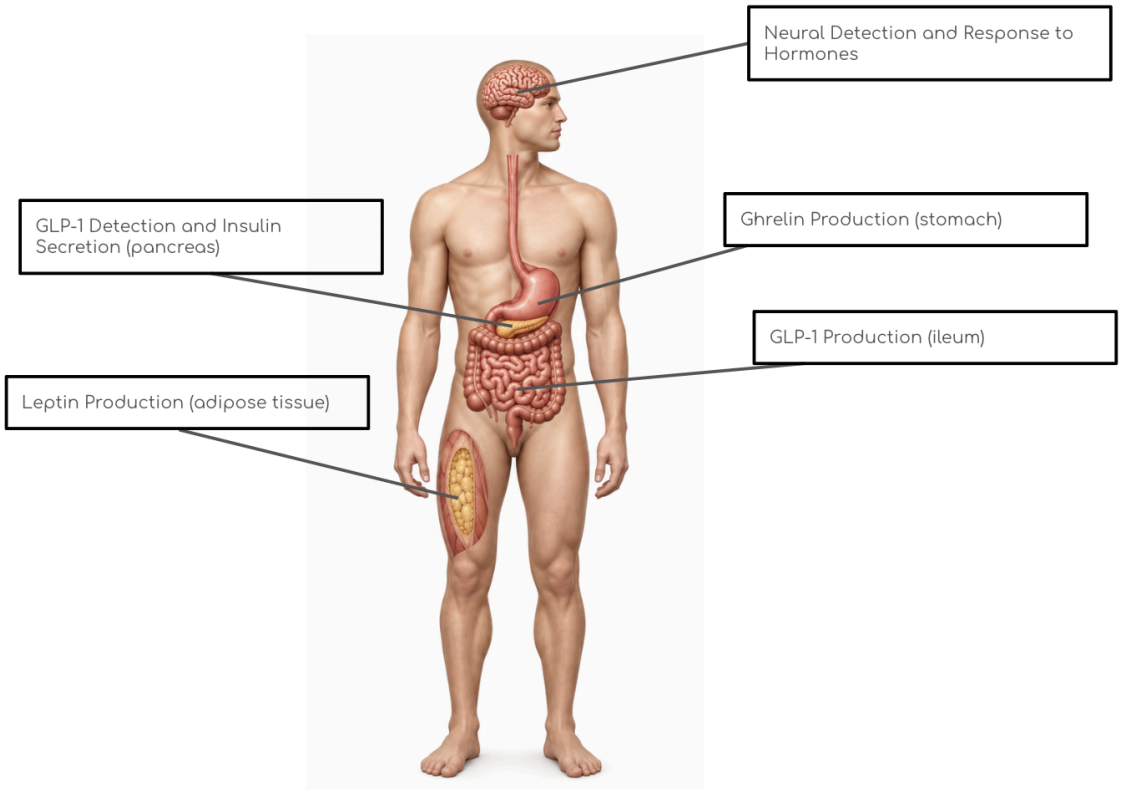
Since GLP-1 is a very short-lived signaling molecule, there arises a biological need for long-term energy regulation. Leptin is produced by adipose (fat) cells when levels of stored fat are high. It is important to note that leptin is not instantaneously produced after a meal, but instead is used to regulate appetite and body fat levels in the long run. In terms of appetite control, we will simplify leptin's more complex nature and interactions to a simple negative feedback loop: fat cells grow in response to a consistent surplus of food, releasing ghrelin to lower appetite. Leptin is known as an anorexigenic hormone (appetite suppressing). Leptin mainly acts on the accurate nucleus, the region of the brain responsible for appetite control. Here, leptin activates

anorexigenic proopiomelanocortin (POMC) neurons, and deactivates orexigenic protein/neuropeptide Y (AgRP/NPY) neurons. Thus, a “don’t eat” signal is sent across the body. When one becomes dangerously underfed, leptin levels plummet in the body, and the body goes into starvation mode, cutting back on unnecessary energy usage such as reproduction.

Ghrelin

Ghrelin stimulates hunger–usually in response to an empty stomach or meal anticipation around a regular meal time. Unlike GLP-1 and leptin, ghrelin is known as orexigenic (appetite stimulant). PD/D1 cells are specialized cells in the stomach that secrete ghrelin upon an empty stomach. PD/D1 cells are known as endocrine cells, as they produce hormones. Ghrelin has the near opposite effect of leptin when it reaches the accurate nucleus: it deactivates anorexigenic proopiomelanocortin (POMC) neurons, and acivates orexigenic protein/neuropeptide Y (AgRP/NPY) neurons.

Figure 2:Anatomical Diagram of Hunger Signaling



Hunger Entrainment

Eating at regular times each day allows for anticipatory hunger caused in part by spikes in ghrelin around one's normal meal time. There are many hormones and proteins involved in the master clock and peripheral clocks, both of which will be explored further in this chapter. It is important to recognise that beyond the master clock, meal timing is highly entrained by peripheral clocks. Take a person with a constant meal schedule into consideration. Shift every one of their meals 2 hours forward for a month. After a month, they will be trained to be hungry at their new meal times. If they go to bed and wake up at the same time, their melatonin levels

and cortisol levels will be mostly unchanged. Take this case study for example: Wehrens et al. took 10 men and placed them on a constant sleep/wake cycle and gave them three meals a day at the same time. They then shifted each meal time forward by five hours. Their master clock markers such as melatonin and cortisol stayed the same, but their natural daily peak in insulin shifted by about 5 hours to match meal timing. There are, of course, exceptions to this general rule, and the connection between these two clocks is still under heavy research.

Hormones and hunger signaling

Leptin and Ghrelin - a Closer Look

Ghrelin levels in the body oscillate depending on time since last meal and time of day. When Ghrelin reaches the bloodstream, it binds to G-protein coupled receptors in the hypothalamus, more specifically the arcuate nucleus. Ghrelin's binding stimulates hunger signals. The cells that release ghrelin contain PER1 and PER2 proteins, which are clock proteins that are also found in the SCN. In the stomach, their concentrations peak in the evening which is when ghrelin production bottoms out, and bottom out in the afternoon, which leads to a ghrelin spike. This is assuming a regular meal schedule (breakfast, lunch, dinner). These PER1 and PER2 proteins in the stomach work mostly independently of CLOCK and the SCN, so exposure to sunlight and having a constant sleep schedule is less important than eating at regular times for a regular hunger rhythm.

Because leptin is produced by white adipose tissue–specifically when fat tissue storage increases–those who chronically undereat or attempt to lose weight can become ravenously

hungry– leptin levels plummet following an adipose tissue shrinkage, which then leads to intense hunger to maintain energy homeostasis. The opposite is true for a healthy individual with sufficient fat storage–leptin secretion increases and appetite is more controlled.

GLP-1 and the Gut Microbiome

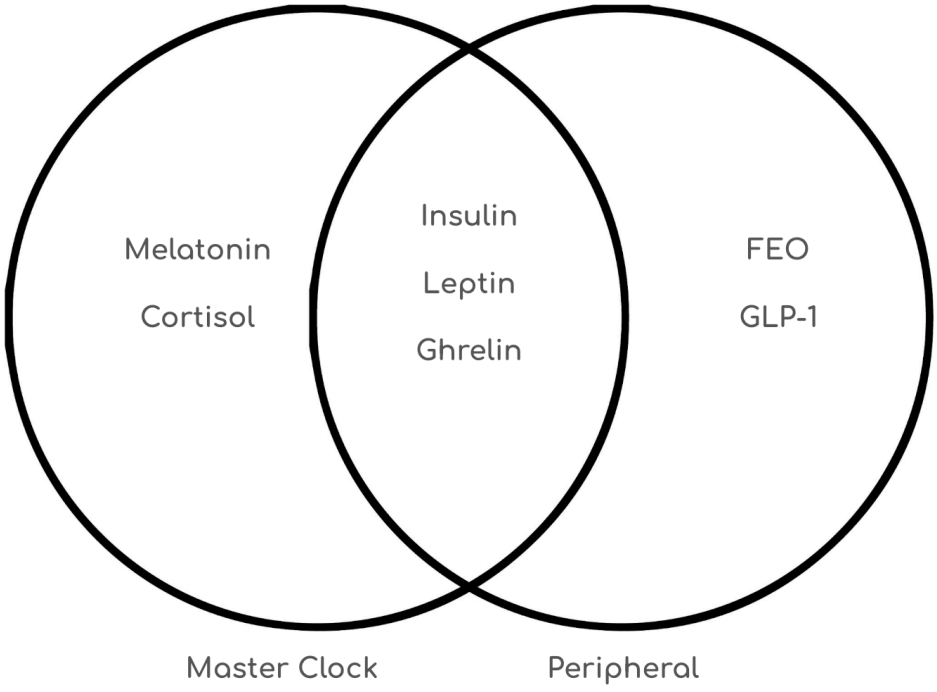
GLP-1 secretion in the gut follows a circadian rhythm, but the exact timekeeping mechanism in L-cells is not well-known. However, it is known that GLP-1 levels are most likely to peak in the morning and afternoon following a meal, and L-cells become less active towards the evening, meaning secretion is relatively lower following a meal. *Akkermansia* and *Lactobacillus* are two species in the gut microbiome known to increase GLP-1 production, and dysbiosis (a gut microbiome imbalance) is associated with a decrease in GLP-1 production.

Food entrainable oscillator (FEO) and Peripheral Clocks

Similar to GLP-1, the food entrainable oscillator (FEO) is a somewhat mysterious topic in mammalian biology, and its mechanism is not well understood. Studies on rats on a strict feeding schedule demonstrates a peak in physical activity and energy expenditure 2-4 hours before a meal. This is known as food-anticipatory activity (FAA). When these rats were then deprived of food, for days straight, they continued to demonstrate this rhythmic physical activity 2-4 hours before their routine meal times, demonstrating that routine meal timing rather than food itself is the reason behind FAA. The FEO runs independent of the master clock in the SCN. Continuous light deprivation or lesion of the SCN (both of which ensure that the master clock doesn't function) both demonstrated no effect on FAA while mice were kept on a constant feeding schedule. Thus, the FEO is known as a peripheral clock, but the phenomena's function and

location in the body is still unknown. With all hormones and phenomena mentioned above, most of the mechanisms of association between them and specific circadian rhythms are not fully understood and under heavy research. Figure 3 demonstrates the associations with these hormones and which type of clock they follow. However, it is important to not that some hormones lie on a spectrum based on how influenced they are by the master clock and peripheral clocks. Take cortisol for example, it is placed in the master clock end of the venn diagram. It can act as an important regulator of peripheral clocks, but it is mainly unaffected by meal timing.

Figure 1: Circadian Cycles and Phenomena - Master Clock and Peripheral Clocks



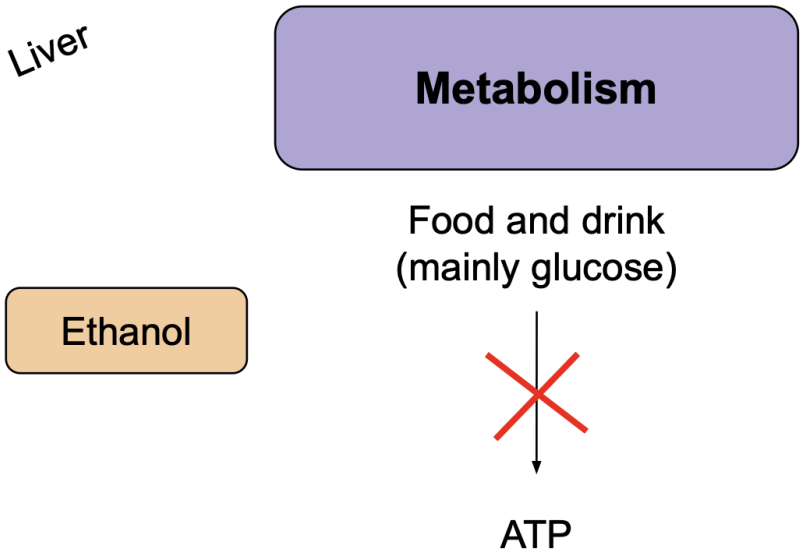
Alcohol's Effect on the Metabolic Cycle

Introduction to Metabolism

Every living organism is constantly performing metabolism. This is when energy is made and broken down in a cycle. The human's metabolic process heavily relies on the circadian rhythm. When someone is active, catabolism occurs in which energy is broken down and subsequently, when one is at rest, energy is made in the process of anabolism. This rhythm keeps humans healthy and alive. When it is broken, however, a cascade of negative effects disrupt the body.

Alcohol is one of many factors that harm the metabolic cycle. The CDC conducted a survey in 2018 inquiring adults about drinking. They found 66.3% of adults consumed at least one alcoholic beverage in the past year. Furthermore, 5.1% of adults were considered heavy-drinkers (7+ alcoholic drinks per week for women and 14+ for men). (Boersma et al., 2020)This is a drastic amount of people who have disrupted their metabolic cycle in extremely harmful ways.

According to the National Institute of Health, metabolism "consists of a series of reactions that occur within cells of living organisms to sustain life" (Judge & Dodd, 2020). Simply, it is the breakdown of food and drink into energy. The preferred source of energy for humans is glucose. Catabolism breaks this sugar down to make ATP which provides energy for the body's vital processes. Consequently, when alcohol is consumed, the body prioritizes breaking down ethanol over glucose. The process of converting glucose is put on hold, inhibiting the production of ATP. In fact, the liver processes alcohol at around one drink per hour, hence the common rule of having a maximum of one-drink-per-hour to not build up ethanol in the body.



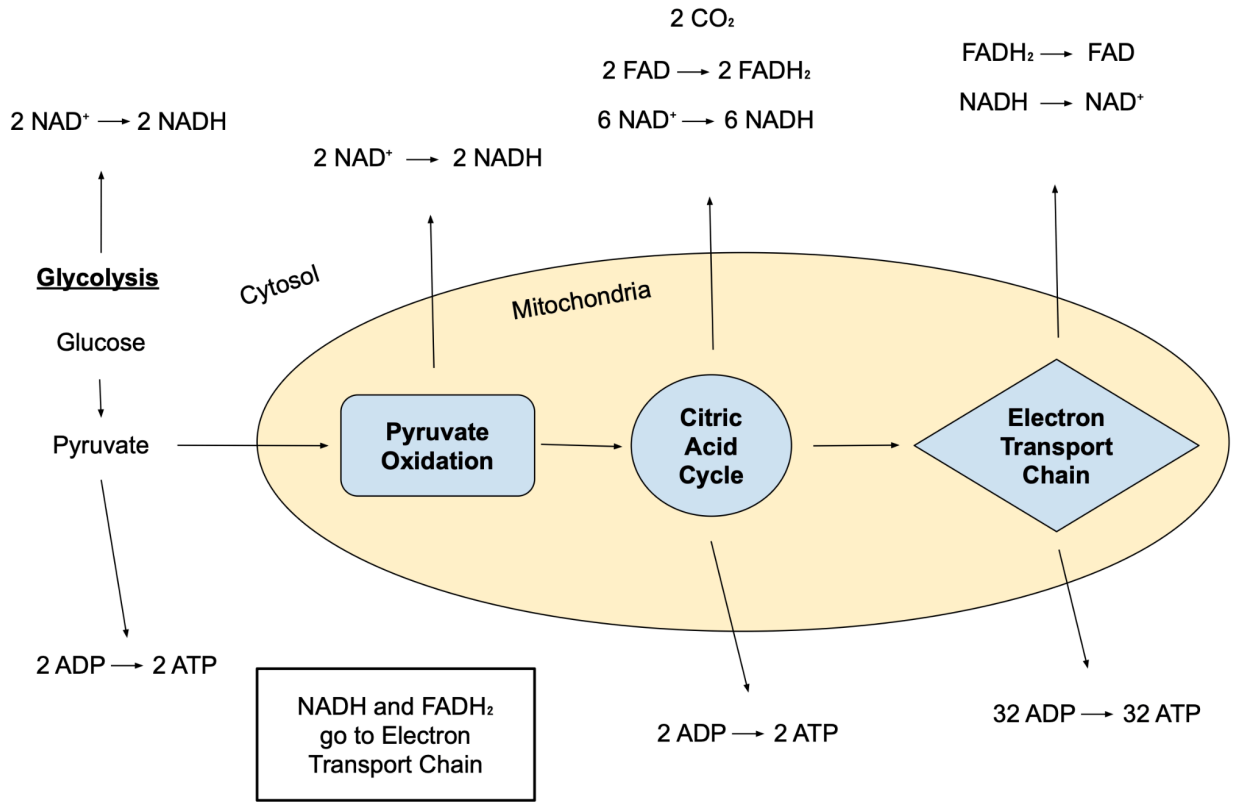
Alcohol's Effect on Metabolism

Metabolism occurs in the liver. It breaks down food and drink into ATP which provides energy for the body. The preferred source of energy is glucose. When ethanol is introduced to the system, this process is blocked. The liver promotes breaking down ethanol over food and drink (including glucose). Consequently, the production of ATP is blocked.

Metabolic Cycle Through Molecular Biology

Metabolism contains many complex molecular processes. The main metabolic process is aerobic respiration, used by countless organisms such as animals, plants, and even some bacteria. Aerobic respiration is a catabolic process that breaks down nutrients (mainly the preferred sugar, glucose) into energy. There are four main stages in aerobic respiration. The process starts off in the cytosol, the gel-like fluid surrounding the cell's organelles. Here, the first stage, Glycolysis, occurs. Glycolysis splits glucose into 2 pyruvates. The second step, Pyruvate Oxidation, occurs in the mitochondria, an organelle in the cell. The 2 pyruvates are transformed into Acetyl CoA. The third step, Citric Acid Cycle, turns Acetyl CoA into CoA. The final step involves the Electron Transport Chain. This process creates a substantial amount of ATP energy using the

byproducts from the previous steps. Metabolic process of aerobic respiration uses the four stages of Glycolysis, Pyruvate Oxidation, the Citric Acid Cycle, and the Electron Transport Chain to convert glucose into energy.



Aerobic Cellular Respiration Process

Aerobic Cellular Respiration's four main stages are Glycolysis, Pyruvate Oxidation, the Citric Acid Cycle, and the Electron Transport Chain. Glucose begins in the cytosol. The first step, glycolysis (left-most blue box), splits glucose into 2 pyruvate, producing 2 ATP and 2 NADH. Next, pyruvate is transferred to the mitochondria where Pyruvate Oxidation occurs. This transforms pyruvate into Acetyl CoA, utilizing 2 NAD+ and releasing 2 NADH. The third step, Citric Acid Cycle, turns Acetyl CoA into CoA. This process releases 2 ATP, 6 NADH, 2 FADH2, and 2CO2. Finally, the Electron Transport Chain occurs. Using the energy products released (FADH2 and NADH) from the previous steps, 32 ATP is created.

Circadian Rhythm of Metabolism

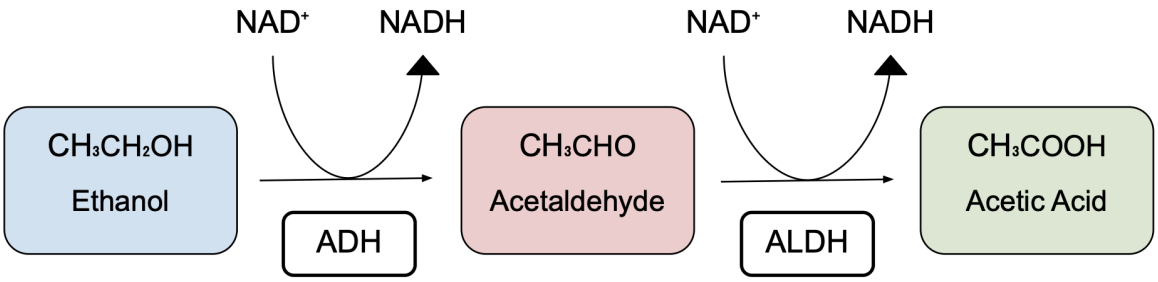
While the metabolic cycle depends on circadian rhythms, circadian rhythms also rely on metabolism. When one is active, catabolism occurs and alternatively, when resting, the body begins anabolism. These follow a 24-hour circadian cycle (Sato & Sato, 2023). The circadian rhythm controls metabolism through the suprachiasmatic nucleus and organs. These use core proteins to signal events, acting as the body's internal clock. They trigger catabolic pathways during the daytime and anabolic pathways in the evening (Ma & Morrison, 2023). So, the circadian rhythm controlled by the suprachiasmatic nucleus and organs determines when metabolism occurs.

On the other hand, metabolism controls the body's circadian rhythms. The circadian clock can sense the molecules that metabolism uses. Then, it alters clock proteins to follow the fluctuating cycle of metabolism. For example, as more NAD+ byproducts are created, a clock protein is modified. So, the amount of energy of a cell alters the molecular clock.

Alcohol Breakdown Through Molecular Biology

When alcohol is consumed, ethanol catabolism is promoted over glucose, offsetting the metabolic cycle. Ethanol catabolism relies heavily on the conversion of NAD+ to NADH. The addition of an electron to NAD+ creates NADH. These are molecules that act as electron carriers and can provide energy for reactions. Firstly, the enzyme Alcohol Dehydrase (ADH) takes over in the cytosol. ADH gives an electron to NAD+, turning it into NADH, providing the energy necessary for the reaction. ADH breaks down ethanol into a toxic chemical, acetaldehyde. This is a known carcinogen capable of harming protein and DNA. Acetaldehyde is transferred to the mitochondria in which the enzyme Aldehyde Dehydrogenase (ALDH) is contained. Acetaldehyde is broken down into acetic acid by the ALDH. Once again, another NAD+

molecule gains an electron, becoming an NADH and providing energy for the reaction. Acetic acid is less toxic and further gets split into CO2 and H2O which are secreted through processes such as urine, breath, and sweat (National Institute on Alcohol Abuse and Alcoholism, n.d.).



The Breakdown of Alcohol

When ethanol catabolism is promoted over glucose, the enzyme Alcohol Dehydrase (ADH) breaks down ethanol into a toxic chemical, acetaldehyde. This process utilizes an NAD+, releasing an NADH. Next, Aldehyde Dehydrogenase (ALDH), turns Acetaldehyde into acetic acid. Once again, ALDH uses another NAD+ molecule and releases an NADH.

The Issues Alcohol Causes in the Metabolic Cycle

There are countless issues that arise through the breakdown of ethanol. Firstly, alcohol disturbs the crucial ratios of molecules in metabolism. Aerobic respiration prefers a one-to-one ratio of NADH to NAD+. However, the breakdown of ethanol causes more NADH than NAD+ to be created. The build-up of NADH is harmful in the Citric Acid Cycle. The high levels of NADH signals the cell that it has plenty of energy. This, in turn, inhibits the citric acid cycle even though in reality, there isn't enough ATP produced. When the citric acid cycle is slowed, the Acetyl-CoA isn't used. Consequently, the liver uses the excess Acetyl-CoA for fatty acid synthesis. This is one of the triggers for fat accumulation from alcohol.

The previous aerobic respiration stage of glycolysis is affected because when the citric acid cycle slows, glycolysis slows down. The slow use of glucose then also slows down the process to create new sugars. This is why blood sugar can drop to dangerous levels with alcohol.

With the inhibition of the processes of aerobic respiration, the final stage of the electron transport chain can't produce sufficient ATP for the body. There are countless harmful effects of decreased ATP production such as organ damage, muscle fatigue, and lethargy.

Alcohol's Effect on Metabolism's Circadian Rhythm

Alcohol-induced metabolic stress can alter countless timing signals sent from the brain. Firstly, the high NADH and low NAD+ makes the Citric Acid Cycle think energy is abundant. So, glucose breakdown is halted. Normally, the liver can turn up energy creation during active hours and slow it down during rest. But, with the slowed breakdown of molecules, the liver's internal clock is shifted (Davis et al., 2018).

Alcohol impairs the sleep-wake schedule, altering the circadian rhythm. The concentration of melatonin and cortisol determines sleep times. Melatonin is higher in the evening while cortisol is low to aid sleep. Moreover, when alcohol is consumed, melatonin secretion is decreased and cortisol patterns are altered. This shifts the circadian rhythm of the sleep-wake cycle (Meyrel et al., 2020). Metabolism's timing depends on the sleep-wake cycle, so it is modified with the alteration of these hormones.

Lastly, alcohol impacts the body's temperature. Alcohol shuts down thermoregulatory control, so the body will take on the temperature of the environment instead of its stable 98.6 degrees Fahrenheit. A fluctuation of temperature can decrease the rate of metabolism. This is because the enzymes that are crucial to breaking molecules down can't perform efficiently in

alternate temperatures (Wasielewski & Holloway, 2001). As mentioned earlier, the circadian clock doesn't just control the metabolic cycle, but the metabolic cycle can determine the body's rhythms as well. This slow breakdown of molecules then sends signals to core clock proteins that alter the circadian rhythm (Myers, 1981).

Long-Term Health Effects of Unstable Circadian Rhythm of Metabolism

Firstly, the liver is physically harmed from excess alcohol consumption. The liver's internal clock may fall out of phase, making it susceptible to tissue injury, scarring, and even cancer.

Another physical harm of alcohol's effect on metabolism is fat accumulation. The liver promotes breaking down ethanol so it doesn't break down fat. The accumulation of fat comes not only from the halt in the breakdown of fat, but also the excess Acetyl-CoA that the Citric Acid Cycle can't use. Acetyl-CoA is then synthesized into fatty acids, accumulating more fat in the body. This can cause Alcohol-Induced Liver Disease. As the liver becomes severely inflamed and cells die, healthy liver tissue gets replaced with scar tissue. The liver's ability to function continuously declines with more alcohol consumption, offsetting the liver's internal clock.

A weak clock within the liver then weakens other organs as well. Specifically, the gut lining can be destroyed, allowing toxins to enter. This causes the condition known as "leaky gut" which can cause digestive issues, fatigue, and even joint discomfort (Breus, 2018).

There are countless other issues that alcohol induces on the circadian rhythm of metabolism. Every impact causes a cascade of harmful effects throughout the body. An altered glucose concentration can change blood sugar, fat accumulation can harm the liver, which can further break down the gut lining. Circadian rhythms hold utmost importance as the clocks

dictate one's life. Going against these rhythms cause harmful effects to the body. Excessive alcohol consumption is only one out of thousands of possible damages to the circadian rhythm.

UNIT AND CHAPTER

Unit: Background / Ultradian Rhythms / **Circadian Rhythms** / Infradian Rhythms

Chapter X: Chronotypes

CONTRIBUTORS

- Jack Lou
    - Subject Matter Expert – 'Chronotypes'
        - Sections 1, 2, 3, 4 and 5
    - Figures 1, 3, 5
- Semaj Cunningham
    - Copyeditor
    - Peer Reviewer

LEARNING OBJECTIVES

After reading this chapter, students should be able to …

1. Understand the 3 main chronotypes and how chronotypes affect someone's life
2. Explain differences in body temperature rhythms between morning and evening types
3. Identify your own chronotype and reflect on how it has shaped your mood, daily functioning, and habits.
4. Propose solutions to health consequences due to chronotype-schedule misalignment

CHAPTER OUTLINE

- What is a chronotype? What are the different variants of chronotypes? (~0.5 page)
    - Subtitles: Chronotype's origin
        - Section: Explaining chronotype from an evolutionary lens
        - Section: when chronotype was discovered and who proposed it
    - Subtitle: The 3 different chronotypes
        - Section: Morning
        - Section: Intermediate
        - Section: Evening
        - Section: What does having different chronotypes imply for society?
- How does chronotype affect hormones and circadian physiology, and how do these affect athletic performance? (2 pages)
    - Subtitle: What is a hormone; which ones affect athletic performance?

        - Section: Cortisol
        - Section: Melatonin
    - Subtitle: Hormone's relation to chronotypes
        - Section: hormone secretion timing
        - Section: Examples from paper#2
- How do athletes adapt/compromise to a chronotype that doesn't favor their training schedule? (1 page)
    - Subtitle: short-term performance decrease from mismatch
        - Section: Athletes not performing at peak because timing of competition/race
    - Subtitle: Adaptation
        - Section: study that has shown that training at specific times can alter your body's timing for peak performance
- Case studies of athletes of differing chronotypes (2.5 pages)
    - Subtitle: Chronotypes in action
        - Section: Morning type
        - Section: Intermediate type
        - Section: Evening type

CHAPTER WRITTEN CONTENT

Learning Objectives

1. Describe the 3 main chronotypes and how chronotypes may affect someone's life
2. Explain differences in body temperature rhythms between morning and evening types
3. Identify your own chronotype and reflect on how it has shaped your mood, daily functioning, and habits.
4. Propose solutions to health consequences due to chronotype-schedule misalignment

Introduction and Background of Chronotypes

Modern research has defined chronotype as "a biological surrogate for an individual's phase of entrainment." In other words, it is a measurable proxy of an individual's internal clock's alignment with external factors (Roenneberg, 2012). Chronotypes have extensive impacts in

daily life- it affects our preferred bedtime, preferred wakeup time, and even productivity and health.

How did chronotypes come to be? From an evolutionary lens, chronotypes evolved because a tribe would require at least one person awake to keep everyone safe because sleep is a state of vulnerability to predators. This is known as the sentinel hypothesis: group animals have asynchronous sleep times because at least one person would need to stay attentive to the surroundings. Samson et al. (2017) found that 99.8% of tribe members in a hunter-gatherer group were awake between the time the first member went to sleep and the last woke up. This study on a tribe's sleep pattern shows how modern differences in our chronotypes may be caused by natural selection, working as a group defense mechanism humans evolved to counteract the susceptibility to danger during sleep.

The Biological Mechanism Behind Differences in Chronotypes

The first trace of research regarding chronotypes originated in 1900 when O'Shea published his observations on differences between peak intellectual and physical timings in the book *Aspects of Economy*. However, chronotypes were only relevantly established as biological mechanisms in 1976 by Horne & Ostbërg. Set out with the aim of revising the concepts of "morningness" and "eveningness", the two researchers created a 19-question multiple choice questionnaire about sleep preferences. The questionnaire included questions such as "If you went to bed at 11:00 PM at what level of tiredness would you be?" and "When you have no commitments the next day, what time do you go to bed compared to your usual bedtime?". Each answer had a score assigned that would cumulate to a total, with a lower score indicating an evening preference and higher score indicating a morning preference. After allowing their participants to complete the questionnaire, they were grouped into 3 categories: morning,

intermediate, and evening type. The result of the participants' questionnaires directly correlated with body temperature, validating the science of chronotypes.

[Insert Figure 1 Add-in]

This graph displays a clear biological distinction between morning and evening types. Starting from 9am, the morning types started their day with a higher body temperature and maintained the upper end until about 8pm, where the trendlines for morning and evening types intersected and the temperature dynamic reversed. While the study showed that both morning and evening types both peaked in body temperature during the evening (7:32PM for morning type and 8:40PM for evening type), morning types had higher body temperatures to start the day along with an earlier temperature peak and decline. On the other hand, evening types had lower body temperatures to start the day, but their peak and decline in body temperature was later in the evening (Horne & Ostbërg, 1976). Replications of this experiment have consistently found that morning types have higher body temperatures in the morning and subsequent earlier peaks and declines compared to evening types. Horne and Ostbërg's research has been instrumental in paving the way for more chronotype research after their paper was published. Their 19-item questionnaire served as a standard for future researchers to use in assigning their own participants into chronotype groups (Baehr et al., 2000; Bailey & Heitkemper, 2001; Duffy et al., 1999; Kerkhof & Van Dongen, 1996). The morning, intermediate, and evening types for chronotypes are still the most relevant categories today.

Chronotypes and Physical Performance

Muscle temperature has direct impacts on performance- for every 1 degree celsius increase in body temperature, performance increase by approximately 3.5% (Wilson et al., 2025). Body temperature means a higher muscle temperature, because blood carries core heat directly to

the skeletal tissues. Therefore, the level of baseline body temperatures may have a direct impact on an athlete's performance (Dykstra, 2024). Chronotypes have been found to differ in onset of body temperature by Horne & Ostberg's study and its replications. Body temperature may explain the relationship between chronotype and exercise, because chronotype is associated with body temperature timings which could in turn affect exercise. However, research in this field is currently limited and the causation between body temperature and chronotypes cannot be fully established. There may only be speculation that chronotypes have specific differences in time-of-day related physical performance levels.

Chronotypes, Social Jetlag, and Mental Disorders

Social Jetlag is the difference between an individual's preferred sleep schedule and their actual sleep schedule that is caused by societal factors such as work, school, or family. The most common workday in the U.S starts at 8AM, with only marginal differences in work start times across many other countries such as Japan, China, UK, and Germany (Torpey, 2015). For instance, the most common workday starts at 8:30AM in Japan but 9AM in China. Chronotypess' relationship with our work-life schedules are related to mental health outcomes. For example, Zou et al. (2022)'s narrative review has found that evening chronotypes are more likely to have mental disorders, while being a morning type is a protective factor against mental disorders. Moreover, a Dutch correlational study found that the evening chronotype was not only correlated with worse reported mood in the morning, but was also associated with major depressive disorder (Antypa et al., 2015). A plausible explanation for these findings is sleep deprivation. Compared with morning types, both evening and intermediate types have a greater risk for insomnia- evening types are 247% and intermediate types 61% more likely to have insomnia than morning types. Insomnia is the condition of difficulty falling asleep, staying

asleep, or having good quality sleep (National Heart, Lung, and Blood Institute, 2022). Since evening types have delayed internal rhythms compared to morning types, it is harder for them to adjust to a misaligned sleep schedule—inducing insomnia-like symptoms and increasing the risk of developing insomnia. In brief, evening types are at a greater risk for having physical, mental, and social issues in daily life because of circadian misalignment (a.k.a social jetlag).

How do Chronotypes Affect Responses to Work Schedules?

[Insert case study add-in]

This case study displays the significant implications chronotypes have on our life. Although Patient F's story does not establish causation between chronotypes and mental illnesses, it is an illustration of how her chronotype affected her response to a schedule change in life. Patient F's experience brings up more links between chronotypes and different aspects of life, such as sleep, mood, and daily functioning.

Conclusion

Chronotypes are a measure of how intrinsic rhythms align with environmental factors; it relates to body temperature, responses to schedule changes, physical performance at specific times, and sleep in general. We can all take time to reflect on how our own chronotypes impact how we interact with this world, along with the steps that can be taken to improve it. Chronotypes' relation with societal timings raises a broader question: are early start times for work and night shifts benefitting society, or is it degrading individual health? Anyhow, the most important takeaway from this chapter should be that sleep should never be compromised- all chronotype-related issues arise from sleep, so always give yourself a good night's sleep.

CHAPTER ADD-INS

Add in #1

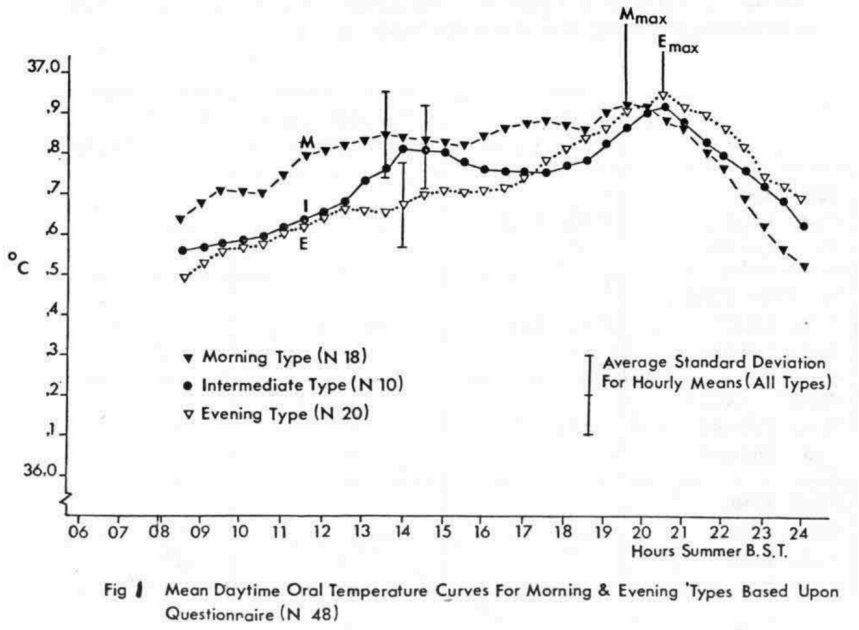


Fig 1 Mean Daytime Oral Temperature Curves For Morning & Evening Types Based Upon Questionnaire (N 48)

Figure 1: Mean Daytime Oral Temperature Curves For Morning & Evening Types Based Upon Questionnaire. *By using a sextic polynomial model, Horne & Ostbërg fitted a least-squares curve and found the time when every individual's temperature peaked. Using those fitted curves, they computed a point for every hour from 9AM to 11PM. Each point on the 3 different trendlines corresponds to the average of the all individual graph temperatures for that specific group at that hour. Significant differences were found between the morning type and evening type, while the intermediate type showed no significant differences with either group.*

Add in #2

When a 30-year-old woman brought herself to a health clinic, she was displaying symptoms of stress,

depression, urge to sleep, and anxiety. At the age of 19, she was diagnosed with bipolar disorder but managed to graduate from university while undergoing treatment. After almost a complete recovery from her mental disorder, she worked morning and afternoon shifts which allowed her to function regularly. However, after switching to night shifts months before she presented herself at the clinic, her life was off balance and was overwhelmed with mental and physical health issues. The clinic diagnosed her as an extreme morning type using the morning-eveningness 19-item questionnaire, and suggested she quit her night shifts to recover a normal life. After 3 days without working night shifts, her mental state improved (Meyrer et al., 2009).

Case Study: Patient F's Chronotype-Work schedule balance.

GLOSSARY

Term	Definition
Chronotypes	a biological surrogate for an individual's phase of entrainment
morningness	those who show extreme preferences for daytime activity; in these individuals, peak performance and alertness is associated with the early-morning hours
eveningness	extreme preferences for nighttime activities; in these individuals, heightened alertness and peak performance is linked to the evening hours
Social Jetlag	the difference between an individual's preferred sleep schedule and their actual sleep schedule that is caused by societal factors such as work, school, or family
Major depressive disorder	a common and serious mental health condition defined by a persistent low mood and a loss of interest or pleasure in daily activities

Insomnia	the condition of difficulty falling asleep, staying asleep, or having good quality sleep

REFERENCES

Antypa, N., Vogelzangs, N., Meesters, Y., Schoevers, R., & Penninx, B. W. J. H. (2016). Chronotype associations with depression and anxiety disorders in a large cohort study. *Depression and Anxiety, 33*(1), 75–83. https://doi.org/10.1002/da.22422

Baehr, E. K., Revelle, W., & Eastman, C. I. (2000). Individual differences in the phase and amplitude of the human circadian temperature rhythm: With an emphasis on morningness–eveningness. *Journal of Sleep Research, 9*(2), 117–127. https://doi.org/10.1046/j.1365-2869.2000.00196.x

Bailey, S. L., & Heitkemper, M. M. (2001). Circadian rhythmicity of cortisol and body temperature: Morningness–eveningness effects. *Chronobiology International, 18*(2), 249–261. https://doi.org/10.1081/CBI-100103189

Duffy, J. F., Dijk, D.-J., Hall, E. F., & Czeisler, C. A. (1999). Relationship of endogenous circadian melatonin and temperature rhythms to self-reported preference for morning or evening activity in young and older people. *Journal of Investigative Medicine, 47*(3), 141–150. https://doi.org/10.1177/108155899904700310

Dykstra, L. (2024, February 16). *JustInTimeMedicine*. https://www.justintimemedicine.com/curriculum/6935

Kerkhof, G. A., & Van Dongen, H. P. A. (1996). Morning-type and evening-type individuals differ in the phase position of their endogenous circadian oscillator. *Neuroscience Letters, 218*(3), 153–156. https://doi.org/10.1016/S0304-3940(96)13140-2

Meyrer, R., Demling, J., Kornhuber, J., & Nowak, M. (2009). Effects of night shifts in bipolar disorders and extreme morningness. *Bipolar Disorders, 11*(8), 897–899. https://doi.org/10.1111/j.1399-5618.2009.00767.x

National Heart, Lung, and Blood Institute. (2022, March 24). *Insomnia: What is insomnia?* https://www.nhlbi.nih.gov/health/insomnia

Roenneberg, T. (2012). What is chronotype? *Sleep and Biological Rhythms, 10*(2), 75–76. https://doi.org/10.1111/j.1479-8425.2012.00541.x

Samson, D. R., Crittenden, A. N., Mabulla, I. A., Mabulla, A. Z. P., & Nunn, C. L. (2017). Chronotype variation drives night-time sentinel-like behaviour in hunter–gatherers. *Proceedings of the Royal Society B: Biological Sciences, 284*(1858), Article 20170967. https://doi.org/10.1098/rspb.2017.0967

Torpey, E. (2015, October). *Careers for night owls and early birds*. U.S. Bureau of Labor Statistics. https://www.bls.gov/careeroutlook/2015/article/night-owls-and-early-birds.htm

Wilson, C. J., Nunes, J. P., & Blazevich, A. J. (2025). The effect of muscle warm-up on voluntary and evoked force-time parameters: A systematic review and meta-analysis with meta-regression. *Journal of Sport and Health Science, 14*, Article 101024. https://doi.org/10.1016/j.jshs.2025.101024

Zou, H., Zhou, H., Yan, R., Yao, Z., & Lu, Q. (2022). Chronotype, circadian rhythm, and psychiatric disorders: Recent evidence and potential mechanisms. *Frontiers in Neuroscience, 16*, Article 811771. https://doi.org/10.3389/fnins.2022.811771

UNIT AND CHAPTER: Circulation and the Cardiac Cycle

Unit: Background / Ultradian Rhythms / Circadian Rhythms / Infradian Rhythms

Chapter X: Heart Clocks

CONTRIBUTORS

- Ada Langford
    - Subject Matter Expert
    - Sections 1.A, 1.B, 1.C, Section 4, Section 5, Section 6
- Daniel Winter
    - Peer Reviewer
- Ryan Nordahl
    - Subject Matter Expert
    - Sections 1.D, Section 2, Section 3
- James Cunningham
    - Peer Reviewer
- Jacob Li
    - Peer Reviewer

LEARNING OBJECTIVES

After reading this chapter, students should be able to …

1. Explain how the heart pumps blood periodically and why it changes throughout the day (Sympathetic/Parasympathetic vs Diurnal Blood Pressure/Heart Rate).
2. Be able to examine how circadian clocks work with heart clocks.
3. Analyze how long-term health, including conditions like aging, cardiovascular and metabolic issues, and lifestyle affect the heart's clock.
4. Be able to examine how different electrical and mechanical problems affect the heart and why.
5. Be able to compare and contrast sympathetic and parasympathetic control
6. Be able to talk about heart function with increased endurance training
7. Sample
8. Sample

CHAPTER OUTLINE

- Section

    - Subsection
- Section
    - Subsection

CHAPTER WRITTEN CONTENT

Section 1: How the Heart Pumps Blood

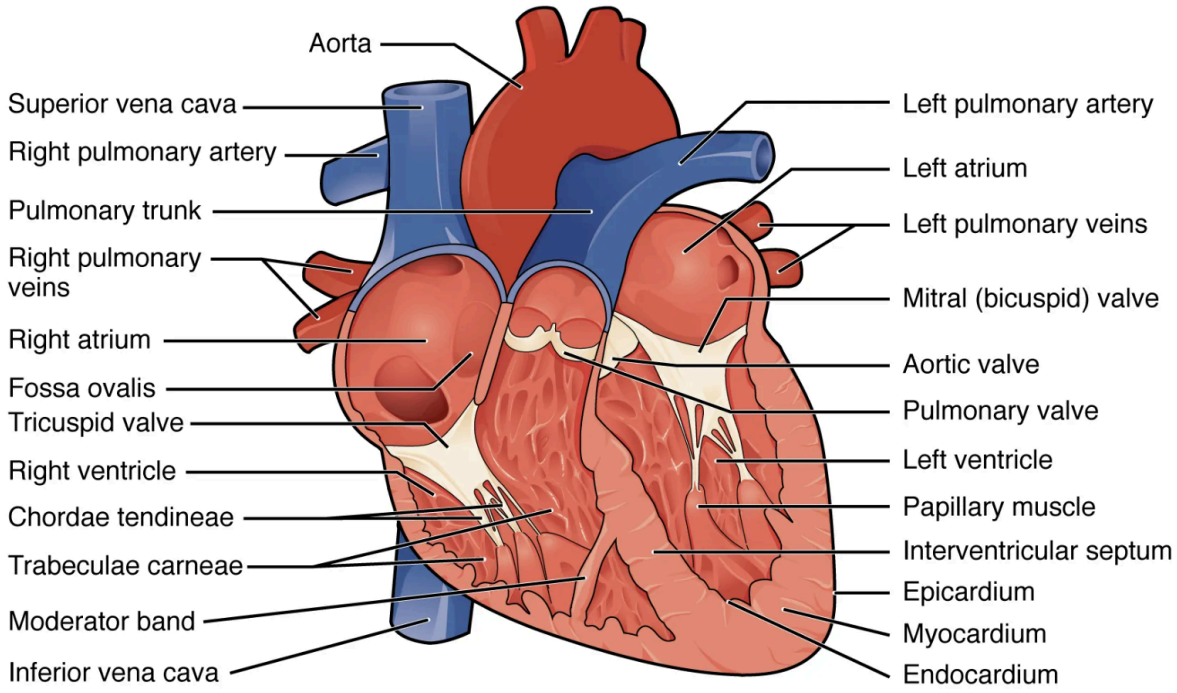
Every day even though you don't think about it, your heart is constantly beating. How is this you may ask, why does the heart know when to speed up or slow down without thinking about it. The simple answer is that your heart, In this chapter, you will learn about how the heart works, reacting to both extrinsic and intrinsic factors.

Section 1.A: Mechanical Components/Blood Flow

The heart is an incredibly complicated muscular organ that circulates blood through the lungs and throughout the body, helping make sure that oxygen and nutrients are delivered to cells and that waste products are removed. In a human, the heart is made up of four hollow chambers: the **right atrium**, the **right ventricle**, the **left atrium**, and the **left ventricle** (Ansari). The right atrium receives deoxygenated blood (blood without oxygen) from the body and sends it below to the right ventricle. This blood is then pumped back to the lungs in order for the carbon dioxide to be removed and for the blood to become oxygenated again.

The oxygenated blood from the lungs then travels to the left atrium, which pumps the blood below to the left ventricle, which then pumps it to the rest of the body. The heart also maintains the direction of blood flow throughout the body using **valves** to prevent blood from flowing backwards (**backflow**). The **tricuspid valve** lies between the right atrium and right ventricle and prevents backflow into the atrium. Similarly, the **pulmonary valve** sits between the

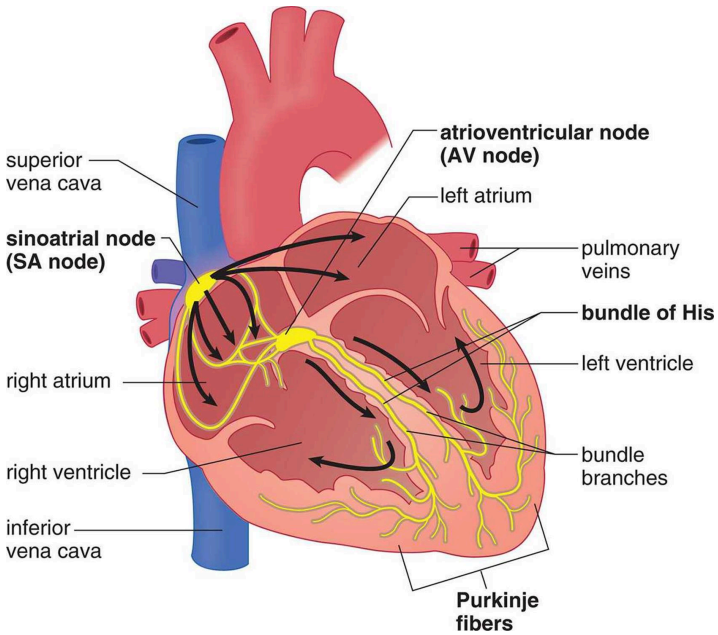
right ventricle and the **pulmonary artery**, the main blood vessel that sends blood from the heart to the lungs. The **mitral (bicuspid) valve** also controls blood flow between the **left atrium** and **left ventricle**. Lastly, the **aortic valve** sits between the **left ventricle** and **aorta** (main blood vessel that sends blood from the heart to the rest of the body). The valves open and close as necessary to allow blood to flow while preventing backflow, and are mostly controlled by pressure differences between the chambers.



Heart's Mechanical Function: Deoxygenated blood from the body travels through the superior and inferior vena cava into the right atrium. After passing through the tricuspid valve, the blood then flows into the right ventricle, where it is pumped through the pulmonary valve and into the pulmonary artery. After the blood passes through the lungs, it travels back to the heart via the pulmonary veins into the left atrium. Next, the blood passes through the mitral valve and into the left ventricle, where it is pumped through the aortic valve and into the aorta to send blood to the rest of the body.

Section 1.B: Electrical Rhythm

The heart also has a self-generating electrical system that ensures the ventricles and atria contract and release at reasonable times to maintain regular heartbeat patterns and circulation. One important component of this system is the **sinoatrial (SA) node**, which is located in the upper wall of the right atrium (Chan-Dewar). The SA node is a small collection of electrical cells that act as the primary pacemaker of the heart, generating electrical impulses at a rate of 60-100 times per minute that tell the atria to contract and push blood into the ventricles. Next, the **atrioventricular node (AV) node**, located in the lower back region of the heart, receives the signal from the SA node and briefly delays its passage by 0.1 seconds until the ventricles can fully fill with blood before they contract. After the impulse passes through the AV node, it then travels down the **Bundle of His**, a collection of specialized heart muscle cells that transmits the impulse to the ventricles. **Purkinje fibers** receive impulses from the bundle branches and deliver them quickly throughout the ventricles, which ensures coordinated contraction of the ventricles to pump the blood out of the heart.



Heart's Electrical Function: The electrical signal travels from the SA node to the AV node, then down the Bundle of His and Purkinje fibers to ensure the heart follows a coordinated contraction pattern (Diagram of the heart's electrical conduction system)

Section 1.C: Circadian Clocks

Over a regular 24 hours, heart rate is the highest during the day and lowest during the night, when we sleep. During the night, not only is the gap between beats longer, but also the conduction through the AV node and ventricles are slower (Black). When a study forced people on an unnatural non-24-hour schedule, their heart rates still followed the roughly 24-hour circadian schedule, with low heart rates during the times they should have been sleeping (Thosar). However, they still had lower heart rates during actual artificial sleep times than during waking hours. This suggests that heart rate fluctuations are due to a mixture of sleep patterns and an internal circadian clock. The body's master circadian clock, the suprachiasmatic nucleus (SCN) of the hypothalamus, tries to maintain the roughly-24 hour cycle with heart rate.

Section 1.D: Extrinsic Factors for Heart Clocks

There are many extrinsic factors that cause heart clock variation, including environmental factors and lifestyles factors, which will be discussed later in the chapter. Over time, your body has evolved to take certain environmental information and use it to prepare for what it thinks comes next. With heart rate and the environment, extrinsic factors are typically tied to the circadian rhythm and sleep. These factors include temperature, time of day, light exposure, and altitude, all of which are directly related to the body's circadian rhythm. For time of day (related to light exposure), heart rate decreases late at night and spikes early in the morning as the body wakes up.

Section 2: Sympathetic vs Parasympathetic Control

The nervous system has two controls on the heart and how quickly it beats: the sympathetic nervous system and the parasympathetic nervous system. The sympathetic nervous system speeds up the heartbeat, whereas the parasympathetic nervous system slows it down, both through chemical signalling on the Sinoatrial (SA) node or the natural pacemaker of the heart.

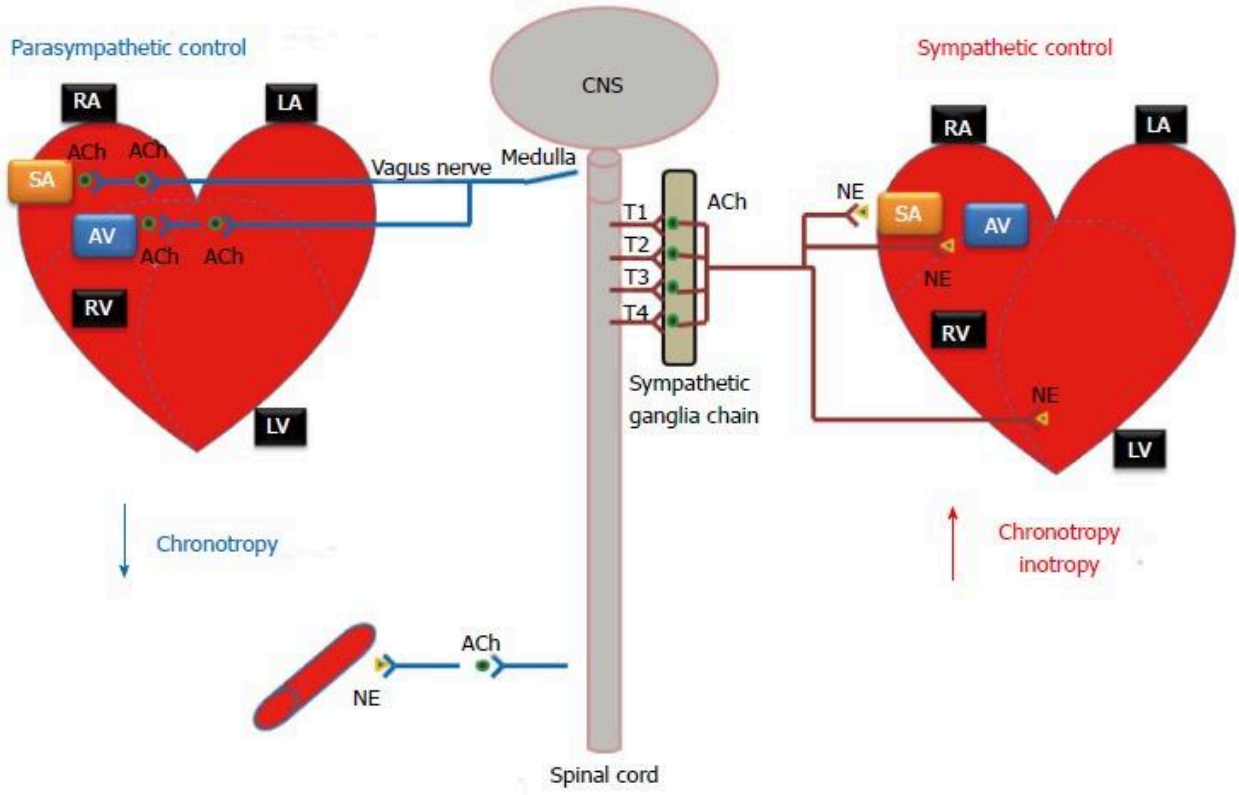


Figure 2: Diagram of the parasympathetic and sympathetic nervous system
Gordan, R., Gwathmey, J. K., & Xie, L. H. (2015). Autonomic and endocrine control of cardiovascular function. *World journal of cardiology*, *7*(4), 204–214. https://doi.org/10.4330/wjc.v7.i4.204

In this image is the parasympathetic and sympathetic nervous system. On the right is the parasympathetic nervous system which relaxes the body. In this diagram the central nervous system (CNS) sends a message from the medulla through the vagus nerve to the Sinoatrial

(SA) and the atrioventricular (AV) node via Acetylcholine(ACh) a neurosignal to slow down the heart. On the right side is the depiction of the pathway of the sympathetic nervous system in which the brain through the sympathetic ganglia chain creates a signalling cascade that releases neuroepinephering(NE) into the SA and AV nodes to speed up the heart.

Section 2.A: Sympathetic Nervous System

The sympathetic nervous system is located in the thoracic spinal nerves and is a group of 12 nerves that sends signals via neurons and chemical signals to the heart. The sympathetic nervous system causes the heart to go into a flight-or-fight response, stimulating the SA node to increase heart rate. There are several different reasons that would explain the need for the stimulation of the sympathetic nervous system, including both extrinsic and intrinsic factors (displayed in the table). All of these factors stem from the body needing more oxygen and glucose sent to different muscles and organs.

Section 2.B: Parasympathetic Nervous System

The parasympathetic nervous system is mainly located in the brainstem with the Vagus nerve being the main signalling pathway. The parasympathetic nervous system is tied to the slowing down of the heart. Similar to the sympathetic nervous system, it uses chemical signals on the SA node to change the heart rate. The parasympathetic nervous system is tied to rest and relaxation and conserves energy when high output is not needed.

Section 3: Heart Function with Increased Endurance Training

Section 3.A: What is Endurance Training?

Endurance, or aerobic training is the process of repeatedly subjecting the heart to exercise at levels lower than the person’s maximal heart rate. Running, cycling, and swimming are common examples. A single session of endurance training places an acute load on the

cardiac muscle, which when the body is subjected to over repeated sessions over weeks, causes the load to add stimulus which causes the heart to adapt. This is the central idea of the chapter: trying to understand how the heart can adapt to cycles of habit.

Section 3.B: Structural and Functional adaptation

Regular endurance training leads to changes in the heart itself. The chambers, particularly the left ventricle, can enlarge and fill with more blood between beats, so each contraction ejects a larger volume of blood. This volume per beat is called stroke volume. Because each beat moves more blood, the heart can deliver the same amount of blood with less strain on the heart. Where this can be seen is in your resting heart rate, or your heart rate when you are sleeping in the most restful position. You can notice this also through taking five minutes and sitting on a couch or chair not doing anything then taking your heart rate, notice how it drops.

Section 3.C: Autonomic Nervous System Adaption

As noted earlier in the parasympathetic and sympathetic nervous system section, the heart is controlled by these two systems to either speed up the heart or slow it down. Endurance tends to shift the resting balance toward greater parasympathetic influence which contributes to the lower resting heart rate as described above.

Section 3.D: Training the Clock as Well as the Heart

Many body functions, including heart rate and blood pressure, follow rhythms. As stated priorly in this chapter the heart rate and blood pressure rhythms are influenced by zeitgebers which allow for the outside world to create rhythms, a process called entrainment. Light is the best-known zeitgeber, but physical activity is a nonphotic zeitgeber. In a sense, exercise can be used to signal the body what the time is.

If exercise takes place at about the same time each day, the same logic may apply. A regular session becomes a predictable daily event, and the cardiovascular system will start anticipating it, raising heart rate and sympathetic activity slightly in advance, so the heart is already ready to speed up when the workout begins. This anticipatory effect is a plausible consequence of entrainment.

Research does show that exercise timing can shift the heart's daily rhythm "Morning or evening exercise: effects on the heart rate circadian rhythm above the Arctic Circle.". In one study participants performed the same routine in the morning or the evening, and evening exercise during a dark winter period delayed the peak of the heart rate rhythm compared to morning exercise and with a control variable which suggests that exercise is a zeitgeber.

Section 4: Heart Long-term Health Impacts

Section 4.A: Aging

A number of factors, including aging, cardiovascular and metabolic conditions, lifestyle, and environment, can affect the heart's adaptability and strength. For example, the maximum oxygen used by the body's cells per minute during intensive exercise, also known as **V_{O_2}max**, drops by about half between the ages of 20 and 80 for both men and women (Strait). This decrease in maximum oxygen is because two separate measures each decline by about 25% as we age: the amount of blood pumped per minute (also known as **cardiac output**), and the amount of oxygen taken out of the blood by cells (Strait). The first measure, the amount of blood pumped per minute, is equivalent to the heart rate multiplied by the amount of blood pushed out with each contraction. Since the amount per contraction stays roughly the same as we age, the drop in cardiac output is mostly due to the drop in the maximum heart rate. While healthy aging does not generally change the resting heart rate, the maximum heart rate declines significantly.

In fact, peak heart rate during exhaustive exercise falls by 25% for both men and women between ages 20 and 80 (Strait). The heart's maximum heart rate decreases because of various aging-related changes to the SA node. For instance, fat can build up around the SA node and impede its ability to send electrical signals throughout the heart (Strait). Additionally, fewer than 10% of the number of SA node cells seen in younger adults remain by age 75 (Strait).

As we age, the electrical signal within each heart cell lasts longer, which makes each contraction take longer and limits how fast the heart can contract. Contraction is also less efficient due to certain walls of the heart getting thicker and more round as we age in order to cope with stiffer arteries (Strait). The aging heart also becomes less responsive to adrenaline, a hormone that usually signals the heart to contract faster and more completely during exercise or stress. Since older hearts relax slower, the fast suction that younger hearts rely on to bring blood into the heart is weaker in older hearts. But since older hearts beat slower during exercise anyways, there is usually more time for the blood to trickle into the heart so it can get just as full. However, when the older heart needs to beat at maximum capacity and recycle as much blood as possible to ensure enough oxygen circulation, it is not able to compensate.

Section 4.B: Cardiovascular and Metabolic Conditions

In addition to age, cardiovascular and metabolic conditions can also affect the heart's natural rhythm. In obesity, there is extra blood in the body in order for oxygen and nutrients to reach the extra body fat (Wiley). This extra blood means that the heart has to pump more every minute, which means it is working harder even at rest. Obesity also activates certain systems in the body that raise blood pressure (Wiley). Since the heart has to pump more blood under higher pressure, its muscle walls respond by thickening, which lowers the amount of space left for blood to take up and makes it more difficult to fill the heart. Additionally, obesity can cause fat to build up in the heart, which can lead to **scarring** (Wiley). The fat would release chemical signals that can cause inflammation and damage, which is repaired by the body by creating the

stiff scar tissue. This stiff scarring can make it harder for the heart to relax and fill with blood as easily. These factors all contribute to higher heart failure rates in obese people, mostly due to the heart not being able to compensate for the added strain. In fact, it was found that the risk of developing heart failure increases on average by 5% in men and 7% in women for every 1-unit BMI increase (Wiley).

Section 4.C: Lifestyle Choices

Heart clocks can also be significantly affected by lifestyle choices, such as sleeping schedules and smoking. As described earlier, the heart follows a roughly 24-hour cycle, controlled by the body's SCN (Chellappa). When the SCN's circadian clock falls out of alignment, meaning that the body's internal clock does not match the timing of sleep and daylight patterns, the heart clock can also become less healthy. Shift work, for example, has been shown to be linked to higher rates of various cardiovascular diseases. One study explored what happens when healthy, usual day-shift workers underwent three day-shifts and three night shifts. They found that the **circadian misalignment** due to the night shift work increased blood pressure during awake hours, and that the healthy decrease in systolic blood pressure that normally occurs during sleep was dampened. It was also found that their **parasympathetic activity**, which slows the heart and protects it from working too hard, was significantly decreased during awake hours (Chellapa). When shift workers (instead of primarily day-workers) were similarly put through the same circadian alignments versus misalignment study, it was found that their blood pressure was also elevated, especially during sleep (Chellappa). This suggests that even people who regularly do shift work do not adapt their heart clocks to be robust to circadian misalignment. One possible reason for the lack of ability for humans to adjust to shift work's shifted sleep schedules is that there is not enough time for the internal clock (which uses slow-changing melatonin and cortisol levels) to adapt. And since the internal clock heavily regulates the heart clock, the heart clock can also be messed up.

Smoking can also heavily disrupt the heart's rhythm, since it disrupts the **autonomic nervous system**, which controls heart rate. The autonomic nervous system includes the **sympathetic nervous system (SNS)**, which increases the heart rate, and the **parasympathetic nervous system (PNS)**, which lowers the heart rate (Dinas). The heart's rhythm depends on a balance between the SNS and PNS' activity. **Heart rate variability (HRV)**, which is the variation in time between consecutive heartbeats, shows how activated one nervous system is compared to the other. PNS acts quicker than SNS, so when it is activated the heart changes a lot beat-to-beat, resulting in higher HRV. Generally, a low HRV means a less healthy, less adaptable heart and has been linked to higher risk of heart disease. Studies have shown that both short-term and long-term smoking increase sympathetic activity and decrease parasympathetic activity, thus decreasing overall HRV. Nicotine is thought to be the main cause of decreased overall HRV, because it increases the release of certain chemicals that activate the SNS, which might make the heart more vulnerable to dangerous **arrhythmias**, or irregularities in heart patterns (Dinas). A study found that mice that were exposed to smoke for three days were more likely to get arrhythmias like atrial fibrillation and tachycardia (Dinas).

	Mechanical Differences	Electrical And Autonomic Differences	Effect on the Heart's Clock
Healthy Heart	● Not too stiff ● Walls aren't too thick or too weak ● Fast refill time	● Working SA node ● Signals can be processed through the heart cells quickly	● Normal sinus rhythm ● Can easily adapt to exercise
Heart in Old Patients	● Stiffer walls ● Slower refill time, can't keep up during intense exercise	● Fewer SA node cells ● Signals last longer in each heart cell ● Less	● Lower maximum heart rate

		responsive to adrenaline	
Hearts in Obese Patients	• Increased blood pressure • Scarring leads to stiffer walls	• Pumps more blood per minute	• Increased risk for heart failure
Hearts in Patients with Circadian Misalignment	• Increased blood pressure	• Decreased parasympathetic activity	• Higher rates of heart disease
Hearts in Patients Who are Exposed to Smoke	• Low HRV	• Decreased parasympathetic activity	• Higher risk for heart arrhythmias

Section 5: Rhythm Disruption

Section 5.A: SA Node Dysfunction

Disruptions, failures, or changes in parts of the heart can also affect heart rhythm. For example, when the SA node, which is the primary pacemaker of the heart, fires too slowly, it is called **sinus bradycardia**. Sinus bradycardia is often caused by a more active parasympathetic system, and it can even be normal in athletes. A more serious problem happens if the SA node ages or if it loses blood supply, which can cause **sinus node dysfunction**. There are three main types: in **sinus pause**, the SA node fires late; in **sinus arrest**, it fails to fire at all; and in **SA exit block**, it fires but the signal cannot leave the node (Desai). These failures cause the heart to beat too slowly or completely skip beats, which means less blood reaches the brain and patients can feel dizzy or even faint. These patients usually need an artificial pacemaker to keep the heart in normal sinus rhythm (Desai).

Section 5.B: Electrical Conduction Problems

In an **AV block**, the electrical signal that usually passes through the AV node is delayed or stopped. In first-degree AV block, all signals go through but they are abnormally delayed (Desai). First-degree AV block can happen if parasympathetic nervous system is increased, conduction through the AV node is somehow delayed, or due to certain medications. In a second-degree AV block, some heartbeats get dropped when the signal from the atria fails to pass through the AV node to the ventricles, meaning they never contract (Desai). This skipped beat can be after a gradually lengthening delay, known as **Mobitz I**, or at random, known as **Mobitz II** (Desai). For both first and second degree AV blocks, people can often be asymptomatic. If they do have symptoms, they can experience dizziness or even fainting, similar to SA node dysfunction (Desai). In a third-degree or complete AV block, no signals go through and the atria and ventricles have to beat independently. The ventricles need to rely on a much slower, unreliable backup rhythm. Since third degree causes a significant decrease in heart rate, it can lead to significant **bradycardia** (slow heart rate), **hypotension** (low blood pressure), and even heart failure (Desai).

Section 5.C: Arrhythmias

Arrhythmias are when the heart's beating pattern is abnormal, and the different types are divided according to whether it is beating slower or faster than normal. Bradyarrhythmias are when the heart beats slower than 60 beats per minute, and tachyarrhythmias are when it beats faster than 100 (Desai). Tachyarrhythmias are further divided depending on what part of the heart starts it: **supraventricular** if it starts above the AV node, or **ventricular** if it starts below it (Desai). The most common arrhythmia is **atrial fibrillation**, a type of supraventricular arrhythmia, when many small electrical signals loop in circles across the atria instead of going through one organized path. Atrial fibrillation is marked by the atria quivering instead of

squeezing all at once. **Ventricular arrhythmias** are usually a lot more dangerous than supraventricular arrhythmias, because they are what drives the blood to the body. For example, **Ventricular Tachycardia** is a large cause of sudden cardiac-related deaths in the US (Desai).

Section 5.D: Heart Failure

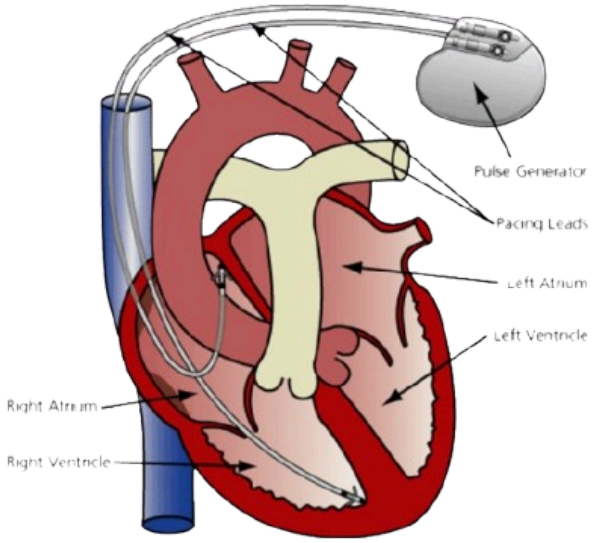
Heart failure is when the heart cannot fill with or push out blood properly or fast enough. The leading cause of heart failure is **ischemic heart disease** (Shams), when the heart muscle is continuously not recieving enough blood flow to function well. Heart failure is classified by the percentage of blood the heart holds that it actually pushes out each beat, called **ejection fraction**: reduced is when it pumps out 40% or less, mildly reduced is between 41-49%, and preserved is 50% or more (when the pump seems to work but heart failure still occurs) (Shams). Heart failure with reduced or mildly reduced ejection fraction is usually due to heart cell damage, such as in ischemia, which means the heart cannot squeeze as hard (Shams). This weak squeezing causes blood to back up in the body and pile up in the lungs, which can cause further problems. In heart failure with preserved ejection fraction, the problem is usually due to the ventricles being too stiff and unable to relax enough to fill up properly between beats (Shams).

Section 6: Supporting Heart Rhythms Artificially

Section 6.A: Supporting Electrical Rhythm

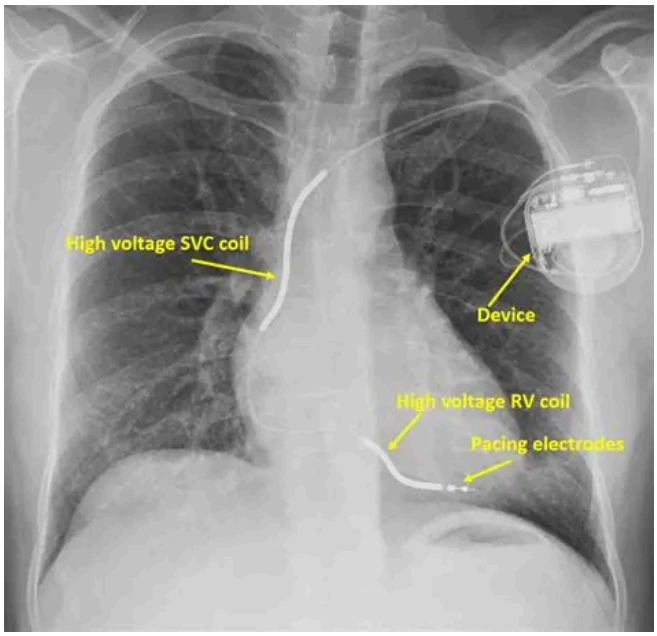
When a heart's natural rhythm fails, some medical devices can help restore regular rhythm. For example, a **pacemaker** can be used when the heart beats too slowly due to bradycardia from sinus node dysfunction or AV block (Sahu). Pacemakers today have three main components: a **pulse generator** for creating small electrical signals, **leads** to send the pulses to the heart, and **sensing circuitry** to see if the heart needs artificial help (for example, if

it is beating too slow) (Sahu). The leads are usually threaded through a small incision under the collarbone, then fastened into the heart muscle with small hooks or screws (Sahu). Once inserted, doctors can change the pacing rate, voltage, and pulse length of the electrical pulses (Sahu). **Single-chamber pacemakers**, which stimulate either the atrium or ventricle, are the most common. In comparison, **dual-chamber pacemakers** coordinate both the atria and ventricles in order to help produce a more natural heart rhythm (Sahu). Modern pacemakers can also imitate how the heart rate increases during exercise.



Pacemaker: A pacemaker's leads are threaded into the heart chambers via a vein (Diagram of a heart with a pacemaker)

For more dangerous arrhythmias like ventricular tachycardia and ventricular fibrillation, doctors instead use an **implantable cardioverter defibrillator (ICD)** (Sahu). An ICD helps prevent sudden cardiac death by looking for (usually rare) dangerous rhythms and deliver high-energy shocks to "restart" the heart and restore normal rhythms (Sahu). An ICD has a battery, a capacitor that stores and delivers electrical charge, and a microprocessor that senses the heart's rhythm (Sahu). ICDs can also dually function as a pacemaker (Sahu).

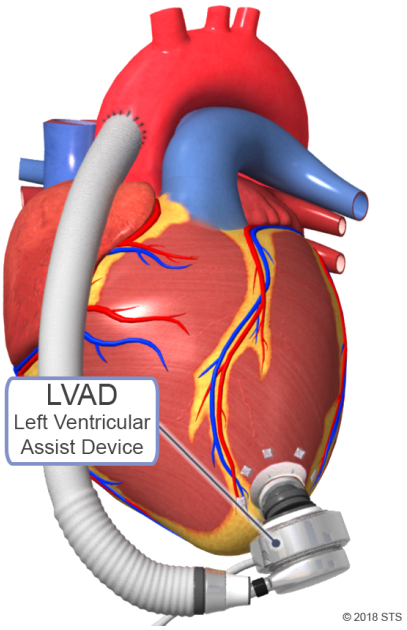


ICDs: A radiology photo of an ICD and its placement relative to the heart. (X-ray showing placed ICD)

Section 6.B: Supporting Mechanical Function

When the heart becomes too weak to pump blood properly, **mechanical circulatory support (MCS) devices** are used (Braga). There are two main types: temporary devices, which support the heart for hours to weeks, or permanent devices, which can be used for months or years (Braga). One example of a temporary MCS device is the **venoarterial extracorporeal membrane oxygenation (VA-ECMO)**. In VA-ECMO, blood is drawn from the right atrium, pumped to an oxygenating membrane outside the body, and then returned to the aorta (Braga). However, it can only be used for a few days to weeks and has some complications such as vessel damage, blood clots, and increased pressure in the left ventricle (Braga). On the other hand, a **left ventricular assist device**, or **LVAD**, is used for long-term support (Braga). The HeartMate 3 is the most widely used LVAD, and it is implanted inside the body, taking blood from the tip of the left ventricle and pumping it into the aorta (Braga). LVADs still carry some risks, such as infection or bleeding, but they can support a patient permanently or until they get

a heart transplant (Braga). HeartMate 3 was shown to have 2-year survival rates ranging from 79% to 83.4%, which is very close to survival rates after actual heart transplants (Braga).



LVAD: LVADs mechanically pump blood from the tip of the left ventricle to the aorta (Left ventricular assist device connected to heart)

CHAPTER ADD-INS

Exercise 1 (Practical– it might fit in better with Ryan's part with the Sympathetic vs Parasympathetic Control):

Maya is a student who wore a fitness tracker during one day to record how her heart rate changes. She found the following:

Time	Activity	Heart Rate (BPM)
3:00 AM	Sleeping	53
11:00 AM	Taking a math test	102
2:00 PM	Break	86
5:00 PM	Running	134

a) Please explain each heart rate reading and identify which nervous system pathway is being activated.
b) Try measuring your heart rate at different times throughout the day. What time was your heart rate lowest or highest? How can you explain your data based on the reading?

Answer: When Maya is sleeping, her heart rate is low in comparison to when she is awake, which follows normal patterns. When taking a math test or running, [Ryan's part]. Maya's BPM during a break represents her normal, awake, resting heart rate.

When Maya is sleeping, her heart rate decreases due to the effects of the parasympathetic nervous system acting upon the heart to create a "restful" state. At 11:00 AM, her heart rate increases due to the extrinsic factor of the stressful test

Exercise 2: Starting from the right atrium, trace the path blood takes through the heart and name each major chamber, valve, and vessel until it reaches the right atrium again. Mark where the blood is oxygenated and where it is deoxygenated.

Answer: right atrium → tricuspid valve → right ventricle → pulmonary valve → pulmonary artery → lungs → pulmonary veins → left atrium → mitral valve → left ventricle → aortic valve → aorta → body → superior and inferior vena cava → right atrium

Blood is deoxygenated up until the pulmonary artery. It becomes oxygenated in the lungs and stays oxygenated until the aorta carries it to the body and it becomes deoxygenated after again.

Exercise 3: A older person's tricuspid valve does not close all the way, so some blood leaks back into the right atrium each time the right ventricle contracts. Why would this valve problem be especially problematic for this older person when they try to exercise?

Answer: Since blood leaks back into the right atrium, less blood is pumped forward in the system. To make up for this during exercise, a younger heart might be able to pump faster, but an older heart cannot increase the speed enough.

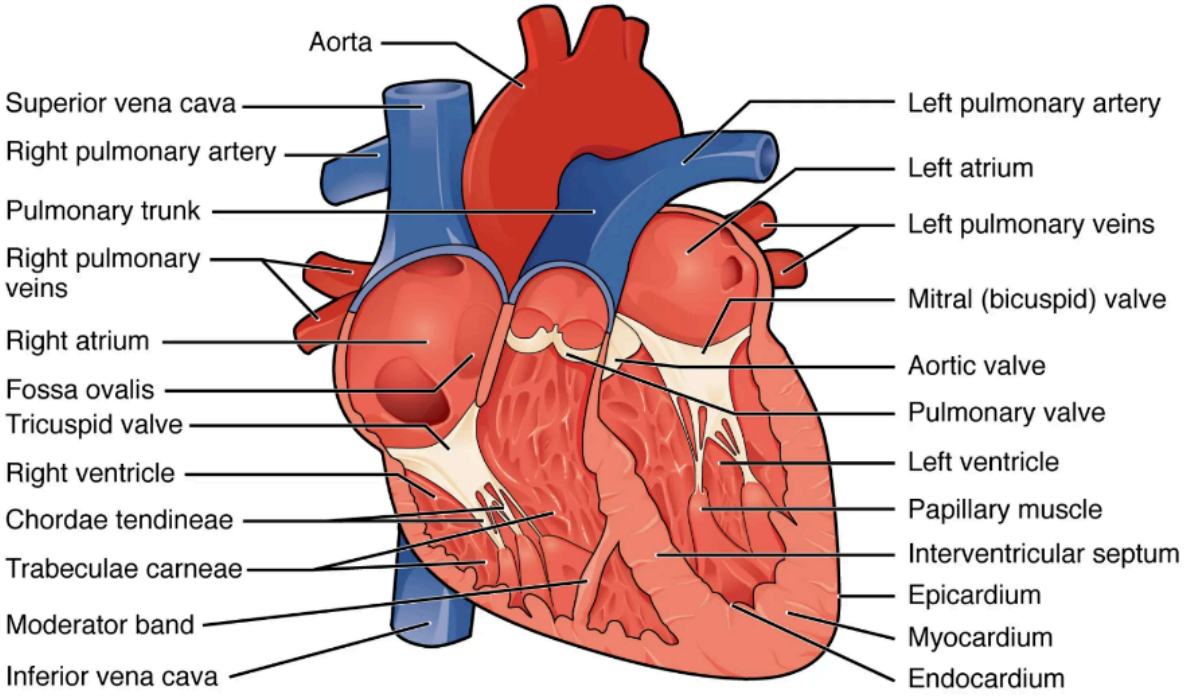
Exercise 4: A person has atrial fibrillation, which is a condition where the atria quiver instead of contracting properly. Why does this person notice more symptoms during exercise than when they are resting?

Answer: A slower heartbeat gives a lot of time for blood to flow into the ventricles on its own, so the atria not contracting properly matters less. But during exercise, the faster heartbeat leaves less time for the blood to passively fill, so the atria's inability to contract properly leads to larger issues.

Exercise 5: Why might an older heart have a more difficult time coping with the added strain of obesity than a younger heart?

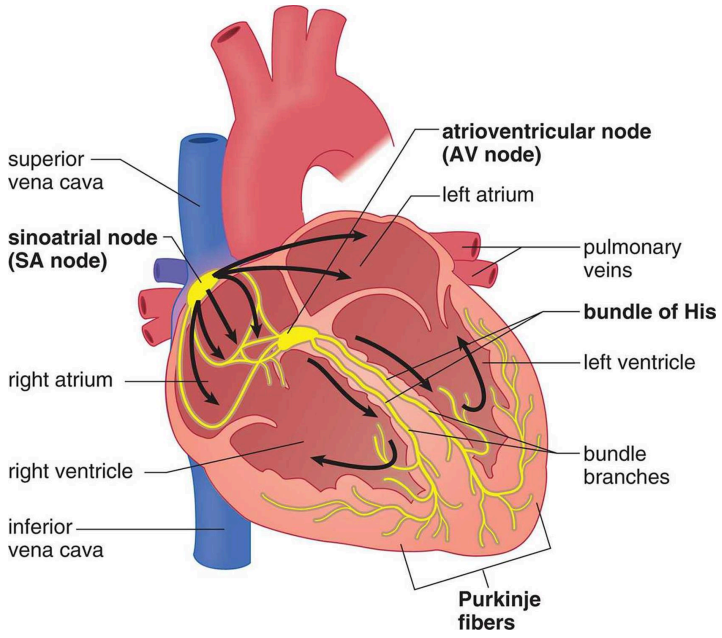
Answer: Obesity means that the heart needs to pump more blood every minute, with thicker and stiffer walls. A younger heart might be able to still increase its output by speeding up, but an older heart is slower and not able to keep up as much.

Figure 1 - Heart's mechanical function



- Description: Heart's Mechanical Function: Deoxygenated blood from the body travels through the superior and inferior vena cava into the right atrium. After passing through the tricuspid valve, the blood then flows into the right ventricle, where it is pumped through the pulmonary valve and into the pulmonary artery. After the blood passes through the lungs, it travels back to the heart via the pulmonary veins into the left atrium. Next, the blood passes through the mitral valve and into the left ventricle, where it is pumped through the aortic valve and into the aorta to send blood to the rest of the body.
- Attribution J. Gordon Betts, Kelly A. Young, James A. Wise, Eddie Johnson, Brandon Poe, Dean H. Kruse, Oksana Korol, Jody E. Johnson, Mark Womble, Peter DeSaix. (2022). *Anatomy and Physiology 2e.* OpenStax. https://openstax.org/books/anatomy-and-physiology-2e/pages/19-1-heart-anatomy

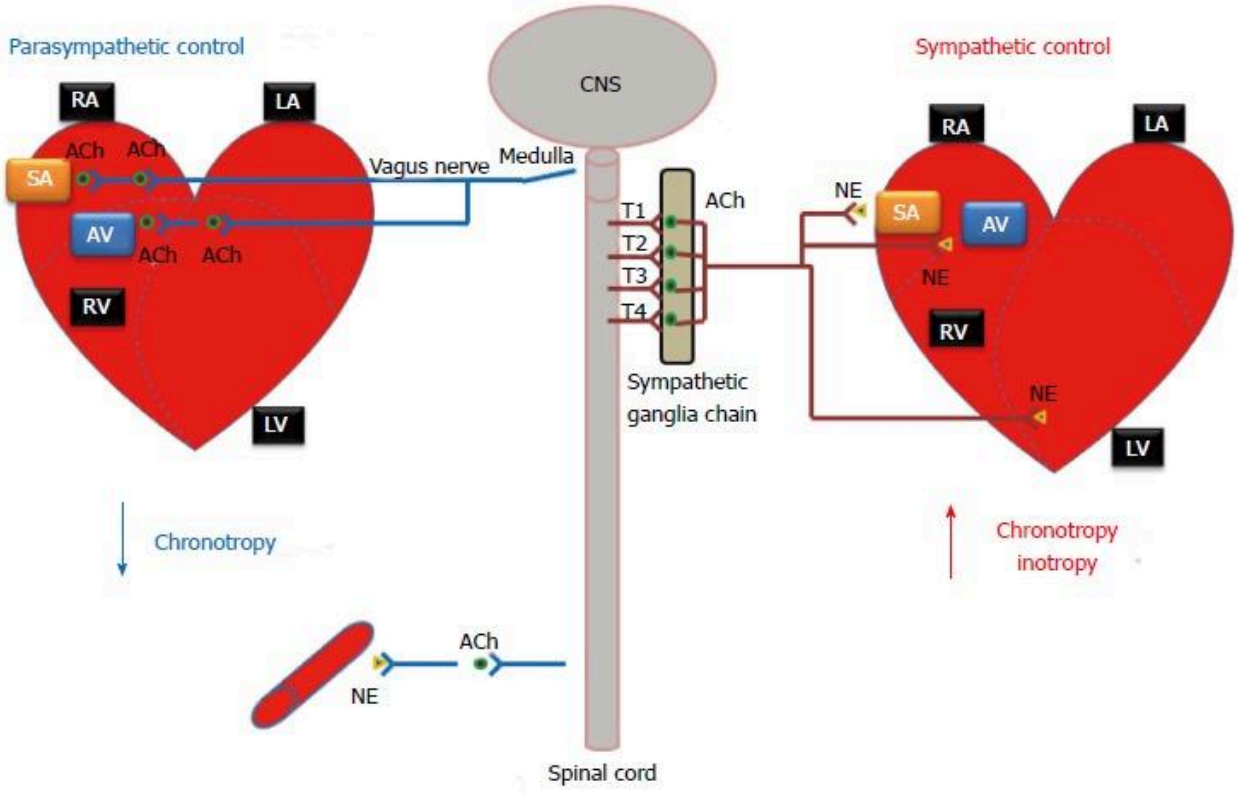
Figure 2 - Heart’s Electrical Function



-
- Description: The electrical signal travels from the SA node to the AV node, then down the Bundle of His and Purkinje fibers to ensure the heart follows a coordinated contraction pattern (Diagram of the heart’s electrical conduction system)

Diagram of the heart’s electrical conduction system. (n.d.). [Diagram]. In Bing Images. Retrieved October 16, 2026, from https://tse1.mm.bing.net/th/id/OIP.ZiFz-gOShkQWBfsfwOz_EgHaGj?r=0&rs=1&pid=ImgDetMain&o=7&rm=3

Figure 3 - Diet, Exercise, and Sleep’s Role in Neuroplasticity



Description: In this image is the parasympathetic and sympathetic nervous system. On the right is the parasympathetic nervous system which relaxes the body. In this diagram the central nervous system (CNS) sends a message from the medulla through the vagus nerve to the Sinoatrial (SA) and the atrioventricular (AV) node via Acetylcholine(ACh) a neurosignal to slow down the heart. On the right side is the depiction of the pathway of the sympathetic nervous system in which the brain through the sympathetic ganglia chain creates a signalling cascade that releases neuroepinephering(NE) into the SA and AV nodes to speed up the heart.lol

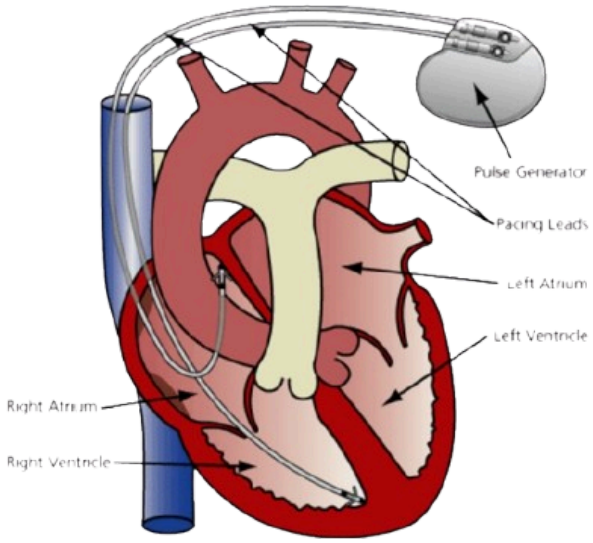
Attribution:Gordan, R., Gwathmey, J. K., & Xie, L. H. (2015). Autonomic and endocrine control of cardiovascular function. *World journal of cardiology*, *7*(4), 204–214. https://doi.org/10.4330/wjc.v7.i4.20

Figure 4 - table of differences between different hearts

	Mechanical Differences	Electrical And Autonomic Differences	Effect on the Heart's Clock
Healthy Heart	● Not too stiff ● Walls aren't too thick or too weak ● Fast refill time	● Working SA node ● Signals can be processed through the heart cells quickly	● Normal sinus rhythm ● Can easily adapt to exercise
Heart in Old Patients	● Stiffer walls ● Slower refill time, can't keep up during intense exercise	● Fewer SA node cells ● Signals last longer in each heart cell ● Less responsive to adrenaline	● Lower maximum heart rate
Hearts in Obese Patients	● Increased blood pressure ● Scarring leads to stiffer walls	● Pumps more blood per minute	● Increased risk for heart failure
Hearts in Patients with Circadian Misalignment	● Increased blood pressure	● Decreased parasympathetic activity	● Higher rates of heart disease
Hearts in Patients Who are Exposed to Smoke	● Low HRV	● Decreased parasympathetic activity	Higher risk for heart arrhythmias

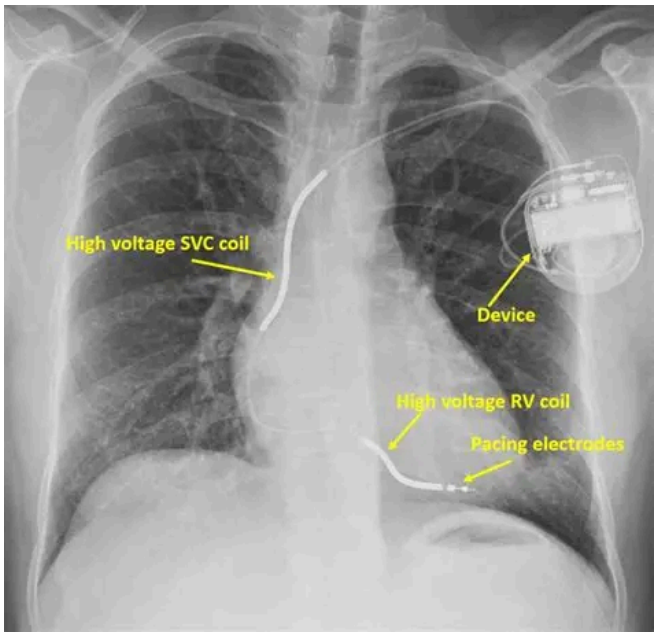
- author: Ada
- Description: table of hearts in people with different conditions, comparing their mechanical, electrical & autonomic, and heart clock differences

Figure 5 - pacemaker diagram



- Description: “Pacemaker: A pacemaker’s leads are threaded into the heart chambers via a vein” (Diagram of a heart with a pacemaker)
- Attribution: *Fig. 1. pacemaker environment pact illustration of BLESS, because it is...* (2025). ResearchGate. https://www.researchgate.net/figure/Pacemaker-Environment-pact-illustration-of-BLESS-because-it-is-a-simple-safety-critical_fig1_290614924

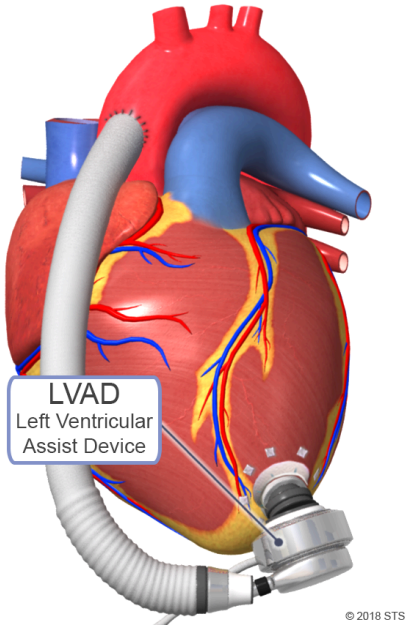
Figure 6 - ICD x-ray image

- 

    - Description: ICDs: A radiology photo of an ICD and its placement relative to the heart. (X-ray showing placed ICD)

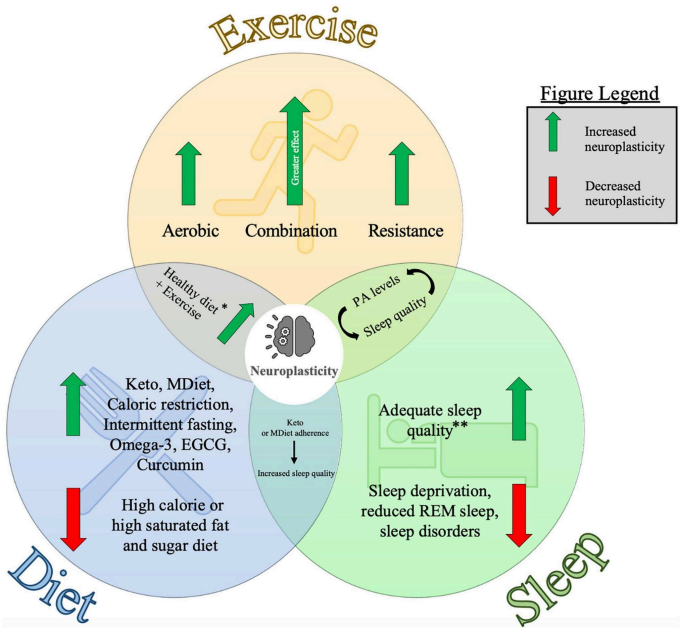
X-ray showing placed ICD. (n.d.). [Photograph]. In Atrial Fibrillation: Resources for Patients. Retrieved October 17, 2026, from https://a-fib.com/wp-content/uploads/2021/05/Implantable_cardioverter_defibrillator_chest_X-ray-By-Gregory-Marcus-MD-with-arrow-250-pix-wide.pn

figure 7 - LVAD



- Description: “LVAD: LVADs mechanically pump blood from the tip of the left ventricle to the aorta (Left ventricular assist device connected to heart)”
- Attribution:

Left ventricular assist device connected to heart. (n.d.). [Illustration]. In Pinterest. Retrieved October 17, 2026, from https://i.pinimg.com/originals/0d/21/4b/0d214bb00ffd1684641d5ff47ea3dee2.png



-
- Figure Description
- Attribution (if applicable)

GLOSSARY

Term	Definition

REFERENCES

Ansari, M. A. (2021). The human anatomy of heart: A review literature. International Journal of Emerging Technologies and Innovative Research, 8(10), f236–f253. http://www.jetir.org/papers/JETIR2110534.pdf

Cardiovascular System Function (2024). Blogspot.Com. https://4.bp.blogspot.com/-jcLaf5BquTM/XIuPCkq3ftI/AAAAAAAAAAk/ONVDgmsBpJI9UZ2h7j7KLN9KucMi1iLowCLcBGAs/s1600/CARDIOVASCULAR+SYSTEM+FUNCTION+-+Info+Dear.jpg

Sahu P, Acharya S, Totade M (October 02, 2023) Evolution of Pacemakers and Implantable Cardioverter Defibrillators (ICDs) in Cardiology. Cureus 15(10): e46389. doi:10.7759/cureus.46389

Dinas, P. C., Koutedakis, Y., & Flouris, A. D. (2013). Effects of active and passive tobacco cigarette smoking on heart rate variability. International Journal of Cardiology, 163(2), 109–115. https://doi.org/10.1016/j.ijcard.2011.10.140

Desai DS, Hajouli S. Arrhythmias. [Updated 2023 Jun 5]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK558923/

Shams P, Malik A, Chhabra L. Heart Failure (Congestive Heart Failure) [Updated 2025 Feb 26].

In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK430873/

Diagram of the heart's electrical conduction system. (n.d.). [Diagram]. In Bing Images. Retrieved October 16, 2026, from https://tse1.mm.bing.net/th/id/OIP.ZiFz-gOShkQWBfsfwOz_EgHaGj?r=0&rs=1&pid=ImgDetMain&o=7&rm=3

Chellappa, S. L., Vujovic, N., Williams, J. S., & Scheer, F. A. J. L. (2019). Impact of Circadian Disruption on Cardiovascular Function and Disease. Trends in endocrinology and metabolism: TEM, 30(10), 767–779. https://doi.org/10.1016/j.tem.2019.07.008

Chan-Dewar, F. (2012). The cardiac cycle. Anaesthesia & Intensive Care Medicine, 13(8), 391–396. https://doi.org/10.1016/j.mpaic.2012.05.010

Black, N., D'Souza, A., Wang, Y., Piggins, H., Dobrzynski, H., Morris, G., & Boyett, M. R. (2019). Circadian rhythm of cardiac electrophysiology, arrhythmogenesis, and the underlying mechanisms. Heart rhythm, 16(2), 298–307. https://doi.org/10.1016/j.hrthm.2018.08.026

Thosar, S. S., Butler, M. P., & Shea, S. A. (2018). Role of the circadian system in cardiovascular disease. Journal of Clinical Investigation, 128(6), 2157–2167. https://doi.org/10.1172/jci80590

Strait, J. B., & Lakatta, E. G. (2012). Aging-associated cardiovascular changes and their relationship to heart failure. Heart failure clinics, 8(1), 143–164. https://doi.org/10.1016/j.hfc.2011.08.011

Wiley, T. M. P., Poirier, P., Burke, L. E., Després, J.-P., Larsen, P. G., Lavie, C. J., Lear, S. A., Ndumele, C. E., Neeland, I. J., Sanders, P., & St-Onge, M.-P. (2021). Obesity and Cardiovascular disease: a Scientific Statement from the American Heart Association. Circulation, 143(21), 984–1010. https://doi.org/10.1161/cir.0000000000000973

X-ray showing placed ICD. (n.d.). [Photograph]. In Atrial Fibrillation: Resources for Patients. Retrieved October 17, 2026, from https://a-fib.com/wp-content/uploads/2021/05/Implantable_cardioverter_defibrillator_chest_X-ray-By-Gregory-Marcus-MD-with-arrow-2

50-pix-wide.png

Left ventricular assist device connected to heart. (n.d.). [Illustration]. In Pinterest. Retrieved October 17, 2026, from https://i.pinimg.com/originals/0d/21/4b/0d214bb00ffd1684641d5ff47ea3dee2.png

Chen, L., & Yang, G. (2015). Recent advances in circadian rhythms in cardiovascular system. *Frontiers in pharmacology*, *6*, 71. https://doi.org/10.3389/fphar.2015.00071

As a source about the diurnal changes of heart rate and blood pressure. The SCN as a master clock for all of these changes.

Shaffer, F., McCraty, R., & Zerr, C. L. (2014). A healthy heart is not a metronome: an integrative review of the heart's anatomy and heart rate variability. *Frontiers in psychology*, *5*, 1040. https://doi.org/10.3389/fpsyg.2014.01040

Heart rate variability with the sympathetic and parasympathetic nervous system.

Michael, S., Graham, K. S., & Davis, G. M. "Cardiac Autonomic Responses during Exercise and Post-exercise Recovery Using Heart Rate Variability and Systolic Time Intervals—A Review." *Frontiers in Physiology.* https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5447093/

How physical activity uses both the sympathetic and parasympathetic nervous system for speeding up and slowing down the heart rate.

Gledhill, N., Cox, D., & Jamnik, R. (1994). Endurance athletes' stroke volume does not plateau: major advantage is diastolic function. *Medicine and science in sports and exercise*, *26*(9), 1116–1121.

Adaptations undergone by the heart after endurance training.

Cleveland Clinic. (2023, November 8). *Blood flow through the heart*. Cleveland Clinic. https://my.clevelandclinic.org/health/articles/17060-how-does-the-blood-flow-through-your-heart

Calogiuri, G., Weydahl, A., Beldo, S., & Montaruli, A. (2010). Morning or evening exercise: Effects on the heart rate circadian rhythm above the Arctic Circle. *[Journal name, volume(issue), pages]*. https://air.unimi.it/handle/2434/171264

Hower, I. M., Harper, S. A., & Buford, T. W. (2018). Circadian rhythms, exercise, and cardiovascular health. *Journal of Circadian Rhythms, 16*(1). https://doi.org/10.5334/jcr.164

Vitale, J. A., Bonato, M., La Torre, A., & Banfi, G. (2019). Heart rate variability in sport performance: Do time of day and chronotype play a role? *Journal of Clinical Medicine, 8*(5), Article 723. https://doi.org/10.3390/jcm8050723

Calogiuri, G., Weydahl, A., Beldo, S. *et al.* Morning or evening exercise: effects on the heart rate circadian rhythm above the Arctic Circle. *Sport Sci Health* **6**, 9–16 (2010). https://doi.org/10.1007/s11332-010-0090-x

Fig. 1. pacemaker environment pact illustration of BLESS, because it is... (2025). ResearchGate. https://www.researchgate.net/figure/Pacemaker-Environment-pact-illustration-of-BLESS-because-it-is-a-simple-safety-critical_fig1_290614924

UNIT AND CHAPTER

Unit: Ultradian Rhythms
Chapter X: Neuronal Activity

CONTRIBUTORS

- Daniel Winter
    - Subject Matter Expert – ‘Neuronal Activity’
- Jacob Li
    - Peer Reviewer
- Ada Langford
    - Peer Reviewer

LEARNING OBJECTIVES

After reading this chapter, students should be able to …

1. LO1 (Remember): Define spike and LFP
2. LO2 (Understand): Describe how brain activity repeats in rhythms (of different speeds)
3. LO3 (Apply): Use breathing as an example of how a body rhythm can affect brain rhythm
4. LO4 (Analyze): Compare how scientists pull spikes and rhythms out of recordings and the limitations of these methods
5. LO5 (Understand): Explain one way spiking neural networks use timing that artificial neural networks do not

CHAPTER OUTLINE

- What is Being Measured
- Neural Activity is Rhythmic
- Connection to Slower Body Rhythms
- Getting a Signal from a Recording
- From Biological to Artificial

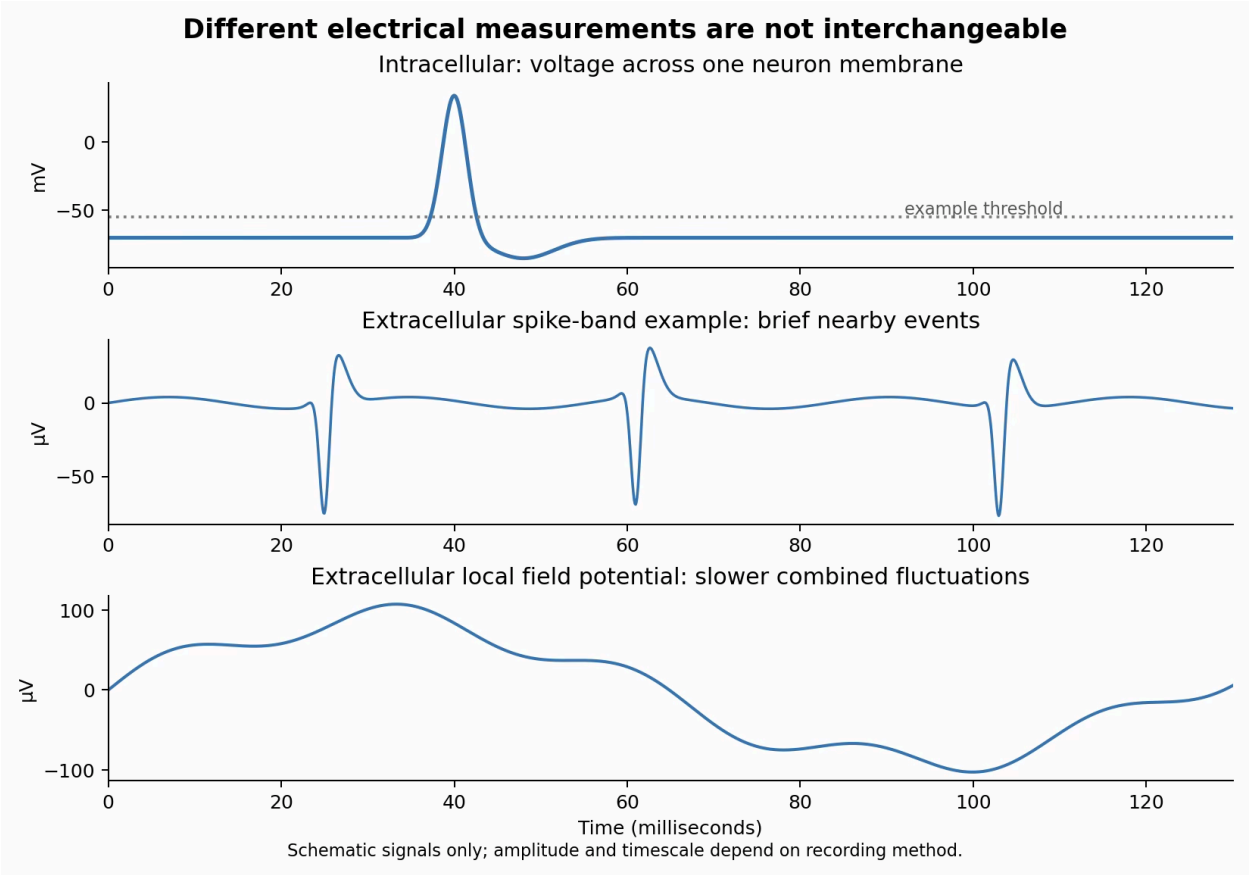
CHAPTER WRITTEN CONTENT

1. **What is Being Measured?**

When scientists put a thin wire (an electrode) into the brain, they record changes in voltage outside the cells, a method called extracellular recording. What results is a noisy wave-like line, containing spikes and local field potentials (LFPs). Telling these apart, and distinguishing extracellular recordings from voltage measured inside a cell, is the first step towards understanding how neurons communicate and how they relate to rhythms.

A spike is an action potential; it's a rapid change in voltage across a cell membrane that neurons use to communicate with one another.A typical neuron has a resting membrane voltage of about -70 mV (relative to the outside of the cell). Stimuli make the inside of the cell less negative until it hits a trigger point (threshold around -55 mV), causing voltage-gated sodium channels to open. This causes **depolarization**, causing the sharp spike up to +30 mV. Sodium channels then inactivate and potassium channels open, letting potassium rush out to drop the voltage back down, a process called **repolarization**. The important thing to note is that the neuron either reaches the threshold and fires an action potential or remains below the threshold and does not fire; exact voltage values vary across neurons.

But these neurons are seldom isolated; on average, a single neuron in the human brain connects directly to about 1,000 to 10,000 other neurons through contact points called **synapses**. To extract information from these neurons, scientists measure electrical signals with electrodes. For example, a local field potential (LFP) reflects the combined transmembrane currents of many neurons, especially synaptic currents, as recorded near an electrode (Buzsáki et al., 2012). Its amplitude also depends on the geometry of those currents. This wave, unlike the spikes generated by single neurons, is smooth and changes much more gradually.



Intracellular action potential versus extracellular spikes and local field potential

Figure A1. Based on: Buzsáki et al. (2012); Rey et al. (2015); OpenStax, Section 12.4.

One challenge of analyzing LFPs is that the same electrode captures both signals. This forces scientists to use clever mathematical techniques to filter out the signals that they are trying to isolate. Since spike frequencies range in the hundreds of hertz while LFPs oscillate at much lower frequencies, their signals are distinguishable. However, spike waveforms leak into the lower frequency range, so what looks like fast LFP activity can be polluted by nearby spiking signals.

Scale is the main difference between spikes and LFPs. A spike tells you what a single neuron is doing, while a local field potential tells you what a group of neurons are doing together; neither is the whole picture. Imagine trying to single out one person's conversation at a

large party. To hear what they are saying, you have to zone out the rest of the noise, and vice versa. If the goal is to hear the volume of the room (which changes over time), you lose the ability to single out individual voices from the crowd.

There is also a relationship between a single neuron's signal and its group's signal. When LFPs follow cyclical patterns, some neurons tend to spike at specific phases in the cycle. The following sections will go into much greater detail on this relationship and the connection between LFP rhythms and those of the rest of the body.

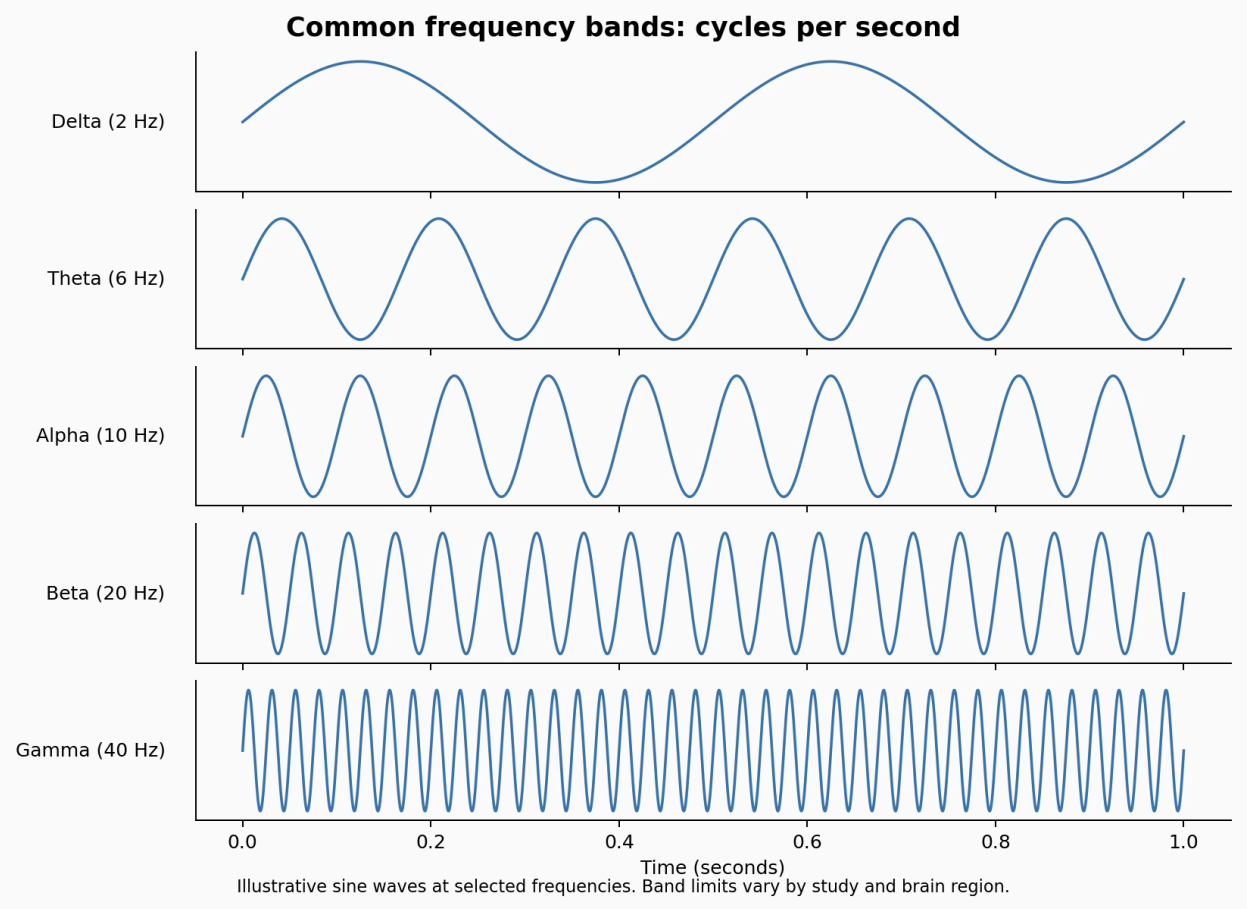
2. Neural Activity Is Rhythmic

A spike tells you about one moment in one particular cell. In comparison, local field potentials follow very different patterns. LFPs often rise and fall in repeated cycles, sometimes for minutes at a time. Some neurons become active and idle together, over and over, and the LFP records that group behavior as a wave. although not all neurons do so. These repeated LFP fluctuations are called oscillations, and researchers use them to understand neural activity.

To study these oscillations, researchers can analyze them as simpler sine-wave components using mathematical techniques. Researchers group these components by how fast they cycle. The unit is hertz (Hz), meaning cycles per second. The common bands are delta (δ: 1 to 4 Hz), theta (θ: 4 to 8 Hz), alpha (α: 8 to 12 Hz), beta (β: 15 to 30 Hz), and gamma (γ: 30 to 100 Hz). Different bands are associated with different patterns of neural activity and can provide clues about the processes occurring at any given time. These band conventions are not universally agreed upon, so different laboratories draw the lines in slightly different places, and different animals run the same functional rhythm at different speeds. Gamma in the olfactory system of rats and mice sits between 40 and 100 Hz, while in insects (the comparable rhythm)

is around 20 Hz (Kay, 2014). The rhythms occur in different species, despite the variations in frequency.

Frequency and period are reciprocals of one another, so a rhythm at 40 Hz completes one cycle every 25 milliseconds, while a 4 Hz rhythm takes 250 milliseconds to complete a cycle. A single spike lasts about a millisecond, meaning dozens of spikes can fit in one slow cycle, so the timing of a spike within a cycle can carry information in some circuits, in addition to how many spikes occur.



Delta, theta, alpha, beta, and gamma compared
Figure A2. Based on: Buzsáki & Draguhn (2004); Kay (2014), Section 3.

For a very long time, it was reasonable to ask if these LFP oscillations do anything meaningful at all. A bridge vibrates when traffic crosses it, but the vibration is not why the bridge

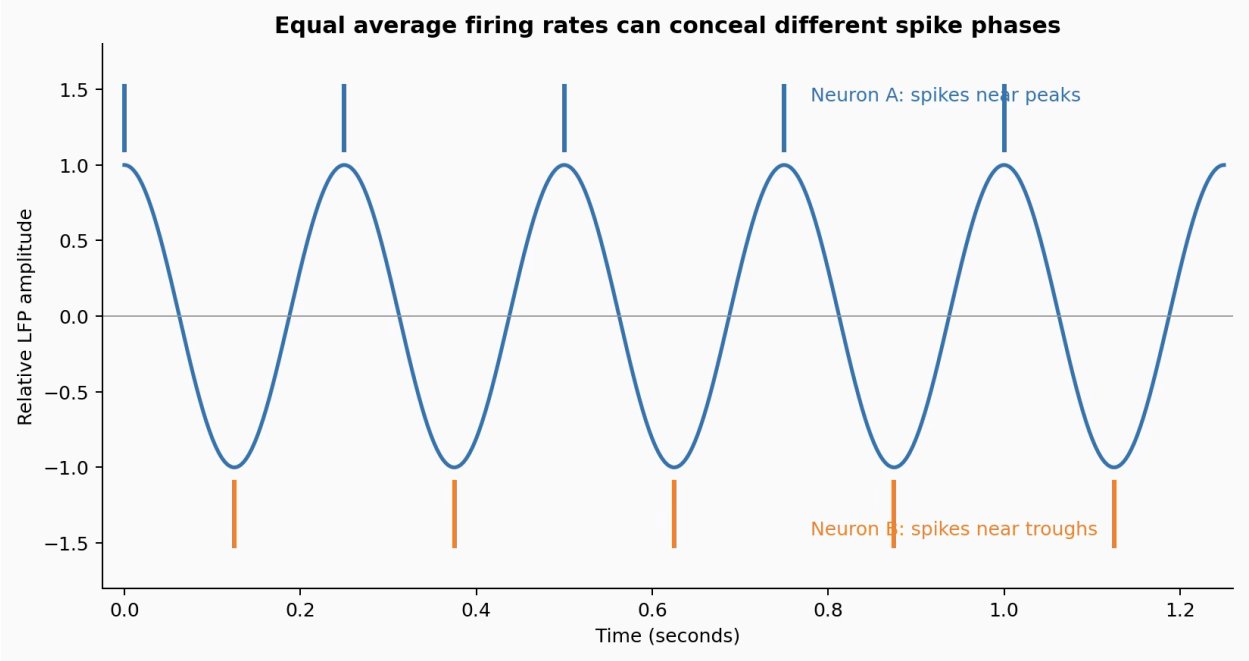
was built. Buzsáki and Draguhn (2004) propose that brain oscillations are not merely that kind of byproduct. Oscillations span five orders of magnitude in frequency and are preserved across species, which suggests they may serve important functions. They also propose that oscillations can bias which inputs reach a population of neurons, temporarily link neurons so they can work together, and support synaptic changes involved in learning. These functions are supported to different degrees across circuits.

Every rhythm in this chapter completes *many* cycles per day (ultradian rhythm). A gamma cycle finishes roughly three and a half million times in the span of a single circadian day, orders of magnitude faster than rhythms measured in minutes and hours rather than milliseconds.

Most claims about what a brain rhythm does are correlational; a rhythm gets stronger during some behavior, and researchers infer a relationship between the two. However, experiments on olfactory gamma provide stronger evidence for a functional contribution. Experiments reviewed by Kay (2014) found that disrupting odor-related oscillations impaired discrimination of similar odors in insects, while mice with enhanced gamma activity showed improved fine discrimination. These findings support a role for oscillatory coordination, although the manipulations can affect other properties of the circuit. Beta oscillations, at 15 to 30 Hz, also appear during odor learning, although their role is less clearly understood (Kay, 2014).

This brings us back to the single neuron. Recall that some neurons tend to fire at a preferred point in the ongoing LFP cycle rather than at random moments. That point in the cycle is called the *phase*, which is the position of a wave relative to one complete cycle. A neuron that tends to fire near the trough of a cycle and one that tends to fire near the peak may have identical firing rates while contributing differently to circuit activity. The group rhythm can

influence spike timing, while individual neurons supply the spikes.



Spikes at different phases of an LFP cycle

Figure A3. Based on: Kay (2014), Figure 5; Buzsáki & Draguhn (2004).

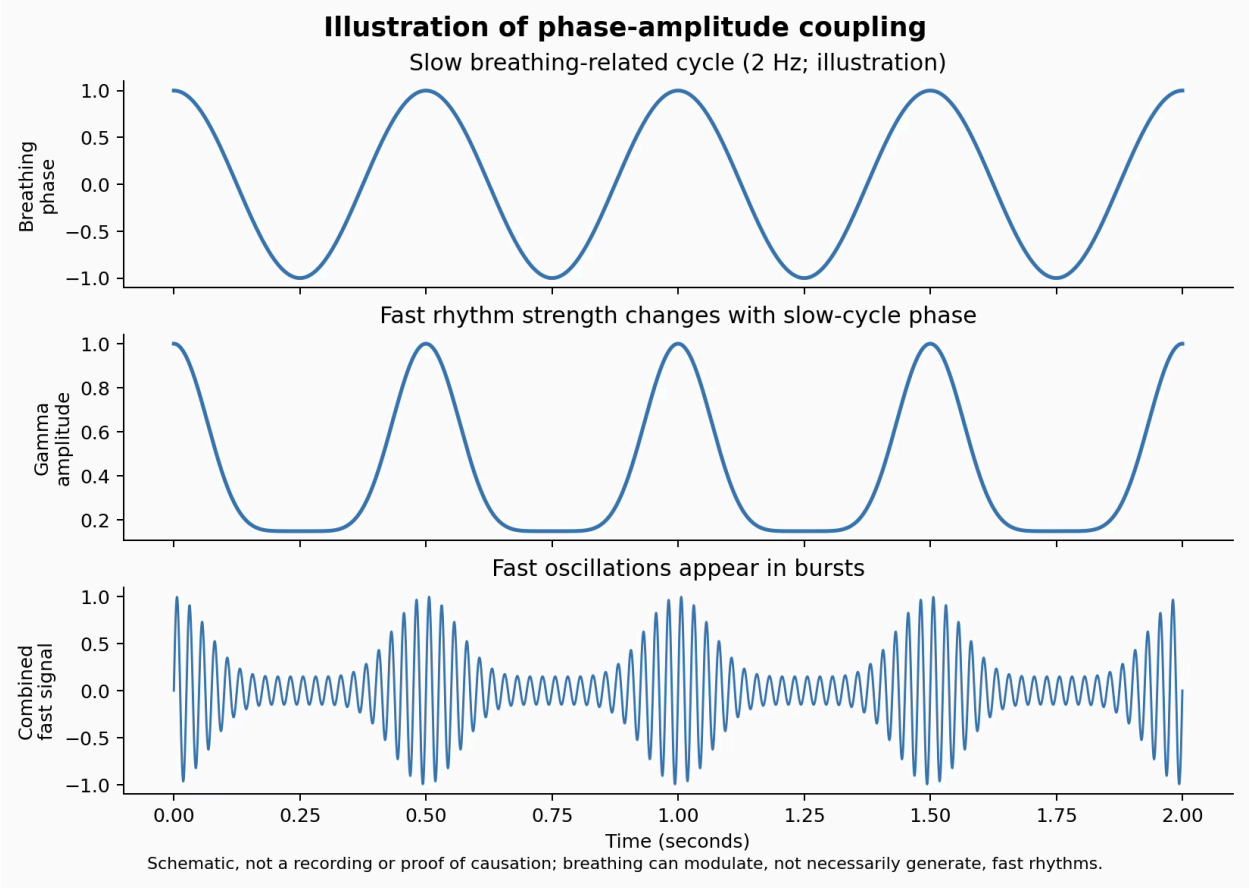
3. The Connection to Slower Body Rhythms

Until now, we have focused on rhythms generated within the brain; but the brain sits inside a body that keeps its own time, and it turns out, the slowest rhythms discussed here are tied to one of the most ordinary processes there is: breathing. Slower body rhythms can influence faster neural activity.

Neural activity that synchronizes with the breathing cycle has been found in a wide range of mammals, humans included (Tort et al., 2025). The surprising part is where this breathing-related neural activity appears. Neural “coupling” to breathing in the olfactory system has been known for a long time, and it makes intuitive sense, since air carrying odor molecules arrives in rhythmic pulses rather than continuously. However, recent work shows that the

coupling extends well past just olfactory and respiratory areas into regions associated with emotion and higher cognition (Tort et al., 2025).

One important pathway begins at the nose, where airflow through the nasal passage leads to waves of sensory activity that travel through the brain, sometimes aligning neuronal activity with the breathing cycle and modulating faster oscillations (Tort et al., 2025). This is an example of phase-amplitude coupling: the phase of a slow rhythm is associated with changes in the amplitude of a faster rhythm.



Breathing-cycle timing and fast oscillation amplitude

Figure A4. Based on: Tort et al. (2025), Figure 1; Kay (2014), Figures 2 and 4.

It is worth noting that breathing can modulate or “entrain” existing brain rhythms; it does not generate every rhythm with which it becomes synchronized. Brain oscillations continue in the absence of nasal airflow. What breathing does is supply a shared reference signal that

distant regions can lock onto, which can help coordinate activity across distant regions, though synchronized activity alone does not prove direct communication.

In rodents, sniffing runs at roughly 4 to 12 Hz, which overlaps with the theta band. A rhythm recorded in the hippocampus at 8 Hz could be hippocampal theta, could reflect respiratory coupling, or both. The two rhythms can occur at similar frequencies without being continuously synchronized (Kay, 2014), so separating them requires recording breathing rather than assuming they are the same. This is a recurring difficulty across chronobiology: two rhythms with the same period are *not* necessarily the same rhythm.

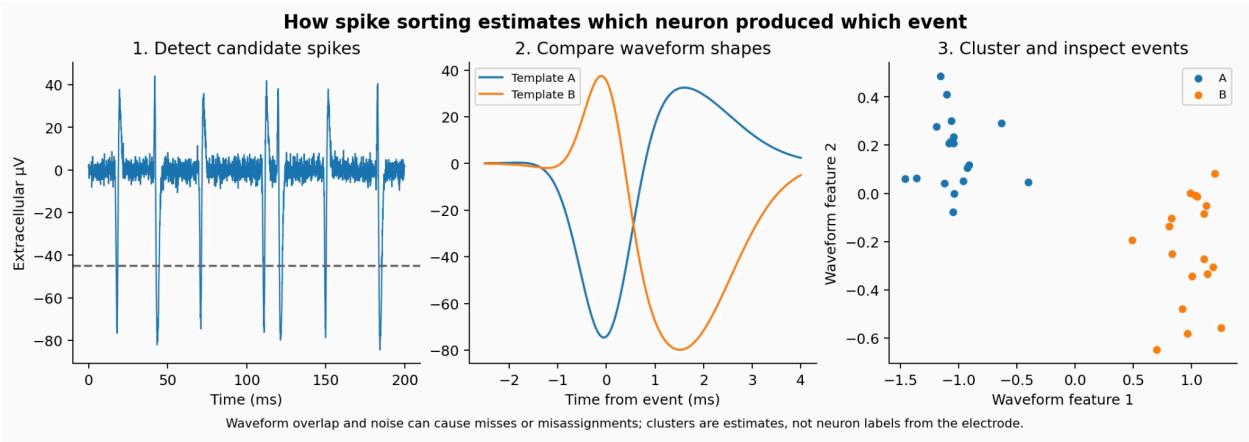
There is an encapsulating principle that generalizes past just breathing. Slower rhythms can coordinate faster ones. Such interactions appear across multiple timescales, including those discussed here. (See Circulation and the Cardiac Cycle)

4. Getting a Signal from a Recording

So far, we assumed that spikes and rhythms are simply ready for researchers to read. In practice, what comes off an electrode is a single messy voltage trace, and getting anything interpretable out of it requires solving two difficult separate problems: figuring out which neuron produced which spike, and figuring out which rhythms are present and when.

An electrode sitting in tissue picks up several neurons at once and nothing in the signal labels them. The first problem is handled by spike sorting. The process occurs in four steps (Rey et al., 2015). First, the recording is bandpass filtered (typically between 300 and 3,000 Hz). Second, the filtered trace is scanned for moments where the voltage crosses a set limit, a technique called thresholding. Third, the shape of each detected waveform is measured. Fourth, waveforms with similar shapes are clustered together and treated as possibly originating from one specific neuron. If a researcher stops after the second step and simply counts every

threshold crossing without assigning it to a cell, the result is multi-unit activity: a record of how much the local population is firing, without knowing which specific cells are responsible for what parts of the signal. The procedure partly rests on the assumption that a given neuron produces a relatively consistent waveform and waveforms differ between neurons because each one sits at a different distance and orientation relative to the electrode tip.



How spike sorting estimates neurons from waveforms

Figure A5. Based on: Rey et al. (2015); Yger et al. (2018), figures.

But that assumption has limitations; sorting accuracy degrades significantly when many neurons sit close to an electrode because their waveforms begin to look too similar (Rey et al., 2015). Two spikes arriving at the same instant can overlap and form a shape that matches neither neuron, making them difficult to classify. And most importantly, neurons that fire rarely are easily lost underneath neighboring neurons that fire more often. This matters when studying sparsely firing cells, including some neurons involved in memory-related tasks (Rey et al., 2015). Sorting errors can bias a dataset toward neurons with more detectable or distinctive waveforms.

The second major problem, actually finding rhythms, is handled by the Fourier transform. The idea is that any complex repeating signal can be rebuilt by adding together simple sine waves of different frequencies, amplitudes, and phases. Formally, the Fourier transform is:

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-i2\pi ft} dt$$

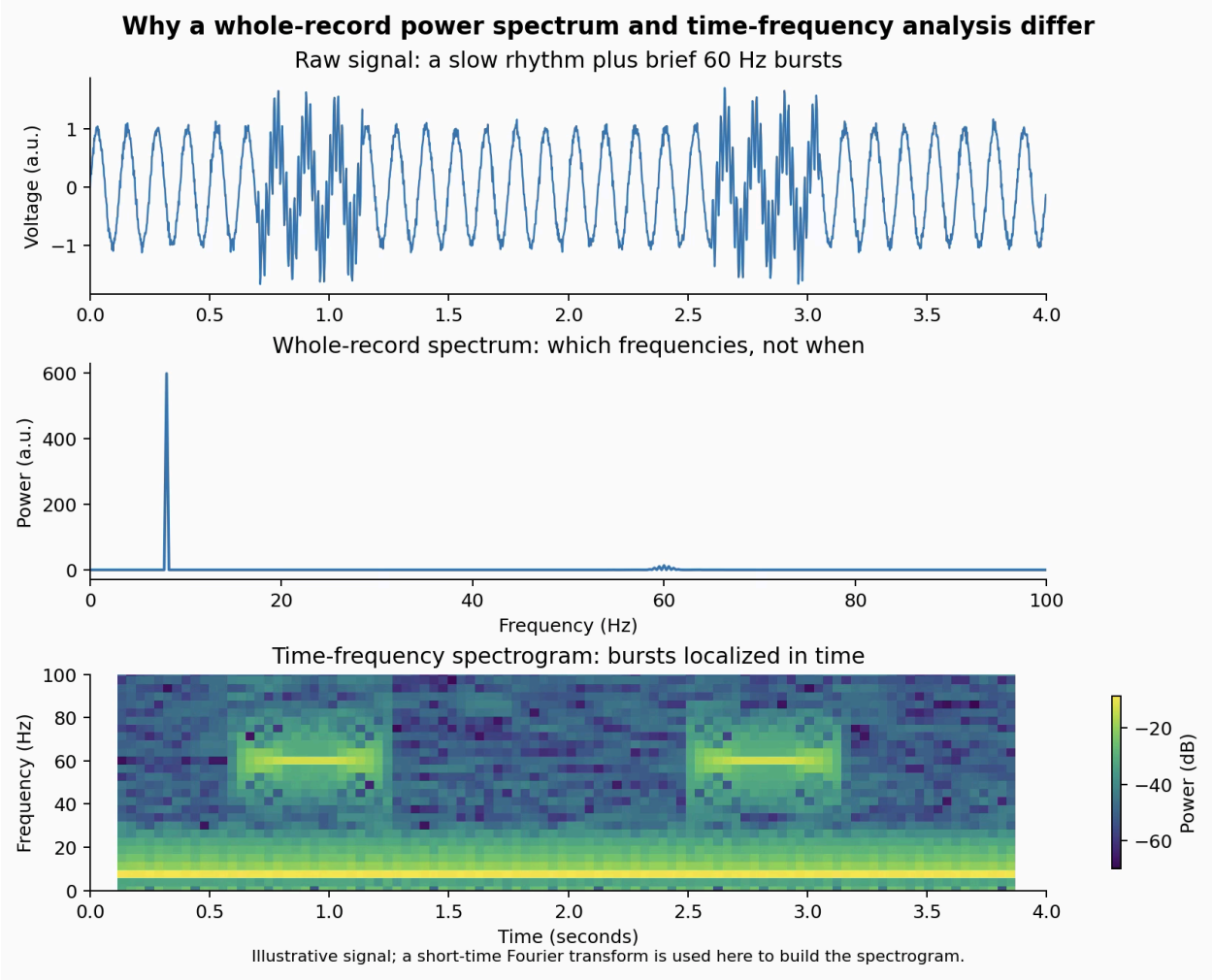
While ostensibly quite intimidating, it asks a single question: how much of the signal x(t) looks like a wave of some frequency f? By integrating the signal by the exponential term, researchers can measure how closely a particular wave matches the signal. Now, ask that question for every frequency and the answer is a power spectrum, where the height at each frequency is:

$$P(f) \propto |X(f)|^2$$

This way, researchers can tell which bands are active from a messy signal. For example, a tall peak at 8 Hz means a strong theta rhythm, while one at 60 Hz means gamma.

The Fourier transform is an incredible tool, but it has one significant limitation. When researchers calculate a single Fourier spectrum for an entire recording, the result summarizes its frequency content without showing when each rhythm occurred. This does not mean the Fourier transform assumes the rhythm is steady. But brain activity does not behave that way. Gamma arrives in short bursts; theta waxes and wanes. When averaged over a long recording,

a brief strong burst and a weak continuous rhythm can produce similar spectra.



Power spectrum versus time-frequency spectrogram

Figure A6. Based on: Kay (2014), Figure 4; Einevoll et al. (2013).

The wavelet transform solves this by replacing the infinite sine wave with a short, localized wave that can be moved and stretched:

$$W(a,b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t)\psi^*\left(\frac{t-b}{a}\right)dt$$

This allows researchers to get information about what is happening at a given frequency *at a given moment*, rather than averaging across an entire recording.

And of course, signal analysis has another limitation. If a signal is sampled f_s times per second, the highest frequency that can be recovered is: $f_{Nyquist} = f_s/2$. Without appropriate filtering before sampling, frequencies above that threshold can “fold back” and reappear as misleading lower frequencies (an artifact called aliasing). A fast spike recorded at too low a sampling rate can falsely be disguised as a slow rhythm, which is why extracellular recordings intended for spike sorting typically use high sampling rates, often exceeding 20,000 Hz, along with anti-aliasing filters.

Finally, there are limits to what an LFP can reveal *no matter* how carefully it is analyzed. The signal reflects population currents, but does not identify which cells within it are responsible. The word “local” is also misleading because currents conduct through brain tissue and are added up at the electrode; some of what is recorded can originate beyond the tissue immediately surrounding the tip (a phenomenon called volume conduction). Herreras (2016) goes even further, arguing that the amplitude of an LFP depends strongly on the topology and spatial arrangement of the sources, not just how much activity is occurring, meaning a weaker but well-aligned source can produce a larger signal than a stronger disorganized one nearby (Pesaran et al., 2018). Truly recovering the underlying neural activity from a field potential generally requires building a model of how the signal was generated in the first place (Einevoll et al., 2013).

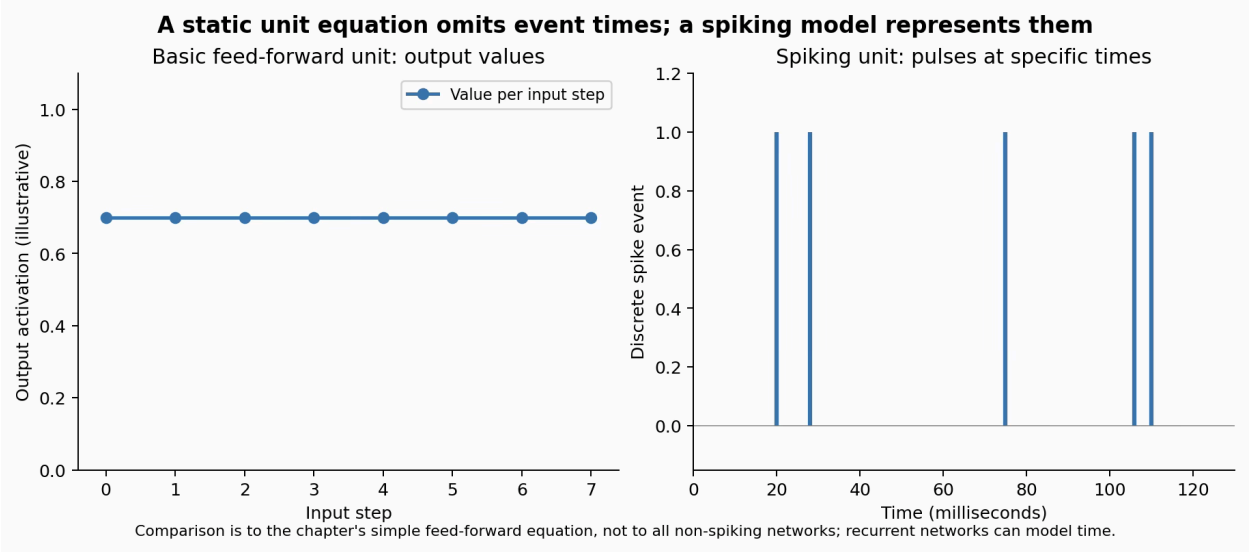
5. From Biological to Artificial

Having seen how much effort goes into recovering timing information from a biological recording, it is informative to turn to a field where engineers have deliberately built that timing in.

Artificial neural networks are computer programs modeled on biological brains. Each unit receives a set of inputs, multiplies each one by a predetermined number called a synaptic weight, sums the results, and passes the total through a function that produces a single output:

$$y = \varphi\left(\sum_i w_i x_i + b\right)$$

Here the x_i are the inputs, the w_i are their weights, b is an offset, and φ is a function that turns those inputs into a meaningful result. But notice what is absent from this basic feed-forward equation: there is no explicit spike time t. The output is a number, not a timed spike event, and nothing in the expression says when a spike happened. Everything the preceding sections established about spike timing, about phase, about a neuron firing at the trough rather than the peak of an ongoing rhythm, is not represented in this particular equation. Other artificial neural networks, however, can process time-dependent inputs.



Basic artificial neuron output versus discrete spike times

Figure A7. Based on: Imam & Cleland (2020).

Models that represent the timing of individual spikes look different. The leaky integrate-and-fire model approximates how membrane voltage accumulates input and leaks toward rest; it is written as a differential equation:

$$\tau_m \frac{dV}{dt} = -(V(t) - V_{rest}) + R_m I(t)$$

The voltage (V) accumulates a current I(t) and is constantly “leaking” back towards its natural resting value. When enough voltage has accumulated to reach a threshold, the model spikes and resets. This is a drastic simplification of the ion channel process that is described at the start of this chapter. The most important feature is still present though; the output arrives as a discrete spiking signal at a specific time.

Spiking neural networks extend this idea across entire systems, with “neurons” communicating with each other through timed pulses (rather than continuous values), allowing them to process information using spike timing. In one form of spike-timing-dependent plasticity (STDP), which changes the connection strength between neurons based on the precise millisecond timing of their electrical impulses, the synaptic weight depends on the interval between a presynaptic and postsynaptic spike ($\Delta t = t_{post} - t_{pre}$) and the sign of that interval. Under a common STDP rule, the connection strengthens if the presynaptic spike comes just before the postsynaptic spike and weakens if it arrives just after, with the size of the change getting smaller as the interval gets longer. If Neuron A fires before Neuron B, the connection becomes stronger under this simplified timing rule. If Neuron A fires after Neuron B, the connection becomes weaker under the same rule. Actual plasticity depends on the circuit and conditions.

At Cornell University, Imam and Cleland (2020) built a working example of a spiking neural network, constructing a circuit based on the architecture of the mammalian olfactory bulb and implementing it on Intel’s Loihi neuromorphic chip (specifically designed for computation with spikes). The algorithm applies spike-timing-dependent plasticity iteratively over sequential gamma-frequency packets. Basically, the system learns to recognize a complex pattern by taking a rapid sequence of timed snapshots, using the split-second arrival order of electrical pulses in each snapshot to adjust and strengthen selected connections while tuning out the noise. Now what can this do? This result describes the tested model rather than all spiking networks. However, the resulting system learned to distinguish odor patterns after limited exposure, even with interference from other odors.

CHAPTER ADD-INS

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-i 2\pi f t} dt$$

$$P(f) \propto |X(f)|^2$$

$$W(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t) \psi^*(\frac{t-b}{a}) dt$$

$$y = \varphi(\sum_i w_i x_i + b)$$

$$\tau_m \frac{dV}{dt} = -(V(t) - V_{rest}) + R_m I(t)$$

1. Intracellular action potential versus extracellular spikes and local field potential

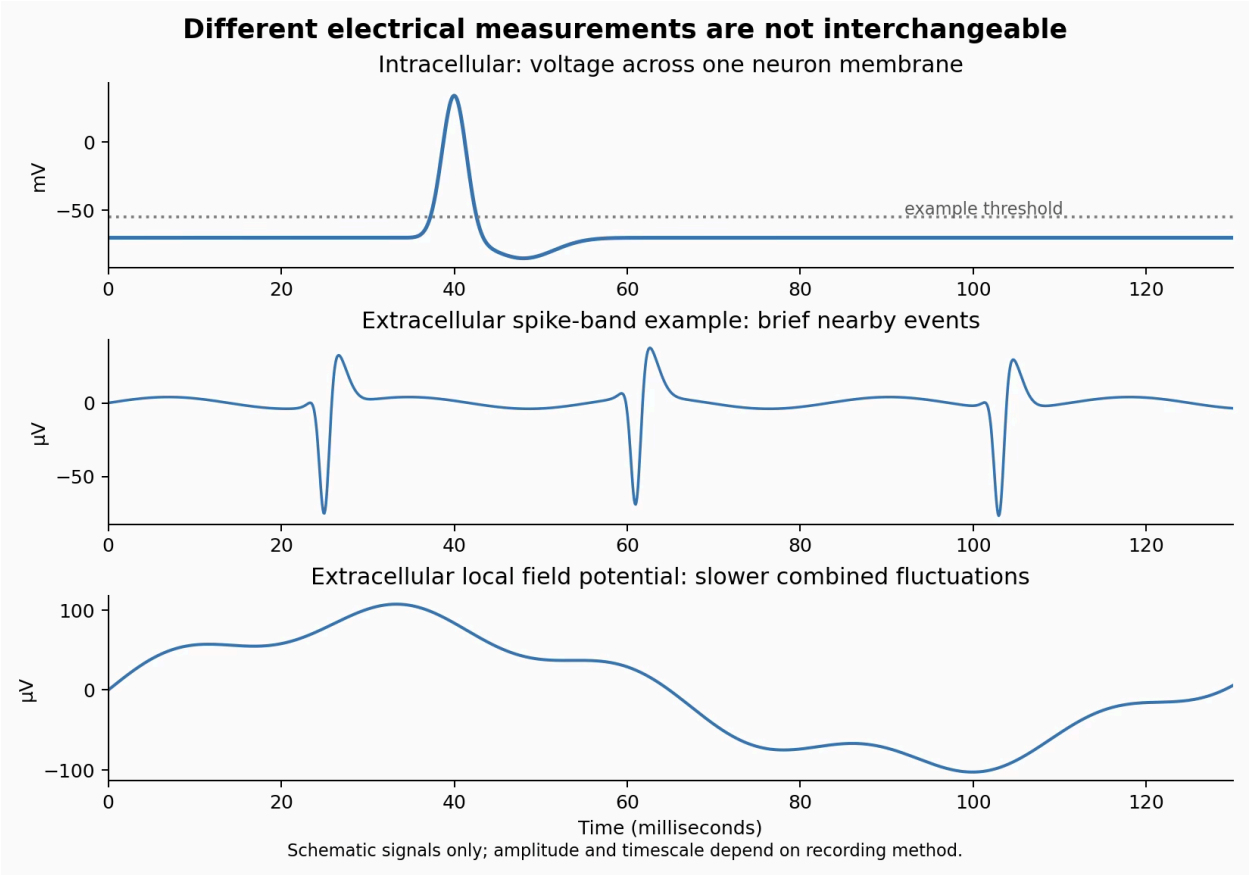


Figure A1. Based on: Buzsáki et al. (2012); Rey et al. (2015); OpenStax, Section 12.4.

2. Delta, theta, alpha, beta, and gamma compared

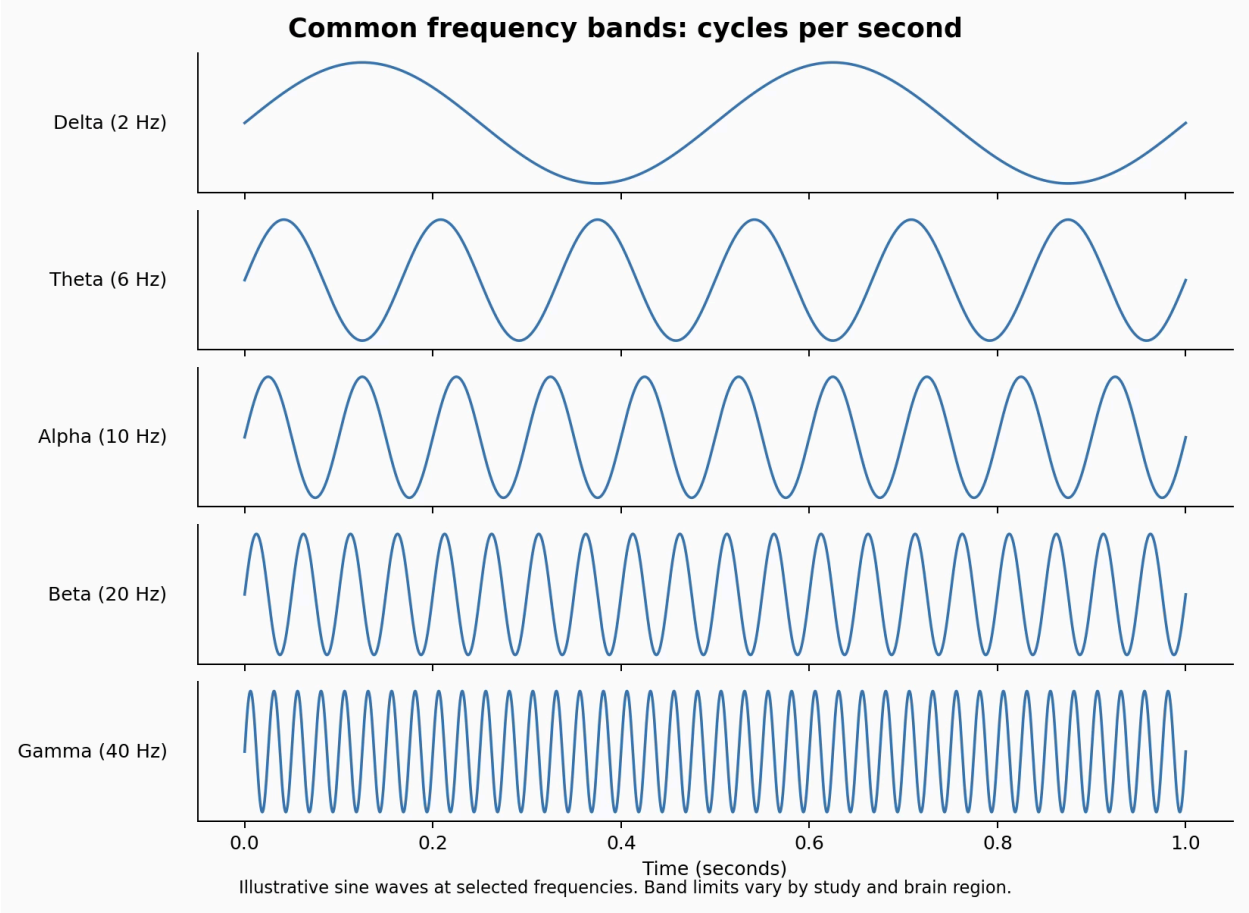


Figure A2. Based on: Buzsáki & Draguhn (2004); Kay (2014), Section 3.

3. Spikes at different phases of an LFP cycle

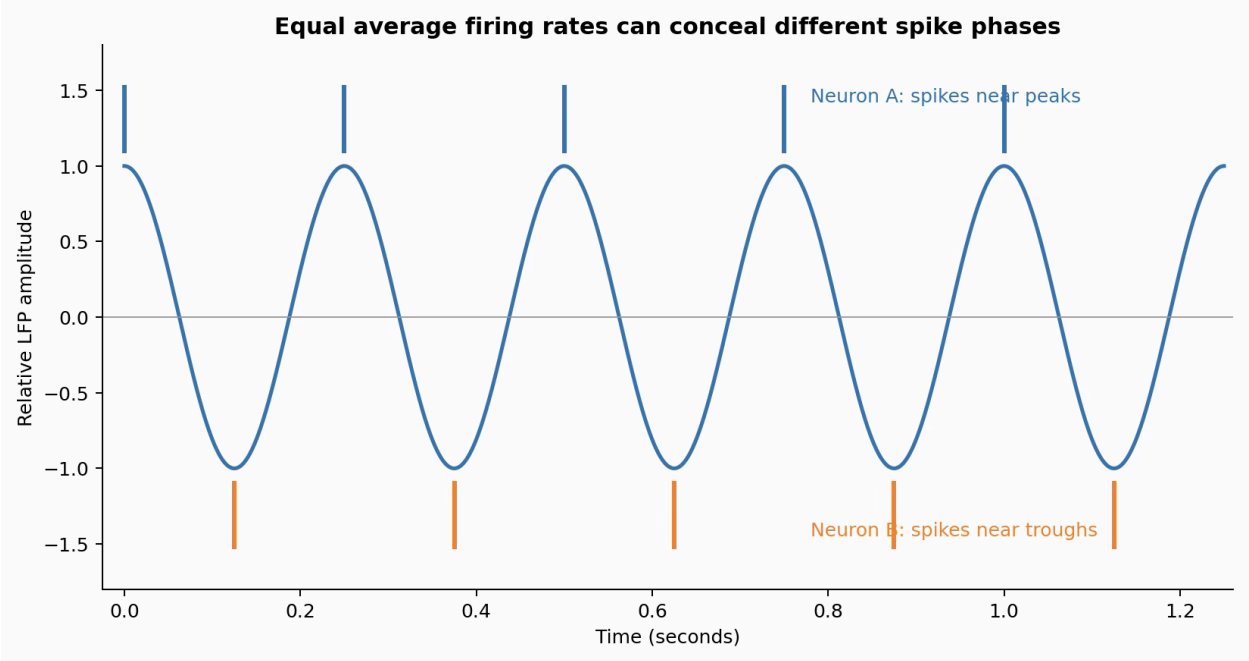


Figure A3. Based on: Kay (2014), Figure 5; Buzsáki & Draguhn (2004).

4. Breathing-cycle timing and fast oscillation amplitude

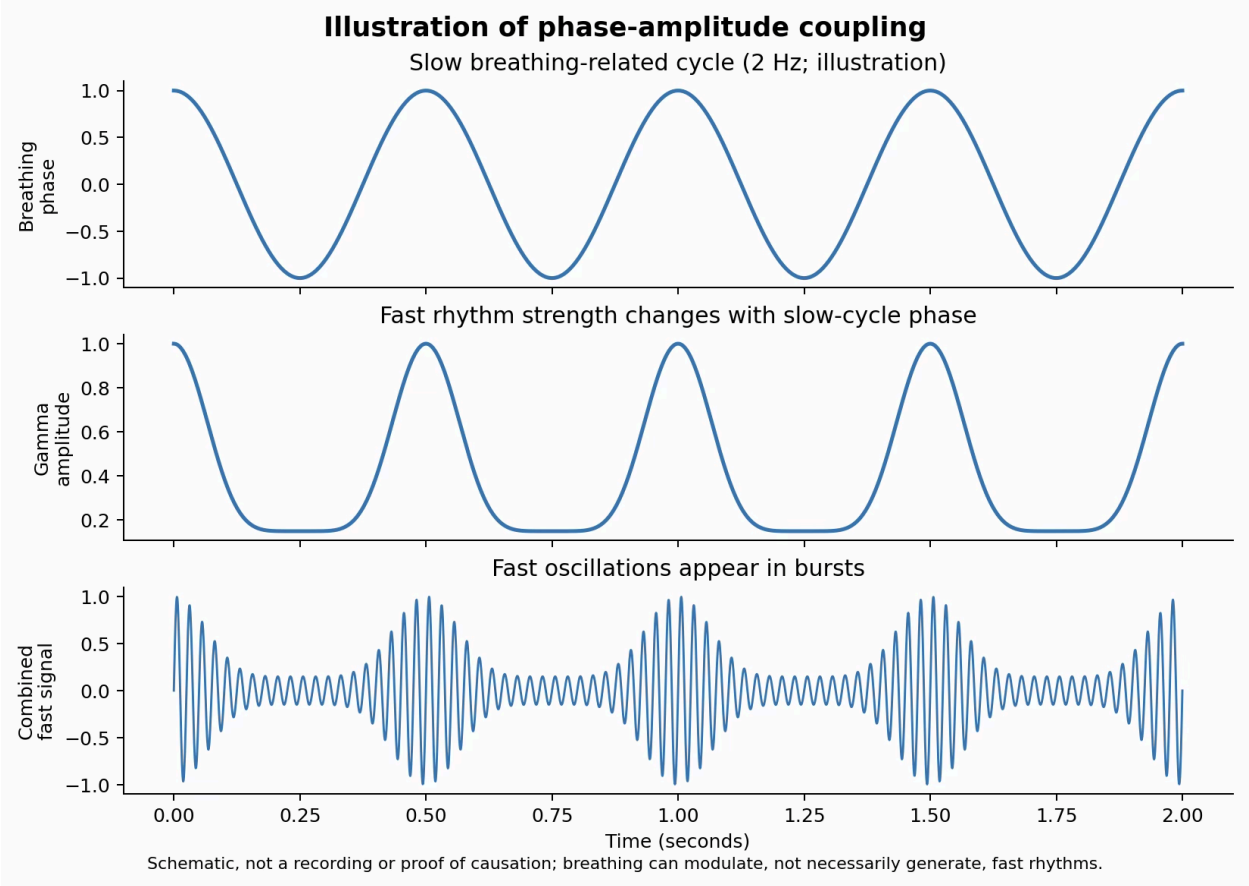


Figure A4. Based on: Tort et al. (2025), Figure 1; Kay (2014), Figures 2 and 4.

5. How spike sorting estimates neurons from waveforms

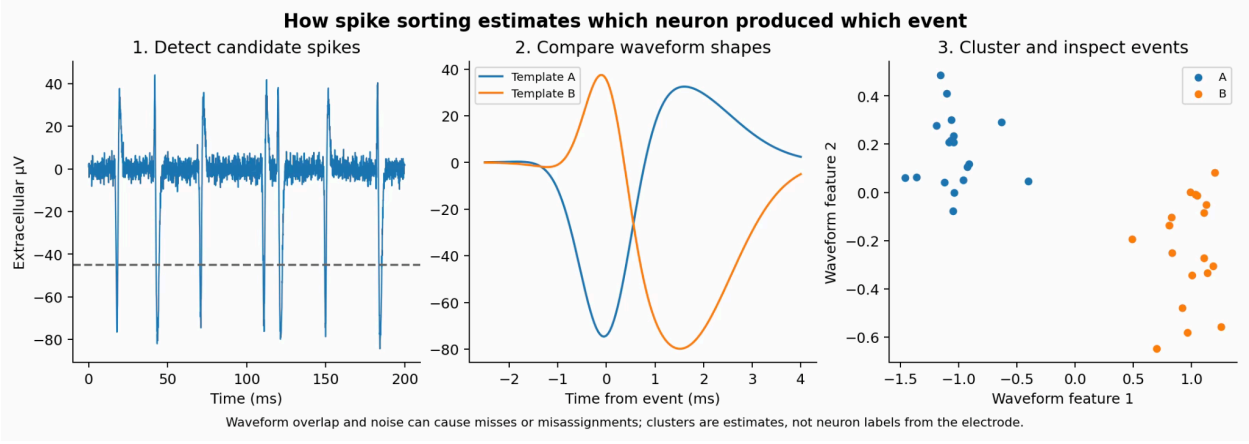


Figure A5. Based on: Rey et al. (2015); Yger et al. (2018), figures.

6. Power spectrum versus time-frequency spectrogram

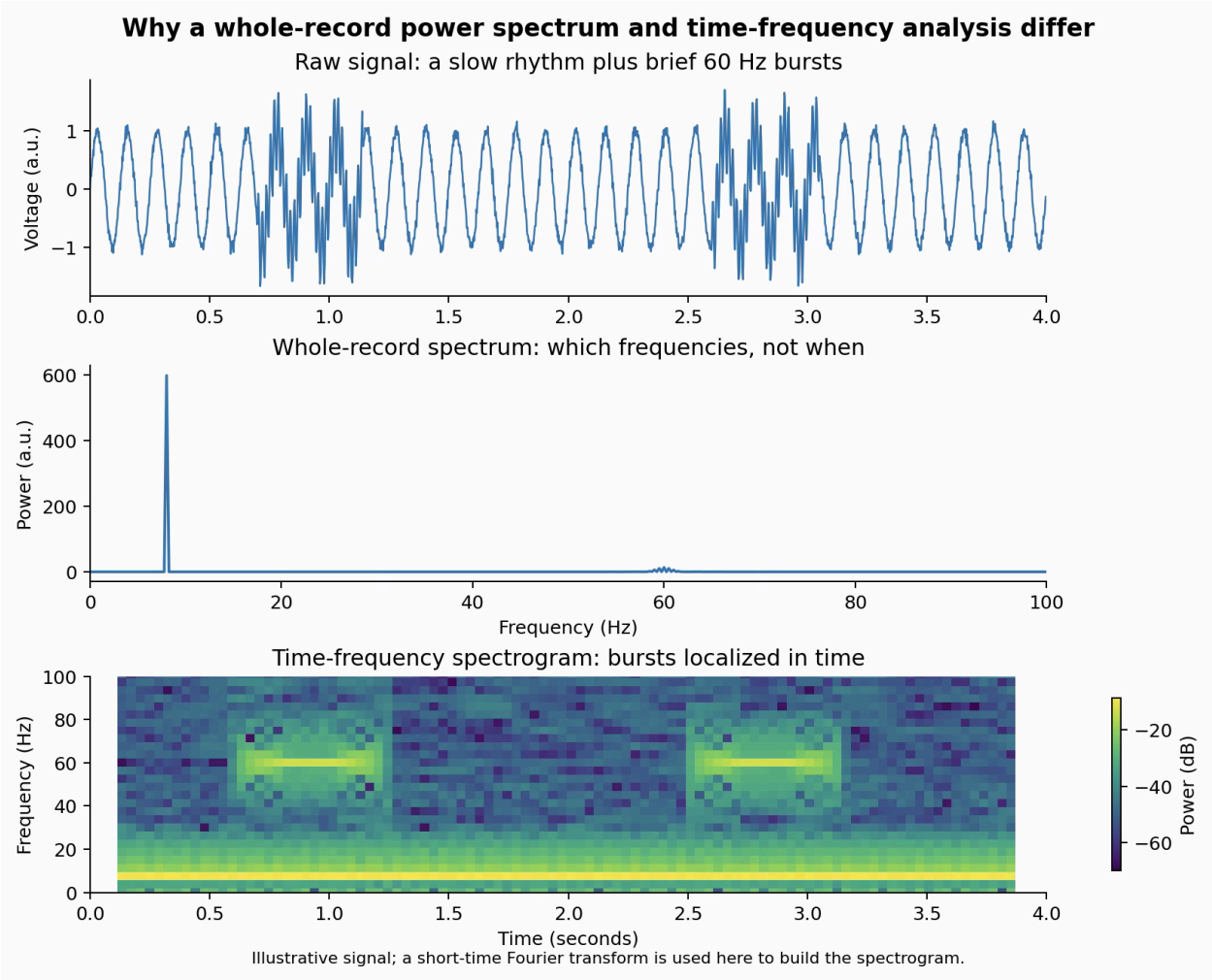


Figure A6. Based on: Kay (2014), Figure 4; Einevoll et al. (2013).

7. Basic artificial neuron output versus discrete spike times

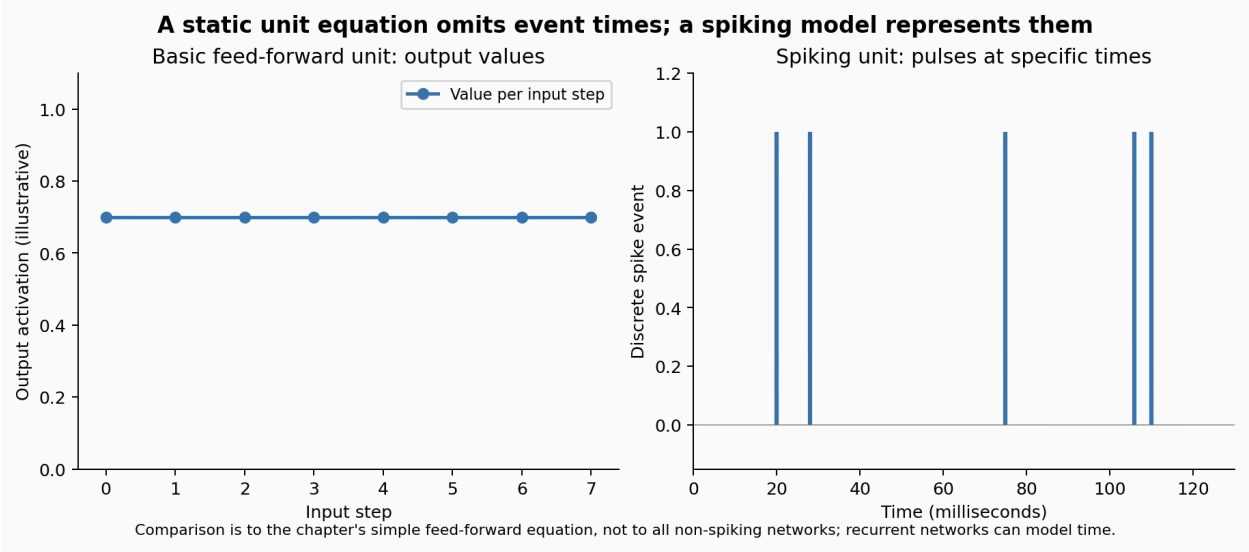


Figure A7. Based on: Imam & Cleland (2020).

Kay (2014), Figure 4

Example recordings and spectrogram illustrating gamma bursts.

Kay (2014), Figure 9

Experimental tests of odor discrimination and gamma oscillations.

Tort et al. (2025), Figure 1

Breathing-related activity at multiple neuronal scales.

Yger et al. (2018), Figures

Examples of spike detection and sorting.

OpenStax, Section 12.4

Annotated neuronal action-potential waveform.

https://www.jezzamon.com/fourier/
GREAT SITE FOR FOURIER DEMONSTRATION

GLOSSARY

Term	Definition
Action potential (spikes)	A rapid reversal of the membrane potential that propagates

	along an axon to transmit information between neurons.
Membrane potential	The difference in voltage between the inside and outside of a cell membrane.
Axon (and hillock)	The axon is the long tail of the neuron that conducts electrical impulses away from itself. The axon hillock is the specialized region of the cell body that connects to the axon where the action potentials are created.
Refractory period	The short time after a neuron fires (action potential) where it cannot fire again in succession, unless a much stronger stimulus is provided.
Extracellular recording	A method of measuring electrical activity from outside the cell membrane (using electrodes).
Intracellular recording	A method of measuring electrical activity in just a single cell (with a microelectrode)
Oscillation	Repeating and rhythmic fluctuations in electrical activity across a group of neurons.
Frequency	The number of cycles (waves) an oscillation completes in a second (measured in Hertz)
Phase	The position of a wave at a specific point with respect to a complete cycle.
Power spectral density	A mathematical measure that shows how the energy of a signal is distributed across different frequencies.
Bandpass filter	A signal processing tool that blocks frequencies outside a certain range, allowing to single out certain wavelengths.
Local field potential	The recorded extracellular electrical signal that reflects the synaptic activity of a local population of cells.
Phase-amplitude coupling	A special type of cross-frequency coupling of a brain signal interaction where the strength (amplitude) of a fast brain wave is controlled by the cycle (phase) of a slow brain wave.
Cross-frequency coupling	An interaction between different frequency bands where neural oscillations interact with each other to process information.
Entrainment	The interaction where an internal brain rhythm is affected by external periodic rhythms (such as sensory stimulus or behavioral cycles).
Spike sorting	A data processing step used to group extracellular signals to

	individual neurons.
Fourier transform	A mathematical tool that is used to break down complex signals into its individual sine wave frequency components, revealing which rhythms are present and how strong they are.
Wavelet transform	A similar tool to the Fourier transform that breaks down a signal into localized wavelets, allowing analysis of how frequency components change over time.
Multi-unit activity (MUA)	The combined activity of several neurons that are recorded by a single electrode.
Nyquist frequency	The highest frequency that can be captured without distortions (half the sampling rate).
Aliasing	Distortions and “ghost frequencies” that occur when not abiding by the Nyquist frequency threshold.
Signal-to-noise ratio	A measure that compares the meaningful neural parts of a signal to background noise.
Thresholding	A processing technique used to separate real neural signals from noise using a specific voltage limit.
Integrate-and-fire model	A model of a neuron that adds up incoming electrical inputs over time until a threshold is reached and then “fires.”
Memristor	An electronic component that mimics biological synapses by “remembering” the amount of charge that previously flowed through it.
Spiking Neural Network (SNN)	An artificial neural network that uses timed pulses (spikes) instead of continuous values to communicate.
Neuromorphic computing	An approach to computer engineering where systems are designed to mimic natural biological processes and structures of the nervous system.
Neuromorphic hardware	Specially designed circuits designed to use brain-like structures to execute neural networks.
Synaptic weights	In an artificial neural network, these are the values that determine how strong the connection between two nodes are (how much they influence one another).
Artificial neural network (ANN)	A computer program modeled after biological brains, where continuous values are created by summing weighted inputs rather than discrete spikes.

Volume conduction	The spread of electrical current through brain tissue, allowing an electrode to record signals generated away from its tip
Spike-timing-dependent plasticity (STDP)	An algorithm used to update the connection model weights of an ANN where the change in a synaptic connection depends on the interval between a presynaptic and postsynaptic spike (and which came first)

REFERENCES

Sources:

Buzsáki, G., Anastassiou, C. A., & Koch, C. (2012). The origin of extracellular fields and currents: EEG, ECoG, LFP and spikes. Nature Reviews Neuroscience, 13(6), 407–420. https://doi.org/10.1038/nrn3241

Buzsáki, G., & Draguhn, A. (2004). Neuronal oscillations in cortical networks. Science, 304(5679), 1926–1929. https://doi.org/10.1126/science.1099745

Einevoll, G. T., Kayser, C., Logothetis, N. K., & Panzeri, S. (2013). Modelling and analysis of local field potentials for studying the function of cortical circuits. Nature Reviews Neuroscience, 14(11), 770–785. https://doi.org/10.1038/nrn3599

Herreras, O. (2016). Local field potentials: Myths and misunderstandings. Frontiers in Neural Circuits, 10, 101. https://doi.org/10.3389/fncir.2016.00101

Imam, N., & Cleland, T. A. (2020). Rapid online learning and robust recall in a neuromorphic olfactory circuit. Nature Machine Intelligence, 2(3), 181–191. https://doi.org/10.1038/s42256-020-0159-4

Kay, L. M. (2014). Circuit oscillations in odor perception and memory. In E. Barkai & D. A. Wilson (Eds.), Progress in brain research: Odor memory and perception (Vol. 208, pp. 223–251). Elsevier. https://doi.org/10.1016/B978-0-444-63350-7.00009-7

Pesaran, B., Vinck, M., Einevoll, G. T., Sirota, A., Fries, P., Siegel, M., Truccolo, W., Schroeder, C. E., & Srinivasan, R. (2018). Investigating large-scale brain dynamics using field potential recordings: Analysis and interpretation. Nature Neuroscience, 21(7), 903–919. https://doi.org/10.1038/s41593-018-0171-8

Rey, H. G., Pedreira, C., & Quian Quiroga, R. (2015). Past, present and future of spike sorting techniques. Brain Research Bulletin, 119, 106–117. https://doi.org/10.1016/j.brainresbull.2015.04.007

Tort, A. B. L., Laplagne, D. A., Draguhn, A., & González, J. (2025). Global coordination of brain activity by the breathing cycle. Nature Reviews Neuroscience, 26(6), 333–353. https://doi.org/10.1038/s41583-025-00920-7

Ultradian Rhythms: Neuronal Activity

1. What is Being Measured?

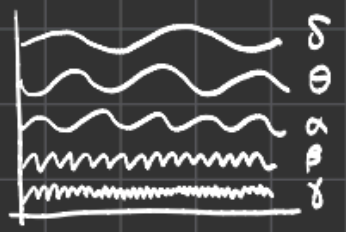
- Electrodes records voltage
- Noisy sine-like line
- Spike: action potential
  - rapid change in Voltage across cell membrane
  - how neurons communicate
  - Resting: -70 mV
  - Stimuli make cell less (-) until -55 mV threshold
    - Sodium channels open → Na^+ ions
      → DEPOLARIZATION (+30 mV)
    - Na channels close & K channels open
      → REPOLARIZATION



- Connected to ~1000 → ~10000 neurons through Synapses (contact pts)
- Local Field Potential (LFP): combined activity (synaptic transmembrane current)
  - smooth & gradual wave
- Problem: same electrodes, both signals
  - Spike interfering leads into hard to see/read the exact LFP
  - Party conversation analogy (zoning out)
- Spikes occur at predictable points in LFP

2. Neural Activity is Rhythmic:

- LFP has a repeated cycle (contractions)
- Oscillations in Hz
  - δ: 1-4 Hz
  - θ: 4-8 Hz
  - α: 8-12 Hz
  - β: 15-30 Hz
  - γ: 30-100 Hz



- Other areas have some comparable rhythms:
  - γ olfactory in rats: 40-100 Hz
  - β in insects: 30 Hz
- 40 Hz is 1 cycle every 25 ms.
- 4 Hz is 1 cycle in 250 ms

- Do they do anything at all?
  - bridge vibrates when traffic goes over it, but that's not why the bridge was built
    - Not necessary
    - 5 orders of magnitude
    - Presumed neural spikes
- Binds which inputs reach a pop. of neurons. Hebbian theory to make synapses
  - LEARNING
- γ happens 3.5 million times a day (estimate)
- Olfactory γ evidence: w/o scent small, with small $\uparrow$
  - air carries odor molecules in rhythmic pulses

3. Connection to Slower Body Rhythms:

- Many models: neural activity synchr. w/ breathing
  - Neural Coupling
  - Slow rhythm controls amplitude of faster one
    - "entrains" not creates
    - slowest reference signal
    - signal times impulse trigger (skin stretch)

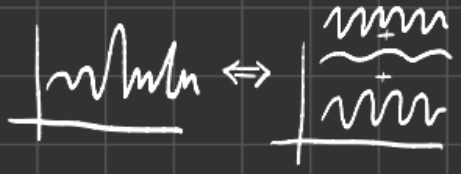
4. Getting a Signal from a Recording:

- 2 Problems:
  1) Which neuron produced which spike?
  2) Which spikes are present & when?
- Electrode picks up many neurons
- Spike Sorting:

MUA (multi-unit activity) [
1) Recording is bandpass filtered (300, 3000 Hz)
2) Filtered trace is scanned for threshold crossings (thresholding)
3) Shape of each detected waveform is measured
4) Waveforms w/ similar shapes are clustered together

- Rests on assumption: consistent waveforms, differ due to noise from electrode
- Sorting degrades when many neurons close to electrode
  - Readily throw neurons out
- Fourier transform:

$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-i2\pi ft}\, dt$$



  - How much of signal $x(t)$ looks like frequency f?
  - Integrate $x(t)$ signal by exponential (-) rotating wave to see alignment.
- Power Spectrum:

$$P(f) \propto |X(f)|^2$$

  - Which bands are active.
  - Problem: $\int_{-\infty}^{\infty}$ and signals are short & changing ∴ avg = bad
- Wavelet transform:

$$W(a,b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t)\, \psi^*\left(\frac{t-b}{a}\right) dt$$

  - Gives normal, not avg
  - Problem, max freq: $f_{Nyquist} = \frac{f_s}{2}$. (Fold back & appear as lower phantom frequency)
- Unsolvable Problems:
  - Local misleading: currents through brain tissue can be seen far away:
    VOLUME CONDUCTION → topology & spatial segment > activity
  - Don't know which cells are responsible

5. Biological to Artificial

- Artificial Neural Nets (ANN) modeled after brain
- Unit receives set of inputs, sums, passes through a function for a single output

$$y = \varphi\left(\sum_i w_i x_i + b\right)$$

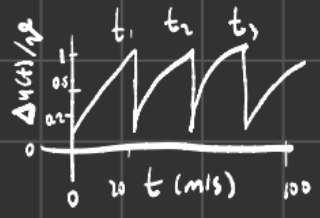
function · inputs · weights · offset

No t!

- Leaky Integrate-and-Fire Model:

$$\tau_m \frac{dV}{dt} = -(V(t) - V_{rest}) + R_m I(t)$$

Voltage · leaking Current



- Simplification of ion channel process
- Discrete spiking signal @ specific time

- Spiking Neural Nets:



- Same ideas, whole systems
  → easier & known
- Spike-timing dependent plasticity: $\Delta t = t_{post} - t_{pre}$
  - if input comes before (presynaptic), connection strengthens (synaptic weight)
  - if after (postsynaptic), synaptic weight weakens (vice versa/effect)
  - backpropagation

- Cornell (Eliasmith & Chalmers 2020)
  - Intel's Loihi neuromorphic chip
  - Improve a complex pathway by using timed synapses ? STDP
  - Tell other apart, even with interference

UNIT AND CHAPTER

Unit: Infradian Rhythms

Chapter 4: Immunity

CONTRIBUTORS

- Ruben Hayrapetyan
  - Subject Matter Expert – 'Immunology and Immune System Functionality Expert'

LEARNING OBJECTIVES

After reading this chapter, students should be able to …

1. Define what the immune system is and how it functions.
2. Relate the functioning of the immune system's circadian rhythm to why various immune functions happen only during certain hours of the day.
3. Describe how the time of day can change the outcome of conditions like sepsis.
4. Interpret seasonal immune differences and how they could impact peoples' daily lives.

CHAPTER OUTLINE

- What is the Immune System and how does it function? (Learning Objective: Define what the immune system is and how it functions)
  - The Innate Immune System
  - The Adaptive Immune System
  - Basic Immune Responses
- Diseases during the rest vs active time
  - Nighttime Symptom flares
  - Morning vaccinations
  - Sepsis during different parts of the day
- Seasonal Cycles within the Immune System
  - Seasonal gene expression

- Seasonal inflammation

CHAPTER WRITTEN CONTENT

The Immune System

The innate immune system is a general, fast-acting first line of defense against any pathogen. It operates through three mediums: physical, chemical, and cellular. The physical barriers include mucus in the nose, and the skin. The mucus traps microbes before they can enter through the nasal passages, and the skin's dry, slightly acidic surface makes it hard for bacteria to colonize. As opposed to the physical barriers, which keep pathogens out, the chemical defenses such as stomach acid, tears and sweat kill pathogens that reach the body's surfaces and openings. For example, stomach acid specifically kills pathogens that are swallowed with food or drinks. Lastly, the innate immune system has cellular defense mechanisms, utilizing cells like macrophages and neutrophils to engulf and destroy pathogens. While fast, these defenses are non-specific and have no memory, and some pathogens have evolved ways to bypass them. For these more complex infections, the adaptive immune system is needed.

The adaptive immune system provides a slower but much more targeted response to specific pathogens. Unlike the innate immune system, the adaptive immune system can take several days to develop upon the first exposure. Moreover, the adaptive immune system retains information about the pathogen for future similar attacks. This is why vaccines inject a harmless form of a pathogen so that the adaptive immune system can develop immunity to it.

The innate and adaptive immune systems' joint work starts when a pathogen enters the body. When the immune system recognizes a threat, innate immune cells release

cytokines (signaling proteins that coordinate the response and alert other immune cells) to the location of the threat. The blood vessels near the infection widen during an immune response so that the fast-acting innate immune system cells reach the site quickly, which also causes redness, warmth, and swelling. While the innate immune system targets what it can, the antibodies and immune cells neutralize or destroy the remaining pathogens and infected cells in a targeted attack. After the attack, anti-inflammatory signals, including cortisol, dampen the immune response, allowing the affected tissue to repair.

Circadian Rhythms in the Immune System

The immune system, while effective at fighting diseases, cannot be on high alert all the time as it would be too taxing for the human body. As a result, the immune system regulates its active hours by surging at night and being less active during the day. Much of this nighttime surge comes from cytokines, the signaling proteins introduced earlier. There is a vast range of cytokines: there are those that are pro-inflammatory, others that reduce inflammation, and some that raise the body's internal temperature. While not quite a one-to-one correlation, cytokines can be linked to symptoms such as fever, aches, tiredness, and inflammation: some cytokines raise the body temperature and cause fatigue while others make nerves more sensitive. Cytokines are balanced by a hormone called cortisol, best known for its relationship with stress, and one of its roles is to suppress immune activity. Furthermore, cortisol has a distinct circadian rhythm, with its lowest levels of production during the first half of the night and its highest around waking.

This was shown in a study by Petrovsky and Harrison (1998), in which blood samples were taken from participants at intervals throughout the day. Then, the immune response of the blood was activated by introducing bacterial endotoxin (the outer shells of bacteria) and tetanus toxoid (a harmless form of tetanus). Cytokine counts and plasma cortisol were measured throughout the day, with plasma cortisol levels indicating how much the immune system was suppressed and cytokine counts showing how active the immune system was. The counts showed that cytokine production surged at night and in the early morning, which ran inversely to the circadian rhythm of plasma cortisol levels. However, an inverse pattern on its own cannot show which of the two is driving the other. To test this, the researchers then artificially raised plasma cortisol levels in the evening, and the cytokine levels were dulled. This indicated that the cytokine rhythm is dependent on the cortisol rhythm. These nighttime cytokine peaks help explain a common phenomenon called nighttime flares, in which one's symptoms worsen during the night and in the early morning.

Beyond nighttime flares, this heightened immune activity also influences the effectiveness of vaccines. Since vaccines are generally harmless antigens that are injected into the body to train the immune system to fight against the harmful pathogens themselves, their success depends on how strongly the immune system responds. So, the time at which they are administered also affects their effectiveness. This is especially relevant for older adults, who tend to have weaker immune responses to vaccination. Long et al. (2016) studied patients over the age of 65 at various general practices in the UK, taking blood samples when they received their flu shots and again one month later to measure antibody levels against the vaccine's flu strains. The trials

found that older adults who received their vaccinations in the morning had higher antibody levels than those who were vaccinated in the afternoon. While intuitively this could be attributed to the increased cytokine production in the early morning, and therefore increased immune activity, researchers found that cytokine levels did not account for the difference. The underlying cause of this difference is still a topic of open research. While part of the mechanism is still unknown, this study shows that the immune system's response to the vaccine depends on the time of day, suggesting a circadian rhythm in its activity.

Despite the generally protective role of the immune system, it is possible, for a large enough threat, to overreact in a poorly controlled manner and to ultimately damage the body's own cells. This state is called sepsis. For sepsis, the circadian rhythm of the immune system is very consequential. This was highlighted when researchers (Halberg et al., 1960) injected mice with bacterial endotoxin at different times to drive them into a state of sepsis. Since the mice's cortisol levels were at their lowest during their daytime rest phase, the mice that were injected during the day were at a much higher risk for their immune systems to overreact as opposed to the mice that were injected at night.

Seasonal Rhythms in Immunity

These daily rhythms are not the only cycles that shape immune activity. Over the course of a year, the immune system is also impacted by a longer infradian rhythm that follows the seasons. One way to see this is by following the body's gene expression as it varies across seasons. Gene expression is how the body controls protein production, reducing the production of some proteins while amplifying the production of others. When a particular gene is active, it means that the body is allowing for the protein encoded by that gene to be produced. This applies to

the immune system because the production of its cells and proteins, such as cytokines, depends on certain genes being active. As a result, seasonal differences in gene expression lead to differences in how the immune system functions.

To investigate this more closely, Dopico et al. (2015) tested whether genes like BMAL1, a gene that is known to follow a circadian rhythm, also follow an infradian rhythm, specifically aligned with the seasons. Instead of running an experiment themselves, the researchers analyzed large existing datasets of blood samples from people around the world. The analysis showed that about a quarter of the tested genes changed their expression seasonally. To further affirm that the finding was seasonal, the researchers tested blood samples from the northern hemisphere and the southern hemisphere separately and found that the shifts in gene expression were reversed, which indicates that the gene expressions matched local seasons. Researchers also observed that in the wintertime, immune-related genes shifted towards a more pro-inflammatory state which is represented by a higher number of monocytes, the white blood cell involved in inflammation, lower activity of BMAL1, which is known to suppress inflammation, and higher levels of proteins that are produced in response to cytokine firing. These changes suggest a more reactive immune system in the winter, possibly because the body evolved to prepare for the higher number of infections during that season.

The variation of the immune system is not solely represented by gene expression. Liu and Taioli (2015) aimed to provide a large-scale baseline showing that the immune system is not stagnant throughout the seasons. Using CDC survey data with blood counts from surveys between the years 1999 and 2012, they found that white blood cells and neutrophils, cells that play a part in inflammation, were also higher in the winter and spring than in the summer and fall. Moreover, this difference was stronger in children than adults, which the authors suggest may be because a developing immune system is more sensitive to seasonal changes. The researchers also suggest that this

more inflammatory winter state may help the body defend against the colds and flu that are common during those months. Together with the findings discussed earlier, it is evident that the immune system not only depends on time of day, but also time of year.

CHAPTER ADD-INS

Cells of the Immune System

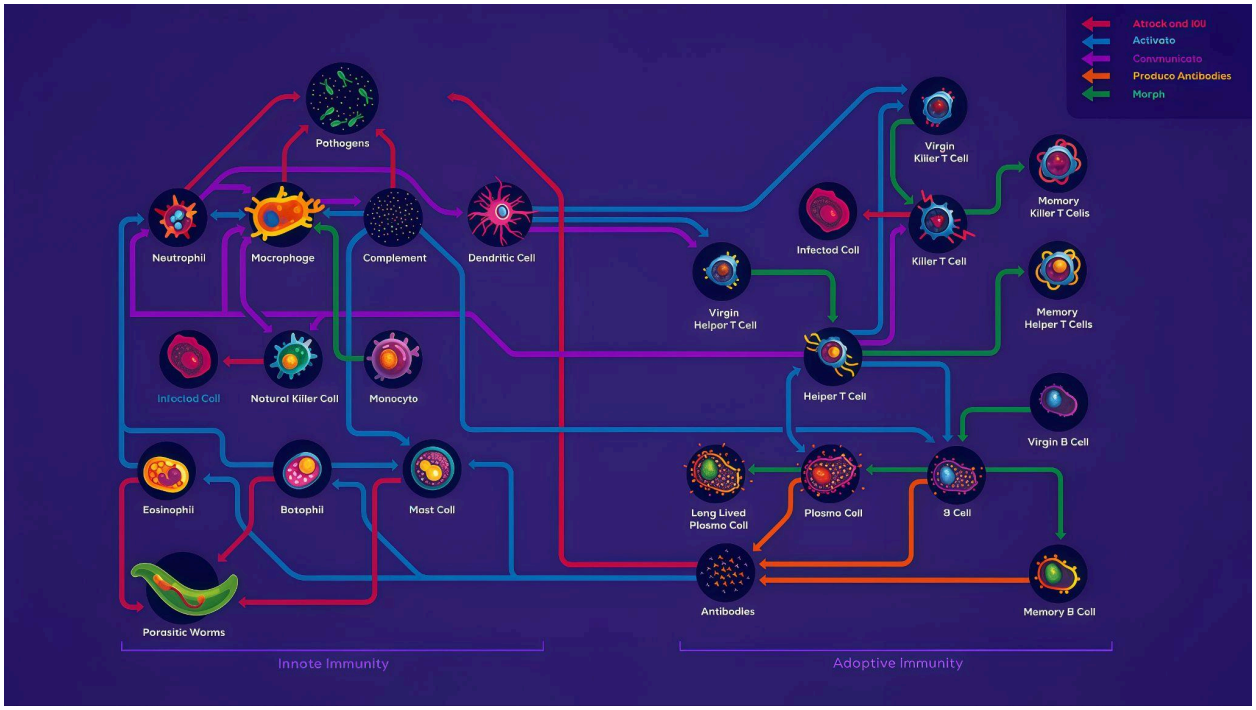


Image: Kurzgesagt – In a Nutshell, from Immune by Philipp Dettmer (2021).

This figure depicts many of the cells involved in the body's daily battles against pathogens, with innate immune cells on the left and adaptive immune cells on the right. This figure originally comes from Kurzgesagt's Immune (2021), and its colored arrows show how each cell acts. Imagine a pathogen invading the body. The first responders on the scene are the macrophages and neutrophils, which engulf the invaders, while the macrophages also send signals for more neutrophils if needed. Complement proteins attack the pathogen directly, and natural killer cells destroy any of the body's own cells that have been infected to stop the disease from spreading. If the innate immune system cannot defeat the pathogen on its own, dendritic cells pick up pieces of it and carry them to the helper T cells. These cells coordinate the attack, activating killer T cells and B cells, which turn into plasma cells and produce antibodies. These antibodies tag the pathogens so other cells can find and destroy them. Once the infection has been

dealt with, some T and B cells become memory cells, allowing for the adaptive immune system to fight this disease much more effectively in the future.

Franz Halberg: Father of Chronobiology

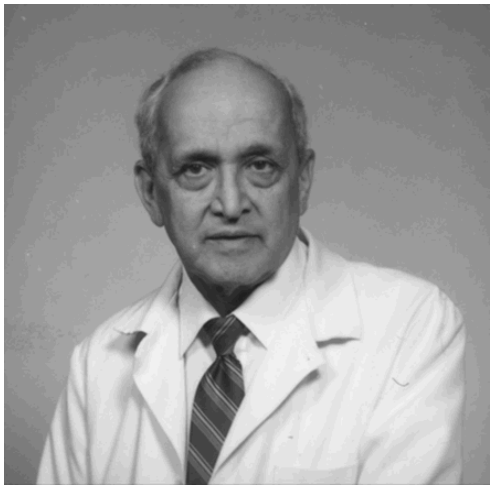


Photo: Halberg Chronobiology Center, University of Minnesota.

Franz Halberg was a Romanian-born scientist considered to be one of the founders of chronobiology. He first began his experiments in 1949 at the University of Minnesota, where he would spend another 60 years. He sought to prove that biological rhythms are internal rather than just passive responses to the environment. In 1959 he coined the term circadian and only a year later, Halberg showed that these rhythms also apply to the immune system. He did this by conducting an experiment where he induced mice into sepsis in different parts of the day. It was observed that whether a mouse survived depended on the time of day, a phenomenon that could be attributed to the immune system. Halberg went on to argue that medical treatments should be timed to the body's internal rhythms, which is still an open field of research today.

Hormones Around the Clock

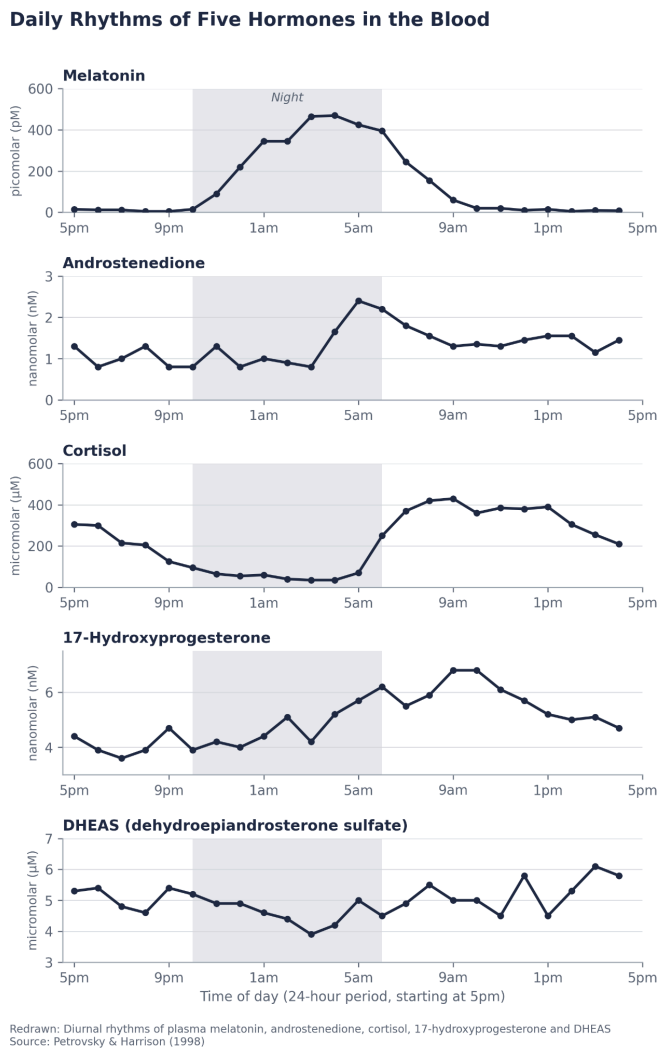


Figure: Adapted from Petrovsky & Harrison (1998).

The immune system's increased activity at night is closely tied to the body's hormones, many of which rise and fall in accordance with a circadian rhythm. This figure shows the levels of five hormones in blood plasma, the liquid part of the blood, measured every hour over 24 hours. The data came from Petrovsky & Harrison's study, which examined how immune activity changes over the course of a day. The grey band denotes the period of time considered to be "nighttime." Melatonin has a clear circadian rhythm, staying close to zero all day, rising at night to a peak around 4am, and falling back down in the morning. Cortisol, the hormone associated with stress, shows the opposite

pattern, being lowest at night and highest in the morning around 9am. Because cortisol suppresses immune activity, its low point at night allows the immune system to be most active. The research also proposes that melatonin may support immune activity, meaning its nighttime high boosts immune activity. Androstenedione and 17-hydroxyprogesterone also rise in the early morning since their release is correlated with cortisol's. As opposed to the first four, DHEAS does not have a strong circadian rhythm showing that not all hormones do.

GLOSSARY

Term	Definition
Cytokine	A protein released by immune cells that coordinate inflammation and the body's response to a potentially harmful infection.
Antibodies	Proteins made by B-cells which bind to specific antigens and mark them for destruction
Sepsis	A state in which your body's immune system overreacts to an infection and starts harming your own cells. This is life threatening in most cases.
Neutrophil	A very common type of white blood cell which is a first-responder to infections and destroys pathogens by absorbing them.
Plasma Cortisol	The concentration of the stress hormone cortisol in blood. This suppresses immune activity and cytokine production.

REFERENCES

Nikolai Petrovsky & Leonard C. Harrison (1998) The Chronobiology of Human Cytokine Production, International Reviews of Immunology, 16:5-6, 635-649, DOI: 10.3109/08830189809043012

Joanna E. Long, Mark T. Drayson, Angela E. Taylor, Kai M. Toellner, Janet M. Lord, Anna C. Phillips, Morning vaccination enhances antibody response over afternoon vaccination: A cluster-randomised trial, Vaccine, Volume 34, Issue 24, 2016, Pages 2679-2685, ISSN 0264-410X, https://doi.org/10.1016/j.vaccine.2016.04.

Dopico, X., Evangelou, M., Ferreira, R. et al. Widespread seasonal gene expression reveals annual differences in human immunity and physiology. Nat Commun 6, 7000 (2015). https://doi.org/10.1038/ncomms8000

HALBERG F, JOHNSON EA, BROWN BW, BITTNER JJ. Susceptibility rhythm to E. coli endotoxin and bioassay. Proc Soc Exp Biol Med. 1960 Jan;103:142-4. doi: 10.3181/00379727-103-25439. PMID: 14398944.

Liu B, Taioli E. Seasonal Variations of Complete Blood Count and Inflammatory Biomarkers in the US Population - Analysis of NHANES Data. PLoS One. 2015 Nov 6;10(11):e0142382. doi: 10.1371/journal.pone.0142382. PMID: 26544180; PMCID: PMC4636256.

Dettmer P. Immune: A Journey into the Mysterious System That Keeps You Alive. 1st ed. New York: Random House; 2021. ISBN: 9780593241318.

Halberg Chronobiology Center. Franz Halberg: In Memoriam [Internet]. Minneapolis (MN): University of Minnesota; 2013 [cited 2026 Oct 16]. Available from: https://halbergchronobiologycenter.umn.edu/franz-halberg/memoriam

UNIT AND CHAPTER

Unit: Infradian Rhythms

Chapter X: Hair Growth/Shedding: Impact of Stress and Potential Therapeutic Approaches.

CONTRIBUTORS

- Kaleb Yoon
    - Subject Matter Expert – 'Hair growth/shedding'
        - Sections 1, 2 and 4
- John Liu
    - Peer Reviewer
- Jack Lou
    - Peer Reviewer

LEARNING OBJECTIVES

After reading this chapter, students should be able to …

1. Describe and distinguish the hair growth/shedding phases
2. Compare and contrast the hair shedding cycles of human and non-human mammals.
3. Design an experiment that would provide the impact of stress on disruption in hair shedding/growth cycle.
4. Defend why or why not the medical therapy cycle is better than nutritional support for treating hair cycle disruptions.

CHAPTER OUTLINE

- **1. Introduction & Basics of Hair Growth and Shedding (~2 pages)**
- 1.1 Definitions of Hair; cycles of degeneration and regeneration. (~1 page)
    - Definitions of key phases in hair growth/shedding
        - **1.1.1 Learning Objective 1:** Describe and distinguish the hair growth/ shedding phases
            - Subtitle: Hair Growth Phases
            - Section: Definitions of Phases
- 1.2 Hair cycle in other animals, mammals & non-mammals (~1 page)

    - How does the hair shedding/growth cycle look in a human-like animal like a chimpanzee?
    - **1.2.1Learning Objective 2:** Compare and contrast the hair shedding cycles of human and non-human mammals.
        - Subtitle: Hair Shedding Cycle in mammals
        - Section: Difference in human & mammal hair cycles.
- **Factors that impact the hair cycle (~1.5 pages)**
- 2.1 Impact of stress on the hair cycle (~1.25 pages)
    - 2.1.2 Cortisol Release (~1 page)
        - **2.1.3 Learning Objective 3:** Design an experiment that would provide the impact of stress on disruption in hair shedding/growth cycle.
            - Subtitles and sections:
                - Subtitle: Disruption in hair growth/shedding cycle
                - Section: Impact of cortisol release
- **Potential Treatments & Interventions (~1.5 pages)**
- 3.1 Nutritional Support and Medical Therapies (~0.5 page)
    - Appropriate timing of chemical treatment
    - Use of nanotechnology to treat hair shedding from stress
    - **3.1.1 Learning Objective 4:** Defend why or why not the medical therapy cycle is better than nutritional support for treating hair cycle disruptions. (~1 page)
        - Subtitle: Treatment for disruptions in hair cycle
        - Section: Nutritional Support and Medical Treatment

CHAPTER WRITTEN CONTENT

Chronobiology of Hair Growth/Shedding

Hair is an essential component of mammals, serving many biological functions, like thermoregulation, protection, and sensory perception. Across mammals, hair provides multiple functions that benefit their survival. It may help regulate their body temperature, protect against physical injury and UV radiation, or even provide camouflage against predators. Some mammals have specialized hairs like vibrissae, allowing certain animals to detect the near environment

through touch. Although humans possess less body hair than other primates, hair continues to serve important functions (Yesudian, 2011).

1. Basics of Hair Growth and Shedding

1.1.1 Learning Objective 1: Describe and distinguish the hair growth/ shedding phases

The hair follicle undergoes continuous cycles of growth, degeneration, rest, and regeneration. Natarelli et al. describe four stages of this process: anagen, catagen, telogen, and exogen. During the anagen phase, also known as the growth stage, cells in the follicle divide and produce the hair shaft. The follicle also extends deeper into the skin, as the hair becomes longer. This stage is essential, as the duration a follicle remains in this stage helps cells determine how long hair can grow. During the catagen phase, also known as the transition stage, active production of the hair shaft ends, and the lower portion of the follicle shrinks. Then the follicle enters telogen, the resting phase, during which a fully formed hair remains present without growing. Finally, in the exogen phase, or the shedding phase, the old hair is released and falls out, and new hair begins to grow underneath it. This cycle repeats again and again, gradually and chronologically. (Natarelli et. al., 2023)

(INSERT ADD-IN #1)

1.2 Hair Cycle in Other Non-Human Animals.

1.2.1 Learning Objective 2: Compare and contrast the hair-shedding cycle of humans and non-human mammals.

Although all of the mammals have hair, there is a variation in the type of hair cycle. Mammals like humans demonstrate **asynchronized** hair growth, meaning that follicles do not

enter four phases —anagen, catagen, telogen, and exogen — at the same time, allowing follicles to cycle independently. This prevents humans from simultaneous shedding across the entire scalp, and allows consistent hair coverage over time. Other mammals can demonstrate more **synchronized** hair growth; for instance, large portions of a coat may change seasonally. When follicles enter a shedding and replacement period around the same time, they might undergo the process of molting. Molting is a coordinated period of hair loss and replacement coat, frequently influenced by environmental seasonal cues (Vogel, 2026).

Chimpanzee hairs, like human hairs, grow through transitions and stages, with individual follicles not being synchronized (Project Chimps, 2026). Chimpanzees do not normally undergo the seasonal molt, unlike other mammals. Instead, their hair grows to its maximum length, a concept known as **terminal length**. Terminal length means hair's reachable maximum length, which is determined by the duration of anagen. Chimpanzees have a short anagen period, leading to shorter terminal length than humans, lasting only from a few weeks to months. It helps them to regulate and maintain their short hair length (Project Chimps, 2026; Vogel, 2026).

2. Factors that Impact the Hair Cycle

The hair cycle is not a completely independent bodily process; there are even factors that cause changes and disruptions in cycles. Several factors are associated with transitions in the cycle, including inflammation, hormones, stress, nutritional deficiency, sleep quality, and medications. (Natarelli et. al., 2023) These internal and biological changes impact every stage of their hair cycle.

2.1 Stress and Disruption of the Hair Growth/Shedding Cycle

Stress is a physiological response to a perceived or physical challenge. A certain degree of stress can support the survival of species by impacting their cardiovascular activity, facilitating memory usage, mobilizing supplementary energy, alertness, and immune signalling (Smeets, 2011). Repeated stress, however, may disrupt body maintenance. Hair follicles can be affected in the neuroendocrine pathway by influencing the timing of regeneration. Stress is often associated with changes in **cortisol**, a steroid hormone that is produced by the adrenal glands (Thom, 2016). Due to its nature, cortisol is commonly used as a biomarker to determine and investigate physiological stress levels.

One method of evaluating the impact of stress on the hair cycle is by using an animal model in a controlled environment. Animals may provide relevant information to humans by investigating their biological processes in labs, while resolving the ethical concerns for human subject experiments. A well-designed experiment would include an independent variable, a dependent variable, and a control group. These components allow researchers to change one factor and determine whether that change was effective through comparison with biological response measurements.

(INSERT ADD-IN #2)

2.1.1 Learning Objective 3: Design an experiment that would reveal the impact of stress on disruption in hair shedding/growth cycle.

Now design the follow-up research that tests whether changes in the hair cycle are followed by stress level. Conduct research where animals are not exposed to harmful stress.

1. State a leading question and hypothesis

    a. Predict how lower or higher cortisol levels relate to hair growth and the shedding cycle.
2. Specify your variables and groups
    a. Identify the manipulated intervention and control constant factors.
    b. Explain group assignments.
3. Evaluate causation
    a. Explain why the previous experiment does not support causation.
    b. Explain why it is challenging to isolate a variable to associate it with direct causation.

3. Potential Treatment and Intervention

3.1 Nutritional Support and Medical Therapies

As hair plays an important role in the aforementioned beneficial functions, humans have been researching ways to treat severe hair shedding. Disruptions in hair shedding can be addressed through nutritional support and medical therapies, but the effectiveness of these approaches depends on the cause of the hair loss. Ideally, hair follicles require sufficient nutrients and sufficient biological signalling to maintain normal hair growth, but deficiencies and imbalances may cause interference in follicular activity (Gokce, Nuriye, et al., 2022; Guo & Katta, 2017; Natarelli, Nicole, et al., 2023).

Nutritional support for the hair cycle refers to the maintenance of a diet to correct deficiencies that may interfere with hair growth. As micronutrients like iron, zinc, and vitamin D contribute to the biological processes of hair follicle development, deficiencies may contribute to increased hair shedding and reduced hair growth. Therefore, correcting the nutritional deficiency

may help promote healthier follicular activity. However, although nutritional support makes it healthier, excessive supplementation does not really improve hair growth and may even produce adverse side effects (Guo & Katta, 2017).

Medical therapies involve treatments designed to address specific biological mechanisms targeted for hair loss. For instance, minoxidil is a commonly used medication that increases blood flow and hair-follicle activity. It promotes the transition from telogen to anagen, which allows hair follicles to move into hair production. Finasteride, which inhibits 5-alpha-reductase, an enzyme that converts testosterone into dihydrotestosterone (DHT), is also an example of medical treatment for the hair growth cycle. DHT contributes to follicular miniaturization, which causes hair follicles to produce thinner hairs. By reducing DHT, finasteride can help to maintain hair growth in individuals who experience hair loss (Natarelli et al., 2023).

Although both nutritional support and medical therapies seem to contribute to improving follicular health, neither of them is a universally effective intervention. Nutritional support is more relevant when the patient is going through nutrient deficiencies, whereas medical therapies are more appropriate when involving specific hormonal mechanisms.

3.1.1. Learning Objective 4: Evaluate and defend which treatment is more appropriate.

1. Consider and create a specific scenario where nutritional support would be more appropriate. Why is it more appropriate than medical therapy?
2. Consider and create a specific scenario where medical therapy would be more appropriate. Why is it more appropriate than nutritional support?

CHAPTER ADD-INS

Add-in #1

Phases	*What happens*	*Typical Duration of Human Scalp Hair Phases*
Anagen (growth)	Cells in the follicle divide and produce the hair shaft; the follicle also extends deeper into the skin, as the hair becomes longer.	About 2~7 years
Catagen (transition)	Hair growth ends, and the lower portion of the follicle shrinks.	About 2~3 weeks
Telogen (resting)	The hair follicle remains inactive while the old hair is retained.	About 2~4 months
Exogen (shedding)	The old hair sheds and the next hair begin to emerge	Overlaps with Telogen

Figure 1: Stages of Hair Growth*. The hair growth cycle consists of four stages: anagen, catagen, telogen, exogen. This figure highlights the essential process during each stage, and approximate duration for human scalp hair growth. Each phase varies in their duration, while exogen overlaps with telogen.*

Add-in #2

Can Stress Disrupt the Hair-Growth Cycle?

What did scientists ask?

Stress activates the hypothalamic-pituitary-adrenal (HPA) axis, which regulates the release of cortisol. As cortisol is believed to be incorporated into growing hair, the hair samples may provide an indicator for long-term HPA activity. Novak et. al. (2014) conducted research on whether rhesus macaques with a high rate of hair loss also had a higher hair cortisol concentration than other macaques with normal hair growth.

Method

They investigated 99 adult rhesus macaques from three different U.S. primate research centers. For the experimental group, they selected rhesus macaques with more than 30% hair loss and a comparison group with less than 5% hair loss. During veterinary examination, the team photographed each macaque and used ImageJ software to calculate the percentage of the body affected by hair loss. For each macaque, a small hair sample was collected from the nape of their neck and was analyzed for cortisol. The researchers also considered other factors, including age, body weight, sampling month, and the facilities.

Result

From female macaques from three centers, mean hair cortisol was $72.4 \pm 6.1\ pg/mg$ for the experimental group, compared to $54.4 \pm 2.4\ pg/mg$ for the comparison group. The association between HPA and hair loss was significant except for the third facility. Although this study establishes the relationship between HPA activity and hair loss, it does not support a direct relationship and did not consider stages of hair cycles: anagen, catagen, telogen, and exogen or shedding rates.

Variables

Terminologies	Application
Independent Variable	Observed hair loss (>30% vs. <5%)

Dependent Variable	Hair Cortisol Concentration
Control/ Comparison Group	Macaques with less than 5% hair loss compared with those that showed more than 30%
Controlled Variable	The team standardized hair-sampling location and accounted for age, body weight, timing of collection, and facility.
Animal Model	Rhesus Macaques (non-human primate)

Novak, M. A., Hamel, A. F., Coleman, K., Lutz, C. K., Worlein, J., Menard, M., Ryan, A., Rosenberg, K., & Meyer, J. S. (2014). Hair loss and hypothalamic-pituitary-adrenocortical axis activity in captive rhesus macaques (*Macaca mulatta*). *Journal of American Association for Laboratory Animal Science*, *53*(3), 261–266.

GLOSSARY

Term	Definition
Anagen Phase	Growth Phase of the hair cycle; the hair follicle rapidly divides, the hair shaft lengthens, and the follicle remains deeply embedded into the skin.
Catagen Phase	Transition phase of hair cycle; the lower part of the hair follicle shrinks and separates from the blood supply.
Telogen Phase	The resting phase of the hair cycle; the hair follicle is more inactive, and the fully formed hair remains in the follicle, without growing.

Exogen Phase	Shedding phase of the hair cycle; The old hair is released and falls out, and new hair begins to grow underneath it.
Hair Follicle Niche	A microenvironment surrounding hair follicle stem cells, including extracellular matrix, blood vessels, nerves, signaling molecules, and other factors that regulate cell-stem regeneration and growth.
Mammals	Animals that have hair or fur at some point in their lives, which help with insulation, sensation, and protection.
Asynchronized	Hair follicles cycle independently, so all the different hairs are in different growth phases at the same time. Asynchronized growth is more typical and helps prevent large amounts of hair from shedding at once.
Synchronized hair growth	Groups of hair follicles that enter the same growth/shedding cycle at the same time (synchronously). Common in many animals that go through seasonal coat changes.
Molting	A seasonal periodic shedding and replacement of hair or fur. Molting can result in a synchronized change in hair growth cycle for mammals.
Terminal Length	Biologically controlled maximum length a hair can reach, determined by how long the follicle remains in anagen
Cortisol	A steroid hormone produced by adrenal glands. Regulates the body's response to stress and immune activity. Cortisol is usually used as an indicator for stress, as its level often increases during physical or psychological stress.
Animal model	A non-human animal that is used in scientific research to study biological processes, treatments, responses, etc that may be relevant to humans.
Independent Variable	The factor that the researcher changes or manipulates in an experiment.

Dependent Variable	The factor that is measured or observed to determine the effect of the independent variable.
Control Group	A group that does not receive the experimental treatment and receives standard conditions. Provides a baseline to compare with the experimental group.
HPA-Axis	A neuroendocrine system involving the hypothalamus and adrenal glands that regulates the body's physiological response to stress, including cortisol release.
DHT	A hormone produced from testosterone by the enzyme 5-alpha-reductase. DHT can contribute to hair follicle miniaturization and hair loss.
Follicular Miniaturization	A process in which hair follicles progressively shrink and produce thinner hairs, often involved with the effects of DHT in hair loss.
Nutritional Support	Use of diet and nutrients to help maintain health, support body functions, and correct nutritional deficiencies.
Medical Therapy	Treatment used to improve or cure a medical condition, which involves medication, procedures, or interventions.
Nanoparticle	An extremely small particle, measured in nanometers (10^-9 m), that can be designed to carry drugs or substances and target specific cells or tissues.
Controlled Drug Release	A drug-delivery method to release medication at a regulated rate, time, or location to maintain the effect of medication while reducing the risk of unwanted side effects.
Micronutrient Supplementation	The use of vitamins, minerals, and other nutrients in small amounts to correct or prevent deficiencies, including supplementation of iron, zinc, and vitamin D.

REFERENCES

Correia, Ana S., et al. "Nanotechnology-Based Strategies for Hair Follicle Regeneration and Targeting." *Nanomaterials*, vol. 13, no. 12, 2023, article 1888. *MDPI*, https://doi.org/10.3390/nano13121888.

Gokce, Nuriye, et al. "An Overview of the Genetic Aspects of Hair Loss and Its Connection with Nutrition." *Journal of Preventive Medicine and Hygiene*, vol. 63, no. 3, suppl. 3, 2022, pp. E228–E238, https://doi.org/10.15167/2421-4248/jpmh2022.63.253.2765.

Guo, Emily L., and Rajani Katta. "Diet and Hair Loss: Effects of Nutrient Deficiency and Supplement Use." *Dermatology Practical & Conceptual*, vol. 7, no. 1, 2017, pp. 1–10. https://doi.org/10.5826/dpc.0701a01

Natarelli, Nicole, et al. "Integrative and Mechanistic Approach to the Hair Growth Cycle and Hair Loss." *Journal of Clinical Medicine*, vol. 12, no. 3, 2023, article 893. *PubMed Central*, https://doi.org/10.3390/jcm12030893.

Novak, M. A., Hamel, A. F., Coleman, K., Lutz, C. K., Worlein, J., Menard, M., Ryan, A., Rosenberg, K., & Meyer, J. S. (2014). Hair loss and hypothalamic-pituitary-adrenocortical axis activity in captive rhesus macaques (*Macaca mulatta*). *Journal of American Association for Laboratory Animal Science*, *53*(3), 261–266.

Randall, Valerie A., John P. Sundberg, and Michael P. Philpott. "Animal and *In Vitro* Models for the Study of Hair Follicles." *Journal of Investigative Dermatology Symposium Proceedings*, vol. 8, no. 1, 2003, pp. 39–45, https://doi.org/10.1046/j.1087-0024.2003.00804.x.

Smeets, T., Otgaar, H., Raymaekers, L., Peters, M. J. V., & Merckelbach, H. (2011). Survival processing in times of stress. *Psychonomic Bulletin & Review*, *19*(1), 113–118. https://doi.org/10.3758/s13423-011-0180-z

Thom, Erling. "Stress and the Hair Growth Cycle: Cortisol-Induced Hair Growth Disruption." *Journal of Drugs in Dermatology*, vol. 15, no. 8, 2016, pp. 1001–1004.

Vogel, Lauren. "Does Chimpanzee Hair Change with the Seasons?" *Project Chimps*, edited by Ali Crumpacker, 1 Jan. 2026, projectchimps.org/chimpanzee-hair-seasonal-change/. Accessed 17 Sept. 2026.

Yamanashi, Yumi, Naruki Morimura, Yusuke Mori, Misato Hayashi, and Juri Suzuki. "Cortisol Analysis of Hair of Captive Chimpanzees (*Pan troglodytes*)." *General and Comparative Endocrinology*, vol. 194, 2013, pp. 55–63. *PubMed*, https://doi.org/10.1016/j.ygcen.2013.08.013.

Yesudian, P. . Human hair – An evolutionary relic? *International Journal of Trichology*, *3*(2), 69, 2011, https://doi.org/10.4103/0974-7753.90799

UNIT AND CHAPTER

Unit: Infradian Rhythms

Chapter X: Breeding and Migration

CONTRIBUTORS

- Sam Ding
    - Subject Matter Expert – 'Breeding and Migration'
        - All the Sections
    - Figures

LEARNING OBJECTIVES

After reading this chapter, students should be able to …

1. Understand how animals use internal circannual clocks and environmental factors to keep track of time
2. Recognize and demonstrate the association between breeding and circannual rhythm
3. Understand the hourglass model that decides breeding schedule
4. Analyze the sensory mechanisms and behavioral stages of animal migration
5. Evaluate how human activities disrupt breeding and migration
6. Assess solutions like lighting reduction or bird safe architecture

CHAPTER OUTLINE

- How Annual Biological Clocks Work
    - Circannual Rhythms and Proof
    - Measuring Daylength and Synchronization
- Breeding Timeline Across Species
    - Interval Timer vs Circannual Clocks

    - Melatonin Duration Determines Breeding Time
- Animal Migration and Navigation
    - Physiological Preparation for Migration
    - Navigational Strategies
    - The Three Phases of a Navigational Task
- Modern Environmental Threats
    - Climate Change and Phenological Shift
    - Artificial Light at Night (ALAN) Alters Nocturnal Migration
    - Bird Safe Architecture

CHAPTER WRITTEN CONTENT

Unit: Infradian Rhythms Chapter: Breeding and Migration

Circannual Rhythms and Synchronization

In the natural world, many animals face challenges that require tracking time throughout the year. From reproducing in warm seasons to migrating across continents, organisms rely on circannual rhythms, internal biological clocks that repeat roughly once a year. Scientists have wondered whether these annual or seasonal behaviors are just passive adaptations to the weather, or whether animals actually possess an internal biological clock that can track time without environmental factors.

A classic study conducted by Dittami and Gwinner (1985) validated the existence of circannual rhythms. Gwinner kept African stonechats, small insect-eating birds found in East Africa, in constant conditions, maintaining a steady room temperature and an unchanging cycle

of 12 hours of light and 12 hours of dark. Even though the seasonal cues were completely removed, the birds still underwent feather molting and testicular growth on schedule to prepare for reproduction. Because the cycles measured in the lab lasted roughly 10-11 months rather than exactly 365 days, their timing would gradually drift away from the natural seasons. Therefore, these cycles are not simply determined by external seasonal changes, but generated endogenously even in constant conditions (Dittami & Gwinner, 1985).

Often, biological clocks require synchronization. Because internal clocks are not exactly one year long, animals rely on environmental cues called zeitgebers to synchronize time. The most reliable way for animals to track time is photoperiodism, which is the difference in daylength during different seasons (Gwinner, 2003). Animals use a hormone called melatonin that is only synthesized during the night to measure daytime (Paul et al., 2008). In the summer, the longer daytime results in a shorter duration of melatonin secretion, while the shorter daytime in winter will result in longer duration of melatonin secretion. A classic experiment in Siberian hamsters reveals that the daily duration of melatonin secretion, rather than total melatonin amount, is the key signal that identifies the day length (Carter & Goldman, 1983). When nights become shorter in spring, the duration of melatonin secretion drops below a certain threshold. After this critical point, the animal's brain will realize the arrival of spring, and reset the circannual clock to ensure it stays synchronized across seasons. Moreover, the accuracy of the circannual clock helps establish the breeding timeline (Paul et al., 2008).

Internal Timing of Seasonal Breeding

In order to provide a good living condition for the offspring to survive, animals need to time their breeding so that their children are born during the warm months of spring. However, they do not keep track of time in the same way. We previously discussed circannual clocks that

cycle roughly once a year, even without noticeable seasonal changes. Some animals also utilize seasonal interval timing, which functions similarly to an hourglass (Paul et al., 2008; Wood & Loudon, 2014). One cycle of an interval timer lasts several months rather than a whole year, and each cycle is activated by a seasonal cue. Unlike a circannual clock, an interval timer cannot repeat the cycle on its own. The start timing is triggered by environmental cues, mainly seasonal changes (Paul et al., 2008; Wood & Loudon, 2014).

The Siberian hamster is an example of an animal that utilizes an interval timer. Under short photoperiods (i.e. short daytime) during winter, the size of gonads (reproductive organs) of male Siberian hamsters will shrink, decreasing the tendency of reproductive activity. After several months, their gonads will start to grow again, even if the daylight time is unchanged in a lab condition. This phenomenon indicates that the hamsters are not simply relying on external conditions, but are "governed by a process with hourglass properties designated an interval timer" (Paul et al., 2008, p. 342). This shift between short and long days is like turning over an hourglass. When the hamster's gonad responds to the arrival of winter, it's like flipping the hourglass and the "sand" starts to run. After the "sand" has run out several months later, the growth of gonads will be triggered automatically. Just like we need to flip the hourglass to start another cycle, the interval timer won't reset itself without exposure to long days (Paul et al., 2008).

By contrast, the golden-mantled ground squirrel illustrates how a circannual clock differs from an interval timer. The squirrels utilize an internal circannual clock, and their reproductive cycle continues without exposure to change in daylength. This is because these squirrels hibernate in dark underground conditions for months, and don't have access to daylight cues (Paul et al., 2008). In an experiment by Pengelley and Fisher (1957), the squirrels were kept in

constant laboratory conditions for years. They still hibernated in autumn and winter, and remained active in spring and summer without experiencing changes in daylength. Both systems can help animals measure reproductive time, but only the circannual clock can repeat its annual cycle indefinitely under constant conditions.

Melatonin Secretion and Breeding Time

Melatonin can help synchronize the internal clock of animals, whether it's a circannual clock or an interval timer. Moreover, it will further affect the release of a hormone called gonadotrophin-releasing hormone (GNRH). GNRH then influences hormones from the pituitary gland that activate the gonads. Then, the reproductive activity of animals increases (Wood & Loudon, 2014).

To ensure their offspring are born during warm seasons, animals with different pregnancy lengths need to adjust their reproductive time to different seasons. For example, small mammals like the Siberian hamster with a short pregnancy period can begin breeding in spring. Larger mammals like sheep and deer have to begin mating in autumn to give birth in the warm spring, because their pregnancy time is longer. As Figure 1 (see chapter add in) suggests, long days (LD) activate reproduction in Siberian hamsters, while short days (SD) activate reproduction in Sheep (Wood & Loudon, 2014). Together, these examples explain that breeding timeline is determined by both the duration of melatonin secretion affected by daylength and the time needed for pregnancy.

Migration Timing and Navigation

In addition to breeding, migration behavior also requires animals to determine the right time of year. Without the correct timing, animals can face countless consequences, such as failing to escape harsh winter, arriving late at breeding sites, or missing the abundant food

sources that are only available in certain months. Gwinner's experiment on songbirds highlights the timing mechanisms that help migratory birds know when to leave. Songbirds are normally only active during the day, but migrate at night to avoid predators. During migration periods, the researchers kept these birds with constant daylength and temperature, but they still became unusually active at night, fluttering their wings in the cage. This phenomenon is called migratory restlessness, and it indicates migratory readiness and validates the existence of an internal circannual clock. Seasonal changes in daylength can help synchronize the circannual clock to ensure the accuracy of migration timing (Gwinner, 2003).

Once birds determine when to migrate, the birds need to determine the correct direction to their destination. One way of navigating is using the relative position of the Sun. However, the Sun is always moving across the sky, so the birds need to change their angle between their path and the Sun to keep flying in the same direction. Luckily, the birds can estimate the time of day using their circadian clocks, and use this time information to adjust the angle as the Sun moves (Mouritsen, 2018). For example, a bird in the Northern Hemisphere flying south would see the Sun on its left side in the morning, ahead at noon, and on its right side in the afternoon. In an experiment conducted by Guilford and Taylor (2014), the circadian clocks of pigeons were shifted, and the birds' choice of direction was affected under sunny conditions. Thus, the circadian clock can help migratory birds maintain a correct direction within a day.

Besides using the Sun as a navigational cue, birds might also use stars or the Earth's magnetic field. The navigation process can be divided into three stages (Mouritsen, 2018). First, broad cues like celestial cues or magnetic cues help animals travel to the correct region. Then, landmarks and olfactory differences help them narrow down the home area. Finally, specific sights or smells help them locate a specific site.

Overall, the internal biological clocks help birds know when to migrate and how to migrate. However, since their internal clocks are synchronized with seasonal cues, a shocking change in seasonal conditions may cause them to migrate at the wrong time.

Modern Environmental Threats

Although internal biological clocks can help animals track seasons, their timing may not match a changing environment. Animals follow the similar seasonal schedule every year. When climate change shifts the timing of seasonal events, such as breeding or plant growth, this change is called a phenological shift. However, different species may adjust at different rates, creating a mismatch between food supply and the time animals arrive.

A study by Both et al. (2006) uses pied flycatchers, migratory birds that feed on caterpillars, to examine this problem. In the Netherlands, climate change caused the springs to be warmer. As a result, caterpillars reproduce a lot earlier than before, making the food peak for the birds shift earlier. Because the flycatchers were not in the Netherlands before migrating, they didn't notice the warm spring or advance their arrival. By the time the birds arrived, there was not enough food source for the flycatchers and their nestlings. Populations in areas with the earliest food peaks declined by about 90%, while those in areas with the latest food peaks only declined by about 10%. This example shows that climate change can harm animals not only through rising temperatures, but also by disrupting the seasonal timing (Both et al., 2006).

Another example of human disruption is artificial light at night (ALAN). As we discussed earlier, songbirds that migrate at night rely on celestial and magnetic cues to migrate. However, strong artificial light can distract the birds' use of celestial cues to navigate. The illuminated building structures in cities will disorient migratory birds, resulting in circling around the

skyscrapers or even fatal collision with the buildings. Hundreds of millions of birds die annually from collisions with buildings (Van Doren et al., 2017).

Van Doren et al. (2017) studied this problem using the light beams of the "Tribute in Light" 911 memorial in New York City. They employed radar to observe and compare bird behavior when the light beams were on and off. Across the seven nights over seven years, the birds gathered around the light beams and exhibited intense vocal activity when the lights were on. Under illumination, peak bird density near the light installation exceeded 20 times the baseline of the surrounding area every year. However, when the lights were switched off, the birds dispersed and moved away from the site within minutes. This finding suggests that light pollution can disrupt migrating birds, but reducing nighttime lighting during migratory periods can help protect them from collision effectively (Van Doren et al., 2017).

CHAPTER ADD-INS

Explanatory Figures

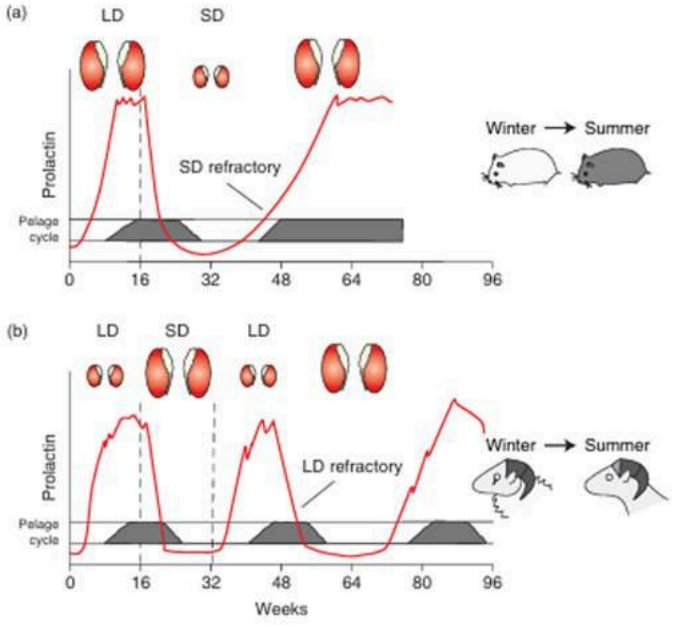


Figure 1. Seasonal changes in testicle size, prolactin and fur color in (a) LD (long day) breeding Siberian hamsters and (b) SD (short day) breeding sheep. From Wood & Loudon (2014).

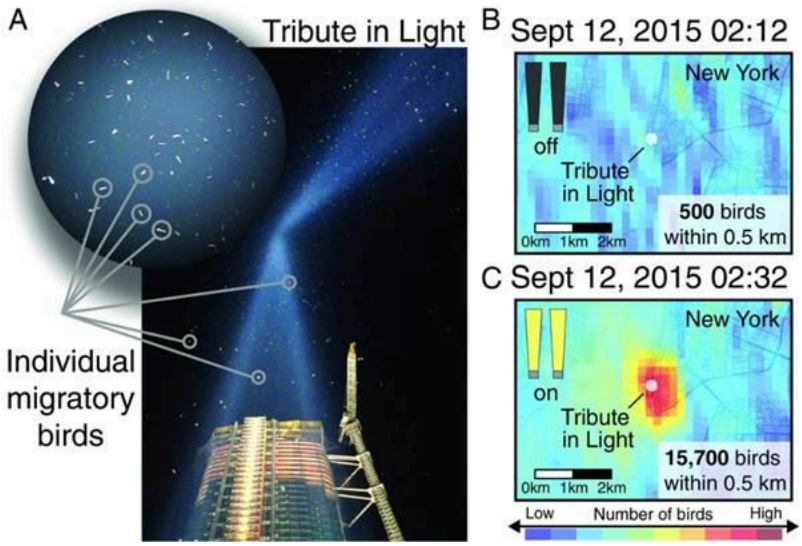


Figure 2. Tribute in Light site, New York City. (A) Migratory birds' behaviors altered by ALAN. Bird density within 0.5 km when lights were off (B) and on (C). From Van Doren et al. (2017).

Exercise 1

Siberian Hamsters use duration of melatonin secretion at night to determine when to reproduce. In an experiment by Carter and Goldman (1983), the pineal glands of juvenile male hamsters were removed to control their melatonin exposure. The researchers gave them the same light and dark schedule and the same amount of melatonin (10 nanograms) each day. However, the infusion duration of melatonin was different. (Female hamsters have a short gestation of less than a month)



Figure 3. Siberian Hamsters with different coat colors in different seasons. From Rendon et al. (2017).

Group	Melatonin amount	duration
A	10 ng	4 hours
B	10 ng	12 hours

1. Which treatment resembles the melatonin signal in summer, and which resembles the signal in winter? Explain the association between duration of melatonin secretion and day length changes.

2. Which group is expected to show testicular development? Explain the logic using the chapter's discussion in seasonal breeding schedules.

3. Why did researchers give both groups the same amount of melatonin every day?

Explanation

1. Group A resembles summer and Group B resembles winter. Melatonin is synthesized in dark. Summer nights are shorter than winter nights, resulting in a shorter duration of melatonin secretion.

2. Group A will show testicular development. Because the gestation period is short, the hamsters will reproduce in spring to make sure their offspring can have good living condition. Therefore, they will increase reproductive activity in spring and summer that have a shorter night length.
3. Keeping the amount equal isolates duration of melatonin secretion's effect from melatonin amount. If Group B receives both longer duration and more amount, we won't know which difference contributes to the change in reproductive activity.

Exercise 2

To protect migratory birds from urban bird collision, two main solutions are employed: (1) "Lights Out" programs that turn off unnecessary building lights during peak migration period. (2) Bird safe architecture using patterned glass curtain on skyscrapers. (Van Doren et al., 2017)

How will both solutions work together to protect migrating birds? Without light prevention, why is glass patterns alone not effective enough to protect birds?

Explanation: Migratory birds rely on star and magnetic cues to navigate. While bird safe patterned glass will prevent direct collision to the building, the birds will still be confused and trapped by urban light pollution (Van Doren et al., 2017). "Lights Out" programs can prevent the birds from being attracted by lights and waste their critical energy to migrate.

GLOSSARY

Term	Definition

Circannual rhythm	An internal biological clock that repeats roughly once a year. Can be kept constantly without outside factors.
Photoperiodism	The ability of animals to measure season changes with daylength.
Melatonin	A hormone produced during darkness by the pineal gland. Animals use the duration of it to track daylength and sense the time of year.
Migration restlessness	Hyperactive behavior shown during animals' normal migration period.
Phenology	The timing of seasonal life-cycle events in creatures in response to environmental conditions.
Zeitgeber	An external environmental cue that synchronizes an organism's internal biological clock.
Diurnal	Active primarily during the daytime
Nocturnal	Active primarily during the nighttime

REFERENCES

Both, C., Bouwhuis, S., Lessells, C. M., & Visser, M. E. (2006). Climate change and population declines in a long-distance migratory bird. *Nature, 441*(7089), 81–83. https://doi.org/10.1038/nature04539

Dittami, J. P., & Gwinner, E. (1985). Annual cycles in the African stonechat *Saxicola torquata axillaris* and their relationship to environmental factors. *Journal of Zoology, 207*(3), 357–370. https://doi.org/10.1111/j.1469-7998.1985.tb04937.x

Guilford, T., & Taylor, G. K. (2014). The sun compass revisited. *Animal Behaviour, 97*, 135–143. https://doi.org/10.1016/j.anbehav.2014.09.005

Gwinner, E. (2003). Circannual rhythms in birds. *Current Opinion in Neurobiology, 13*(6), 770–778. https://doi.org/10.1016/j.conb.2003.10.010

Mouritsen, H. (2018). Long-distance navigation and magnetoreception in migratory animals. *Nature, 558*(7708), 50–59. https://doi.org/10.1038/s41586-018-0176-1

Paul, M. J., Zucker, I., & Schwartz, W. J. (2008). Tracking the seasons: The internal calendars of vertebrates. *Philosophical Transactions of the Royal Society B: Biological Sciences, 363*(1490), 341–361. https://doi.org/10.1098/rstb.2007.2143

Van Doren, B. M., Horton, K. G., Dokter, A. M., Klinck, H., Elbin, S. B., & Farnsworth, A. (2017). High-intensity urban light installation dramatically alters nocturnal bird migration. *Proceedings of the National Academy of Sciences, 114*(42), 11175–11180. https://doi.org/10.1073/pnas.1708574114

Wood, S., & Loudon, A. (2014). Clocks for all seasons: Unwinding the roles and mechanisms of circadian and interval timers in the hypothalamus and pituitary. *Journal of Endocrinology, 222*(2), R39–R59. https://doi.org/10.1530/JOE-14-0141

Carter, D. S., & Goldman, B. D. (1983). Antigonadal effects of timed melatonin infusion in pinealectomized male Djungarian hamsters (*Phodopus sungorus sungorus*): Duration is the critical parameter. *Endocrinology*, *113*(4), 1261–1267. https://doi.org/10.1210/endo-113-4-1261

Pengelley, E. T., & Fisher, K. C. (1957). Onset and cessation of hibernation under constant temperature and light in the golden-mantled ground squirrel (*Citellus lateralis*). *Nature*, *180*(4598), 1371–1372. https://doi.org/10.1038/1801371b0

Rendon, N. M., Amez, A. C., Proffitt, M. R., Bauserman, E. R., & Demas, G. E. (2017). Aggressive behaviours track transitions in seasonal phenotypes of female Siberian hamsters. *Functional Ecology*, *31*(5), 1071–1081. https://doi.org/10.1111/1365-2435.12816

UNIT AND CHAPTER

Unit: Infradian Rhythms

Chapter X: Predator–Prey Population Dynamics with Infradian Rhythms

CONTRIBUTORS

- Binqi Zhan
    - Subject Matter Expert – ‘Predator–Prey Population Dynamics with Infradian Rhythms’’
    - Marketing Specialist
        - Sections 1, 2, 3a, 3b, 3c, and 3d
    - Figures 1, 2, 3
-
    - Copyeditor
    - Peer Reviewer

LEARNING OBJECTIVES

After reading this chapter, students should be able to …

1. Define predators, prey, and infradian rhythms, and explain the interconnected relationship between them.
2. Demonstrate the predator and prey population cycle in the ecosystem and question the changes in their population over time.
3. Differentiate and justify distinct types of factors that contribute to the periodic infradian population cycles in predator–prey systems.
4. Evaluate how abiotic variables affect the stability of populations in the predator–prey cycle.
5. Evaluate how resource and habitat factors contribute to population variation in the predator–prey cycle.
6. Evaluate how community factors affect the stability of populations in the predator–prey cycle.
7. Evaluate how biological and evolutionary variables influence the population change in the predator–prey cycle.

CHAPTER OUTLINE

- What are predators, prey, and infradian rhythms?

- The predator and prey population cycle in the ecosystem
- What factors drive or maintain the periodic infradian population cycles in predator–prey systems?
    - Abiotic factors
    - Resource and habitat factors
    - Community factors
    - Biological and evolutionary factors

CHAPTER WRITTEN CONTENT

Predator–Prey Population Dynamics with Infradian Rhythms

What are Predators, Prey, and Infradian Rhythms?

- Learning Objective: Define predators, prey, and infradian rhythms, and explain the interconnected relationship between them.

In an ecological setting, predators and prey have an interconnected relationship: one hunts the other for survival, while the other hides. As the Earth developed from the earliest stage, the Last Universal Common Ancestor, also known as LUCA, evolved into different organisms. As ancestral animals got exposed to different environmental factors and genetic mutations, they underwent evolutionary changes that caused them to develop different features, appearances, and adaptations over time. Predators, animals that naturally hunt other animals for food, often possess certain traits that assist them in the predation process. For instance, snakes and scorpions produce neurotoxic venoms to subdue prey, cheetahs have specialized body structures that allow them to hunt prey faster, and crocodiles have great ambush tactics. On the other hand, prey, animals that are caught and killed by predators for food, have also developed adaptations that allow them to maintain survival. For example, skunks have foul-smelling chemicals in their internal body systems, turtles have shells to prevent physical

damage from predators, and nocturnal animals are specialized in avoidance strategies to hide away from others.

Infradian rhythms are connected to the predator–prey cycle in many different aspects. First, the evolutionary changes and genetic mutations that both species undergo are classified as infradian rhythms, as these are continual cycles that last longer than a 24-hour period. In addition, seasonal variation in certain ecosystems causes behavioral or physiological changes in both predators and prey. This aspect contributes to the fluctuation in predator–prey populations. Generally, since the populations of predators and prey change dynamically in a cycle, this allows scientists to collect sample data and discover patterns and trends in species' responses to changes. Many of the behaviors exhibited by different animal species can be classified into the same category according to rationales proposed by ecologists. For example, bears, rats, and turtles all experience the hibernation process in the winter to conserve internal energy, also taking into account that there is a scarcity of food and resources during that season. This behavioral pattern allows scientists to recognize how common hibernation is among different animal species, leading them to classify it as an infradian rhythm instance. From an ecological perspective, infradian rhythms are not only cycles that last more than 24 hours, but they are also one of the vital processes that maintain long-term stability in the ecosystem.

The Predator and Prey Population Cycle in the Ecosystem

- Learning Objective: Demonstrate the predator and prey population cycle in the ecosystem and question the changes in their population over time.

Predator–prey dynamics are natural occurrences when the population sizes of predators and prey in a community are not constant over time, and they sometimes fluctuate in opposite

directions (Ha, 2022). In the ecosystem, this pattern can be seen between many types of predators and prey. One salient example is the change in population size of snowshoe hares and lynxes. Snowshoe hares, scientifically known as *Lepus* americanus, are herbivorous mammals that reside in the boreal forests of North America (National Park Service, 2021). In terms of infradian rhythms, snowshoe hare populations cycle in eight- to eleven-year periods, meaning that the population returns to its original level in eight to eleven years of time. Additionally, the species has evolved some adaptations in its living environment that allow it to have survival advantages over others. For instance, snowshoe hares have extremely fast agility: they can run up to 27 miles per hour, allowing them to escape from slow-moving predators (Kline, 2024). Moreover, snowshoe hares are prey for various predators due to their rapid reproductive rate: they breed at about one year of age and have two to three litters per year (Alaska Department of Fish and Game, n.d.), with one to nine young per litter. Hence, the prevalence of the *Lepus* americanus in the ecosystem makes it prone to being targeted by many animals, one being lynxes. Since Canadian lynxes, scientifically named *Lynx* canadensis, are predators of the snowshoe hare, Canadian lynxes also predominantly dwell in northern North America (Fish & Wildlife Service, n.d.). They are mid-sized boreal forest carnivores and are also classified as mammals. Snowshoe hares are the primary food source of the lynx, suggesting that the population cycles of these two species are closely linked.

Figure 1 presents the predator–prey dynamics of the two studied species. The population of snowshoe hares initially increases due to their high reproductive rates. This increase in prey population attracts more predators to the area and allows predators to have more food available for growth and reproduction, causing the population of lynxes to rise accordingly. When the population of predators increases, more prey would be physically consumed by predators, which lowers snowshoe hare population size. This decline in snowshoe hare population level contributes to a scarcity of food resources for lynx, thereby gradually

diminishing the predator's population size. This pattern repeats over and over again, which forms an infradian cycle in the ecosystem. Moreover, snowshoe hares and lynxes are not the only two species in which their population fluctuates repeatedly: most of the predator and prey populations resemble the population pattern which can be generalized into the predator–prey dynamics diagram shown in Figure 2.

Furthermore, it is important to note that this generalized population graph involves both top-down and bottom-up controls (Figure 3). When a population is mainly limited by predators, top-down control influences and exerts forces on the population size. In this case, when the population of predators rises, the predation pressure significantly limits the prey population, leading to a diminishing number of prey in the area. Conversely, bottom-up control is when the population is heavily dependent on the availability of resources or the bottom level of the food chain. For instance, when the prey population peaks, this indicates that predators have abundant food sources, which allows a significant increase in the population of predators. However, when the prey population reaches its trough, that signifies less food availability for predators, which in turn causes a considerable drop in predator population.

What Factors Drive or Maintain the Periodic Infradian Population Cycles in Predator–Prey Systems?

Learning Objectives:

- Differentiate and explain distinct types of factors that contribute to the periodic infradian population cycles in predator–prey systems.
- Evaluate how abiotic variables affect the stability of populations in the predator–prey cycle.
- Evaluate how resource and habitat factors contribute to population variation in the predator–prey cycle.

- Evaluate how community factors affect the stability of populations in the predator–prey cycle.
- Evaluate how biological and evolutionary variables influence the population change in the predator–prey cycle.

Although most predator–prey population dynamics resemble the generalized pattern shown in Figure 2, there are many instances where prey and predators resist changes in population through their evolved adaptations, environmental changes, and habitat structures. Both prey and predators evolve through natural selection, in which certain favorable traits allow species to survive longer compared to other less well-adapted organisms in the environment. Hence, it is important to consider the external factors that contribute to the maintenance of the periodic infradian population cycle in predator–prey systems, including but not limited to abiotic factors, resource and habitat factors, community factors, and biological and evolutionary variables.

a) Abiotic Factors

Abiotic factors contribute to population variation among predator and prey populations. Abiotic factors are the non-living components of the environment that influence the growth of organisms (BiologyOnline, 2023). These include the impact of sunlight, water, and temperature on the population size of species over time. One notable case study on how abiotic factors contribute to population change in prey is when Berg et al. investigated how rain serves as a density-independent factor for the population of aphids, a type of prey, in 2008. The study concluded that rain removed aphids from shoots and ears of wheat by 57% and 25%, respectively (Figure 4). These data later translated into the statistic that rain independently reduces aphid populations by 27%, serving as a significant mortality factor for aphid populations. Heavy rainfall also causes more aphids to be exposed to the ground, which is

where their predators, ground-dwelling predatory arthropods, reside. This unexpected rise in prey exposure on the ground increases the arthropods' consumption of aphids, which results in the prey population being considerably lower than the predator population in that specific time period. Hence, the decrease in prey population is not solely due to predation: extreme weather conditions are not negligible when evaluating population fluctuations among predators and prey.

Similarly, another research study also analyzed rainfall as a climate factor in predator–prey dynamics from various perspectives. It first outlines that when there is no rainfall, species experience drought in their environment. Low rainfall levels could potentially cause weaker body systems in prey, which makes them vulnerable to predators (Morin et al., 2021). Hence, this would increase predator population in the short run, but the predator population level would drop in the long run due to prey scarcity. From another point of view, high rainfall can contribute to vegetation growth, causing vegetation to become extremely dense in the area where predators and prey dwell. This rise in vegetation allows herbivorous prey to increase their growth rate and reproduce more frequently. Additionally, the dense vegetation can mutually help both the predators and prey in terms of survival and predation. For instance, dense vegetation assists predators like lions, leopards, and lynx to execute ambush attacks and tactics on prey, increasing predator population levels and diminishing prey population size. Conversely, densely formed vegetation can also help prey to conceal themselves from predators, which maintains the survival rate of prey. Hence, climate factors like rainfall do not universally benefit prey or predators in terms of population growth. It heavily depends on how predators and prey respond to the changes in abiotic factors.

b) Resource and Habitat Factors

Resource and habitat factors also have a significant impact on the population density of predators and prey. Resources in the ecosystem can include food, mates, and decomposers.

Specifically, food quantity serves as the main source of survival for snowshoe hares in the boreal forests of North America. As previously mentioned, snowshoe hares go through an approximate 10-year population cycle, in which nutrient abundance helps the prey increase its population density after a huge decline. Krebs et al. conducted an 8-year field experiment in a one km^2 forest block in the Yukon to investigate how manipulating food supply, predator availability, and soil nutrient level would affect the population of snowshoe hare. While increasing fertilizer use in the control area helped the growth of plant biomass, it did not display a significant change in the population density of the targeted prey. Conversely, with a constant supply of supplemental food to the studied area, the population of snowshoe hare tripled during the cyclic peak and decline (Figure 5), revealing that food resource availability is the paramount factor that promotes population growth of the prey.

Furthermore, habitat structures of prey also play a pivotal role in the predator–prey dynamics. In 2025, Lennox et al. researched different ways in which prey's habitats help them maintain survival in densely predator-populated environments. One notable instance from the research focuses on predator sculpin (*Cottus gobio*) and salmon (*Salmo salar*) eggs as the prey. In the ecosystem, habitat refers to the place in which a group of related organisms lives. Habitat is also commonly used as the area of refuge, and predation success for sculpin on salmon eggs is heavily dependent on how effectively the habitat can be used as refuge. Specifically, salmon eggs reside under substrates, which are layers of materials covering the bottom of a riverbed. When the substrates are smaller in size, it increases the protection layer density and reduces the accessible gap for predators to reach prey in the water, fully covering the salmon eggs for protection. Hence, the population level of salmon eggs is greatly affected by their habitat structure, making it a substantial factor to consider when evaluating population fluctuations of salmon eggs and sculpin.

c) Community Factors

The relationships of species in an ecosystem can be simplified into a complex, interconnected food web, where the diagram shows “what-eats-what” in an ecological community. Oftentimes, a predator does not have only one specific prey that they hunt for survival. Predators can also switch their target prey in order to maintain their dominant position in the ecosystem. For instance, tigers not only search for zebras as a food source, but they also have other prey to look for, including but not limited to wildebeests, buffaloes, and impalas. Interestingly, this mechanism, also known as predator switching, allows for the coexistence between different prey types (Archibald et al., 2023) by rendering sufficient time for the other prey species to grow in population size before the predator hunts for them. Hence, this ultimately stabilizes the food web model and predator–prey cycle by preventing the extinction of a certain species in the ecosystem.

d) Biological and Evolutionary Factors

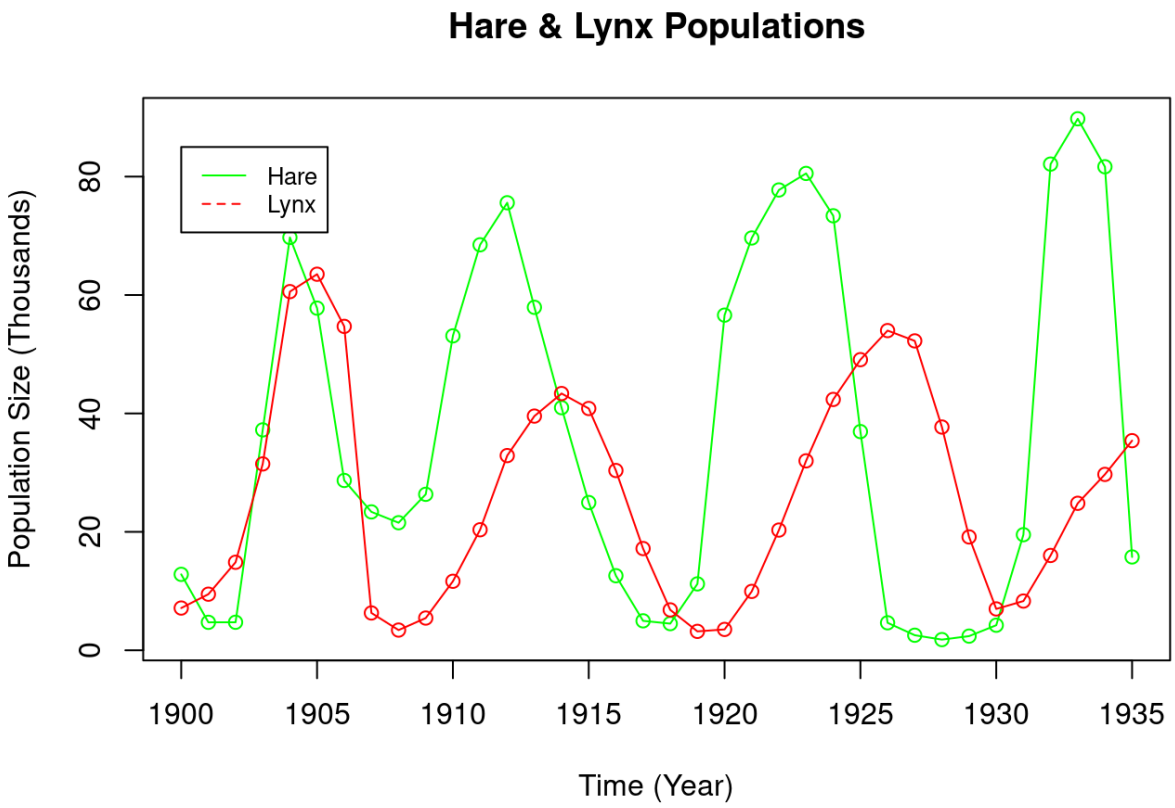
Biological factors can be defined in many different ways. From an evolutionary aspect, biological factors that contribute to population variation can include animals evolving different traits that make them more adapted to the environment and less prone to being hunted by predators compared to other species. For example, chameleons can easily change the color of their skin to match the color of their background, confusing predators and allowing them to survive longer. From a different perspective, there are also other molecular biological processes that take place in the predator–prey cycle. One significant example is an observational experiment carried out by Sheriff et al. in 2009. Sheriff et al. found that when maternal snowshoe hares were under predation risk, their endocrine system gradually secreted a type of stress hormone called glucocorticoids. Simultaneously, it was also observed that the reproductive rate of snowshoe hares decreased constantly when they were under the pressure of being hunted by predators. Hence, the researchers concluded that higher stress levels,

indicated by elevated glucocorticoid levels, were associated with lower reproduction in snowshoe hares, justifying the precipitous decline in prey population level when the predator population size surged.

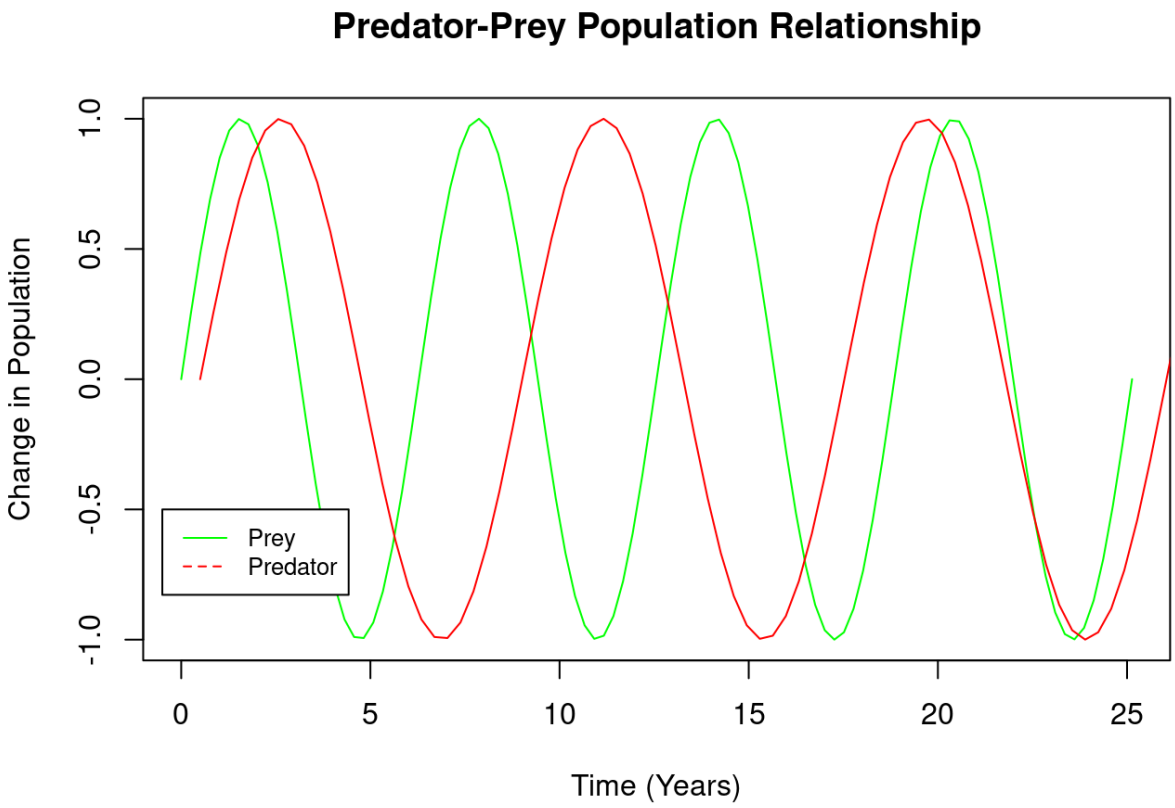
Conclusion

In conclusion, the predator–prey cycle is a complicated yet continuous process that is prevalent in all ecosystems. The population sizes of both species fluctuate due to a range of reasons, ranging from environmental factors all the way to molecular biological factors. In order for the ecosystem to sustain and function properly, predators will keep hunting for prey, and prey will keep hiding away from the predators. This process will cause more evolutionary changes to occur in the ecosystem, and further research will be conducted to analyze the intricate system of predator–prey dynamics.

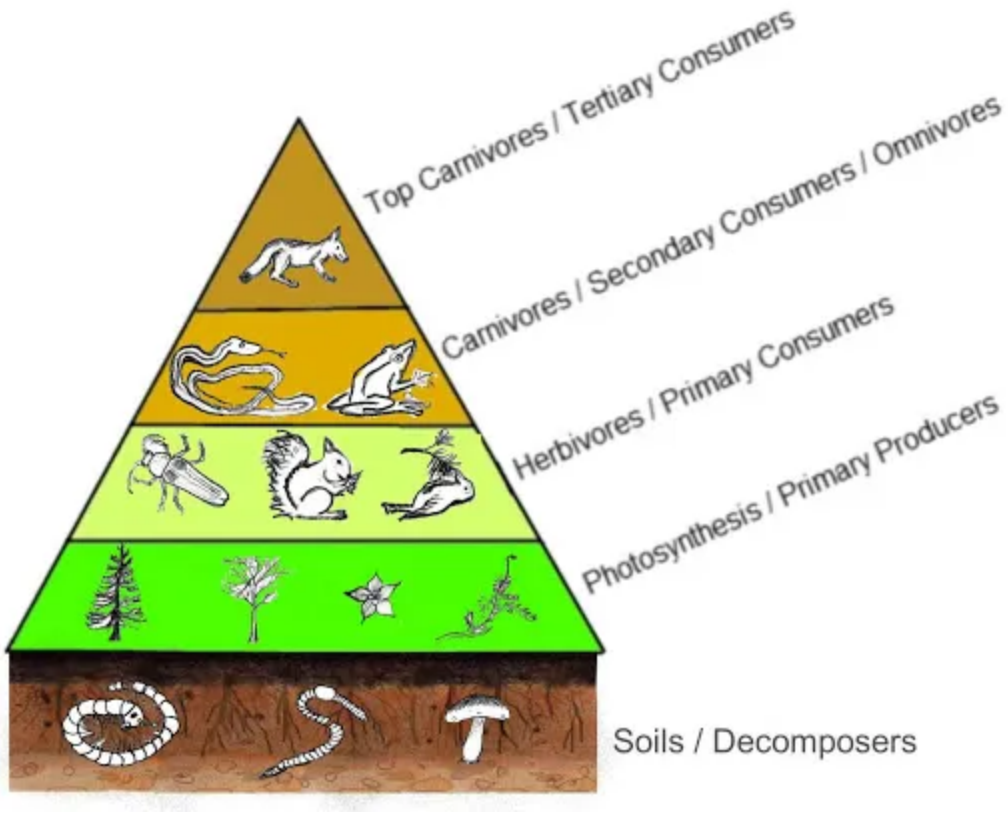
CHAPTER ADD-INS



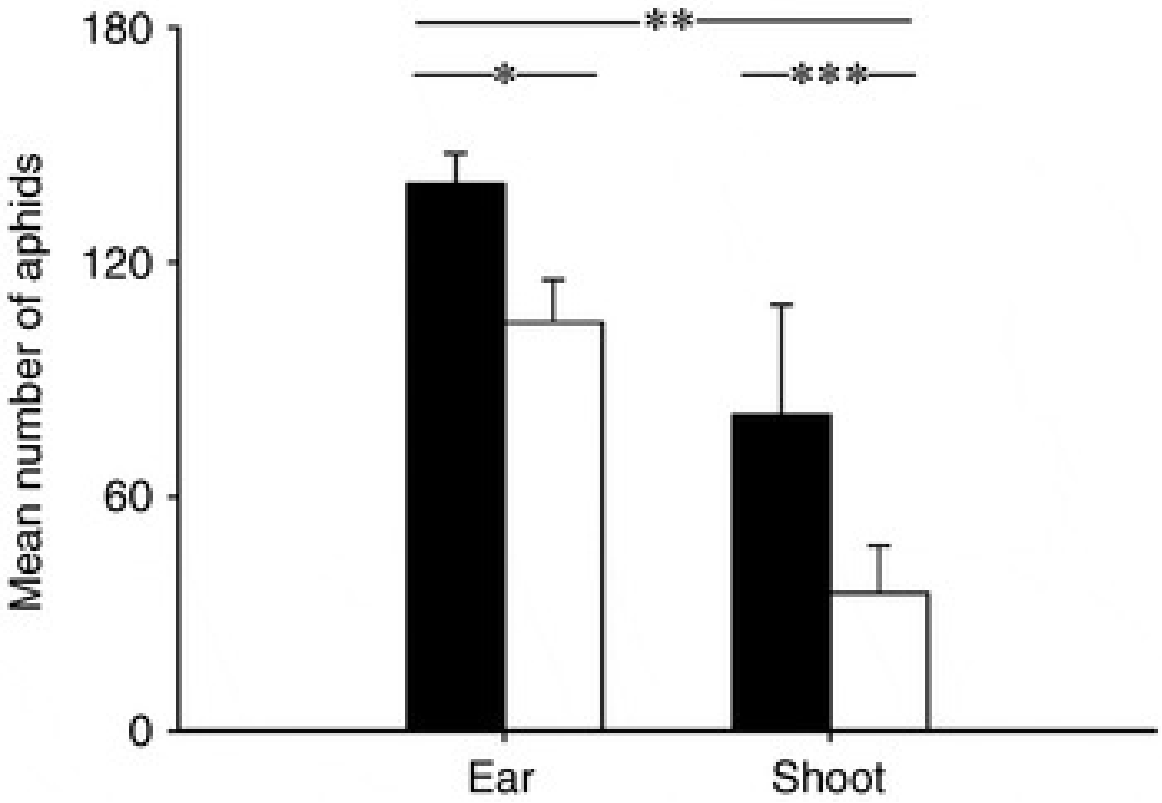
- Figure 1: Diagram displaying the changes in population between hare and lynx from 1900 to 1935. (Roditis, 2022)



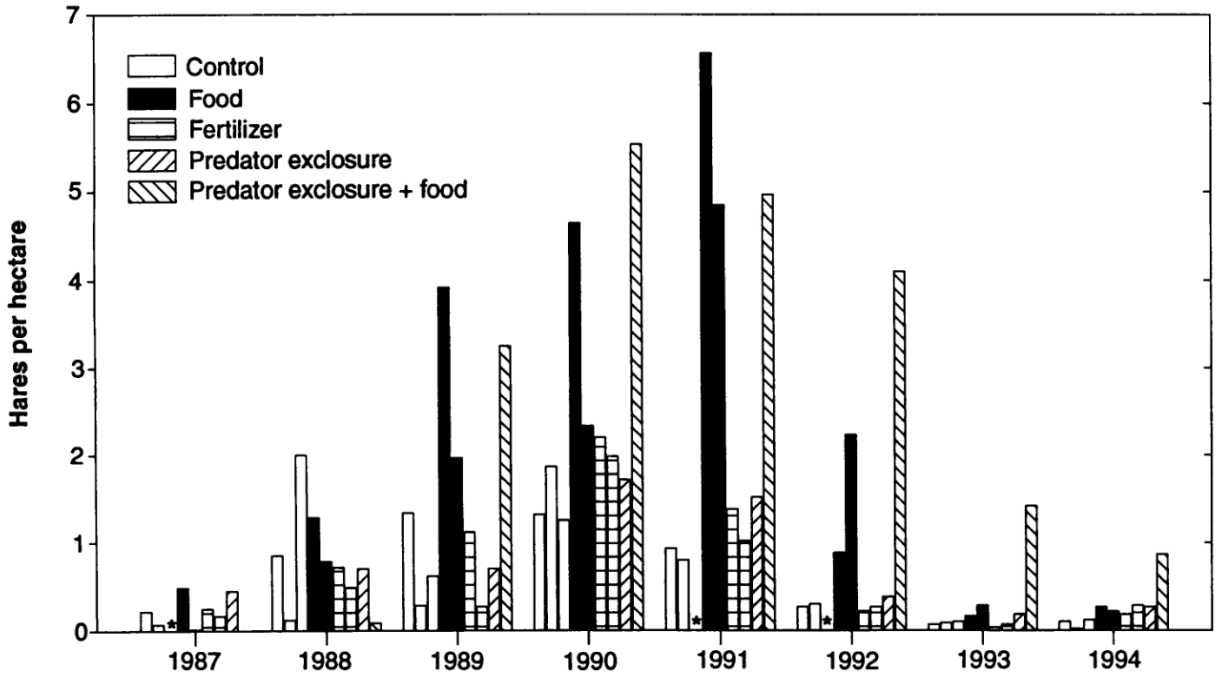
- Figure 2: Predator–prey population relationship graph (Roditis, 2022)



- Figure 3: Top-down and bottom-up control diagram (Babu, n.d.)



- Figure 4: Bar graph presenting the mean number of aphids per ear and shoot before (black bars) and after rain (white bars) (Berg et al., 2008).



- Figure 5: Graph showing that food quantity is the main factor that drives the increase in snowshoe hare population (Krebs et al., 1995)

GLOSSARY

Term	Definition
Predators	An animal that naturally hunts other animals for food.
Prey	An animal that is caught and killed by another (predator) for food.
Infradian rhythms	A biological cycle that occurs for longer than a 24-hour period.

REFERENCES

Alaska Department of Fish and Game. (n.d.). *Snowshoe Hare Species Profile, Alaska Department of Fish and Game*. Retrieved October 10, 2026, from https://www.adfg.alaska.gov/index.cfm?adfg=snowshoehare.printerfriendly

Archibald, K. M., Sosik, H. M., Moeller, H. V., & Neubert, M. G. (2023). Predator switching strength controls stability in diamond-shaped food web models. *Journal of Theoretical Biology*, *570*, 111536. https://doi.org/10.1016/j.jtbi.2023.111536

Babu, S. (2018, April 26). The top-down vs bottom-up control in an ecosystem. *Eco-Intelligent*. https://eco-intelligent.com/2018/04/26/the-top-down-vs-bottom-up-approach-in-an-ecosystem/

Berg, K. von, Traugott, M., Symondson, W. O. C., & Scheu, S. (2008). Impact of abiotic factors on predator-prey interactions: DNA-based gut content analysis in a microcosm experiment. *Bulletin of Entomological Research*, *98*(3), 257–261. https://doi.org/10.1017/S0007485308006007

BiologyOnline. (2023, September 26). Biotic factor—Definition and Examples—Biology Online Dictionary. *Biology Articles, Tutorials & Dictionary Online*. https://www.biologyonline.com/dictionary/biotic-factor

Fish & Wildlife Service. (2026, September 8). *Canada Lynx (Lynx canadensis) | U.S. Fish & Wildlife Service*. https://www.fws.gov/species/canada-lynx-lynx-canadensis

Ha, M. (2020, July 21). *6.1.1.1: Predation*. Biology LibreTexts. https://bio.libretexts.org/Bookshelves/Ecology/Environmental_Science_(Ha_and_Schleiger)/02%3A_Ecology/2.03%3A_Communities/2.3.01%3A_Biotic_Interactions/2.3.1.01%3A_Trophic_Interactions/2.3.1.1.01%3A_Predation

Kline, E. (2024, December 12). *Snowshoe Hare Adaptations: Nature's Winter Survival Experts*. BeCause Tees.

https://becausetees.com/blogs/articles/snowshoe-hare-adaptations-natures-winter-survival-experts-because-tees

Krebs, C. J., Boutin, S., Boonstra, R., Sinclair, A. R. E., Smith, J. N. M., Dale, M. R. T., Martin, K., & Turkington, R. (1995). Impact of Food and Predation on the Snowshoe Hare Cycle. *Science*, *269*(5227), 1112–1115. https://doi.org/10.1126/science.269.5227.1112

Lennox, R. J., Kambestad, M., Berhe, S., Birnie-Gauvin, K., Cooke, S. J., Dahlmo, L. S., Eldøy, S. H., Davidsen, J. G., Hanssen, E. M., Sortland, L. K., Shea, D., Nilsen, C. I., Skoglund, H., Velle, G., Hellström, G., Lundberg, P., Junge, C., Ferter, K., & Wiik Vollset, K. (2025). The role of habitat in predator–prey dynamics with applications to restoration. *Restoration Ecology*, *33*(3), e14354. https://doi.org/10.1111/rec.14354

Morin, A., Chamaillé-Jammes, S., & Valeix, M. (2021). Climate Effects on Prey Vulnerability Modify Expectations of Predator Responses to Short- and Long-Term Climate Fluctuations. *Frontiers in Ecology and Evolution*, *8*. https://doi.org/10.3389/fevo.2020.601202

National Park Service. (2021, October 28). *Snowshoe Hare (U.S. National Park Service)*. https://www.nps.gov/articles/snowshoe-hare.htm

Roditis, P. (2022, May 10). *RPubs—Ecology: A Look at Predator-Prey Relationships & Carrying Capacity*. https://rpubs.com/Proditis/900694

Sheriff, M. J., Krebs, C. J., & Boonstra, R. (2009). The sensitive hare: Sublethal effects of predator stress on reproduction in snowshoe hares. *Journal of Animal Ecology*, *78*(6), 1249–1258. https://doi.org/10.1111/j.1365-2656.2009.01552.x

Sullivan, J. (1995). *Lepus americanus, snowshoe hare | US Forest Service Research and Development* [Text]. https://doi.org/10.2737/feis-species-review-leam

Vermont Fish & Wildlife Department. (n.d.). *Snowshoe Hare | Vermont Fish & Wildlife Department*. Retrieved October 10, 2026, from

https://www.vtfishandwildlife.com/learn-more/vermont-critters/mammals/snowshoe-hare